UNIVERSIDAD POLITÉCNICA DE MADRID
Escuela Técnica Superior de Ingenieros Industriales



Convergencia de Arquitecturas Software, Robótica Colaborativa y Aprendizaje Automático en la Industria 5.0. Métodos y Aplicación

T E S I S D O C T O R A L

Presentada para optar al título de Doctor por:

Rubén Alonso Hernández
Ingeniero en Informática

Madrid, 2025





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Bajo la dirección de:
Dr. Rodolfo E. Haber Guerra
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avoa, abuela, amama, avó, abuelo, aitite

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Abstract

The digitalization of the industrial sector is currently undergoing a transformation driven by the convergence of two revolutions. On one hand, the Fourth Industrial Revolution, where short innovation cycles, interoperability, information and automation play a fundamental role. On the other hand, the emerging Fifth Industrial Revolution which places technological and operational resilience, a human-centric approach and sustainability as the strategic pillars for industrial development and the industries of the future.

If this context is combined with advances in artificial intelligence, and its potential to transform the industry, a scenario of change emerges in which many companies lack both the resources and the necessary knowledge for its implementation.

This work aims to study, from a theoretical-practical perspective, the key elements of Industry 5.0 within a narrative that integrates case studies where human and technology are interconnected. Consequently, the goal is to observe the impact of their cooperation on environmental sustainability, economic viability and workforce adaptation needs.

The thesis consists of five articles that present five case studies, ranging from the analysis of pilot line trends in relation to the challenges of the new industrial paradigm to the study for labor force adaptation to industry requirements. Additionally, it also includes various examples of methodologies and solutions applied to the industry, detailing how these can be connected to a human-centric approach and their impact on waste reduction in quality control processes.

In terms of resilience and technology, the findings reveal a trend towards including components related to human factors, sustainability, artificial intelligence, automation and robotics in pilot lines. Furthermore, their use is highlighted as a training mechanism, a test-before-invest facilitator and as a method for analyzing sustainability indicators. Additionally, an hybrid risk-goal analysis methodology is presented that enables the definition of interoperable software platforms for these pilot lines.

Regarding the human factor, it is demonstrated that it is possible to develop capacity building and training recommendation systems based on occupational databases to detect skill gaps and training needs, allowing workers and companies to adapt to new industrial activities.

From a broader perspective, the quality control studies are presented as an example where technology, under operator supervision, enables the generation of synthetically generated images of defective components to train artificial intelligence models. This approach eliminates the need for manually manufacturing these parts, thus reducing waste. Additionally, it is demonstrated that it is more effective to train classification models by increasing the data volume with synthetic parts rather than using only manually made defective parts.

As a complement to these case studies, an analysis of strengths and limitations is presented, leading to the conclusion that Industry 5.0 should be understood as an intelligent industry, based on innovative technologies and machines, supported by skilled and resilient workers and driven by objectives of competitiveness, sustainability, scalability and quality.

Resumen

La digitalización del sector industrial se encuentra en una etapa de transformación caracterizada por la convergencia de dos revoluciones. Por un lado, la denominada Cuarta Revolución Industrial, en la que los tiempos cortos de innovación, la interoperabilidad, la información y la automatización desempeñan un papel fundamental. Por otro lado, la emergente Quinta Revolución Industrial, que pone énfasis en la resiliencia tecnológica y operativa, la centralidad en el ser humano y la sostenibilidad como ejes estratégicos del desarrollo industrial, y de la industria del futuro.

Si a este contexto se suman los avances en inteligencia artificial, y su potencial para transformar la industria, se obtiene un escenario de cambio en el que muchas empresas no disponen ni de los recursos adecuados ni del conocimiento necesario para su implementación.

Esta tesis surge con el propósito de estudiar, desde un enfoque teórico-práctico, los elementos clave de la Industria 5.0. Para ello, parte de una narrativa que integra casos de estudio donde se interrelacionan el ser humano y la tecnología. De este modo, se pretende observar el impacto de su cooperación en la sostenibilidad ambiental, económica y en cuanto necesidad de adaptación laboral.

La tesis se compone de cinco artículos que presentan cinco casos de estudio, abarcando desde el análisis de las tendencias de las líneas piloto en función de los desafíos del nuevo paradigma industrial hasta el estudio de técnicas de capacitación de la fuerza laboral para adaptarse a los nuevos requisitos de la industria. También se incluyen diversos ejemplos de metodologías y soluciones tecnológicas aplicadas a la industria, detallando cómo estas pueden relacionarse con la centralidad en el ser humano y su impacto en la reducción de residuos en los procesos de control de calidad.

En lo que respecta a la resiliencia y la tecnología, los hallazgos revelan una tendencia a incluir componentes relacionados con el ser humano, la sostenibilidad, la inteligencia artificial, la automatización y la robótica en las líneas piloto. Además, se destaca su uso como herramienta formativa, facilitadora de la validación previa a la inversión y como método de análisis de indicadores de sostenibilidad y ciclo de vida. Asimismo, se presenta una metodología de análisis de riesgos que permite definir plataformas de software interoperables sobre estas líneas piloto.

En cuanto al factor humano, se demuestra que es posible desarrollar sistemas basados en bases de datos ocupacionales para detectar necesidades de capacitación, permitiendo a trabajadores y empresas adaptarse a nuevas actividades industriales.

Desde una perspectiva más amplia, los casos de estudio relacionados con el control de calidad se presentan como un ejemplo donde la tecnología, con supervisión del operador, permite generar sintéticamente imágenes de piezas defectuosas para entrenar modelos de inteligencia artificial, evitando la fabricación manual de estas piezas y, por tanto, reduciendo residuos. A su vez, se demuestra que es más eficaz entrenar modelos de clasificación incrementando el volumen de datos con piezas sintéticas que utilizando exclusivamente piezas defectuosas fabricadas a mano.

Como complemento a estos casos de estudio, se presenta un análisis de fortalezas y limitaciones, que lleva a la conclusión de que la Industria 5.0 debe ser entendida como una industria inteligente, fundamentada en tecnologías y máquinas innovadoras, respaldada por trabajadores capacitados y resilientes, y apoyada por objetivos de competitividad, sostenibilidad, escalabilidad y calidad.

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Abreviaturas y acrónimos

3D Tres Dimensiones

AMQP Advanced Message Queuing Protocol

BERT Bidirectional Encoder Representations from Transformers (Representación de Codificador Bidireccional de Transformadores)

CPU Central Processing Unit (Unidad Central de Procesamiento)

CSIC Consejo Superior de Investigaciones Científicas

DBL Digital Building Logbook (Libro de Edificio Digital)

DENIM Digital intelligence for collaborative ENergy management in Manufacturing

DOI Digital Object Identifier (Identificador de Objeto Digital)

ESCO European Skills, Competences, Qualifications and Occupations (Habilidades, Competencias, Cualificaciones y Ocupaciones Europeas)

GAMHE Group of advanced Automation of Machines, Highly complex processes and Environments

GPT Generative Pre-training Transformer (Transformador Generativo Pre-entrenado)

IA Inteligencia Artificial

IEEE Institute of Electrical and Electronics Engineers (Instituto de Ingenieros Eléctricos y Electrónicos)

IFAC International Federation of Automatic Control (Federación Internacional de Automatización)

IoT Internet of Things (Internet de las Cosas)

ISCO International Standard Classification of Occupations (Clasificación Internacional Uniforme de Ocupaciones)

IT Information Technology (Tecnologías de la Información)

KITT4SME platform-enabled KITs of arTificial intelligence FOR an easy uptake by SMEs

MQTT Message Queuing Telemetry Transport

NGSI Next Generation Service Interface

O*NET Occupational Information Network

PYME Pequeña y Mediana Empresa

RGPD Reglamento General de Protección de Datos

SARA Software Architecture Review and Assessment

STAR Safe and Trusted Human Centric Artificial Intelligence in Future Manufacturing Lines

TIC Tecnologías de la Información y la Comunicación

UPM Universidad Politécnica de Madrid

VICA Volatilidad, Incertidumbre, Complejidad y Ambigüedad

YOLO You Only Look Once

ZDM Zero Defect Manufacturing (Fabricación Cero Defectos)

Capítulo 1

Introducción

La Industria 4.0 se define como la cuarta revolución industrial (Lasi et al., 2014), una revolución basada en tiempos cortos de innovación y desarrollo, más flexibilidad y eficiencia de recursos. Impulsada principalmente por la automatización, la digitalización y la interacción entre componentes, tiene como objetivo aumentar la competitividad de las empresas industriales en general, y con ellas la de toda la industria manufacturera (Ferrer et al., 2018; Mohammed et al., 2022).

La idea de la Industria 4.0 va de la mano de soluciones y tecnologías, las más importantes de las cuales están asociadas a la interoperabilidad, la unificación de los flujos de trabajo digitales con los reales (sistemas ciberfísicos), las capacidades en tiempo real y el análisis de datos para la toma de decisiones (Mohamed, 2018).

Sí la Industria 4.0 se caracteriza por tiempos de producción más cortos, mayores expectativas de calidad y un deseo de integrarse con proveedores y clientes, la nueva Industria 5.0 hereda estos puntos y pone el foco en la sostenibilidad y, sobre todo, sitúa al trabajador como un elemento clave (Xu et al., 2021).

Las tecnologías disruptivas, la interconectividad, la robótica y la interacción humano-máquina han evolucionado hacia la robótica colaborativa, los humanos en el bucle y una mayor observabilidad y explicabilidad. Esta nueva revolución ha llegado en un momento de gran avance en temas relacionados con la inteligencia artificial, que está contribuyendo a diversos procesos y operaciones de la industria, al tiempo que pone de manifiesto una mayor necesidad de adaptación a las nuevas innovaciones tecnológicas.

El estudio de KPMG *Thriving in an AI World* (Krishna et al., 2021) ya reflejó esta tendencia en 2021 en diferentes industrias. Este estudio menciona, entre otros, el deseo mayoritario de los líderes empresariales de adoptar tecnologías de inteligencia artificial, y cómo de estos mismos líderes, el 73 % de los entrevistados detallan cómo el COVID-19 ha tenido un impacto en la aceleración de la adopción de la inteligencia artificial en su empresa.

Todas estas tecnologías requieren plataformas y soluciones de software que faciliten su implementación y, de alguna manera, simplifiquen la complejidad de los requisitos funcionales y no funcionales asociados a los nuevos sistemas de las industrias del futuro.

Estos factores y tecnologías forman parte del conjunto de herramientas utilizadas para abordar los múltiples desafíos a los que se enfrenta la industria. En Kumar et al., 2020 se definen quince retos principales para la Industria 4.0. Centrándonos en los principales retos técnicos, es posible identificar preocupaciones sobre el elevado coste inicial (Reto 3), la falta de recursos para investigación y desarrollo en tecnologías de la Industria 4.0 (Reto 6), la falta de infraestructuras basadas en tecnologías de la información, tanto de software como de hardware (Reto 9), la falta de conocimiento y mano de obra formada (Reto 10), o el miedo al fracaso de las tecnologías de la Industria 4.0 (Reto 13).

Estos desafíos son particularmente significativos en el ámbito de las pequeñas y medianas empresas (PYMEs) (Gladysz et al., 2023), la columna vertebral de muchas industrias globales, cuyo tamaño y capacidad a menudo las hacen menos capaces de hacer frente a los desafíos, y, en las cuales el impacto puede ser crítico si fracasan. A su vez, esta situación se ha vuelto más compleja cuando la inteligencia artificial es una variable de la ecuación. Por un lado, promete mejores tiempos, precisión y resultados, así como mejores decisiones, pero al mismo tiempo complica aún más los desafíos anteriores (Sana et al., 2023).

De hecho, no todos estos avances están teniendo un impacto positivo en la industria. En múltiples casos se está notando que las empresas, especialmente las pequeñas y medianas, no pueden seguir el ritmo de los avances en inteligencia artificial y, sobre todo, que estos avances requieren conocimientos que sus empleados no necesariamente tienen.

Esta tendencia se puede observar en estudios realizados por consultoras globales. Por ejemplo, Deloitte informa, en su estudio sobre la inteligencia artificial en las empresas manufactureras en China, que la adopción de la inteligencia artificial no ha cumplido con las expectativas (en más del 90 % de los casos). Las razones son, entre otras, los obstáculos y limitaciones organizacionales y de infraestructura, la falta de experiencia de la fuerza laboral, los problemas con la recopilación de datos y la complejidad involucrada en la implementación de soluciones de inteligencia artificial (Deloitte, 2020). Por supuesto, todo esto está alineado con los retos mencionados anteriormente.

Estas barreras son similares en otras economías del mundo. En Estados Unidos, como se detalla en el Índice Global de Adopción de Inteligencia Artificial de IBM 2022 (IBM, 2022), la complejidad, la dificultad de integración, los desafíos relacionados con los datos y la falta de habilidades se citan de manera similar. Además, una cuarta parte de los encuestados cita la falta de plataformas como una de las barreras. Europa sigue una tendencia similar, como se detalla en el informe de Eurostat "Uso de la inteligencia artificial en las empresas" (Eurostat, 2022). Un punto interesante de este informe se refiere a dónde, de qué fuentes, estas pequeñas y medianas empresas adquieren tecnologías de inteligencia artificial. Si bien predominan las soluciones de aprendizaje automático o profundo comerciales "listas para usar", hay un gran número de soluciones que se crean internamente, o son soluciones comerciales o de código abierto que requieren expertos internos para modificarlas y adaptarlas a las condiciones específicas del caso de uso de la PYME.

Esta situación se agrava todavía más cuando entran en juego diferentes requisitos no-funcionales y sobre todo las necesidades de integrar distintas soluciones software. De hecho, los autores de Khan et al., 2023 citan la privacidad y la seguridad de los datos entre las cuestiones relevantes

en las que las PYMEs deben emprender esfuerzos de digitalización. También mencionan el concepto de interoperabilidad en las plataformas como un aspecto crítico y desafiante que debe abordarse.

En Hansen y Bøgh, 2021, por ejemplo, después de analizar el estado y aceptación de la inteligencia artificial y el Internet de las Cosas (IoT) en las PYMEs, los autores remarcan la complejidad del uso de la inteligencia artificial, dado que "las PYMEs carecen de los conocimientos y recursos para utilizar la tecnología, a pesar de que en apariencia es beneficiosa para ellas", y hacen una recomendación clara, totalmente coherente con los objetivos de esta tesis: "la investigación futura debe centrarse en simplificar las soluciones basadas en inteligencia artificial para las PYMEs y, por lo tanto, hacerlas más directamente aplicables a ellas". Los autores de Bunte et al., 2021 también están de acuerdo con la afirmación anterior y sugieren que el uso de la inteligencia artificial debe simplificarse, a la vez que se deben proporcionar formas de medir los beneficios para que las nuevas soluciones digitales basadas en inteligencia artificial tengan una base sólida para respaldar las decisiones.

Al haber mencionado el Internet de las Cosas, es de relevancia comentar que poco a poco se está viendo más aceptación y despliegue de estas tecnologías. Quizás, junto con el uso de la computación en la nube, este es el primer paso que las PYMEs están dando hacia la Industria 4.0 (Alexopoulos et al., 2022). Frameworks como FIWARE están facilitando esta adopción (Cirillo et al., 2019). Además de utilizar estándares para la representación digital de objetos del mundo real y la gestión de información contextual (NGSI), el middleware FIWARE facilita la interoperabilidad de componentes al admitir múltiples agentes y protocolos de IoT para redes y mensajería, como MQTT o AMQP (Araujo et al., 2019).

Por su parte, las plataformas de orquestación de contenedores permiten la composición dinámica de componentes y operaciones entre ellos. Esto facilita la gestión de aplicaciones complejas y permite superar los desafíos y obstáculos mencionados anteriormente (Liu et al., 2022). Estas plataformas sirven también para agilizar el desarrollo y la integración, y permiten proporcionar soluciones para abordar requisitos complejos de seguridad, escalabilidad o interoperabilidad. En el caso de los sistemas basados en componentes, confiar en plataformas maduras que admitan la orquestación de módulos facilitan a las PYMEs, e incluso a los proveedores de inteligencia artificial, centrarse en el diseño y desarrollo de estos componentes. Esto les permite tener la certeza de que la plataforma digital se basa en tecnologías conocidas, al tiempo que saben que su solución puede desplegarse en plataformas con características similares.

La necesidad de adaptación y despliegue de aplicaciones innovadoras, junto con el conocimiento de las plataformas donde van a funcionar estos, así como una cierta explicabilidad para entender el rendimiento de los modelos de inteligencia artificial, están impactando en las plantillas de las PYMEs y en cómo cooperan los sistemas con los trabajadores.

Este es uno de los puntos clave de la Industria 5.0 a través de la integración de las características humanas en el bucle. Es decir, al situar al ser humano como un elemento clave (*human-centric*), se proporciona información y autonomía. Esta centralidad humana es una estrategia que requiere cooperación, confianza y retroalimentación/comunicación fiable entre las entidades implicadas, y apoya la idea de una transformación en la que el trabajador no solo no es

sustituido sino empoderado (Cimino et al., 2023).

El concepto de Industria 5.0 fue formalizado por la Comisión Europea en el año 2021 (Breque et al., 2021), si bien ya existía actividad en foros y redes sociales (Rada, 2017) que discutía sobre el concepto de una quinta revolución industrial, partiendo de una mayor cooperación entre humanos y máquinas y un enfoque de reducción de residuos. La centralidad en el ser humano, junto con la resiliencia y la sostenibilidad son los pilares de esta Industria 5.0, y de la transición verde y digital impulsada por la Comisión Europea.

La idea de sostenibilidad no solo se basa en la protección del medio ambiente, sino que también tiene como objetivo reducir los residuos y desarrollar procesos circulares, lo que a su vez tendrá un impacto positivo en los costes de producción. Para ello, esta idea se beneficia del uso de nuevas tecnologías para facilitar funcionalidades, ajustar procesos y adaptar ciclos de producción. Esta sostenibilidad también vendrá de aprovechar y escalar las plataformas de software que puedan orquestar soluciones que integren la inteligencia artificial y proporcionen al operador toda la trazabilidad y observabilidad necesarias para tomar decisiones informadas y cooperar con las soluciones técnicas.

En particular, el control de calidad es uno de los procesos que puede tener un impacto directo en la sostenibilidad. No solo a nivel de proceso, haciéndolo más rápido y eficiente, sino también a nivel de resultados, evitando la generación de residuos y reduciendo el consumo de energía en ciclos no productivos (Castaño et al., 2020). Para ello, enfoques como la fabricación cero defectos (ZDM) son clave para lograr la máxima calidad y el menor desperdicio, evitando imperfecciones y fallos. Las plataformas compatibles con ZDM proporcionan funciones de detección y reparación a través de medidas preventivas y predictivas. La irrupción de la inteligencia artificial también está colaborando en estas tareas, con la creciente disponibilidad de modelos predictivos o de detección de defectos.

A menudo, para desarrollar módulos de inteligencia artificial o para configurarlos adecuadamente para la detección de defectos y fallos, es necesario realizar ciclos y ciclos de generación de datos, lo cual impacta no solo en tiempos, sino también en el consumo energético y la generación de residuos. La falta de datos es especialmente significativa en casos de inspecciones basadas en visión artificial donde el entrenamiento de los modelos requiere, en muchos casos, tener imágenes de piezas correctas y defectuosas, y, por lo tanto, se requiere tiempo y múltiples ciclos de clasificación.

Estos casos se pueden abordar mediante técnicas de aumento de datos y generación de imágenes sintéticas. A través de diversas técnicas, desde el procesamiento de imágenes hasta el uso de modelos de inteligencia artificial, se pueden generar grandes volúmenes de datos de calidad similar a las imágenes originales utilizando solo una pequeña colección de imágenes como fuente. Esto reduce tiempo y costes, y también facilita otras tareas como probar diferentes modelos de inteligencia artificial y comparar resultados.

Todos estos elementos y áreas de investigación mencionados anteriormente, en especial cuando se combinan todos ellos, requieren de validación en diferentes entornos de laboratorio, para ser posteriormente validados operativamente. Las líneas piloto son una herramienta muy interesante para este propósito. Parte de la investigación realizada en esta tesis ha sido validada sobre la línea piloto GAMHE 5.0. GAMHE 5.0 es la línea piloto de la iniciativa

"Industria y Sociedad 5.0" que tiene como objetivo desarrollar estudios y soluciones desde lo social y económico hasta la optimización de fábricas de alto nivel.

En resumen, para conseguir desarrollos compatibles con esta nueva Industria 5.0, se ha detectado la necesidad de entender el ecosistema Humano-Maquina-Sostenibilidad y proporcionar entornos colaborativos que permitan a las empresas industriales y a la fuerza laboral, adaptarse a entornos rápidamente cambiantes y desplegar soluciones innovadoras de forma sencilla. También se ha observado la necesidad de proporcionar soluciones flexibles, por ejemplo, para generar datos que puedan ser utilizados para entrenar modelos o incluso para evaluarlos. Ambas brechas son especialmente significativas en casos como la inspección de calidad de los paneles termoaislantes, o en la inspección de piezas de automóvil, con el objetivo de que la producción de estas piezas pueda alcanzar un punto de cero defectos.

1.1 Motivación y marco de trabajo

Para detallar la motivación inicial de la tesis es preciso partir de una serie de puntos o premisas que sitúen el estado actual de la materia.

Se parte de los siguientes puntos:

- La industria todavía no ha asimilado las características de la Industria 4.0 y ve como ya se sugiere un nuevo modelo.
- Existe una diferencia notable en cuanto al nivel de digitalización de la industria, y sobre todo de las capacidades de ésta y de sus trabajadores para adaptarse.
- Se observa una complejidad añadida debido a las múltiples herramientas y soluciones disponibles, a la vez que cambian los escenarios y surgen nuevas posibilidades, pero también nuevos problemas.
- Los sistemas industriales se pueden beneficiar de líneas piloto y gemelos digitales, que faciliten las pruebas antes de invertir y la toma de decisiones.
- La inteligencia artificial y el aprendizaje automático avanzan rápidamente ofreciendo posibilidades que no se esperaban a corto plazo. A su vez surgen una serie de nuevos proveedores con ofertas no siempre maduras y probadas.
- Las políticas, especialmente en la Unión Europea, fomentan la competitividad a la vez que la sostenibilidad, el uso ético de la inteligencia artificial, y la importancia del elemento humano.
- Paradigmas como "Manufactura de cero defectos" o "Centricidad Humana", modelos de arquitecturas para la industria, o tendencias como el *Green Computing* y *Green IT*, enriquecen y pueblan el ecosistema de conceptos industriales.

De esa situación surge la motivación de esta tesis, que toma como hilo conductor la innovación en distintos conceptos software y hardware, para elaborar una narrativa que integra los principales elementos de la Industria 5.0: la interrelación entre el humano, la resiliencia y tecnología y la sostenibilidad.

En concreto la motivación surge del interés en las industrias del futuro, cuál es el estado de éstas y cómo se adaptan a las nuevas tecnologías. Otro punto importante de la motivación es el interés en enfoques mixtos teórico-prácticos. Especialmente con relación a cómo estas premisas anteriores permiten tanto analizar el estado del arte y diseñar metodologías o arquitecturas, como ser objeto del desarrollo de pruebas de concepto y de validación de resultados.

El entorno del centro de automática y robótica de la UPM-CSIC[1], la competencia de sus equipos de investigación, la variedad de sus actividades, junto a la línea piloto GAMHE 5.0, y la colaboración en proyectos de I+D Europeos, proporcionaba un ecosistema idóneo para este trabajo, que como se ha mencionado anteriormente cubre diferentes aspectos tecnológicos siempre desde la narrativa de la Industria 5.0.

1.2 Objetivos

Según recoge el profesor H.T Kung en Kung, 1987, Allen Newell, reconocido profesor e investigador de Carnegie Mellon, clasifica las tesis en diferentes tipos según lo que aportan. Estos tipos abarcan desde la apertura de una nueva área de investigación hasta el desarrollo de una nueva herramienta, pasando por la presentación de datos empíricos o la resolución de problemas o cuestiones que llevan tiempo siendo discutidas entre la comunidad científica.

En lo que a los objetivos se refiere, esta tesis se identifica principalmente con dos de los que Kung y Newell mencionan. Concretamente con la "exploración de un área de investigación" y con la "validación experimental".

En concreto, el **objetivo principal** de esta tesis es **estudiar, desde una perspectiva teórico-práctica, los elementos clave del paradigma Industria 5.0, mediante una narrativa que integra casos de estudio donde se interconectan el factor humano, la tecnología y la sostenibilidad**.

Este objetivo se articula mediante la investigación y diseño, desarrollo y validación de métodos o estrategias que van desde el diseño de arquitecturas o el impacto en la seguridad y privacidad, hasta técnicas concretas de cero defectos en entornos robóticos colaborativos, reducción de residuos o capacitación de los trabajadores.

Partiendo del objetivo general mencionado anteriormente, existe una serie de objetivos más específicos que justifican la realización de esta tesis y están relacionados con proveer conocimiento, métodos y técnicas para poder contestar a una serie de preguntas que pueden surgir en el momento que se evalúe el estado de la industria y el impacto "de" y "sobre" la

[1]GAMHE 5.0 team and infrastructure has been involved in different activities of this thesis. Their work was partially supported by the project "Digitalization of Power Electronic Applications within Key Technology Value Chains (PowerizeD)", ID 101096387, funded by the European Union HORIZON FP and Chips JU, and PCI2022-135004-2 funded by MICIU/AEI/10.13039/501100011033 by the "European Union NextGenerationEU/PRTR". Their activities are also partially supported by the project "Self-reconfiguration for Industrial Cyber–Physical Systems based on Digital Twins and artificial intelligence. Methods and application in Industry 4.0 pilot line", Spain, grant ID PID2021-127763OB-100, and supported by MICIU and NextGenerationEU/PRTR. Their work was also partially supported by the European Commission through the Horizon Europe project "Multi-Modal and Multi-Aspect Holistic Human-Robot Interaction (FORTIS)", grant ID 101135707.

tecnología, el humano y el entorno. Un ejemplo de estas preguntas son las siguientes:

- ¿Cómo se puede simplificar la composición de soluciones software para la industria?
- ¿Cómo puedo evaluar las implicaciones de privacidad y seguridad de las plataformas industriales?
- ¿Son las líneas piloto útiles para abordar algunos de los retos de la nueva industria?
- ¿Qué pasa si mi PYME no tiene capacidad para seguir una metodología compleja de seguridad y privacidad?
- ¿Qué implicaciones tiene la Industria 5.0 para una PYME manufacturera?
- ¿Cómo se sitúan las ideas de cero defectos, centricidad humana, reducción de residuos, junto con la inteligencia artificial, la automática y robótica y la IoT en entornos industriales?
- ¿Son verdaderamente útiles las técnicas de aumento de datos en los contextos de control de calidad y cero defectos asistidos por aprendizaje profundo?
- ¿Tienen aplicabilidad los enfoques de aumento de datos en contextos de robótica colaborativa?
- ¿Cómo impacta el control de calidad en la generación de residuos?
- ¿Cómo se puede capacitar al humano en una industria rápidamente cambiante?

Por lo tanto, los **objetivos específicos** se podrían resumir en los siguientes:

- Analizar el estado actual de las líneas piloto y evaluar cómo se alinean con los principios de la Industria 5.0, en términos de resiliencia, centricidad humana y sostenibilidad. Estudiar el papel de las líneas piloto en la validación de procesos y reducción de incertidumbre. Presentar la línea piloto GAMHE 5.0 como caso de estudio, mostrando sus enfoque centrado en la automatización y la robótica colaborativa, y discutir las ventajas y desventajas de las líneas piloto a pequeña escala en el contexto de las industrias del futuro.
  - Método principal: Análisis del estado del arte de las líneas de automática y robótica (Sección 2.2.1).
  - Artículo: Automation and Robotics Pilot Lines in the Context of Industry 5.0 (Sección 3.1).
- Explorar métodos para facilitar la adopción de tecnologías innovadoras, como el aprendizaje automático, simplificando plataformas de software complejas mediante la orquestación de componentes. Desarrollar y validar un marco práctico, basado en riesgos y objetivos, para analizar e impulsar la seguridad y la privacidad. Demostrar el proceso seguido en el proyecto industrial KITT4SME para definir las características de seguridad clave de la plataforma modular desarrollada en el proyecto.
  - Método principal: Adaptación de la taxonomía de riesgos (Sección 2.2.4).

  - Artículo: Interoperable Software Platforms for Introducing Artificial Intelligence Components in Manufacturing: A Meta-framework for Security and Privacy (Sección 3.2).

- Proponer un método para identificar automáticamente las habilidades y capacidades en currículos y relacionarlas con descripciones de empleo o de actividades laborales, utilizando bases de datos ocupacionales y técnicas de aprendizaje profundo para medir la similitud semántica. Mejorar el emparejamiento entre ofertas laborales y candidatos, o entre trabajadores y actividades, y proporcionar recomendaciones tanto para empleadores como para empleados o personas en busca de trabajo. Evaluar dos escenarios: i) dado un currículo, identificar las ocupaciones de O*NET que mejor coincidan con el perfil del candidato; ii) dado el currículo de un candidato y un conjunto de descripciones de puestos o actividades, identificar cuál de los puestos es el más adecuado para el candidato.
  - Método principal: Transformadores y similitud semántica entre entidades O*NET y texto (Sección 2.2.2).
  - Artículo: A Novel Approach for Job Matching and Skill Recommendation Using Transformers and the O*NET Database (Sección 3.3).
- Estudiar cómo se puede mitigar la falta de datos de piezas defectuosas mediante la generación artificial de imágenes para entrenar modelos de clasificación. Demostrar cómo el uso de plataformas de software interoperables, junto con técnicas de aumento de datos, segmentación y clasificación, mejoran el control de calidad industrial. Validar, a través de casos en el sector automotriz y en la inspección de paneles termoaislantes, cómo este enfoque contribuye a una detección temprana de defectos, mejora los tiempos de monitorización y control y reduce la cantidad de piezas defectuosas, alineándose con los principios de calidad y sostenibilidad de la Industria 5.0.
  - Método principal: Aumento de datos y generación de imágenes sintéticas (Sección 2.2.3).
  - Artículos: Early Defect Detection in Thermo-Insulating Panels: A Case Study on Quality Inspection (Sección 3.5) y Extending Asset Lifespan Through Data Augmentation-Assisted Quality Control (Sección 3.4).

Estos objetivos específicos permiten a su vez:

- Descubrir y estudiar la situación actual. En concreto, y como se ha mencionado, en el sector industrial y especialmente en cuanto a pequeñas y medianas empresas se refiere.
- Identificar los principales problemas, lagunas y deficiencias. Entendiendo, las implicaciones de las nuevas tecnologías y los nuevos paradigmas y las dificultades en su adopción.
- Entender los elementos que componen el estado actual del área. En particular, analizando los elementos clave de la industria 5.0. Siendo estos el humano, la resiliencia tecnológica y la sostenibilidad en general, y la reducción de consumo y generación de residuos en particular.

1.3 Adelanto de la hipótesis

Al tratarse de una tesis que engloba artículos de carácter teórico, centrados en el análisis del estado del arte, y artículos más prácticos, con investigaciones empíricas, se ha optado por una tesis de hipótesis múltiples.

Estas hipótesis convergen en la idea global de que el avance hacia una industria 5.0 depende significativamente de la comprensión del contexto de colaboración entre tecnología y humano, y del análisis del impacto de esta cooperación en la sostenibilidad ambiental, económica y el bienestar laboral.

Las hipótesis son las siguientes:

1. **Las líneas piloto mejoran significativamente la transición de la investigación a la producción a escala industrial y facilitan una industria más adaptable, centrada en el ser humano y sostenible.**

   Esta hipótesis parte de la visión de que las líneas piloto de robótica colaborativa son herramientas esenciales para la experimentación y para acelerar la innovación. Además, en un contexto tecnológico rápidamente cambiante, permiten adaptarse a una transición de una Industria 4.0, basada en las tecnologías avanzadas, a una Industria 5.0, en la que estas tecnologías avanzadas se potencian gracias a la centricidad humana y fabricación sostenible.

2. **Las pequeñas y medianas empresas manufactureras pueden acelerar su transformación digital adoptando arquitecturas de software basadas en componentes y entendiendo de manera temprana los posibles riesgos de seguridad y privacidad.**

   Esta segunda hipótesis nace de la idea de que las PYMEs industriales tienen dificultades para adaptarse a los sistemas complejos, a la nueva oferta de inteligencia artificial y a la interoperabilidad necesaria a la hora de integrar distintos componentes. Un enfoque basado en plataformas modulares, que enmascaren las complejidades, junto con marcos para identificar, evaluar y mitigar riesgos de seguridad y privacidad, puede facilitar esta adaptación.

3. **Integrar datos estructurados de ocupaciones laborales con modelos basados en aprendizaje profundo para extraer habilidades semánticamente relevantes de textos no estructurados permite el descubrimiento de necesidades formativas y la alineación de las capacidades de los trabajadores con las actividades laborales.**

   Por su parte, esta hipótesis se centra en la necesidad de estrategias adecuadas para que el trabajador pueda adaptarse a los requisitos del nuevo paradigma industrial. A su vez, estas estrategias deben facilitar que las empresas puedan analizar la información proporcionada por el trabajador para detectar los perfiles más adecuados para una tarea.

4. **Las imágenes sintéticas podrían utilizarse como reemplazo o complemento en entornos donde no hay suficientes piezas defectuosas para entrenar una**

inteligencia artificial, reduciendo, a su vez, tiempos, materiales y residuos derivados de fabricar las piezas físicas defectuosas.

En entornos donde se requiere entrenar modelos de clasificación con datos balanceados, y donde generar los datos de entrenamiento implica esfuerzos añadidos y genera residuos, disponer de estrategias de generación de imágenes sintéticas puede facilitar el entrenamiento de los modelos sin la necesidad de las piezas físicas.

1.4 Idea inicial para la validación de la hipótesis

La validación de la hipótesis se realiza mediante varios artículos publicados que presentan casos de estudio prácticos y el análisis de sus resultados. En ellos se revisa el estado del arte, se contextualiza el ámbito relacionado (tecnología, humano u otros aspectos clave) y se presenta el desarrollo y pruebas realizadas. Por último, se detallan los resultados, que posteriormente se analizan y discuten.

En concreto, esos artículos, se centran en los temas siguientes:

- Estudio de las líneas piloto, en especial aquellas basadas en robótica colaborativa, como elemento clave para "evaluar-antes-de-invertir", como elemento formativo y de mejora de capacidades de los trabajadores, y como medio de conceptualización de medidas sostenibles y de reducción de residuos. (Ámbito: tecnología, humano, sostenibilidad)
- Análisis de la problemática de las PYMEs al adaptarse a las nuevas tecnologías software y al desarrollo basado en inteligencia artificial, presentando un diseño e implementación de una arquitectura genérica para los sistemas digitales industriales, e incluyendo un análisis de privacidad y seguridad en plataformas interoperables, adaptado a las características de estas empresas. (Ámbito: tecnología)
- Formación y desarrollo de capacidades del humano en industria 5.0. Estudio de las bases de datos ocupacionales y cómo las nuevas tecnologías de aprendizaje automático, utilizando bases de datos fiables para reducir alucinaciones, pueden servir para la mejora de las capacidades de los trabajadores. (Ámbito: tecnología, humano)
- Inspección de calidad de tapones de aceite en automoción, basada en inteligencia artificial y aumento de datos, como medio para alcanzar las ideas de ceros defectos de fabricación, pero también con el objetivo de reducir los residuos derivados de la generación de piezas defectuosas en el entrenamiento de los modelos de clasificación. (Ámbito: tecnología, sostenibilidad)
- Inspección de calidad de paneles termoaislantes, basada en plataformas software interoperables, aumento de datos e inteligencia artificial. Validación en una línea piloto de automática y robótica colaborativa. Objetivos de detección temprana de piezas defectuosas para su eliminación o reprocesado. (Ámbito: tecnología, sostenibilidad)

Los dos primeros casos de estudio están relacionados con la tecnología. El tercer artículo está más relacionado con el humano y como éste puede mejorar y adquirir habilidades para alinearse con las nuevas tendencias en la industria. Los dos últimos tienen una componente

tecnológica y, también, un impacto en lo relativo a la sostenibilidad, dado que se reduce la generación de residuos y el consumo energético.

1.5 Contribución

Si partimos desde unas contribuciones más genéricas a unas más concretas, se podría decir que se contribuye globalmente presentando una visión teórico-práctica de múltiples de los elementos, conceptos y técnicas que incluye la industria 5.0. Una visión que no se queda simplemente en definiciones y que trata de enlazar todos estos conceptos en un narrativa coherente y apoyada por casos de estudio validados.

Si entramos en el detalle de las contribuciones de cada caso específico, el artículo relacionado con las líneas piloto presenta un análisis exhaustivo del estado actual de éstas. Para ello estudia ampliamente la literatura y el estado del arte analizando cómo la líneas se sitúan en el contexto de la industria 4.0 y cómo se relacionan éstas con los aspectos de automática y robótica, inteligencia artificial y con conceptos de la Industria 5.0, como el humano y la sostenibilidad. En el artículo también se presenta la línea piloto desplegada en GAMHE y se contextualizan una serie de trabajos realizados en esa línea piloto. En el artículo se contribuye discutiendo los beneficios y los potenciales problemas de estas líneas piloto, comparándolas a su vez con otras formas de validar antes de invertir, de formar técnicos o de desarrollo de pruebas de concepto de nuevas tecnologías.

Respecto al artículo de arquitecturas interoperables es de interés mencionar la plataforma orquestable ligera que se ha utilizado en múltiples de los casos de uso. Esta plataforma sigue las ideas clave del proyecto KITT4SME, que tiene como objetivo facilitar la creación de aplicaciones complejas en entornos industriales mediante el uso de componentes. Esta plataforma se detalla no solo a alto nivel, sino también mediante sugerencias concretas de herramientas y mediante la justificación de las decisiones técnicas. Está plataforma se complementa con un diseño y la demostración de un metamarco que se puede utilizar para definir los requisitos de seguridad y privacidad de las plataformas interoperables para la industria. En concreto, este metamarco permite identificar y analizar los problemas críticos de seguridad y privacidad dentro de las plataformas interoperables basadas en componentes de software, al tiempo que proporciona información a los desarrolladores del ecosistema de fabricación y a los proveedores de inteligencia artificial sobre cómo abordar el análisis de riesgos de seguridad y privacidad de sus soluciones o despliegues.

Sobre esta plataforma se han desarrollado una serie de componentes. Uno de ellos es un módulo de aumento de datos y generación de datos sintéticos, que ha contribuido a la confirmación, validada en dos casos industriales, de que los datos generados sintéticamente pueden ser válidos como reemplazo de imágenes de piezas reales en situaciones en que éstas no estén disponibles. No solo eso, sino que también se ha demostrado cómo, gracias al entrenamiento de modelos inteligencia artificial con estos datos aumentados, se han conseguido mejoras de tiempo en procesos de control de calidad e incluso detección temprana de defectos, reduciendo también la generación de residuos.

En lo relativo al humano, la contribución está relacionada con demostrar que es posible unir

las capacidades del Procesamiento del Lenguaje Natural y el aprendizaje automático con las bases de datos ocupacionales. El objetivo es el de conceptualizar sistemas que incluyan las capacidades de los modelos de lenguaje, todavía propensos a alucinaciones, junto con datos validados y aceptados como son las bases de datos O*NET o ESCO (Base de datos europea de Habilidades, Competencias, Cualificaciones y Ocupaciones), permitiendo así identificar deficiencias en habilidades y facilitar el desarrollo de capacidades. En términos prácticos, el objetivo es facilitar al humano el acceso a puestos de trabajo de la industria analizando cómo de lejos están sus capacidades de los requisitos básicos de un puesto específico. A su vez, se pretende permitir que el operador pueda detectar actividades para las que está más capacitado. Todo eso acompañado de un sistema de recomendación sugiriendo actividades formativas. Dentro de la contribución, es importante mencionar el sistema híbrido basado en bases de datos bien conocidas y en modelos basados en transformadores que permite el análisis de currículos y descubrimiento de carencias formativas. Siempre con el punto de vista de que los nuevos paradigmas de industria y la evolución de la inteligencia artificial va a implicar nuevas actividades y por tanto la necesidad de actualización de habilidades e incluso de reconversión profesional.

1.6 Fuentes y orígenes del contenido de la tesis

En esta sección se pretenden listar otros artículos, capítulos y otras publicaciones en las que el autor ha contribuido, y en las que ha desarrollado parte de la investigación presentada en esta tesis. Los artículos cubren el espectro de la Industria 5.0, y se alinean con los conceptos mencionados en este capítulo de introducción.

Para hacer más sencillo al lector la vinculación entre publicaciones y temáticas, para cada artículo se detallan las palabras clave, la relación con la tesis y la contribución del autor.

- **Publicación:** *A Novel Knowledge Plug-in for Incorporating Information about Employability from the O* NET Database into Conversational Agents*. Rubén Alonso; Danilo Dessì; Antonello Meloni; Diego Reforgiato Recupero. (2024). IEEE Access.
  - **Palabras clave:** Bases de datos ocupacionales, agentes conversacionales, modelos de lenguaje, asistentes virtuales, experiencia de usuario, interacción humano-robot
  - **Relación con la tesis:** Bases de datos ocupacionales, capacitación, centricidad humana, interacción humano-máquina, aprendizaje automático
  - **Contribución del autor:** Bases de datos ocupacionales, facilitación de la validación, conclusiones
  - **DOI:** http://dx.doi.org/10.1109/access.2024.3423359
- **Publicación:** *A Seamless ChatGPT Knowledge Plug-in for the Labour Market*. Rubén Alonso; Danilo Dessì; Antonello Meloni; Marco Murgia. Diego Reforgiato Recupero. Giuseppe Scarpi (2024). IEEE Access.
  - **Palabras clave:** Inteligencia artificial, bases de datos ocupacionales, agentes conversacionales, experiencia de usuario, interacción humano-robot

  - **Relación con la tesis:** Bases de datos ocupacionales, capacitación, centricidad humana, aprendizaje automático
  - **Contribución del autor:** Bases de datos ocupacionales, facilitación de la validación
  - **DOI:** http://dx.doi.org/10.1109/access.2024.3485111

- **Publicación:** *Defining Valuable Data and Stakeholder Engagement in Digital Building Logbooks: A Framework for Effective Data Storage*. Andrea Ibba; Sun Ah Hwang; Diego Reforgiato Recupero; Rubén Alonso; Rosamaria Olivadese (2024). European Conference on Computing in Construction.
  - **Palabras clave:** Construcción, cuadernos de bitácora digitales, marco de datos, digitalización en la construcción
  - **Relación con la tesis:** Arquitecturas software, gestión de datos, plataformas digitales, interoperabilidad
  - **Contribución del autor:** Alcance del DBL, tipos y esquemas de datos
  - **DOI:** https://doi.org/10.35490/ec3.2024.207

- **Publicación:** *An Early Decision Support Tool for Energy-Focused Renovation of Residential Buildings*. Beñat Arregi; Amaia Castelruiz; Enrico Prataviera; Angelo Zarrella; Pierre Bourreau; Hugo Viot; Müge Yüksel Çetin; Massimo Fuccaro; Rubén Alonso; Tatiana Armijos-Moya et al. (2024). International Conference on Smart and Sustainable Technologies (SpliTech)
  - **Palabras clave:** Construcción, sistemas de soporte a la decisión, ciclo de vida, sostenibilidad
  - **Relación con la tesis:** Sistemas software, sostenibilidad, otras industrias
  - **Contribución del autor:** Estado del arte de los sistemas de soporte a la decisión.
  - **DOI:** https://doi.org/10.23919/splitech61897.2024.10612484

- **Publicación:** *Architecture of a Software Platform for Affordable Artificial Intelligence in Manufacturing*. Vincenzo Cutrona; Giuseppe Landolfi; Rubén Alonso; Elias Montini; Andrea Falconi; Andrea Bettoni (2024). Capítulo del libro *Artificial Intelligence in Manufacturing*
  - **Palabras clave:** Plataformas software, inteligencia artificial en la industria, casos de uso
  - **Relación con la tesis:** Industria 5.0, arquitecturas software, inteligencia artificial, sistemas componibles, robótica colaborativa
  - **Contribución del autor:** Arquitectura de alto nivel
  - **DOI:** https://doi.org/10.1007/978-3-031-46452-2_6

- **Publicación:** *Integrating Knowledge into Conversational Agents for Worker Upskilling*.

Rubén Alonso; Danilo Dessí; Antonello Meloni; Marco Murgia; Reforgiato Recupero Diego (2024). Capítulo del libro *Artificial Intelligence in Manufacturing*

  - **Palabras clave:** Agentes conversacionales, capacitación de los trabajadores, procesamiento de lenguaje natural
  - **Relación con la tesis:** Centricidad humana, capacitación de los trabajadores, interacción humano-máquina, aprendizaje automático
  - **Contribución del autor:** Centricidad humana y habilidades, competencias y ocupaciones
  - **DOI:** https://doi.org/10.1007/978-3-031-46452-2_15

- **Publicación:** *Improving Digital Twin Experience through Big Data, IoT and Social Analysis: an architecture and a case study.* Rubén Alonso; Riccardo Locci; Diego Reforgiato Recupero (2024). Heliyon Cell Press
  - **Palabras clave:** Gemelo digital, Big Data, Internet de las cosas, análisis social
  - **Relación con la tesis:** Gemelos digitales, visualización de datos, interoperabilidad
  - **Contribución del autor:** Concepto, prueba de concepto inicial, entornos 3D navegables
  - **DOI:** https://doi.org/10.1016/j.heliyon.2024.e24741

- **Publicación:** *The worker profiler: Assessing the digital skill gaps for enhancing energy efficiency in manufacturing.* Silvia Fareri; Riccardo Apreda; Valentina Mulas; Ruben Alonso (2024). Technological Forecasting and Social Change
  - **Palabras clave:** Evaluación de las competencias, competencia ecológica, análisis de patentes, ESCO, reconocimiento de entidades, Industria 4.0
  - **Relación con la tesis:** Industria 4.0/5.0, capacitación y análisis de competencias, trabajadores, bases de datos ocupacionales
  - **Contribución del autor:** Software para el análisis de competencias
  - **DOI:** https://doi.org/10.1016/j.techfore.2023.122844

- **Publicación:** *Towards the definition of a European Digital Building Logbook: A survey, Rubén Alonso, Rosamaria Olivadese, Andrea Ibba, Diego Reforgiato Recupero. (2023). Heliyon.*
  - **Palabras clave:** Diario digital de la construcción, digitalización en la construcción, sistemas de gestión. Construcción
  - **Relación con la tesis:** Digitalización, gestión de datos, interoperabilidad, otras industrias
  - **Contribución del autor:** Concepto, arquitectura, gestión de los datos
  - **DOI:** https://doi.org/10.1016/j.heliyon.2023.e19285

1.7 Esquema y distribución de la tesis

La tesis comienza con el capítulo de Introducción (Capítulo 1) que concluye en esta sección de distribución de la tesis. Tras este capítulo se presenta la información contextual y los materiales y métodos utilizados (Capítulo 2).

Posteriormente se presentan los artículos incluidos en este compendio (Capítulo 3), detallando para cada uno de ellos información básica sobre el editor y dónde acceder a la versión pública. También se incluye el resumen en castellano.

Por último se presenta una discusión (Capítulo 4), donde se ponen en valor los resultados y se analizan críticamente. Esta sección precede a la conclusión (Capítulo 5), que además de profundizar en los resultados, presenta la evaluación de la hipótesis, resume la aportación al campo y adelanta el trabajo futuro.

Capítulo 2

Materiales y métodos

2.1 Materiales y origen de la información

Dada la amplitud de los conceptos de la Industria 5.0, y por tanto de la tesis, son múltiples y muy diversos los materiales utilizados. En todos los artículos presentados se ha partido siempre de un análisis exhaustivo del estado del arte y la literatura científica. Esto es especialmente significativo en el caso de estudio relacionado con las líneas piloto y en el análisis de las implicaciones de seguridad y privacidad en las plataformas para la digitalización de pequeñas y medianas empresas.

En el caso de las líneas piloto, tal y como se menciona en el artículo, se realizaron búsquedas en bases de datos de literatura científica y en buscadores de información científica. En concreto, se utilizaron ScienceDirect, Scopus, OpenAlex y Google Scholar. Los resultados fueron analizados en función de su relación con la automática y la robótica y los conceptos de Industria 5.0, siguiendo una metodología propia de búsqueda y análisis del estado del arte.

En el caso del diseño de plataforma simplificada y el análisis de seguridad y privacidad, se siguió un proceso similar. En este caso se contaba con una serie de referencias que incluían metodologías y marcos de seguridad estándar o bien conocidos. Estos se utilizaron como base, analizando su aplicación al caso concreto de las pequeñas industrias. La mayor parte de los marcos utilizados estaban orientados a la seguridad organizacional, lo cual hacía más compleja su aplicación directa en pequeñas empresas. Esta limitación impulsó el trabajo en una adaptación para definir un nuevo marco más ligero y adaptable, mucho más centrado en arquitecturas software.

También en este artículo se evaluaron dos métodos distintos para el análisis de las necesidades de privacidad y seguridad en plataformas interoperables desplegadas en entornos industriales. Uno de los métodos estaba basado en el análisis de riesgos, mientras que el otro estaba basado en los objetivos. Se trabajó de modo que ambos convergieran y se desarrolló un metamarco que permite el análisis desde las dos perspectivas. Este metamarco metodológico fue utilizado en el proyecto KITT4SME para definir los requisitos de seguridad, tal y como se detalla en el artículo.

El concepto de Industria 5.0 se cataloga en ocasiones dentro del ámbito de industrias del futuro, donde, si bien el entorno manufacturero y la automática y robótica colaborativa tienen un peso significativo, se observa que otras industrias también tienen impacto. En concreto, industrias que tradicionalmente estaban poco digitalizadas, como por ejemplo, el sector de la construcción, son ya casos de uso en los que la automática y robótica, las fábricas móviles o las líneas piloto para producción a escala reducida, son elementos clave del proceso constructivo y de renovación. Si bien esto no ha tenido un impacto directo en los métodos utilizados en la tesis, sí que ha hecho que incluso se participe y se encuentre inspiración en actividades relacionadas con la construcción, las infraestructuras, los sistemas de soporte a la decisión o la capacitación y formación general de los trabajadores.

Otra de las fuentes de datos ha sido la línea piloto GAMHE y las investigaciones realizadas sobre ella. La línea piloto GAMHE desplegó algunos de los componentes desarrollados en el proyecto KITT4SME y se utilizó para realizar estudios de aumento de datos y generación de datos sintéticos. También ha sido fuente de información para casos de estudio en Industria 4.0 y 5.0, como se analiza en el artículo sobre líneas piloto.

Por último, es importante mencionar que parte de los materiales se originan en las actividades realizadas en dos proyectos de investigación financiados por la Comisión Europea en programas relacionados con la transformación digital y las industrias del futuro. Concretamente, los proyectos KITT4SME y STAR han servido como fuente de información y contexto para la realización de los estudios de esta tesis. Estos dos proyectos se detallan a continuación.

2.1.1 El proyecto KITT4SME

KITT4SME es un proyecto de investigación financiado por la Comisión Europea que tiene como objetivo proporcionar kits de inteligencia artificial sobre plataformas orquestales para su uso por parte de las PYMEs. Uno de los objetivos del proyecto es proporcionar kits de hardware y software que puedan construirse sobre una plataforma modular y adaptable, lo que facilitará la integración de componentes de inteligencia artificial en los sistemas de producción.

El proyecto constaba de cuatro pilotos experimentales planificados en cuatro PYMEs manufactureras diferentes, supervisados por un mentor técnico y que involucraban al menos un proveedor de tecnología. El alcance de estos cuatro pilotos, que también se utilizan como entornos relevantes para la validación, se puede resumir de la siguiente manera:

- **Piloto en la industria del moldeo por inyección:** con un objetivo de implementar el concepto de mutualismo en un entorno de trabajo adaptable, donde los humanos y las máquinas complementan sus capacidades para optimizar rendimientos de fabricación y mejorar la satisfacción de los trabajadores.
- **Proyecto piloto en el ámbito de los sistemas de calidad:** mejorando las técnicas de inspección tradicionales con las características innovadoras de la inteligencia artificial, y a su vez retener y reutilizar el conocimiento experto.
- **Proyecto piloto en el campo de las herramientas e instrumentos manuales de precisión:** introduciendo un control de calidad digitalizado basado en inteligencia artificial y estableciendo un proceso para identificar y caracterizar las mejores prácticas

mediante la combinación de diferentes fuentes de datos.

- **Piloto en el sector de material eléctrico:** asegurando la calidad y la entrega a niveles satisfactorios bajo condiciones VICA (volatilidad, incertidumbre, complejidad y ambigüedad) que requieren cero defectos, cero averías y fácil reconfigurabilidad.

Este ecosistema proporcionó los requisitos y la información inicial para definir la plataforma de referencia y el metamarco, así como la oportunidad de aplicarlo y evaluarlo. La aplicación del metamarco permitió identificar las necesidades de seguridad y privacidad en la plataforma, introducir elementos que mitigaran los riesgos y generalizar un enfoque para plataformas modulares similares. Los componentes de aumento de datos también se han validado en pilotos de este proyecto, tal y como se detalla en uno de los artículos presentados.

2.1.2 El proyecto STAR

STAR es también un proyecto financiado por la Comisión Europea orientado a la investigación en inteligencia artificial segura, confiable y centrada en las personas para las líneas manufactureras del futuro.

Uno de los objetivos de STAR era desarrollar nuevas tecnologías basadas en inteligencia artificial que permitieran adquirir conocimiento para tomar decisiones oportunas y seguras en entornos dinámicos e impredecibles. También estudiar potenciales problemas de seguridad en los sistemas basados en inteligencia artificial y explorar soluciones para que estos sistemas puedan enfrentarse a adversarios sofisticados y sean capaces de mantener la continuidad operacional frente a ataques.

El proyecto desarrolló y validó diferentes sistemas en varios pilotos en Países Bajos, Alemania y Portugal. Entre ellos caben destacar:

- **Sistemas de aprendizaje activo** para la aceleración la adquisición de conocimiento.
- **Sistemas de realidad simulada** que optimizan el aprendizaje por refuerzo en escenarios de colaboración entre humanos y robots.
- **Sistemas de inteligencia artificial y aprendizaje automático explicables** para aumentar la transparencia de los sistemas industriales y mejorar la confianza.
- **Gemelos digitales** centrados en las personas para detección de fatiga y sobresfuerzo.
- **Técnicas avanzadas para la navegación óptima de robots móviles** y la detección de zonas de seguridad en plantas industriales.
- **Mecanismos de ciberdefensa** contra ataques sofisticados dirigidos a la inteligencia artificial que opera con datos industriales.

Siendo STAR un proyecto de la línea de financiación relacionada con el liderazgo en tecnologías facilitadoras e industriales, y dado que la centricidad humana en la industria era uno de los puntos principales del proyecto, un elemento central era entender tanto al operador en el puesto de trabajo como su relación con el impacto que las nuevas tecnologías van a tener en su actividad diaria.

Por ese motivo se investigó en arquitecturas de redes neuronales basadas en técnicas de aprendizaje profundo que permitan a un operador interaccionar de manera natural y obtener recomendaciones en cuanto a desarrollo de capacidades, cambio de ruta profesional o necesidades de formación.

Como resultado de esta investigación se trabajó en agentes conversacionales, que utilizaban, por un lado, tecnologías de aprendizaje automático y, por otro, bases de datos de información laboral y ocupacional en dos casos de uso. El primero como agente de información, para guiar al operador en la búsqueda de actividades laborales y obtener más información sobre las necesidades en cuanto a habilidades, tecnologías y tareas. El segundo mediante un agente conversacional que, a modo de entrevista, conversaba con el operador y obtenía la información necesaria para comparar el currículum y la experiencia del trabajador con una línea base y proporcionarle recomendaciones laborales y formativas.

Este proyecto proporcionó el entorno adecuado para investigar los conceptos de centricidad humana y confianza en la Industria 5.0. También facilitó el desarrollo de ideas para la capacitación de trabajadores, búsqueda de necesidades formativas y adecuación al puesto de trabajo que fueron validadas en el proyecto y publicadas en sendos artículos científicos. Entre estos artículos se incluye el enfoque para el mapeo de puestos de trabajo y habilidades utilizando redes neuronales y bases de datos ocupacionales, que se presenta en el capítulo 3.

2.1.3 Software y Hardware

Si entrar en detalle, dado que se detalla en cada artículo del capítulo 3, en esta sección se pretende hacer un pequeño resumen de los elementos más significativos en cuanto a hardware y software.

En cuanto al hardware, el elemento principal es todo el conjunto englobado dentro de la línea piloto GAMHE del Centro de Automática y Robótica de la UPM-CSIC. Como se detalla en 3.1, esta línea piloto es una línea flexible y modular de automatización y robótica colaborativa, que sirve como exponente de las líneas piloto para la Industria 5.0 y facilita casos de estudio relacionados con la tecnología, el ser humano y la sostenibilidad.

En concreto, esta línea piloto incluye una mezcla de sistemas heredados y nuevas tecnologías, que integra desde máquinas de control numérico hasta brazos robóticos y robots móviles de última generación. Estas máquinas, integradas mediante protocolos de comunicación basados en publicación-subscripción y mediante una cinta transportadora, y acompañadas de sistemas de sensores y visión por computador la posicionan como un elemento relevante para investigaciones como las incluidas en esta tesis. En concreto, el artículo 3.1, elementos del artículo 3.2 y el estudio 3.5 que analiza la calidad y sostenibilidad de un sistema de clasificación basada en aumento de datos para paneles termoaislantes utilizados en el sector de la construcción, se han beneficiado de esta línea piloto.

En lo que al software se refiere, se han utilizado diferentes herramientas de código abierto, junto con *scripts* y *notebooks* elaborados en Python. Quizás el artículo más centrado en el software sea el que presenta la plataforma interoperable para la industria (artículo 3.2), y en él se detallan los principales elementos.

Sistemas basados en Linux, sistemas de gestión de despliegues como Kubernetes, mallas de gestión de microservicios como Istio, junto con una colección de bases de datos (PostgreSQL, MariaDB), sistemas de despliegue y monitorización continua (Argo CD) y decenas de bibliotecas Python para el aumento de datos, generación de imágenes sintéticas e interoperabilidad han sido los principales elementos para el desarrollo de los componentes software.

También es importante mencionar el uso de BERT (*Bidirectional Encoder Representations from Transformers*, Representación de Codificador Bidireccional de Transformadores) como técnica basada en redes neuronales para entender la relación entre palabras y sus significados y facilitar así las actividades de lenguaje natural que permiten al operador interaccionar con las bases de datos ocupacionales.

2.1.4 Bases de datos Ocupacionales: O*NET y ESCO

Como el autor de esta tesis menciona en Alonso, Dessí, Meloni, Murgia et al., 2024, para los trabajos relacionados con la capacitación del trabajador se han utilizado las bases de datos ocupacionales O*NET y ESCO. Ambas bases de datos han sido desarrolladas por iniciativas respaldadas por agencias gubernamentales y proporcionan información actualizada sobre ocupaciones, habilidades y competencias. La información incluida en estas bases de datos permite tanto a los trabajadores como a las empresas obtener respuestas e información sobre las necesidades del mercado laboral.

O*NET es la base de datos de información ocupacional de referencia en los Estados Unidos. Esta base de datos es el resultado del programa de la Red de Información Ocupacional del Departamento de Trabajo de los Estados Unidos y la agencia de Administración de Empleo y Capacitación, y desarrollado por el Centro Nacional para el Desarrollo de O*NET.

A través de esta base de datos y otras herramientas, O*NET tiene como objetivo contribuir a la comprensión de las trayectorias profesionales y ocupacionales en Estados Unidos y apoyar el desarrollo de la fuerza laboral. La base de datos se revisa y actualiza periódicamente mediante entrevistas con organizaciones y empleados. Los datos obtenidos se refinan con la revisión y el conocimiento de los expertos, para publicar cuatro versiones cada año. Esta actualización periódica no solo permite comprender los cambios en el mercado laboral, sino también estar al día respecto a las necesidades emergentes o los cambios en las habilidades requeridas.

La base de datos incluye una serie de conceptos asociados a cada ocupación. Estos conceptos consisten en habilidades, conocimientos, actividades, tecnologías o tareas, entre otros. Actualmente incluye miles de vínculos entre las ocupaciones y los conceptos mencionados anteriormente (por ejemplo, habilidades o tecnologías). El procesamiento de esta información y de los vínculos permite caracterizar las ocupaciones, calcular las distancias entre ellas, detectar lagunas o establecer posibles itinerarios formativos.

ESCO (*European Skills, Competences, and Occupations*) es una base de datos multilingüe que soporta, entre otras cosas, todas las lenguas oficiales de la Unión Europea e incluye la correspondencia entre capacidades y cualificaciones. Lo primero es útil para adaptar los resultados a las necesidades específicas de una región determinada y, sobre todo, para llegar a un mayor número de usuarios potenciales. Lo segundo facilita estandarizar la información

relacionada con las ocupaciones, habilidades y calificaciones en diferentes países y sectores.

ESCO se publica con menos frecuencia que O*NET, y su última actualización fue una versión menor en 2022. Sin embargo, abarca más de 3.000 ocupaciones y más de 13.000 habilidades asociadas a esas ocupaciones. La última versión de ESCO también incluye una revisión de la correspondencia entre ESCO e ISCO-08, un índice que organiza y categoriza las ocupaciones en todo el mundo.

Es igualmente relevante destacar los esfuerzos por encontrar interconexiones. Estas interconexiones son métodos y herramientas esenciales para mapear algunas de estas bases de datos y sistemas de información ocupacional. Estos mapeos permiten la interoperabilidad y la comparación entre habilidades y ocupaciones en diferentes ubicaciones. En los últimos años se ha hecho un esfuerzo por buscar la interoperabilidad entre O*NET y ESCO, dada la importancia de ambas.

En 2022, ESCO publicó el primer informe ESCO, 2022 que detalla el mapeo entre O*NET y ESCO. Este mapeo se desarrolló utilizando un enfoque híbrido basado en modelos de inteligencia artificial basados en Representación de Codificador Bidireccional de Transformadores (BERT) (Devlin et al., 2018), cuyos resultados fueron refinados posteriormente por un grupo de expertos.

2.2 Metodología

Se entiende la tesis como teórico-práctica con un compendio de artículos que tratan de cubrir dos fases interconectadas: la primera, una revisión histórica y técnica del área dentro de la Industria 4.0/5.0. La segunda, el diseño y desarrollo de pruebas de concepto y experimentos que validan la idea inicial de cada artículo. Al entender así la tesis, se plantean al menos dos métodos complementarios entre sí para desarrollarla.

Por un lado, para las actividades de exploración del área se considera un método descriptivo en el cual, tras una revisión de la literatura asociada al ámbito de investigación, se clasifica y detalla la situación actual en el ámbito de las industrias del futuro. Este método descriptivo proporciona una base para la selección de las técnicas más adecuadas para, por ejemplo, evaluar el estado actual de las líneas piloto en el contexto de la Industria 5.0 o para adaptar los marcos de seguridad y privacidad a pequeñas empresas manufactureras.

Para la elaboración de las pruebas de concepto y casos de estudio se plantea una metodología basada en el método científico. Se parte de la observación y del estudio del área para diseñar e implementar conceptos que respondan a la pregunta de investigación o hipótesis de cada trabajo. Así, en los artículos relacionados con el mapeo de capacidades de los operadores y en los casos de estudio de control de calidad, se presenta cómo se ha validado la investigación. Incluso, en artículos más descriptivos se presentan ejemplos de validación y discusión de resultados, o se detalla cómo se aplicó la investigación a actividades reales.

Es importante mencionar que se ha mantenido una orientación hacia la publicación científica, no solo como elemento de diseminación de información, sino también como herramienta de validación de las actividades realizadas. Los artículos presentados en el compendio, junto con

los otros elaborados durante el periodo de doctorado y presentados en la sección 1.6, han partido de trabajos colaborativos enmarcados en actividades de proyectos de investigación donde se ha aplicado el método teórico-práctico anteriormente mencionado.

Estos métodos basados en el análisis del estado del arte, definición del contexto y metodología, presentación de la solución, validación de resultados y discusión no dejan de ser métodos generales y habituales en la investigación. En cuanto a métodos específicos, a continuación se detallan los métodos más significativos utilizados en los artículos presentados en el compendio. Para cada uno de ellos, además de proporcionar detalles sobre su aplicación, se incluye un párrafo a modo de conclusión donde se resume por qué era necesaria la aplicación del método y para qué se ha sido relevante.

2.2.1 Análisis del estado del arte de las líneas de automática y robótica

La metodología aplicada para seleccionar la literatura relevante consistió la búsqueda del término "pilot line" (línea piloto) en diversos índices académicos y motores de búsqueda. En concreto, se utilizaron las siguientes fuentes de datos: ScienceDirect, Scopus, OpenAlex y Google Scholar.

En ScienceDirect, se realizó la búsqueda utilizando la frase "pilot line", entre comillas, para asegurar que tanto el singular como el plural fueran incluidos. Esta búsqueda produjo 1.559 resultados (a fecha del 9 de diciembre de 2024). En Scopus, se llevó a cabo una búsqueda por "título del artículo, resumen, palabra clave" para el término "pilot line". Dado que esta búsqueda no distingue entre singular y plural, también se incluyeron búsquedas para "pilot lines". El resultado en Scopus fue un total de 912 publicaciones.

También se utilizó en el análisis el catálogo de artículos científicos abiertos, OpenAlex. Para OpenAlex, los términos de búsqueda empleados fueron "pilot line | pilot lines" para incluir tanto la forma singular como plural utilizando el operador "OR". En este caso, OpenAlex devolvió 6.444 publicaciones.

Por último, se utilizó Google Scholar debido a su amplia cobertura. El término de búsqueda utilizado fue "pilot line | lines" para asegurar que tanto la forma singular como plural fueran incluidas. La búsqueda retornó un resultado que Google Scholar estima en aproximadamente 15.900 publicaciones.

El resultado inmediato de la aplicación de esta metodología es la observación de que el interés en las líneas piloto ha permanecido relativamente estable en los últimos años, con ligeros aumentos en el número de publicaciones. En OpenAlex, hubo un pico en 2019, que disminuyó en 2020 pero luego volvió a aumentar gradualmente, superando el nivel de 2019 en 2023. En Google Scholar, el nivel de publicaciones permaneció relativamente similar entre 2019 y 2021, con un crecimiento en 2022. Desde 2022, los números han aumentado ligeramente. En la figura 2.1, también presentada en el artículo 3.1, se muestra la evolución del numero de publicaciones.

Una vez obtenida la lista de artículos y dada la cantidad de menciones a líneas piloto en diferentes contextos, se refinó la búsqueda utilizando términos relacionados con los concep-

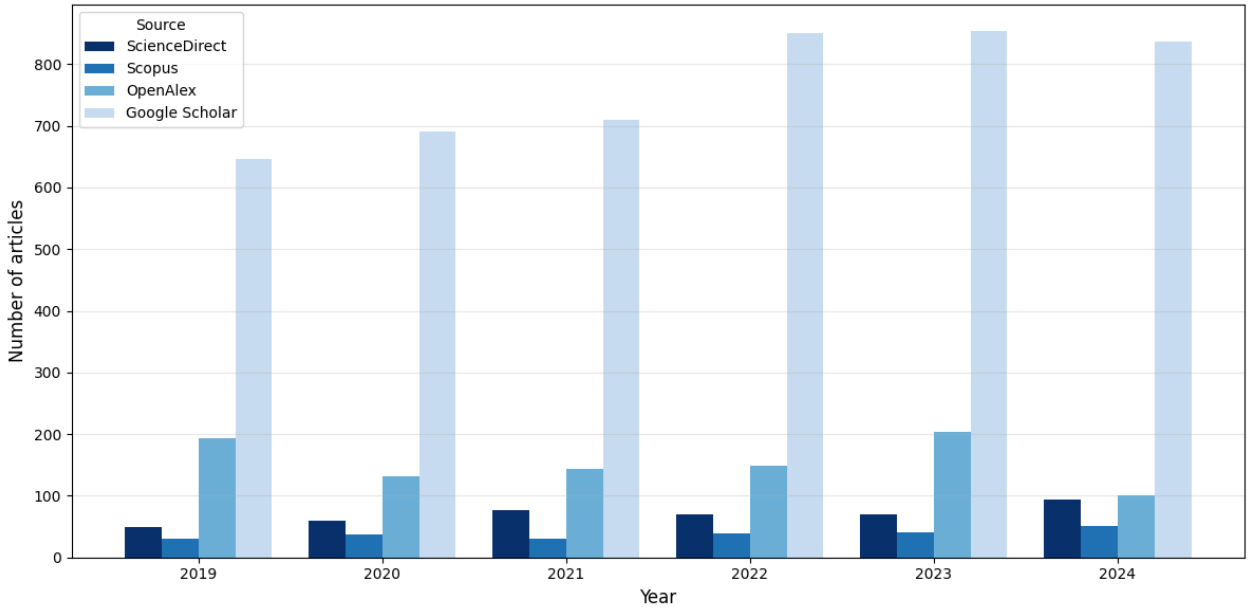


Figura 2.1: Publicaciones científicas que incluyen el concepto de línea piloto

tos que se trataban de analizar. Específicamente, términos como "industria", "robótica", "automática", "manufactura" o "inteligencia artificial".

Durante la selección de artículos, se priorizó tanto mantener variedad en diferentes áreas donde se utilizan las líneas piloto como seleccionar publicaciones altamente relacionadas con los conceptos de Industria 4.0/5.0.

Las varias docenas de artículos resultantes se caracterizaron usando una metodología propia de análisis del estado del arte basada en la extracción de los temas principales, el objetivo del artículo, su contribución y resultados, las deficiencias que detectan y, por último, su relevancia. Cada artículo también se mapeó y clasificó según los siguientes parámetros: Industria 4.0, automática y robótica, inteligencia artificial, factor humano e impacto ambiental o estudio de la sostenibilidad. Todos los resultados fueron nuevamente clasificados posteriormente en grupos en función de la relevancia e impacto, incluyendo en el análisis final del artículo aquellos con mayor interés y peso.

Metodologías similares a esta son habituales en el análisis del estado del arte y en la elaboración de artículos de revisión bibliográfica. Su universalidad y versatilidad son los principales motivos para su uso. Los sistemas de búsqueda de las principales bases de datos de artículos científicos permiten filtrar la información de una forma rápida y precisa. A su vez, el refinado de la literatura obtenida mediante extracción estructurada de información clave facilita organizar la información relevante y simplifica la comparación y clasificación de los artículos.

La relevancia de esta metodología y su aplicación se evidencia en que proporciona un punto de partida y una base sólida para entender el contexto de las líneas piloto. También ha permitido identificar tendencias, disponer de contexto y alinear la investigación con el estado del arte. Por último, ha facilitado comprender cómo se vinculan la automatización, la robótica o la inteligencia artificial con los conceptos de sostenibilidad y centralidad en el ser humano.

2.2.2 Transformadores y similitud semántica entre entidades O*NET y texto

Se parte de la idea de desarrollar un sistema que automáticamente busque similitudes entre las características de los trabajadores o candidatos y las ocupaciones. En concreto, las fuentes de datos son:

- **Información del candidato (R)**: Información en formato de texto que represente a un trabajador o candidato. Por ejemplo, en formato de currículum o biografía breve.
- **Ocupaciones O*NET (J)**: Ocupaciones y entidades descritas en la base de datos ocupacional O*NET, incluyendo habilidades, conocimientos, actividades laborales, calificaciones, tareas y herramientas relacionadas con esas ocupaciones.

Para ello se considera una función $f : R \rightarrow J$ que, para cada currículum o biografía $r \in R$ devuelva una lista de ocupaciones $\{j_1, j_2, ..., j_n\} \subset J$ ordenadas por relevancia con respecto al perfil del candidato. La relevancia se calcula utilizando la similitud semántica entre los elementos extraídos en el currículum (R) y la base de datos ocupacional (J). Esto se realiza para cada categoría de entidades de O*NET (entre otras, las ya mencionadas habilidades, conocimientos, actividades laborales, calificaciones, tareas y herramientas).

Para la realización de estos cálculos se utiliza la librería SentenceTransformers (Reimers y Gurevych, 2019). Esta librería está optimizada para generar representaciones vectoriales (*embeddings*) de frases y textos de manera eficiente. Esas representaciones se pueden comparar, por ejemplo, mediante la similitud del coseno, para realizar tareas como el agrupamiento en clústeres, la búsqueda semántica o la recuperación de información. Se seleccionó el modelo *all-mpnet-base-v2* por ser uno de los modelos con mayor rendimiento y estar optimizado para tareas de nivel de frase, y en particular para las comparaciones entre pares.

La información proporcionada por los trabajadores o candidatos se procesa para extraer frases, nombres (por ejemplo, comunicación o matemáticas) o sintagmas nominales (por ejemplo, operador de máquina o trabajo en equipo).

Estos nombres o sintagmas nominales se comparan con entidades O*NET, mientras que las frases se comparan con descripciones incluidas en la base de datos ocupacional. En ambos casos se utilizan los transformadores anteriormente mencionados. Concretamente, se trata de buscar la entidad O*NET con mayor similitud semántica para cada frase, nombre o sintagma nominal. Una vez calculados los valores, se ordenan por relevancia y se selecciona el de mayor grado de similitud. Si este supera un umbral predefinido, se aumenta la puntuación para esa ocupación. Por último el sistema calcula la puntación total y la normaliza en función las entidades asociadas a una ocupación en O*NET.

Este método facilita obtener la alineación de los textos con ocupaciones y se adapta a la perfección a diferentes casos de uso, por ejemplo, el descubrimiento de la ocupación más adecuada en caso de que un trabajador desee cambiar de actividad o incluso encontrar al trabajador más adecuado para la realización de una nueva tarea.

El porqué del uso de esta metodología basada en transformadores y similitud semántica está relacionado con la capacidad de los transformadores para ir más allá de la búsqueda de palabras,

capturando significado y contexto. También, por su capacidad de manejo de sinónimos y variaciones lingüísticas. Por último, su amplia utilización en aprendizaje automático y la variedad y disponibilidad de modelos permiten abordar diferentes tareas de alineamiento semántico.

En cuanto a la utilidad del método, ha demostrado ser útil para cubrir las necesidades de mapeo entre datos estructurados y no estructurados. Además, ha proporcionado versatilidad, facilitando su aplicación a los diferentes escenarios que se han comentado anteriormente. En la parte más relacionada con la tesis y la Industria 5.0, esta metodología permite un análisis del factor humano dentro del paradigma. Especialmente, resulta útil para entender cómo la fuerza laboral deberá adaptarse a las novedades tecnológicas.

2.2.3 Aumento de datos y generación de imágenes sintéticas

En ambos estudios de control de calidad las necesidades están relacionadas con la falta de un volumen significativo de imágenes relevantes para entrenar los modelos de inteligencia artificial. Esta necesidad se abordo mediante el uso técnicas de aumento de datos. El enfoque de aumento de datos se centra en la generación de imágenes sintéticas, y utiliza específicamente tres métodos fundamentales: i) manipulación de imágenes, ii) mezcla de imágenes (*blending*) y iii) generación de patrones. Cada una de estas técnicas se detalla a continuación:

Métodos basados en manipulación de imágenes

El método más tradicional de aumento de datos se basa generalmente en el uso de técnicas de manipulación de imágenes para generar nuevas imágenes sintéticas. Estas técnicas abarcan desde simples rotaciones, reflejos y escalados, hasta modificaciones de la forma de la imagen, incluyendo la repetición de partes de esta, así como variaciones en la luminosidad o el contraste.

En los casos de estudio de control de calidad se utilizaron estas técnicas para generar imágenes como las que se muestran en 2.2. A la izquierda se muestra la imagen original y a la derecha un par de ejemplos con la misma pieza rotada, desplazada, escalada y modificada tonalmente.



Figura 2.2: Imágenes sintéticas generadas mediante técnicas de manipulación de imagen

Métodos basados en mezcla de imágenes

La manipulación por mezcla de imágenes no es más que la aplicación de técnicas de edición para alterar la imagen original. En concreto se combinan píxeles de dos imágenes para generar un resultado integrado. Para esta mezcla o fusión de píxeles, se utilizan lo que se conocen como modos de mezcla: una serie de operaciones bien conocidas que modifican la forma en que estas imágenes se integran.

Algunos ejemplos de estos modos de mezcla son: "oscurecer" (que compara los colores de ambas imágenes y mantiene los más oscuros), "restar" (que resta de la imagen base los colores de la imagen a mezclar) o "*dodge*" (que aclara los colores de la imagen base y reduce el contraste). Como estas, existen decenas de operaciones útiles para mezclar imágenes defectuosas sobre imágenes sin defectos. Por ejemplo, la imagen presentada en 2.3 muestra un ejemplo, que presenta un defecto mezclado utilizando un modo de mezcla de color *dodging* sobre una pieza de referencia.

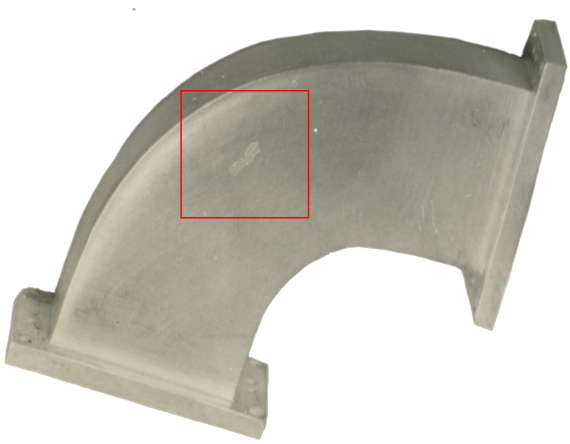


Figura 2.3: Imagen sintética generada mediante mezcla

Métodos basados en generación de patrones

La técnica de generación de patrones consiste en añadir manchas, gotas o salpicaduras a una imagen. Esta técnica juega con la generación automática de estos elementos de forma aleatoria en una región definida, sobre la cual luego se aplican técnicas de manipulación de imágenes (por ejemplo: escalado, rotación, traslaciones) y mezcla (por ejemplo: iluminación, resta). De esta manera se pueden generar imágenes como las mostradas en 2.4, en las cuales se pueden ver salpicaduras generadas artificialmente en las áreas marcadas por el cuadro rojo. Las técnicas pueden combinarse de tal manera que a partir de estas imágenes con salpicaduras podríamos generar cientos de nuevas imágenes mediante las técnicas de aumento de datos y mezcla mencionadas en los puntos anteriores.

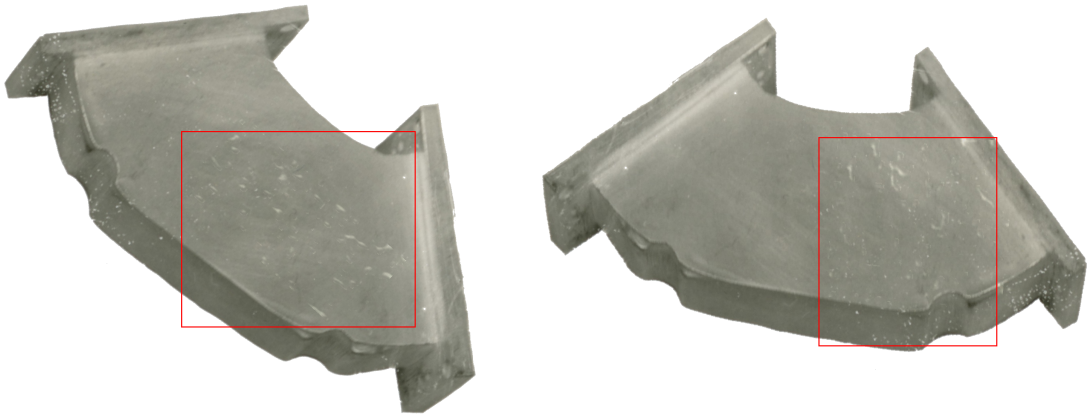


Figura 2.4: Imágenes sintéticas generadas mediante técnicas de generación de patrones

La razón principal para el uso de estos métodos de generación sintética de imágenes radica en la fidelidad respecto a la imagen original y, sobre todo, en el rendimiento. Aunque los modelos generativos son interesantes, todavía el coste de la inferencia es significativo y requieren tiempos de ejecución elevados. El objetivo de poder ejecutar estos métodos como componentes sobre plataformas ligeras o incluso embebidas es también uno de los motivos del uso de estos métodos de aumento de datos. Además, el hecho de que son técnicas tradicionales, bien estudiadas y ampliamente utilizadas en manipulación fotográfica y en edición y diseño de imágenes digitales, permite disponer de una amplia colección de materiales y librerías.

En lo relativo a para qué han sido útiles estos métodos, es importante mencionar que han permitido confirmar en varios casos que, a falta de piezas defectuosas, se pueden elaborar sintéticamente estas. No solo esto, sino que estos modelos de imágenes sintéticas han funcionado sobre plataformas modulares como las mencionadas en la tesis y han sido validados en líneas piloto de automática y robótica colaborativa, así como en empresas de control de calidad. Los métodos han permitido evaluar algoritmos de visión basados en inteligencia artificial y aprendizaje automático, observando cómo se puede realizar una detección temprana reduciendo tiempos de procesamiento de las piezas, energía, tiempo de máquinas y materiales.

2.2.4 Adaptación de la taxonomía de riesgos

En general, los marcos de seguridad están estrechamente relacionados con los riesgos organizacionales, mientras que el metamarco propuesto en el artículo 3.2 se centra en guías y sugerencias orientadas a las plataformas digitales y casos de usos específicos de industria.

Por ese motivo, era importante disponer de un método para filtrar los riesgos, haciendo hincapié en los sistemas software más que en la organización. También era clave reagrupar algunos de estos riesgos, por ejemplo, los que están más orientados a organizaciones. Los riesgos aplicables se han alineado a su vez con marcos de seguridad bien conocidos, como *Secure Controls Framework* o el *Cybersecurity Framework* del Instituto Nacional de Estándares y Tecnología (NIST). La adaptación propuesta se muestra en la tabla 2.1.

Tabla 2.1: Taxonomía adaptada, basada en los riesgos del marco de controles seguros y vinculada a las funciones CSF del NIST

Grupo	Riesgo	Propuesta de adaptación	Función
Control de acceso	Incapacidad para mantener la responsabilidad individual	Incapacidad para mantener la responsabilidad	Proteger
	Asignación incorrecta de funciones privilegiadas	Gestión inadecuada de privilegios	Proteger
	Escalada de privilegios	Gestión inadecuada de privilegios	Proteger
	Acceso no autorizado	Acceso no autorizado	Proteger
Gestión de activos	Pérdida de integridad a través de cambios no autorizados	Pérdida de integridad	Proteger
Continuidad del negocio	Interrupción del negocio	Interrupción de la funcionalidad	Recuperar
	Pérdida de datos / corrupción	Pérdida de información / corrupción o compromiso del sistema	Recuperar
	Reducción de la productividad	Reducción del rendimiento	Proteger
	Pérdida de información/corrupción o compromiso del sistema debido a un ataque técnico	Pérdida de información/corrupción o compromiso del sistema	Proteger
Exposición	Vulnerabilidades no mitigadas	Vulnerabilidades no mitigadas	Proteger
	Compromiso del sistema	Pérdida/corrupción de la información o compromiso del sistema	Proteger
Gobernanza	Prácticas internas inadecuadas	Prácticas inadecuadas de integración de módulos	Proteger
	Prácticas inadecuadas de terceros	Prácticas inadecuadas de integración de módulos	Proteger
Respuesta a incidentes	Incapacidad para investigar / procesar incidentes	Incapacidad para detectar, analizar y responder a incidentes	Responder
	Respuesta inadecuada a incidentes	Incapacidad para detectar, analizar y responder a incidentes	Responder
	Acciones correctivas ineficaces	Incapacidad para detectar, analizar y responder a incidentes	Proteger
Conciencia situacional	Incapacidad para mantener la conciencia situacional	Incapacidad para detectar, analizar y responder a incidentes	Detectar

La tabla muestra los riesgos, agrupados por grupos, tal y como lo hace el *Secure Controls Framework*. También se agrupan por función, como sugiere el *Cybersecurity Framework* (Barrett, 2018). Es importante tener en cuenta que no todos los grupos y riesgos en los marcos descritos anteriormente están representados, ya que algunos no impactan directamente en las plataformas industriales o porque simplemente están más enfocados en la seguridad organizacional y la continuidad del negocio.

En la tabla también se incluye el ajuste y la agrupación de riesgos propuestos. Como se puede observar, los riesgos ajustados se han agrupado en diez categorías principales. Este ajuste se ha realizado centrándose en los riesgos relacionados con el desarrollo y la ejecución del software en lugar de la continuidad del negocio de la organización.

En cuanto a la privacidad y la protección de datos, también se ha considerado la idea de basarse en tres definiciones conocidas de privacidad. Estas son las definiciones *Privacy-by-Design*, *Privacy-by-Default* y C4P (*Cybersecurity For Privacy by Design*). Este trabajo propone agrupar los conceptos de estas tres definiciones en tres principios:

- Privacidad y seguridad por defecto en el núcleo de la aplicación y no como un ”extra”;
- Simplicidad y transparencia para el usuario;
- Privacidad y seguridad a lo largo del ciclo de vida de la solución;

La tabla 2.2 mapea los principios generales de utilizados en la definición del marco de análisis de riesgos con los de los diferentes términos de privacidad anteriores.

Tabla 2.2: Mapeo de los principios de privacidad

Adaptación propuesta	**Privacy-by-Design**	**Privacy-by-Default**	**C4P**
Privacidad por defecto en el núcleo	1) Privacidad proactiva y preventiva 2) Privacidad como configuración por defecto 3) Privacidad incorporada desde el diseño	a) uso de solo los datos estrictamente necesarios para el propósito específico de la solución b) no se requieren acciones para mantener la privacidad	1) Habilitar la privacidad 2) Seguridad preestablecida 3) Mecanismos de seguridad y privacidad en el núcleo
Simplicidad y transparencia	6) Visibilidad y transparencia 7) Respeto a la privacidad de usuario	-	4) Simplificar las cosas 6) Confiar pero verificar 7) Educar a los usuarios
Privacidad durante todo el ciclo de vida	4) Funcionalidad completa 5) Seguridad integral	-	5) Integración durante todo el ciclo de vida

El porqué del diseño y la utilización de este método está relacionado con la necesidad de

simplificar las taxonomías de riesgos y adaptarlas al contexto de las pequeñas empresas manufactureras. Las metodologías de análisis de riesgos y de definición de contramedidas en las que se basa este método son muy extensas y centradas en diferentes ámbitos de continuidad del negocio. En el caso de la tesis, y en el contexto del proyecto KITT4SME, era necesario realizar el análisis a nivel de plataforma. Por ese motivo se simplificó y adaptaron estas metodologías, siempre manteniendo cierto mapeo con ellas.

La relevancia de este método radica en que permite que el análisis lo puedan liderar desarrolladores o departamentos técnicos, partiendo de una serie de objetivos y un conocimiento de los activos. Asimismo, su escalabilidad es importante, ya que puede servir como punto de partida para posteriores análisis de riesgos y continuidad del negocio organizacionales. A nivel de esta tesis, además de para el diseño de las plataformas modulares que han servido de base para los modelos de aumento de datos, ha facilitado la comprensión y los experimentos de resiliencia tecnológica e interoperabilidad.

Capítulo 3

Compendio de Artículos

3.1 Automation and Robotics Pilot Lines in the Context of Industry 5.0

Publicación: *Alonso, R., Sánchez, T. F., Alfaro, D. A., Cruz, Y. J., Villalonga, A., & Castaño, F. (2025). Automation and Robotics Pilot Lines in the Context of Industry 5.0. Applied Sciences, 15(5), 2510.*

- **Resumen:** Las líneas piloto se han convertido en un elemento clave en la industria moderna. Las demandas de resiliencia, centralidad en el ser humano y sostenibilidad en la Industria 5.0, combinadas con las innovaciones tecnológicas, ponen de manifiesto el papel fundamental de estas líneas en el nuevo paradigma industrial. Las líneas piloto facilitan la validación de procesos y la reducción de incertidumbres, lo que las convierte en herramientas esenciales para las industrias del futuro. Este artículo examina el estado actual de las líneas piloto y evalúa su alineación con los principios de la Industria 5.0. Como parte de este análisis, se presenta la línea piloto GAMHE 5.0 como una línea diseñada con un enfoque en la automatización y la robótica, cumpliendo con los requisitos de las industrias del futuro. Este artículo también detalla una serie de estudios de caso realizados en la línea piloto, demostrando su flexibilidad y adaptabilidad. Finalmente, se analizan las ventajas y desventajas de las líneas de pequeña escala en el contexto de la Industria 5.0, enfatizando su importancia en el equilibrio entre tecnología, factores humanos y sostenibilidad.
- **Revista:** Applied Sciences. Feature Papers in Robotics and Automation
- **Editorial:** MDPI
- **DOI:** https://doi.org/10.3390/app15052510
- **Palabras clave:** Líneas piloto, Industria 5.0, Automatización, Robótica, Manufactura sostenible
- **Relación con la tesis:** Aspectos tecnológicos, humanos y de sostenibilidad de la herramientas clave de la industria 5.0.

applied sciences

MDPI

Review

Automation and Robotics Pilot Lines in the Context of Industry 5.0

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Abstract: Pilot lines have become a key element in modern industry. The demands for resilience, human-centricity, and sustainability in Industry 5.0, combined with technological innovations, highlight the critical role of these lines in the new industrial paradigm. Pilot lines facilitate the validation of processes and the reduction in uncertainties, making them essential tools for future industries. This article examines the current state of pilot lines and evaluates their alignment with Industry 5.0 principles. As part of this analysis, the GAMHE 5.0 pilot line is presented as a line designed with a focus on automation and robotics, meeting the requirements of the industries of the future. This article also details a series of case studies conducted on the pilot line, demonstrating its flexibility and adaptability. Finally, the advantages and disadvantages of small-scale lines are analyzed within the context of Industry 5.0, emphasizing their importance in balancing technology, human factors, and sustainability.

Keywords: pilot lines; Industry 5.0; automation; robotics; sustainable manufacturing

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1. Introduction

In its 2025 manufacturing industry sector outlook report (https://www.deloitte.com/us/en/insights/industry/manufacturing/manufacturing-industry-outlook.html (accessed on 27 December 2024)), Deloitte points out that despite financial uncertainty, rising costs and the complex economic and business landscape, the industry continues investing in digital solutions.

Nevertheless, this transition towards digitalization introduces new challenges. On the one hand, there are challenges associated with the rapid pace of technological evolution and the need for a skilled workforce [1]. On the other hand, the complexity of these new technologies, along with issues of interoperability and integration with the industry's existing processes [2], pose significant difficulties.

In recent years, discussions about Industry 4.0, and more recently Industry 5.0, have gained relevance as paradigms to address these and other issues [3]. Specifically, Industry 4.0 aimed to reduce production times and enhance quality, flexibility, and interoperability [4]. Meanwhile, the new Industry 5.0 extends these concepts by focusing on greater collaboration between human operators and machines, while also emphasizing waste reduction and sustainability [5].

The concept of Industry 5.0 is linked to the idea of Society 5.0: a society that extends beyond industrial- or information-based societies to find a balance between technology, economic development, social progress, and environmental sustainability [6]. Likewise,

if Industry 4.0 was primarily focused on digitalization and productivity, the Industry 5.0 paradigm builds on these aspects while adding a social and responsible dimension. Human-centricity, along with resilience and sustainability, are the core pillars of Industry 5.0. Specifically, the human-centric aspect places humans at the center of the production process, guiding them, easing their workload, and supporting decision-making. The resilience aspect refers to a more adaptable and flexible production system that also enhances quality. Lastly, the sustainability aspect focuses on optimizing resources, reducing environmental impact, and minimizing waste.

When discussing sustainability and, in particular, waste reduction, it is often useful to start from simpler and more production-related concepts, for example, "Right First Time" and "Zero Defect Manufacturing". Right First Time (RFT), often referred to as First Time Right, is a production strategy based on quality improvement movements from the 1960s and the Japanese philosophy of continuous quality processes known as Kaizen [7]. The strategy emphasizes the concept that there should be no defects throughout the entire production process, thus avoiding unnecessary rework and waste. In the 1980s, it was used as a slogan in mass-production factories to encourage workers to stop the line and solve problems as soon as they were detected [8].

Zero Defect Manufacturing (ZDM) is another concept related to the quality of processes and products. Although it originated in the 1960s, ZDM remains a trend in Industry 4.0. Typically, ZDM combines both a product-oriented and a process-oriented approach. The product-oriented approach focuses on studying defects in parts and pieces, and finding solutions, while the process-oriented approach focuses on equipment and processes to identify issues in how products are manufactured [9].

Both approaches need to be validated, preferably in non-productive environments before being implemented. This is especially critical in an ecosystem where the offering of solutions and technologies, which are sometimes immature, is broad and constantly evolving. These changes, in hardware, and particularly in software, are becoming even faster and more impactful due to the rapid evolution of AI and robotics [10]. Therefore, it is necessary to have methods, strategies, and tools that help reduce uncertainty on such innovations and serve as a bridge between research and market needs.

Often, Technology Readiness Levels (TRLs) are presented as a measure of maturity and as a scale to evaluate the degree of development of a technology. These levels, which originated from NASA [11], range from identifying basic principles to operating the technology in real-world environments. Between these two extremes, there are several intermediate steps that include scientific risk reduction, proof of concepts, laboratory tests, and validation and demonstration in relevant settings such as pilot lines or pilot plants.

Pilot lines are a relevant element in this advance along the TRL scale, particularly when developing solutions that facilitate quality improvement and waste reduction. A pilot line is a small-scale version of a production system used to test, evaluate, adapt, or even refine products, processes, or technologies. It serves as an intermediate step between research and full-scale production, facilitating the research and development (R&D) and serving as a test-before-invest environment [12], all without disrupting ongoing production processes.

Pilot lines play a crucial role in validating the technical viability and reducing risks. They are also a valuable tool for experimentation during the early stages of product design and for rapid prototyping [13]. Furthermore, pilot lines facilitate the production of proof-of-concept batches, enabling their use in refining the designs. Additionally, they serve as tools for technology adaptation and optimization, and represent a valuable complement in the human–technology–sustainability aspects of Industry 5.0.

In this context, this article aims to understand the usefulness of pilot lines as an intermediate step between research and production, while analyzing the status of pilot lines

in Industries 4.0 and 5.0. It also seeks to detail how the "Group of advanced Automation of Machines, Highly complex processes and Environments" (GAMHE) 5.0 pilot line has been designed and implemented. Additionally, it presents, from an applied perspective, a collection of case studies where this pilot line has been used in the industrial field. To achieve these objectives, the following points are addressed:

- Analysis of the state of the art related to pilot lines in different industries;
- Critical comparison of pilot lines for automation and robotics;
- Design and implementation of the GAMHE 5.0 pilot line;
- Demonstration of the use of the pilot line in several industrial cases studies;
- Discussion on the benefits and possible negative impacts of pilot lines.

These points are detailed in the following sections. Section 2 details the state of the art and provides an extensive review of automation and robotics pilot lines. Section 3 presents the GAMHE 5.0 line and details how it addresses the needs of Industry 5.0. Section 4 provides the information of case studies developed and validated on the pilot line. Finally, Sections 5 and 6 provide the discussion, the conclusions, and future work.

2. Pilot Lines in the Literature

The significant demand for variability and adaptability has led to a shift from mass production to the production of smaller batch sizes tailored to individual customer needs [14,15]. This has led to significant efforts in various sectors to design and implement pilot lines with the goal of staying aligned with technological advancements and meeting customer needs.

As stated previously, pilot lines are scaled-down versions or replicas of installations used in productive processes. These pilot lines may be larger in scale and then referred to as pilot plants. Regardless of their size, the objective of both pilot lines and pilot plants is to reduce unpredictability and risks, and serve as tools for development, testing, adaptation, or improvement of a production environment [16].

The concept of pilot lines originated during the rise of mass production as a method for testing alternatives and optimizing processes without impacting ongoing production. Today, these pilot lines are crucial elements in industry due to the need to adapt to new technologies, meet market requirements for continuous quality and process improvement, and serve as training tools.

This section is aimed to provide an overview of the literature related to pilot lines, starting with general approaches and moving on to those in Industry 4.0 contexts, and finally focusing on those most relevant to automation and robotics. Additionally, the objective is to examine the key features of the analyzed pilot lines and their contributions.

2.1. Methodology

The methodology applied for selecting the relevant literature involved searching for the term "Pilot line" across various academic indexers and search engines. Specifically, the following databases were used: ScienceDirect, Scopus, OpenAlex, and Google Scholar.

In ScienceDirect, the search was conducted using the phrase "pilot line", in quotation marks, to ensure that both singular and plural forms were included. This search yielded 1559 results as of 9 December 2024. In Scopus, an "Article title, abstract, keyword" search was performed for the term "pilot line". Since this search does not differentiate between singular and plural forms, it also included searches for "pilot lines". The result in Scopus was a total of 912 publications.

The bibliographic catalogue of open scientific articles, OpenAlex, was also used in the state-of-the-art analysis. For OpenAlex, the used search terms were "pilot line | pilot

lines" to include both singular and plural forms using the " | " (OR) operator. In this case, OpenAlex returned 6444 publications.

Lastly, Google Scholar was used due to its extensive coverage. The search term employed was "pilot line | lines" to ensure that both singular and plural forms were included. The search returned a result that Google Scholar estimates to be approximately 15,900 publications.

As shown in Figure 1, the interest in pilot lines has remained relatively stable over recent years on both Scopus and ScienceDirect, with slight increases in the number of publications. In OpenAlex, there was a peak in 2019, which declined in 2020 but then steadily increased again, surpassing the level of 2019 in 2023. For 2024, the number of publications is lower due to it being the current year and probably incomplete indexing. On Google Scholar, the level of publications remained relatively similar between 2019 and 2021, with a growth in 2022. Since 2022, the numbers have been slightly increasing.

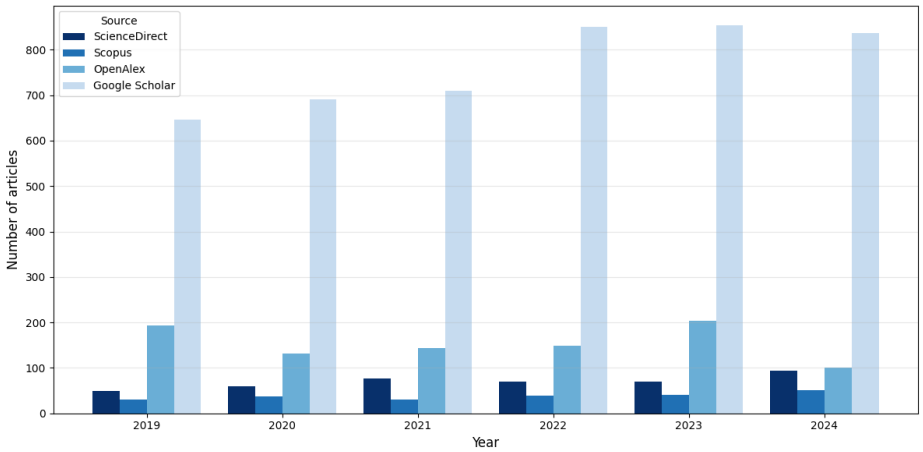


Figure 1. Articles including the term "Pilot Line".

Once the list of articles was retrieved, and given the large number of articles mentioning pilot lines in different ways, the search was refined using terms related to the scope of this article. Specifically, terms such as "industry", "robotics", "automation", "manufacturing" or "artificial intelligence". During the selection of articles, priority was given both to maintaining variety across different areas where pilot lines are used and to selecting publications highly related to Industry 4.0/5.0 concepts.

To identify ongoing projects and pilot line initiatives, the primary sources were the findings from the analysis of state-of-the-art articles and the additional information obtained through general search engine queries.

2.2. Industrial Pilot Lines

Regarding the academic literature, one relevant paper is [17]. In their research on industrial cyber–physical systems, particularly those involving milling processes, the authors identified several challenges such as the difficulty in configuring parameters to guarantee the required surface roughness and the absence of procedures for predicting surface roughness at both micro and macro scales. For this reason, Villalonga et al. developed a machine learning-based system to analyze and configure these parameters, testing and validating it on an Industry 4.0 pilot line. The authors report achieving positive results regarding response times and facilitating optimal settings.

Aligned with pilot line initiatives for Industry 4.0, the authors in [16] present a multi-technology sensorized Digital Twin of a line used to assemble medical devices. Their pilot line includes a linear motion platform and a sensor system, performing assembly steps

while verifying results. The case study addresses the need for accuracy in an assembly process that has a critical point where two components must engage precisely. The pilot line also enabled the experimentation with data augmentation and fault injection, generating large volumes of information from each assembly step, and facilitating the modeling of the Digital Twin. Additionally, it enabled testing processes at speeds valid for high-volume production.

In [18], the authors present a pilot line for computing chip manufacturing, specifically, an Extreme Ultraviolet Lithography (EUV) mask pilot line used at Intel Corporation. This pilot line is characterized by its reliance on a standard toolset, with a particular emphasis on high-definition and high-speed inspection tools. The use of the pilot line is justified through the conduct of studies and validation of approaches to address defect detection and process simplification challenges. In [19], another case study is presented for a line designed to detect and analyze defects on EUV masks, which is a crucial process in small-chip manufacturing. This pilot line is based on electron microscopes or other tools utilizing extreme ultraviolet light and it supports various types of inspection activities. The authors also present the pilot line as useful for benchmarking activities and have designed it in such a way that beta pilot line tools can be replaced with production tools when they become available.

Without leaving the field of electronic components and systems, ref. [20] presents an initiative associated with an innovation project that introduces a multi-stakeholder pilot line including producers of 8-inch silicon carbide (SiC) substrates and equipment developers, as well as SiC process technologists, Research and Technology Organizations (RTOs), and end-users. This consortium introduced the world's first 200 mm silicon carbide pilot line for power technology, targeting applications in the automotive power electronics sector. The research and development activities conducted in this pilot line focused on enhancing competitiveness, energy savings, and reducing CO_2 (carbon dioxide) emissions.

The field of electronics offers many other examples of activities utilizing pilot lines. For instance, in their analysis of trends in microelectronic technologies within the context of Industry 4.0 [21], the authors mention the European Union's "Two-Dimensional Experimental Pilot Line" project as one of the enablers for new research and call for a more mature Industry 4.0 that addresses changing market needs, increased control requirements, and automation advancements.

Likewise, ref. [22] focuses on operational optimization and upscaling pilot lines in the semiconductor industry. The authors integrated logistics simulation with discrete event simulation for production planning and control, demonstrating their solution through a pilot line. The article highlights complexity, investments for machine acquisition and operation, product variety, and short product life cycles as some of the challenges faced by the sector, and where pilot lines can be particularly useful. Their publication also conveys an interesting message that emphasizes the importance of pilot lines: "Due to the capital-intensive equipment required for semiconductor production, companies often use low-volume manufacturing to validate market demand, optimize production processes, and mitigate risks before scaling up to full production". These small-scale activities where technology validation, process optimization, and uncertainty reduction can occur are precisely where pilot lines prove valuable.

The idea of using small-scale and pilot lines for "what-if" analyses while avoiding disruptions on main production lines is discussed in [16]. These "what-if" scenarios also use pilot lines as a tool for generating information to support decision-making and serve as a data source for Digital Twins and dashboards.

Another pilot line, which focuses more on evaluating workstation setups rather than technologies, is presented in [23]. This pilot line facilitates tests aimed at reducing dead

times, unnecessary inventories, and bottlenecks in production processes within the clothing industry. In their work, the authors analyzed variations in efficiency when switching from a conventional production system to one that is modular and aligned with lean manufacturing principles. They suggest using mobile workstations in a modular system, allowing operators to perform multiple tasks, and present positive indicators related to improved production efficiency.

The intersection of lean manufacturing and pilot lines is also explored in the works presented in [24]. The authors adopted an automation and robotics approach in a pilot line for battery module production, specifically focusing on modules that use cylindrical cells. This pilot line incorporates the use of a cell testing system along with a pick-and-place robot, referred to as a "cell loading system" and a pulse-arc welding robot. These elements are used to validate a Six Sigma-based sustainable manufacturing approach. The sustainability in production is associated with the concept of the 6Rs: Rationalize, Reduce, Redesign, Reuse, Repair, and Recycle.

A relevant pilot line in this context is one of those used within the same research project to calculate station processes, supporting the transition from pilot line to production and reducing time-to-market [25]. The authors aimed to address the lack of a well-defined methodology for scaling up pilot lines and present an eight-workstation pilot line designed to demonstrate a virtual modeling tool integrated with an ontology model for calculating station process times.

Following the topic of battery production, an article that covers this subject along with pilot lines and sustainability is presented by von Drachenfels et al. [26]. The article is based on the idea that it is complex to scale environmental impact data measured in pilot lines to full-scale production environments. Specifically, they suggest that while pilot lines are common for increasing understanding of environmental impacts when producing battery cells, scaling these data to large-scale production systems is challenging. This situation becomes even more complicated with emerging technologies, due to which the data at the pilot line level needed for a life cycle assessment (LCA) are scarce. The authors highlight pilot lines as a key element in improving awareness of the environmental impact of various technologies. They conclude by suggesting that, to some extent, pilot lines can be used to estimate production environmental impacts but emphasize the importance of recognizing that certain variables will require specific methodologies.

The environmental impact, along with challenges in profitability and motivating new young employees is addressed in [27]. From an Industry 4.0 perspective and supported by pilot lines, the authors present a flexible, interoperable end-of-line solution specifically designed for the ceramic tile sector. The pilot line contributes to the development of new optimization tools, creation of new machinery, and integration of technical modules into existing solutions. In their article, they also present tests of new waterproofing treatments and an improved process control, detailing outcomes such as a 25% reduction in development time, a 25% decrease in development costs, and a 20% increase in efficiency and production capacity.

In the photovoltaic panel production industry, pilot lines have been used as development platforms and as tools for monitoring single processes [28]. Specifically, pilot production of hundreds of units of high-efficiency solar cells has provided relevant data and insights for future production scaling analysis. Another example is a multi-process, multi-manufacturing-tool pilot line approach for the rapid technological development of flexible and lightweight displays [29], which enabled a rapid and cost-effective method for process development and the fabrication of limited-size Thin-Film Transistor (TFT) arrays. This approach helped break down technological barriers to commercialization, as in many cases detailed previously.

The need to make manufacturing more autonomous by providing capabilities in perception, prediction, optimization, and control has also facilitated the creation of a pilot line and its associated Digital Twin in the field of polymer processing [30]. The pilot line includes an injection molding machine, robotic arms, conveyor belts, and inspection cameras. The Digital Twin is utilized for real-time monitoring, movement optimization (for both the robotic arm and the conveyor belt), and predictive maintenance.

Even in the aerospace sector, there are examples of pilot lines. The European Space Agency (ESA), under the PIOTS project, developed photonic packaging technologies meeting space requirements, as well as the packaging line process controls, and deployed a pilot line for the packaging of photonic integrated circuits [31]. Shifting from aerospace to aviation, this field has also examined the potential benefits of Industry 4.0 and how pilot lines can help mitigate investment risks and uncertainties. Specifically, in [32], the authors note that "no matter how good a technology is, if the companies do not feel that they can get the benefits from their huge investment, they will be reluctant to invest".

Adapting facilities to varying production volumes is also one of the challenges in the automotive sector. For this reason, in [33], the authors built customized low-cost robots for different scenarios. These robots consist of plug-and-produce robotic modules that can be composed in a design platform. In their case studies, they demonstrated the scenario of physical cooperation and task sharing between humans and robots on a car assembly pilot line, underlining improved usability and sensing capabilities.

Lastly, there are several articles in which the pilot line is not centered on a specific sector but rather serves to validate approaches and technologies across various fields. One such article discusses designing reconfigurable modular production systems to improve human–system integration [34]. The pilot line presented spans across diverse industries, including automotive, wood products, bicycles, bearings, and metal manufacturing. Specifically, the article showcases pilot lines equipped with agents supporting artificial intelligence (AI), the semantic web, machine learning, and flexible planning as part of a framework for agile and smart manufacturing.

In [35], the researchers present a study that will be validated across several multi-sectoral pilot lines. The study focuses on the concept of Zero Defect Manufacturing and quality control. The authors begin by acknowledging that legacy IT systems and a low level of digitization make technology integration challenging, and that some new technologies require investment when their benefits are uncertain. To address this challenge, they designed a multilayer Data Quality Management (DQM) architecture to deploy cyber–physical production systems for digitally enhanced quality management, ready to be tested in various pilot lines ranging from automation to healthcare. Similarly, ref. [36] presents a study on AI technologies and applications for production, and highlights the need for test beds, pilot lines, and factory lines as research and demonstration facilities.

Focusing on the generation of AI solutions and models, such as those related to human–AI collaboration, the authors in [37] address challenges associated with changing customer needs, which require agile and flexible strategies within the industry, to present solutions applicable across various sectors—such as food, plastic automotive components, and gear-box machinery. The authors aimed to demonstrate and validate these solutions on reduced-scale production lines.

The ODIN Project has published articles focusing on Open, Digital, Industrial, and Networking pilot lines as a means to research and validate modular and reconfigurable production systems based on robots. Specifically, ref. [38] aims at efficient and sustainable robotics for immediate implementation on the shop floor, detailing challenges related to manufacturing high volumes and diverse parts, operational complexity, and the necessity of flexible monitoring. The article demonstrates safe human–robot collaboration, perception capabilities, decision-making, and planning across various sectors, including automotive, white goods, and aeronautics. Regarding the white goods sector, this is further elaborated in [39], where challenges associated with interoperability with legacy systems, flexibility, and efficiency are extended. The authors present a use case for semi-automated quality control of washing machine control panels, achieving improved results in error detection, logging, and human–robot collaboration.

2.3. Pilot Lines for Workforce Skilling

Pilot lines are not only being deployed as tools for technology development and validation, or for mitigating technical and investment uncertainties. The idea of using pilot lines as a formative element and facilitator for capacity building is discussed in multiple articles.

A relevant perspective is provided by the authors of [40], who emphasize that new technologies are emerging faster than workers' ability to adopt them and use pilot lines for validating one of these technologies. They specifically focus on an educational teleoperation platform as a learning environment for heavy industrial robotics. The challenges identified during the pilot line validation include data transfer issues between the robot and user interface as well as ensuring the safety of laboratory staff while remotely teleoperating the equipment. Additionally, they highlight a challenge applicable to other teleoperated lines used for training: the need for mechanisms to recover from common errors that occur when learning.

A similar case related to education and training is presented in [41]. The authors rely on a human–robot collaboration pilot line designed for diesel engines to present a test case for Virtual Reality-based safety training. The article confirms the usefulness of the pilot line for evaluating the case study and suggests the value of Virtual Reality, at least at the pilot line level, for training operators on safety.

When discussing the digitalization of manufacturing Small and Medium Enterprises (SMEs), cost-related obstacles, as well as challenges associated with hardware and software investments, technological and implementation difficulties, and skills and knowledge gaps are often cited [42]. In their work, the authors summarize the industrial R&D needs for testing before investment and the lifelong learning, evaluating how pilot lines can be helpful. Their research focuses on three different pilot lines: Robolab Tampere, human–robot collaboration (HRC) pilot line, and the Tampere University of Applied Sciences (TAMK) fieldlab. In their analysis, they concluded that skills and technologies related to human–machine collaboration, robotics, cybersecurity, machine learning (ML), and artificial intelligence are the most relevant.

Expanding on the subject of pilot lines for training, ref. [43] emphasizes the role of learning factories in supporting digitalization and the adoption of artificial intelligence. The high demand for automation and new technologies also implies a need for new workforce skills, and pilot lines and learning factories facilitate AI training through practical tasks and applications. Specifically, the article focuses on additive manufacturing, human–robot collaboration, and Digital Twins. The perspective on pilot lines as a training tool is also linked to the concepts of "test-before-investment" and pre-production validation, ensuring that the workforce is adequately trained when investments in technologies for production are made.

Additionally, the article details specific objectives for learning pilot lines and factories, such as (i) strengthening technological and business competencies; (ii) developing AI-based Industry 4.0 applications; (iii) demonstrating and validating Industry 4.0 technologies; and (iv) facilitating the rapid implementation and application of results.

A somewhat different approach to workforce capacity building is presented in [12]. This article addresses the definition of requirements for Reconfigurable Pilot Lines (RPLs), which can be used for skill development and adapted to future training needs. The authors present various surveys conducted with industrial stakeholders and detail cybersecurity, robotics, human–machine collaboration, flexible manufacturing, and digital design and optimization as essential competencies in future industries. On this matter, they mention that, in addition to Learning Factories (LFs) and Teaching Factories (TFs) traditionally used to develop skills, RPLs enable a rapid and efficient way to introduce the workforce to new technologies and acquire knowledge.

2.4. Innovative Activities and Digital Factories

The trends towards supporting the research and development of pilot lines by the European Commission (EC) have facilitated the launch of several projects emphasizing the idea of using pilot lines as elements to bridge the gap between research and industrial production. For example, H2020 Pillar 2: Industrial Leadership defined six Key Enabling Technologies (KETs) in manufacturing (nanotechnology, micro- and nanoelectronics, photonics, advanced materials, industrial biotechnology, and advanced manufacturing systems) and promoted the idea of multi-KET pilot lines [44]. Within the Digital Europe Program, which aims to bring digital technologies and infrastructure to citizens, public administrations, and industry, the EC has specific calls in which pilot lines are fundamental components. The Chips Joint Undertaking, an initiative seeking to strengthen the European semiconductor industry, organized a call for the setup of pilot lines on advanced photonic integrated circuits and another call for proposals on operational activities of these pilot lines during 2024. In this way, they are stimulating innovation through pilot lines and experimental facilities in the field of semiconductors.

Some of the projects detailed in the state-of-the-art review are framed within these calls and similar initiatives. For instance, FAMES (https://fames-pilot-line.eu/ (accessed on 27 December 2024)) is a project and a pilot line aimed at developing more efficient and sustainable semiconductor technologies, or NanoIC (https://www.nanoic-project.eu/ (accessed on 27 December 2024)). The project led by the center for nanoelectronics and digital technology research, IMEC, together with a consortium of companies, tackles the efficiency, affordability, and sustainability in chip manufacturing. Another example is PhotonHub (https://www.photonhub.eu/pilot-lines/ (accessed on 27 December 2024)), which serves as an innovation hub for photonics and covers the concepts of "test-before-invest" and upskilling opportunities discussed throughout this work. PhotonHub presents seven different pilot lines where technologies such as chemical sensors or medical applications can be validated. Also, the AMPLiFII2 (https://gtr.ukri.org/projects?ref=104175 (accessed on 27 December 2024)) project, funded by the UK Research and Innovation program, leverages the concept of pilot lines to validate some of its developments and begin producing battery modules at small scales, but ready for high-scale production.

When analyzing public and private initiatives related to pilot lines, it is possible to find a wide variety with objectives and characteristics adapted to different specific needs. Some examples launched in recent years are the pilot line production services offered by RISE (Research Institutes of Sweden), covering a full addition manufacturing development cycle (https://www.ri.se/en/am-center/our-offer/pilot-line-production (accessed on 27 December 2024)), or the pilot line for manufacturing coatings, including different equipment

and which is adaptable to the specific needs of industrial applications, presented by the SPRI Development Agency and the Tekniker research center (https://bdih.spri.eus/en/pvd-pilot-line-coating-facilities-for-industrial-components/ (accessed on 27 December 2024)).

Regarding the lines most related to automation and robotics, it is worth referencing the pilot line of the Booster Manufacturing Lab at CTAG (Automotive Technology Center of Galicia), which was used in works such as [45]. This line is part of an Industry 5.0-oriented laboratory and consists of two vehicle stations equipped with collaborative robots, mobile robots, and machine vision systems.

In [46], the pilot line from the Bi-Rex Competence Center (Big Data Innovation and Research Excellence) was used. The article presents a reconfigurable mobile robotic manipulator that relies on the hardware and workspaces of the Bi-Rex pilot line. Bi-Rex is presented as a pilot line integrating innovative technologies and traditional solutions within an interconnected ecosystem. The line focuses on the implementation of advanced collaborative robotics, leveraging IoT, Big Data, and AI technologies to support tasks in smart and additive manufacturing. Another infrastructure similar to Bi-Rex is MADE, a competence center focused on Industry 4.0. In [47], MADE integrates training, technology, serious games and Lean 4.0 principles to facilitate the digital transformation of manufacturing companies.

In [48], the Competence Industry Manufacturing 4.0 (CIM4.0) digital factory pilot line is involved in cybersecurity-related research, including testing Secure Quantum Key Distribution over fiber optic networks and using elements of the pilot line as endpoint for secure data communication when connecting with edge nodes. The CIM4.0 digital factory is designed for organizations to test innovative technologies that can improve business competitiveness, with advanced robotics being one of those included in this group of novel technologies.

Improving quality in Industry 4.0 and Zero Defect Manufacturing are concepts being addressed in innovation projects like Qu4lity (https://qu4lity-project.eu/ (accessed on 27 December 2024)). In Qu4lity, multinational companies like Philips present their OneBlade pilot line (https://digitalfactoryalliance.eu/progress-on-the-philips-oneblade-pilot-line/ (accessed on 27 December 2024)) as a tool for enhancing production efficiency and product quality. For achieving these objectives, they use, for example, AI applications for visual quality control and machine learning applied to the movements of robots on the pilot line. The extracted data are exported to dashboards for decision-making and improvement of production processes.

Lastly, it is worth mentioning the human–robot collaboration pilot line at Tampere University, an active line that has been utilized in some of the articles referenced in the state-of-the-art section [40,49]. This line incorporates robots, AR/VR, Digital Twins, and computer vision systems, and serves both as a reconfigurable system for research purposes and as a showroom environment.

2.5. Key Features and Comparison

The most obvious feature of pilot lines is their scale. They are all scaled-down representations of production lines or complete factories. This scale also implies a cost control aspect, as it provides a controlled, cost-efficient, environment to identify improvements and detect issues. When referring to scale, production capacity is also considered. Part of the lines analyzed produce small batches or limited quantities to understand how they will behave in the market or to validate them before larger-scale production.

Not only is the scale common, but the scalability is also a key characteristic. The analyzed lines are, in general, modular, adaptable, or designed to be reconfigurable. This allows for flexibility both in the processes applied and target outcomes, on one hand, and on adaptation to production needs by scaling up and replicating the pilot line, on the other.

Regarding the flexibility, it is worth noting how several of these pilot lines serve general purposes or support various applications. General-purpose lines allow for testing technologies and processes across sectors, for example, human–machine collaboration, data augmentation mechanisms, safety methods, and fault injection approaches. Lines with various applications are designed to address multiple uses cases, usually also covering a few different sectors.

Another common characteristic is their use for risk mitigation and for reducing uncertainty. The concept of "test-before-invest" mentioned multiple times in the state of the art is closely related to this point. A smaller scale implies lower costs and allows for replicating parts of a production process where technologies and approaches can be tested without impacting factory production.

In some cases, generating data is costly; in others, data generation directly produces waste, such as when creating reference datasets for training quality control models. In other instances, the small-scale model enables anticipating how a larger system will behave and, thus, make decisions related to sustainability. The generation of these datasets, the preliminary development of scalable models, and the understanding of critical aspects and monitoring needs are additional capabilities provided by many pilot lines.

Regarding automation and robotics, as well as Industry 4.0 and Industry 5.0 concepts discussed in this article, it is important to note that a significant number of the works analyzed are based on Industry 4.0 or at least linked to manufacturing innovations. In terms of Industry 5.0, there is more diversity. If Industry 5.0 is understood as the integration of technologies, human-centric approaches and collaboration with skilled workforces, and sustainable practices aimed at reducing waste, the coverage varies widely.

Regarding technology, several analyzed lines either have robotic components or include AI agents and models. It is also worth noting that other technologies, like Digital Twins and Big Data, are mentioned across many analyzed works. At the human-centric level, human–machine collaboration and human-in-the-loop approaches are the most common. Within this human element of Industry 5.0, it is important to highlight the use of lines as training and capacity building tools. In terms of sustainability, concepts such as zero defects, quality control, reducing waste, and improving efficiency in time and resource usage are commonly found across the articles reviewed.

Table 1 presents a comparative analysis of the articles reviewed. This comparison shows the primary area of each article. In certain cases, the differences may not be significant, for example, in the semiconductor or electronics sectors. On the other hand, some articles mention generic pilot lines that are not sector-specific. These are marked as "general" to differentiate them from those mentioning multiple sectors, which are labeled with the term "various". The table also highlights the relationship with Industry 4.0 and the key concepts of Industry 5.0, including the connection to automation and robotics (A&R), the integration of AI technologies, and the focus on human-centric approaches and sustainability.

Table 1. State-of-the-art comparison.

Article	Area	I4.0	Industry 5.0			
			A&R	AI	Human	Sustain.
[12]	General	X	X	X	X	X
[16]	General	X	X	X		
[17]	General	X	X	X	X	X
[18]	Semicond.		X			
[19]	Semicond.		X			
[20]	Electronics	X				X
[21]	Microelec.	X	X			
[22]	Semicond.	X				
[23]	Clothing				X	
[24]	General	X	X			X
[25]	Batteries	X				
[26]	Batteries	X				X
[27]	Ceramics	X	X	X	X	X
[28]	Solar					X
[29]	Electronics				X	
[30]	Polymers	X	X			
[31]	Space		X			
[32]	Aviation	X			X	
[33]	Automotive	X	X			X
[34]	Various	X	X	X	X	
[35]	Healthcare	X	X		X	
[36]	General	X	X	X		
[37]	Various	X		X	X	
[38]	Various	X	X	X	X	X
[39]	White g.	X	X		X	
[40]	General	X	X	X	X	X
[41]	Training	X	X		X	
[42]	General	X	X	X	X	
[43]	Training	X	X	X	X	X
[48]	Security	X				
[50]	General	X	X		X	X

3. GAMHE Pilot Line

The GAMHE 5.0 pilot line (Figures 2 and 3) was established thanks to the research and development in artificial cognitive control systems and reference architectures within the framework of Industry 5.0. The system focuses on integrating computer numerical controls (CNCs) for machine tools, including a Deckel Maho DMC 75 V Linear high-speed machining center equipped with a Siemens 840D CNC, and a Kern-Evo ultraprecision micro-machining center featuring a Nano NT laser control and Heidenhain CNC.

To simulate a manufacturing production line, additional components were incrementally incorporated. These include two conveyor belts, a Universal Robots UR5e collaborative robotic arm, and a Han's Robot Elfin 10L-PRO collaborative robotic arm, both used for part handling alongside the machining centers. For inspection and quality control, an Allied Vision Mako G-192 camera was connected to a Raspberry Pi 4 Model B with 8 GB of RAM, enabling the development and testing of artificial vision routines. The system's most recent integration involved an AMR ROSbot XL HUSARION, equipped with a RPLIDAR A2 LiDAR sensor and an Intel RealSense stereoscopic camera, facilitating the transportation of manufactured components across different workstations.

Each element of the pilot line is equipped with an Internet of Things (IoT) node capable of communicating with its respective machine through various industrial protocols. The CNC Deckel Maho uses Open Platform Communications Data Access (OPC DA); the CNC

Kern Evo employs Language for Service and Visualization version 2 (LSV-2); both UR5e robotic arm and the Allied Vision Mako G-192 camera use TCP-IP; finally, EtherCAT is utilized for the Elfin 10L-PRO robotic arm. All these IoT nodes communicate with one another through a central server functioning as an MQTT broker. This configuration enables each machine to interact with other IoT nodes within the pilot line, transmit sensor data, report operational states and alarms, and receive operation commands. The IoT nodes run on two Raspberry Pi 3B+, two Raspberry Pi 4B with Raspbian/Linux, and two MSI Core i3 mini-computers with Windows operating systems.

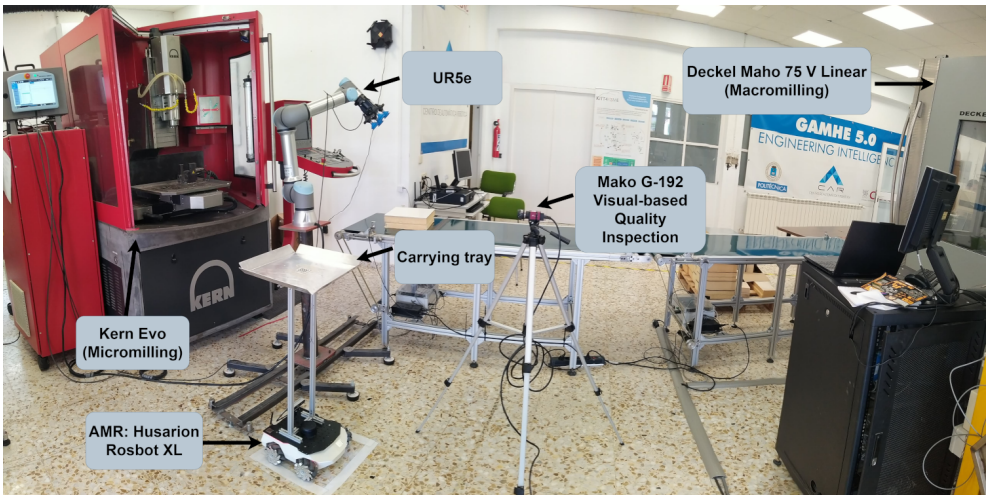


Figure 2. GAMHE 5.0 pilot line I.



Figure 3. GAMHE 5.0 pilot line II.

The GAMHE 5.0 pilot line is distinguished by integrating systems operating under different protocols and from various technological eras, combining early 2000s machining centers with state-of-the-art collaborative robots. Thus, this configuration serves as a case study on implementing digital transformation in established manufacturing plants. The system enables bidirectional communication among all elements of the pilot line, facilitating performance optimization through precise coordination of its various components. Additionally, the versatility of the CNC machines allows for the execution of diverse manufacturing operations, making it possible to adapt the line for producing a wide range

of products. Within this interconnected environment, the pilot line is both flexible and scalable, with the capability to integrate new systems and adapt to different purposes or use cases.

Another notable feature of the pilot line is the integration of artificial vision systems for both inspection and quality control, as well as for enabling robotic arms to interact with workpieces. The UR5e robotic arm is equipped with a conventional camera mounted on its wrist for object detection. Additionally, an Intel RealSense stereoscopic camera was incorporated to the Elfin 10L-PRO robotic arm. For this robot, custom software was developed to identify various workpieces and determine their position and orientation relative to a predefined coordinate system.

Another advantage of the pilot line is its reconfigurability and adaptability to different use cases. For example, a typical setup used in quality control processes is detailed in Figure 4. In this setup, all nodes of the GAMHE 5.0 pilot line (A1, A2, A3, A4, A5, A6, A7, I) are capable of sending and receiving messages using the MQTT protocol. These messages contain status information and operation commands for the respective nodes. The process starts with the first message sent from a human–machine interface to node A1 to pick up the part and position it in node A2. Each node then reports the end of its operation and sends a message to the next node to continue the process or notify the operator of any anomalies. At node I, quality control is performed, resulting in three outputs: (i) O1: the process continues at node A6; (ii) O2: the part is transferred to the storage of damaged parts; and (iii) O3: the part is marked for reprocessing. The output of the inspection stage determines the destination to which AMR A7 should transport the part.

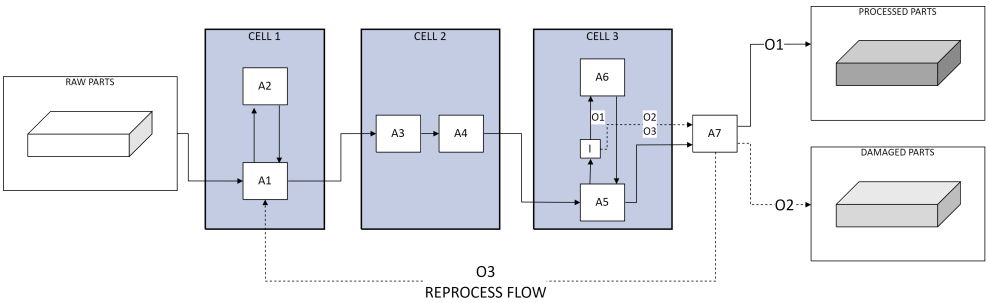


Figure 4. GAMHE 5.0 functional diagram: (A1) Hans Robot Elfin 10L-PRO robotic arm; (A2) Deckel Maho DMC 75V Linear high-speed machining center; (A3) conveyor belt; (A4) conveyor belt; (A5) Universal Robots UR5e robotic arm; (I) Inspection Allied Vision Mako G-192 camera; (A6) Kern Evo ultraprecision micro-machining center; (A7) AMR 460 ROSbot XL HUSARION; (O1) output 1, processed parts; (O2) output 2, damaged parts; and (O3) output 3, reprocessed parts.

3.1. Development and Continuous Improvement of the Pilot Line

Currently, data acquisition software is under development to facilitate the collection of data required for experimental designs. These programs enable real-time data capture and analysis, with the option of local storage at each IoT node or centralized database management. Additionally, they incorporate advanced tools for time-series data analysis and visualization. These data acquisition systems form the foundational infrastructure for developing Industry 5.0 applications such as predictive maintenance, machine and process optimization, quality control, modeling, simulation, and Digital Twin development.

In addition to the automated elements of the line, data are also collected from human operators. Developing a Human Digital Twin (HDT) is a crucial step towards harnessing the full potential of Industry 5.0. In this context, data are gathered from wearable devices that connect and transmit sensor readings via MQTT for processing. The primary objective is to use these metrics to adjust task and resource allocation, minimizing operator fatigue and stress. It is essential to establish correlations between sensor-generated data and

workers' self-assessments of task performance. These Human Digital Twin and human-in-the-loop experiments enable addressing part of the requirements related to human-centricity. Specifically, the information obtained from sensors and wearables enables adapting processes to operator needs. The human-in-the-loop in the pilot line also facilitates the empowerment of the operator in terms of machine supervision and control, which is one of the key elements of the human-centricity. These points, along with adaptive interfaces and explainability in AI models, allow for balancing technology with workforce needs.

3.2. Integration with the Legacy Systems

The GAMHE 5.0 pilot line includes both legacy and modern systems. Some of the older systems, over 15 years old, include the Deckel Maho DMC 75 V Linear high-speed machining center and the Kern Evo ultraprecision micro-machining center. On the other hand, state-of-the-art technologies such as the UR5e robotic arm, ARM ROSbot, and machine vision systems for quality control are also integrated. To ensure seamless communication between these diverse systems, specialized nodes were developed to translate MQTT messages into each system's respective communication protocol.

For example, for the Kern Evo integration (Figure 5), an MSI Core i3 mini-computer with Windows operating systems is used as IoT Node. This node has a software written in C++ with a library for MQTT messaging and a library for communication through the Language for Service and Visualization version 2 protocol of the Heidenhain SDK to execute direct numerical control operations such as starting or stopping a machining program. This node connects directly to the Heidenhain iT530 CNC via an ethernet cable, in parallel, the wireless network adapter is used to connect to the local network of the MQTT Broker.

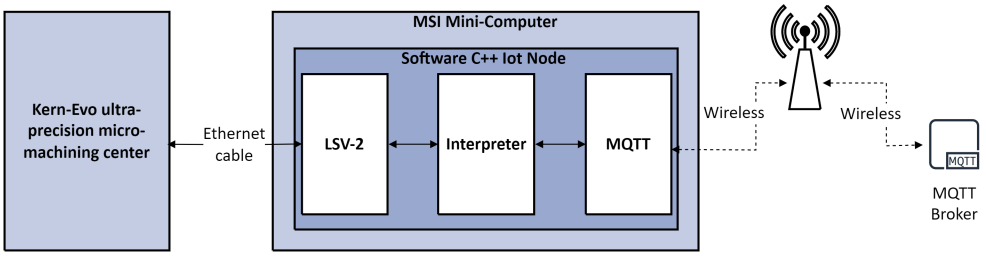


Figure 5. Kern Evo ultraprecision micro-machining center integration.

A similar approach is used to integrate the Deckel Maho DMC 75 V Linear high-speed machining center (Figure 6). In this case, the communication protocol with the machine is OPC DA-integrated in a software written in Python 3.9+ programming language.

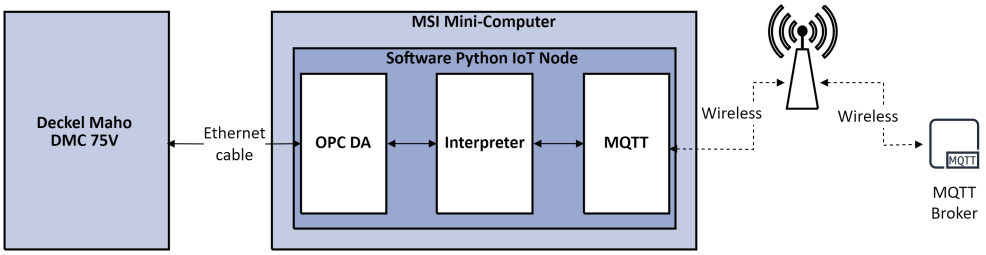


Figure 6. Deckel Maho DMC 75 V integration.

In general, IoT nodes are composed of a machine/system protocol, an interpreter, and the MQTT protocol. In this way, the MQTT protocol provides a common communication layer for all nodes in the pilot line, including those proxying legacy systems.

3.3. Comparative Analysis

Consistent with the majority of lines previously analyzed, GAMHE has a clear focus on Industry 4.0. Due to its generic nature and adaptability, it can be classified as a general pilot line with a clear emphasis on automation. Regarding Industry 5.0, the key technology aspects focus on robotics, artificial intelligence, IoT, and communications. This makes GAMHE similar to the lines discussed in [16,36,50].

In terms of human aspects, GAMHE places significant emphasis on human-centric design, being prepared for validation of human-in-the-loop cases and Human Digital Twins. It also serves as a valid educational tool and is fully replicable as a capacity-building asset, like those mentioned in [40,43].

GAMHE focuses on predictive maintenance and quality control, prioritizing early defect detection and reducing waste at the end of production lines. This aligns the line closely with the examples provided in [24,26]. All these lines highlight sustainability as a key element of the new industrial paradigm.

In terms of hardware, GAMHE shares certain similarities with [30,33,34,38,40]. It aligns with [30] in the use of robots, conveyor belts, and vision systems, and with [40] in its focus on human–robot collaboration, robotic arms, and IoT protocols. With [33,38], it shares similarities in mobile platforms and AMRs (Autonomous Mobile Robots). Like [34], GAMHE focuses on Digital Twins, cyber–physical systems, and the acquisition and generation of data. In turn, GAMHE includes CNC machinery for macro- and micro-milling.

Its modularity and scalability are comparable to those found in [33]. Reconfigurability and flexibility are highly demanded in manufacturing industries and are therefore fundamental requirements when designing lines like GAMHE. This flexibility and scalability enable the line to validate multiple technologies and processes. Additionally, it offers the capability to generate and manage data for subsequent analysis or integration into models, in a similar manner to [26,28].

Lastly, a key aspect of GAMHE 5.0, also highlighted as a key feature in other works [35], is its ability to integrate legacy equipment with innovative technologies. This capability enables the line to support a broad range of devices and technologies and adapt to various use cases.

4. Case Studies

As mentioned previously, flexibility and adaptability are two of the key aspects of GAMHE 5.0. This section presents a summary of relevant case studies that the pilot line has supported in recent years.

4.1. Digital Twin-Based Optimization for Ultraprecision Motion Systems with Backlash and Friction (2019)

The study focuses on developing a Digital Twin for modeling and optimizing an ultraprecision motion system with backlash and friction, commonly found in CNC machining centers. The primary objective is to minimize the maximum positioning error without significantly increasing control effort. The Digital Twin (DT) includes a virtual representation of mechanical and electrical components, models of nonlinearities such as backlash and friction, and integration of a cascaded P-PI control system. Four methods were used to optimize the controller parameters: a standard industrial procedure (FT) and three gradient-free heuristics (SA, GA, and CE). The study demonstrates the DT's effectiveness in improving control system behavior and highlights the superiority of gradient-free heuristics over the FT method, with the cross-entropy (CE) method standing out for its speed and efficiency.

In particular, improvements in the maximum absolute errors of 28% (GA), 26% (CE), and 20% (SA) with respect to the FT method were demonstrated. Considering the integral time absolute error the improvements were 21% (GA), 23% (CE) and 19% (SA). The parameters obtained through the optimization heuristics did not result in an increase of more than 3% in the control effort, as measured by the Integral of the Absolute Control Signal. These results validate the feasibility of DT-based optimization for improving ultraprecision motion systems, paving the way for industrial implementation [51].

4.2. Cloud-Based Industrial Cyber–Physical System for Data-Driven Reasoning: A Review and Use Case on an Industry 4.0 Pilot Line (2020)

The paper describes a cloud-based Industrial Cyber–Physical System (ICPS) designed for smart manufacturing, validated on an Industry 4.0 pilot line. Its main objective was to present a cloud-based data-driven reasoning process for reparametrizing predictive models in edge modules. Specific goals included improving surface roughness quality prediction, implementing a monitoring and prediction system in smart manufacturing scenarios, and surpassing the performance of standalone predictive models. The methodology involved describing the pilot line, generating and training data models, configuring the cloud-based ICPS, and validating the system. The study advances the state of the art by proposing a cloud-based ICPS, demonstrating the effectiveness of reinforcement learning through the Q-Learning method, and validating the system on a pilot line, showing improved prediction accuracy and reduced computational load. For example, in the case of computational load, the latency for the best model selection with the reasoning module was 0.085 s, compared to 1.10 s for the brute-force search. Limitations include the need for adaptation to other processes and the representativeness of the dataset used [17].

4.3. Industry 5.0 Pilot Line Digital Twin (2024)

The study focuses on the design, development, and evaluation of Digital Twins for modeling operator, robot, and interaction behaviors in an experimental pilot line. The main goal was to optimize the production line within the Industry 5.0 context, which advocates a human-centered approach by integrating human intelligence and skills with advanced technologies. The study sought to establish safe and efficient human–robot interactions in a collaborative manufacturing environment. Contributions include integrating Digital Twin technologies with human–robot interaction, creating a virtual space for optimizing collaboration between humans and machines, demonstrating the use of FlexSim and Gazebo for decision-making and resource allocation in manufacturing systems, implementing communication protocols such as TCP/IP, ROS2, and MQTT for IoT nodes, machine tools, and wearable devices, and incorporating physiological data from wearables into human Digital Twins to enhance human–robot interaction and improve worker well-being [52].

4.4. Self-Reconfiguration for Smart Manufacturing Based on Artificial Intelligence (2024)

The study evaluates the integration of an AI-based solution for visual inspection, determines the optimal line parametrization using AutoML, and uses a fuzzy-logic-based reconfigurator to manage performance degradation. The methodology includes deep learning-based visual inspection, automated machine learning (AutoML) workflow automation, and development of a fuzzy-logic-based reconfigurator. The study demonstrates the feasibility of integrating AI solutions for self-reconfiguration in production lines, presents an end-to-end AutoML approach, and develops a fuzzy-logic-based reconfigurator.

In this case study, two segmentation models were trained for the inspection task, with the best one achieving an accuracy of 0.995 and an F1 score of 0.992. This model was deployed in the pilot line for product handling. Additionally, an AutoML approach was used to determie the optimal parametrization of the line, resulting in a model with an

R^2 of 0.963. This led to an expected 55.1% improvement in throughput compared to the training data, which aligned with the values achieved in real production at maximum capacity. Limitations include result generalization and hyperparameter optimization. Overall, the study suggests that self-reconfiguration can significantly improve the efficiency and flexibility of production systems [53].

4.5. Quality Inspection of Prefabricated Panels Made for Construction (2024)

This case study focuses on quality control from a Zero Defects Manufacturing perspective. Specifically, the pilot line is used to inspect the quality of thermal-insulating panels for the construction sector. The system relies on several components of the pilot line, including the manipulator robot, the industrial camera, and the AI IoT node running a segmentation algorithm. In this case, the system is complemented by a human operator and an Autonomous Mobile Robot, which ensures the operator's safety.

The results demonstrated improvements in various aspects related to Industry 5.0. For example, in the technological domain, quality control achieved a validation precision of 98.3% and a recall of 94.7%. Regarding the human aspect, the average operator physical activity time during quality control was evaluated across the three possible panel routes. Specifically, the time the operator was physically involved in the process was analyzed in relation to the total quality control process time. This time decreased from 14.46% to 8.69%, thereby reducing the operator's workload [54].

5. Discussion

Throughout this article, many of the advantages of pilot lines have been presented, and even some disadvantages have been detailed. This section aims to review these advantages and disadvantages, and to analyze the approach from a critical point of view.

The advantages of pilot lines are clear: having an at-scale line where organizations can investigate, test, validate, generate data, and even produce on a small scale is already interesting in itself, particularly because this can be performed without impacting the production or affecting critical processes.

In relation to this, the concept of "test-before-invest" has been mentioned throughout this article. This possibility is also one of the positive points, as pilot lines offer an efficient way to test technology, processes, communication, sustainability, or human operations in controlled environments and can later be applied to production or scaled up to more complex lines.

The reconfigurability and modularity of pilot lines are another advantage. This makes the lines flexible and adaptable to different cases, and also allows for independent component replacement with higher TRL solutions as they become available.

Small-scale production is another strength of this concept. In many cases, small batches need to be produced and it would not be profitable to adapt or reconfigure an entire production line. It also allows for predicting how a larger-scale production might work and evaluating methods of quality and efficiency.

The ability to mitigate risks and clear uncertainties regarding technology is another significant advantage, especially in an environment where advances in artificial intelligence have substantial impacts on different aspects of the production processes. Pilot lines facilitate research and development of new technologies, and most importantly, they allow for validation in environments close to operational ones. Related to this, the ability to generate data enables model training, decision-making, and analysis for efficient scaling.

Another clear advantage is the possibility of using pilot lines for technology transfer and facilitating workers' training on technologies that will soon be deployed in production. Capacity building, upskilling or reskilling workers can benefit from these small-scale

controlled environments. Lastly, pilot lines serve as demonstration and showcase elements, allowing stakeholders to view prototypes and innovations at early stages.

Regarding the challenges faced when scaling-up pilot lines, it is important to mention that while the initial cost may not be as high as implementing the changes on a full production line, costs can still be significant and may not be profitable in all cases. In some instances, considering virtual pilot lines, as presented in [41], or focusing on Digital Twins is a valid alternative.

The costs associated with maintaining and operating the pilot line can be substantial. Given the availability of pilot-line-as-a-service offerings and funded projects for their use, it might be worth exploring these options instead of setting up an in-house solution.

Modularity offers flexibility and helps mitigate the potential issues of obsolescence in rapidly changing technologies. However, a modular and adaptable pilot line may not be as efficient as one specifically designed for a particular process.

Regarding the efficiency, the yield achieved in pilot lines does not always hold in full-scale production. This can be due to the increased complexity of processes or potential bottlenecks when handling large volumes. Additionally, efficiency is also impacted by machine downtimes in production. While these stoppages in production are common for maintenance, they are usually not considered in pilot lines. Therefore, it is important not to assume that production in a pilot line scales linearly to full-scale production.

Finally, a critical point is the reliability and variability of results when moving from small-scale to large-scale production and across different levels of technological maturity. Von Drachenfels et al. highlight this issue, providing an example of the differences in environmental impact analysis results in the battery production sector [26].

Regarding the alignment with the principles of Industry 5.0 and zero defects, pilot lines demonstrate a perfect fit. While pilot lines experienced a surge during the era of Industry 4.0, under the new paradigm of Industry 5.0, they have the potential to become key elements. The Industry 5.0 component related to technology is evident in the relevance of pilot lines. General technologies and manufacturing-specific advancements are constantly evolving. Having an environment for prototyping, interoperability, and validation is crucial for advancing on the TRL scale. The human-centricity promoted by Industry 5.0, which is expected to arrive to production lines, also has an opportunity for testing and validation in pilot lines. In fact, many concepts of human-centricity, human–machine collaboration, and explainability are being evaluated in pilot lines.

Lastly, although some challenges have been mentioned regarding the scalability of certain sustainability analyses from pilot lines to full-scale production lines, it is clear that pilot lines can be used to obtain some indicators and facilitate decision-making. Pilot lines also allow for the development of quality control models and predictive maintenance tools, aligning with zero defects and first-time-right principles, thereby reducing times, consumption, and waste.

It is important to mention that in certain cases, there is little distinction between pilot lines and testbeds. In theory, pilot lines are more focused on industrial environments, emphasizing integration, scaling, and production-oriented viability analysis. Testbeds, on the other hand, are primarily oriented towards experimenting with technologies in somewhat controlled settings. Pilot lines would be at higher Technology Readiness Levels and closer to production, whereas testbeds would be nearer to laboratory validation stages. Due to rapid technological evolution and availability of different technical solutions, pilot lines are often being used for experimentation and quick technological validation. As a result, there is some overlap between the two concepts. Sometimes, there is also some overlapping with research and development showrooms. In principle, pilot lines are focused on validating production-related processes, while showrooms are more oriented

towards showcasing products or proof of concepts for commercial purposes. However, the configurations of these showrooms tend to be scaled-down systems where technologies can be validated.

This overlapping also occurs when talking about competence centers. Both have equipment for research and development, and interest in the skilling of workers. Pilot lines are usually smaller in scale and focus more on validating technologies and processes, while competence centers are typically organizations for driving innovation. In many cases within the competence centers, there are pilot lines as detailed in [43,48].

In general, these overlaps do not reduce the importance of the pilot lines, but rather contextualize them and place them in broader ecosystems.

6. Conclusions

Industry 4.0 is evolving into Industry 5.0. Pilot lines have been one of the main elements of the modern industry. In the paradigm of Industry 4.0, they were key elements due to the needs for shorter production times and interoperability, and the expectations in terms of quality. In Industry 5.0, these needs are expanded to include new technologies, with a focus in human- and sustainability-related objectives.

The interest in pilot lines, their link with Industry 5.0, and understanding how automation and robotics fit in these ecosystems has led us to analyze the state of the art and evaluate different approaches to highlight advantages and disadvantages when designing and deploying lines on a reduced scale.

To this end, articles focused on pilot lines were searched in some of the main academic indexers and search engines, and we obtained thousands of results and observed how the works related to pilot lines increase slightly or are maintained year after year. From these articles, a subset was selected, refining the search by key terms related to Industry 4.0/5.0 concepts. These articles were analyzed, detailed, and compared. Once analyzed, the modular and scalable design followed in the GAMHE 5.0 pilot line was presented, in addition to a series of case studies carried out in the line. Also, the GAMHE line was compared with the ones detailed in the state of the art, to certify certain similarities in terms of approaches and trends.

Finally, the aforementioned advantages and disadvantages of pilot lines are presented, whereby the main advantages are the ability to reduce risks and eliminate uncertainty before investing, and as an element of testing and validation of technologies without impacting main production. Regarding the disadvantages, issues related to the scaling of sustainability indicators and also potential initial, maintenance, and operating costs are mentioned.

The scaling of sustainability indicators is an area of ongoing research, as several of the parameters used in life cycle assessments do not scale linearly [26]. Initial costs and operational costs may be justified by the benefits obtained. Some of these initial costs can be mitigated through the use of modular pilot lines, which can grow based on available funding. Additionally, a mix of models or Digital Twins can be used to replace some components and machines. Lastly, as discussed, pilot-line-as-a-service models can assist in reducing initial costs.

The pilot lines are also framed in the context of Industry 5.0 and quality concepts, such as Zero Defect Manufacturing. Also, the lines are compared with slightly overlapping concepts such as testbeds and competence centers.

The conclusion is that the pilot lines are relevant to adapt the organizations to the new Industry 5.0 and play a key role for validating technologies and processes. These lines could be key to the three pillars of the new industrial paradigm: regarding resilience, by enabling more robust, adaptable, and innovative production; in human-centricity, by

supporting the training of operators and testing approaches where humans are enhanced by machines; and lastly, in sustainability, by focusing on process and energy efficiency, as well as improving quality control processes to avoid waste.

The GAMHE 5.0 line, as a flexible and modular automation and robotics line, is an exponent of the pilot lines for Industry 5.0, and will facilitate new case studies related to technology, humans, and sustainability. Among these future cases, human-centricity and human–machine interaction are some of the activities that the authors are actively involved in. In particular, Human Digital Twins, to analyze data on fatigue, posture, and ergonomics, as well as human-in-the-loop approaches to empower the operator and enhance decision-making, are some of the main aspects planned to be addressed in future works.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
AMR	Autonomous Mobile Robot
A&R	Automation and Robotics
AutoML	Automatic Machine Learning
CE	Cross-Entropy
CNC	Computer Numerical Control
CSIC	Consejo Superior de Investigaciones Científicas. Spanish National Research Council
CTAG	Centro Tecnológico de Automoción de Galicia. Automotive Technology Center of Galicia
DA	Data augmentation
DQM	Data Quality Management
DT	Digital Twin
EC	European Commission
ESA	European Space Agency
EUV	Extreme Ultraviolet Lithography
FP	Framework Program
FT	Fine-Tune
GA	Genetic Algorithm
GAMHE	Group of advanced Automation of Machines, Highly complex processes and Environments
HDT	Human Digital Twin
HRC	Human–Robot Collaboration
I4.0-I5.0	Industry 4.0. Industry 5.0
ICPS	Industrial Cyber–Physical System

ICT	Information and Communication Technology
IP	Internet Protocol
JU	Joint Undertaking
KETs	Key Enabling Technologies
LCA	Life Cycle assessment
LFs	Learning Factories
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
RAM	Random Access Memory
RFT	Right First Time
RPLs	Reconfigurable Pilot Lines
RTO	Research and Technology Organizations
R&D	Research and Development
SA	Simulated Annealing
SiC	Silicon Carbide
SME	Small and Medium Enterprise
TCP	Transmission Control Protocol
TFs	Teaching Factories
TFT	Thin-Film Transistor
TRL	Technology Readiness Level
VR	Virtual Reality
ZDM	Zero Defect Manufacturing

References

1. Kumar, R.; Singh, R.K.; Dwivedi, Y.K. Application of industry 4.0 technologies in SMEs for ethical and sustainable operations: Analysis of challenges. *J. Clean. Prod.* **2020**, *275*, 124063. [CrossRef]
2. Zeid, A.; Sundaram, S.; Moghaddam, M.; Kamarthi, S.; Marion, T. Interoperability in smart manufacturing: Research challenges. *Machines* **2019**, *7*, 21. [CrossRef]
3. Lasi, H.; Fettke, P.; Kemper, H.G.; Feld, T.; Hoffmann, M. Industry 4.0. *Bus. Inf. Syst. Eng.* **2014**, *6*, 239–242. [CrossRef]
4. Yin, Y.; Stecke, K.E.; Li, D. The evolution of production systems from Industry 2.0 through Industry 4.0. *Int. J. Prod. Res.* **2018**, *56*, 848–861. [CrossRef]
5. Leng, J.; Sha, W.; Wang, B.; Zheng, P.; Zhuang, C.; Liu, Q.; Wuest, T.; Mourtzis, D.; Wang, L. Industry 5.0: Prospect and retrospect. *J. Manuf. Syst.* **2022**, *65*, 279–295. [CrossRef]
6. Breque, M.; De Nul, L.; Petridis, A. *Industry 5.0: Towards a Sustainable, Human-Centric and Resilient European Industry*; European Commission, Directorate-General for Research and Innovation: Luxembourg, 2021; Volume 46.
7. Gregoire, C. Understanding and Increasing Right First Time (RFT) Performance in a Production Environment: A Case Study. Ph.D. Thesis, Kansas State University, Manhattan, KS, USA, 2017.
8. Poppendieck, M. Principles of lean thinking. *IT Manag. Sel.* **2011**, *18*, 1–7.
9. Psarommatis, F.; May, G.; Dreyfus, P.A.; Kiritsis, D. Zero defect manufacturing: State-of-the-art review, shortcomings and future directions in research. *Int. J. Prod. Res.* **2020**, *58*, 1–17. [CrossRef]
10. Liu, J.; Jiang, X.; Shi, M.; Yang, Y. Impact of artificial intelligence on manufacturing industry global value chain position. *Sustainability* **2024**, *16*, 1341. [CrossRef]
11. Mankins, J.C. Technology readiness levels. *White Pap.* **1995**, *6*, 1995.
12. Siivonen, J.; Pöysäri, S.; Hakamäki, A.M.; Lanz, M.; Salminen, K.; Ijas, M.; Aho, M.; Nieminen, H. Reconfigurable pilot lines enabling industry digitalization: An approach for transforming industry and academia needs to requirements specifications. *Procedia CIRP* **2022**, *107*, 1226–1231. [CrossRef]
13. Saffaro, F.A.; Trescastro, M.G.; Rodriguez, A.; Schramm, F.; Formoso, C.; Heineck, L. Prototyping contributions for production management in construction. In *Proceedings of the 14th Annual Conference of the International Group for Lean Construction, Santiago, Chile, 25–27 July 2006*; International Group for Lean Construction: Santiago, Chile, 2006; pp. 25–27.
14. Unger, H.; Börner, F.; Müller, E. Context related information provision in Industry 4.0 environments. *Procedia Manuf.* **2017**, *11*, 796–805. [CrossRef]
15. Kletti, J.; Schumacher, J. *Die Perfekte Produktion*; Springer: Berlin/Heidelberg, Germany, 2011; pp. 1–10.

16. Kakavandi, F.; Gomes, C.; de Reus, R.; Badstue, J.; Jensen, J.L.; Larsen, P.G.; Iosifidis, A. Towards developing a digital twin for a manufacturing pilot line: An industrial case study. In *Digital Twin Driven Intelligent Systems and Emerging Metaverse*; Springer: Berlin/Heidelberg, Germany, 2023; pp. 39–64.
17. Villalonga, A.; Beruvides, G.; Castano, F.; Haber, R.E. Cloud-based industrial cyber–physical system for data-driven reasoning: A review and use case on an industry 4.0 pilot line. *IEEE Trans. Ind. Inform.* **2020**, *16*, 5975–5984. [CrossRef]
18. Stivers, A.R.; Yan, P.Y.; Zhang, G.; Liang, T.; Shu, E.Y.; Tejnil, E.; Lieberman, B.; Nagpal, R.; Hsia, K.; Penn, M.; et al. EUV mask pilot line at Intel Corporation. In Proceedings of the 24th Annual BACUS Symposium on Photomask Technology, Monterey, CA, USA, 14–17 October 2004; SPIE: Bellingham, WA, USA, 2004; Volume 5567, pp. 13–22.
19. Chan, Y.D.; Rastegar, A.; Yun, H.; Putna, E.S.; Wurm, S. EUV mask defect inspection and defect review strategies for EUV pilot line and high volume manufacturing. In Proceedings of the Extreme Ultraviolet (EUV) Lithography, San Jose, CA, USA, 22–25 February 2010; SPIE: Bellingham, WA, USA, 2010; Volume 7636, pp. 454–469.
20. Messina, A.A.; Imbruglia, A.; Calabretta, M.; Vinciguerra, V.; Moise, C.C.; Sitta, A.; Enachescu, M.; Roccaforte, F. The "first and euRopEAn siC eighT Inches pilOt liNe": A project, called REACTION, that will boost key SiC Technologies upgrading (developments) in Europe, unleashing Applications in the Automotive Power Electronics Sector. In Proceedings of the 2020 AEIT International Conference of Electrical and Electronic Technologies for Automotive (AEIT AUTOMOTIVE), Catania, Italy, 23–25 September 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 1–6.
21. Wang, Y. Research Progress on Key Technologies of Microelectronics for Industry 4.0. *Acad. J. Sci. Technol.* **2022**, *2*, 4–6. [CrossRef]
22. Preising, H.; Gutmann, D.; Wolf, M. Application of Logistics Simulations for Scaling Up a Semiconductor Pilot Line: A Case Study. In Proceedings of the Conference on Production Systems and Logistics 2024, Honolulu, HI, USA, 9–12 July 2024; publish-Ing.: Hannover, Germany, 2024; pp. 563–572. Available online: https://repo.uni-hannover.de/items/6b6dfc6a-18d6-4868-8592-494c52929b98 (accessed on 22 February 2025).
23. Flores, A.; Aguilar, J.; Choque, R. Implementing of a pilot line of clothing industry under the philosophy lean manufacturing. *Int. J. Mater. Mech. Manuf.* **2014**, *2*, 214–228.
24. Titmarsh, R.; Assad, F.; Harrison, R. Contributions of lean six sigma to sustainable manufacturing requirements: An Industry 4.0 perspective. *Procedia CIRP* **2020**, *90*, 589–593. [CrossRef]
25. Chinnathai, M.K.; Al-Mowafy, Z.; Alkan, B.; Vera, D.; Harrison, R. A framework for pilot line scale-up using digital manufacturing. *Procedia CIRP* **2019**, *81*, 962–967. [CrossRef]
26. von Drachenfels, N.; Engels, P.; Husmann, J.; Cerdas, F.; Herrmann, C. Scale-up of pilot line battery cell manufacturing life cycle inventory models for life cycle assessment. *Procedia CIRP* **2021**, *98*, 13–18. [CrossRef]
27. Raffaeli, R.; Pazzi, L.; Pellicciari, M. Industry 4.0 Solutions as Enablers for the Sustainability of the Italian Ceramic Tiles Sector. *Sustainability* **2024**, *16*, 4301. [CrossRef]
28. Engelhart, P.; Wendt, J.; Schulze, A.; Klenke, C.; Mohr, A.; Petter, K.; Stenzel, F.; Hörnlein, S.; Kauert, M.; Junghänel, M.; et al. R&D pilot line production of multi-crystalline Si solar cells exceeding cell efficiencies of 18%. *Energy Procedia* **2011**, *8*, 313–317.
29. Raupp, G.B.; O'Rourke, S.M.; Moyer, C.; O'Brien, B.P.; Ageno, S.K.; Loy, D.E.; Bawolek, E.J.; Allee, D.R.; Venugopal, S.M.; Kaminski, J.; et al. Low-temperature amorphous-silicon backplane technology development for flexible displays in a manufacturing pilot-line environment. *J. Soc. Inf. Disp.* **2007**, *15*, 445–454. [CrossRef]
30. Singh, M.; Srivastava, R.; Lee, S.; Murray, N.; Qiao, Y.; Devine, D. Digital Twin of Polymer Processing Pilot Line. In Proceedings of the 37th International Manufacturing Conference by Irish Manufacturing Council, Athlone, Ireland, 7–8 September 2021; pp. 7–8.
31. Rotta, D.; Preve, G.; Rodrigo, A.S.; Chiesa, M.; Cucinella, G.; Bogoni, A. The ESA project PIOTS: Setting up a complete pilot line for the packaging of photonic integrated circuits for space applications. In Proceedings of the International Conference on Space Optics—ICSO, Virtual, 30 March–2 April 2020; SPIE: Bellingham, WA, USA, 2021; Volume 11852, pp. 2259–2270.
32. Ghafar, N.; Harun, K.; Lailawati, M.; Rahman, N. Preliminary investigation of Industry 4.0 readiness among small and medium aviation companies in Malaysia. In *Journal of Physics: Conference Series*; IOP Publishing: Bristol, UK, 2020; Volume 1573, p. 012030.
33. Wögerer, C.; Bauer, H.; Rooker, M.; Ebenhofer, G.; Rovetta, A.; Robertson, N.; Pichler, A. LOCOBOT-low cost toolkit for building robot co-workers in assembly lines. In Proceedings of the Intelligent Robotics and Applications: 5th International Conference, ICIRA 2012, Montreal, QC, Canada, 3–5 October 2012; Proceedings, Part II 5; Springer: Berlin/Heidelberg, Germany, 2012; pp. 449–459.
34. Sidorenko, A.; Motsch, W.; van Bekkum, M.; Nikolakis, N.; Alexopoulos, K.; Wagner, A. The MAS4AI framework for human-centered agile and smart manufacturing. *Front. Artif. Intell.* **2023**, *6*, 1241522. [CrossRef]
35. Caccamo, C.; Eleftheriadis, R.; Magnanini, M.C.; Powell, D.; Myklebust, O. A hybrid architecture for the deployment of a data quality management (DQM) system for zero-defect manufacturing in industry 4.0. In Proceedings of the Advances in Production Management Systems. Artificial Intelligence for Sustainable and Resilient Production Systems: IFIP WG 5.7 International Conference, APMS 2021, Nantes, France, 5–9 September 2021; Proceedings, Part III; Springer: Berlin/Heidelberg, Germany, 2021; pp. 71–77.

36. Nolan, A. Artificial intelligence and the technologies of the Next Production Revolution. In *OECD Science, Technology and Innovation Outlook 2018*; Organisation for Economic Co-operation and Development: Paris, France, 2018; pp. 51–74.
37. Alvarez-Napagao, S.; Ashmore, B.; Barroso, M.; Barrué, C.; Beecks, C.; Berns, F.; Bosi, I.; Chala, S.A.; Ciulli, N.; Garcia-Gasulla, M.; et al. Knowledge project–concept, methodology and innovations for artificial intelligence in industry 4.0. In Proceedings of the 2021 IEEE 19th International Conference on Industrial Informatics (INDIN), Palma de Mallorca, Spain, 21–23 July 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 1–7.
38. Makris, S.; Michalos, G.; Kousi, N.; Papavasileiou, A.; Campos, U.E.; Bravo, J.O.; Cueva, H.H.; Siltala, N.; Wang, L.; Suppa, M.; et al. Open-Digital-Industrial and Networking pilot lines using modular components for scalable production–ODIN project approach. *Procedia CIRP* **2022**, *106*, 162–167. [CrossRef]
39. Karagiannis, P.; Xanthakis, V.; Apostolopoulos, G.; Michalos, G.; Makris, S. Quality control of white goods parts using robotic technologies. *Procedia CIRP* **2021**, *104*, 1759–1764. [CrossRef]
40. Pöysäri, S.; Kaarlela, T.; Dianatfar, M.; Lanz, M. Educational teleoperation platform for heavy industrial robotics as a learning environment. In Proceedings of the 13th Conference on Learning Factories (CLF 2023), Reutlingen, Germany, 9–11 May 2023; Elsevier BV: Amsterdam, The Netherlands, 2023.
41. Dianatfar, M.; Heshmatisafa, S.; Latokartano, J.; Lanz, M. Feasibility analysis of safety training in human-robot collaboration scenario: Virtual reality use case. In Proceedings of the International Conference on Flexible Automation and Intelligent Manufacturing, Detroit, MI, USA, 19–23 June 2022; Springer: Berlin/Heidelberg, Germany, 2022; pp. 246–256.
42. Lanz, M.; Latokartano, J.; Järvenpää, E.; Pöysäri, S.; Dianatfar, M.; Siltala, N.; Salminen, K.; Hilman, L.; Siivonen, J.; Aho, M. Development and deployment of services based on D-BEST methodology for robotics and production automation related pilot lines. In *AIP Conference Proceedings*; AIP Publishing: Melville, NY, USA, 2024; Volume 2989.
43. Gyulai, T.; Viharos, Z.J.; Kása, F.; Wolf, P. Upskilling SME Workforce by Learning Factories. In Proceedings of the 13th Conference on Learning Factories (CLF 2023), Reutlingen, Germany, 9–11 May 2023.
44. Butter, M.; Fischer, N.; Gijsbers, G.; de Heide, M.; Hartmann, C. *Horizon 2020: Key Enabling Technologies (KETs), Booster for European Leadership in the Manufacturing Sector*; Publications Office of the European Union: Luxembourg, 2014.
45. Regalado, S.; Pereira Domínguez, A.; Marchi, S.D.; Dacal-Nieto, A.; Triviño-Tonato, E. Design and validation of a knee exoskeleton prototype for squatting operations. *Int. J. Eng. Technol.* **2024**, *16*, 130–135. [CrossRef]
46. Baldassarri, A.; Bertelli, M.; Carricato, M. Design of a Reconfigurable Mobile Collaborative Manipulator for Industrial Applications. *J. Comput. Nonlinear Dyn.* **2023**, *18*, 091006. [CrossRef]
47. Masoero, L.; Moreta Maldonado, C.M. *Proposal for the Implementation of a Lean 4.0 Teaching Factory Inside the New MADE Competence Center of the Politecnico di Milano*; POLITesi, Politecnico di Milano: Milano, Italy, 2019.
48. Corrias, N.; Vagniluca, I.; Francesconi, S.; De Lazzari, C.; Biagi, N.; Menchetti, M.; Lombardi, G.; Scordato, A.; Gionco, V.; Mercinelli, R.; et al. Implementation of Italian industry 4.0 quantum testbed in Turin. *IET Quantum Commun.* **2024**, *5*, 46–51. [CrossRef]
49. Dianatfar, M.; Pöysäri, S.; Latokartano, J.; Siltala, N.; Lanz, M. Virtual reality-based safety training in human-robot collaboration scenario: User experiences testing. In *AIP Conference Proceedings*; AIP Publishing: Melville, NY, USA, 2024; Volume 2989.
50. Lutters, E. Pilot production environments driven by digital twins. *S. Afr. J. Ind. Eng.* **2018**, *29*, 40–53. [CrossRef]
51. Guerra, R.H.; Quiza, R.; Villalonga, A.; Arenas, J.; Castaño, F. Digital twin-based optimization for ultraprecision motion systems with backlash and friction. *IEEE Access* **2019**, *7*, 93462–93472. [CrossRef]
52. Alfaro, D.; Castaño, F.; Cruz, Y.J.; Villalonga, A.; Haber, R.E. Human-Centric Approach for the Digital Twin of Human-Machine/Robot Collaboration in an Industry 5.0 Pilot Line. *Rev. Iberoam. Automática Informática Ind.* **2025**, *22*, 1–5.
53. Cruz, Y.J.; Castaño, F.; Haber, R.E.; Villalonga, A.; Ejsmont, K.; Gladysz, B.; Flores, Á.; Alemany, P. Self-Reconfiguration for Smart Manufacturing Based on Artificial Intelligence: A Review and Case Study. In *Artificial Intelligence in Manufacturing: Enabling Intelligent, Flexible and Cost-Effective Production Through AI*; Springer: Cham, Switzerland, 2024; pp. 121–144.
54. Villalonga, A.; Cruz, Y.J.; Alfaro, D.; Haber, R.E.; Martínez-Lastra, J.L.; Castaño, F. Enhancing Quality Inspection in Zero-Defect Manufacturing Through Robotic-Machine Collaboration. In Proceedings of the 2024 7th Iberian Robotics Conference (ROBOT), Madrid, Spain, 6–8 November 2024; IEEE: Piscataway, NJ, USA, 2024; pp. 1–6.

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3.2 Interoperable Software Platforms for Introducing Artificial Intelligence Components in Manufacturing: A Meta-framework for Security and Privacy

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- **Resumen:** La irrupción de tecnologías avanzadas y el limitado conocimiento de las arquitecturas de software están dificultando que muchas pequeñas y medianas empresas manufactureras puedan seguir el ritmo de lo que se denomina la cuarta revolución industrial (Industria 4.0, Industria del Futuro). Las plataformas de orquestación de contenedores proporcionan capas de simplificación para requisitos clave como la interoperabilidad, la seguridad y la privacidad, y proporcionan mecanismos que permiten a las empresas y a los proveedores de tecnología centrarse en sus funcionalidades y objetivos específicos, en lugar de invertir un tiempo y un esfuerzo considerable en la plataforma subyacente en la que operará la solución. Este artículo se centra en estas plataformas y en los problemas a la hora de desarrollarlas, y propone un metamarco híbrido orientado al riesgo y al objetivo para el análisis de la seguridad y la privacidad. El metamarco utiliza estándares y marcos de seguridad y privacidad bien conocidos como referencia y se puede utilizar para comprender los activos y los requisitos y, en particular, para seleccionar y configurar contramedidas. Para la evaluación práctica del metamarco, se aplicó a un caso real. Este caso muestra cómo se analizaron las necesidades de la plataforma desarrollada en el proyecto KITT4SME para soportar, entre otros, cuatro casos piloto de fabricación y para definir las características clave de seguridad y privacidad que deben introducirse al implementar una plataforma de software para que las pequeñas y medianas empresas la adopten fácilmente.
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Research article

Interoperable software platforms for introducing artificial intelligence components in manufacturing: A meta-framework for security and privacy

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ABSTRACT

The irruption of advanced technologies and the limited knowledge of software architectures are making it difficult for many small and medium-sized manufacturing companies to keep up with what is being called the fourth industrial revolution (Industry 4.0, Industry of the Future). Container orchestration platforms provide layers of simplification for key requirements such as interoperability, security, and privacy, and provide mechanisms that allow companies and technology providers to focus on their specific functionalities and goals, instead of investing considerable time and effort in the underlying platform on which the solution will operate. This article focuses on these platforms and the issues when developing them, and proposes a risk- and goal-oriented hybrid meta-framework for security and privacy analysis. The meta-framework uses well-known security and privacy standards and frameworks as a reference and can be used to understand assets and requirements and, in particular, to select and configure countermeasures. For practical evaluation of the meta-framework, it was applied to a real case. This case shows how the needs of the KITT4SME project platform were analyzed to support, among others, four manufacturing pilot cases and to define the key security and privacy features that should be introduced when implementing a software platform for easy uptake by small and medium enterprises.

1. Introduction

Industry 4.0 is defined as the fourth industrial revolution [23], a revolution based on short innovation and development times, flexibility, and resource efficiency. Mainly driven by automation, digitalization, and the interaction between components, it aims to increase the competitiveness of industrial enterprises in general, and with them the entire manufacturing industry [14,28].

The idea of Industry 4.0, sometimes referred to as Industry 5.0 or bundled in Industries of the Future (IotF) [8], goes hand in hand with solutions and technologies, the most important of which are associated with interoperability, unification of digital with real

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workflows (cyber-physical systems), real-time capabilities and data analytics for decision-making [27]. All these technologies require platforms and software solutions that facilitate their implementation and, in some ways, simplify the complexity of non-functional requirements associated with novel IotF systems.

These drivers and technologies are part of the toolkit used to address the multiple challenges facing the industry. In [22] fifteen main challenges for Industry 4.0 are defined. Focusing on the main technical challenges, it is possible to identify concerns about the high initial cost (Challenge 3), the lack of resources for research and development in Industry 4.0 technologies (Challenge 6), the lack of IT-based infrastructures, both software and hardware (Challenge 9), the lack of knowledge and trained workforce (Challenge 10), or the fear of failure of Industry 4.0 technologies (Challenge 13).

These challenges are particularly acute in the realm of SMEs [17], the backbone of many global industries, whose size and capacity often make them less able to cope with the challenges and whose impact can be more severe if they fail. This is even more true with the advent of Artificial Intelligence (AI), which promises better times, accuracy, and results, as well as better decisions, but at the same time further complicates the above challenges [33]. The authors in [21] cite data privacy and security among the necessary issues in which digitization efforts should be undertaken by small and medium-sized enterprises (SMEs), and mention the issue of platform interoperability as a critical challenging aspect that needs to be addressed.

In [19], for example, after analyzing the state of artificial intelligence and the Internet of Things (IoT) in SMEs, the authors conclude on the complexity of the use of IA as it stands today *where SMEs lack the knowledge and resources to utilize the technology, even though it seemed to be beneficial for them*, and make a clear recommendation that is consistent with the goals of this article: *future research should focus on simplifying AI solutions for SMEs and thus make them more directly applicable to them*. The authors of [9] also agree with the above statement and suggest that the use of AI needs to be simplified, while the solutions should provide ways to measure the benefits so that the new AI-based digital solutions have a solid foundation to support decisions.

As for the Internet of Things, it is admittedly slow to be adopted by SMEs. Together with the use of cloud computing, this is the first step SMEs are taking towards Industry 4.0 [4]. Frameworks such as FIWARE are facilitating this adoption [12]. In addition to using standards for digitally representing real-world objects and managing contextual information (NGSI), FIWARE middleware facilitates component interoperability by supporting multiple IoT agents and protocols for networking and messaging such as MQTT or AMQP [5].

Container orchestration platforms can support dynamic composition and operations and help overcome the challenges and obstacles mentioned above [25]. These platforms can streamline development and integration and provide viable solutions to complex security, scalability, or interoperability requirements. For component-based systems, relying on mature platforms that support component orchestration can allow SMEs or AI providers to focus on the design and development of these components, knowing that the digital platform is based on well-known technologies, while also knowing that their solution can be deployed on component-based platforms with similar characteristics.

To study the needs of the manufacturing industry on its path to Industry 5.0, and, in particular, because these organizations usually struggle to find adequate solutions to support the developments related to the rapid advance of new technologies, especially artificial intelligence, a consortium of organizations teamed up in the KITT4SME project,[1] co-funded by the European Commission.

This consortium, composed of universities, technology providers, and industrial end users, dedicated part of the effort to co-create and develop a reference platform, that could be replicable, customizable, and extendable to several industrial cases, and could serve for the seamless introduction of innovative software components in their production systems.

The security and privacy needs of this reference platform, its clear practical orientation, and the pilot experiences in different fields of the industry have allowed the study and definition of this meta-framework, facilitating its validation and its orientation to be used in platform customizations such as KITT4SME, or other composable platforms for the industry.

The project consists of four main experimental pilots planned in four different manufacturing SMEs, supervised by a coach, and working with a technology provider. The scope of these four pilots, which are also used as a relevant environment for validation, can be summarized as follows:

- **Pilot in the injection molding industry:** aiming at deploying the mutualism concept in a continuously adaptive work environment, where humans and machines complement their capacities to achieve optimized manufacturing performances and unprecedented worker satisfaction.
- **Pilot project in the field of quality systems:** improving traditional inspection techniques with the innovative features of AI and retaining and reusing expert knowledge.
- **Pilot project in the field of precision hand tools and instruments:** by introducing AI-based digitalized quality control and establishing a process to identify and characterize best practices by combining different data sources.
- **Pilot in the electrical equipment sector:** ensuring quality and delivery at satisfactory levels under VUCA (Variability, Uncertainty, Complexity, Ambiguity) conditions that require zero defects, zero Breakdowns, and easy reconfigurability.

This ecosystem provided us with the requirements and initial information to develop the meta-framework, as well as the opportunity to apply and evaluate it. Applying the meta-framework allowed us to identify security and privacy needs in the platform, introduce elements that could mitigate the risks, and generalize an approach for analogous modular platforms.

[1] KITT4SME, Platform enabled kits of AI for an easy uptake by SMEs, GA 952119. https://kitt4sme.eu

This article builds on the Industry of the Future challenges, mentioned in this introduction, and focuses on designing and demonstrating a meta-framework that can be used to define the security and privacy requirements of interoperable platforms for industry. Specifically, the goal of the article is to identify and analyze critical security and privacy issues within interoperable software component-based platforms, while providing insights to manufacturing ecosystem developers and AI technology providers on how to approach security and privacy risk analysis of their solutions or deployments. In line with this goal, the contributions of this paper are:

- The analysis of different methodologies and frameworks for security and privacy risk analysis.
- The adaptation of the frameworks so that they can be applied to the development of software platforms and components.
- The development of a taxonomy of security and privacy risks within digital platforms for industry.
- The design of a method and a meta-framework to design industry-oriented software platforms that consider security and privacy from the beginning.
- The identification of sensitive and confidential information that could flow through the digital platform.
- The evaluation of privacy and security requirements and data sets and understanding the implications.
- The analysis of the security and privacy based on the individual risks of KITT4SME assets.
- The proposition of countermeasures to be investigated and addressed as part of the software platform infrastructure implementation.

Information on these items is presented in the various sections of this article. Section 2 provides contextual information and insights about the ecosystems and pilot projects. Section 3 defines the framework for security and privacy analysis. Section 4 focuses on the evaluation of the framework and describes how the security and privacy issues in the KITT4SME case are addressed. Section 5 identifies the challenges and limitations of the framework and shows how the framework can be applied to other architectures. Finally, Section 6 discusses the relevance of the approach and the results, and Section 7 presents the conclusions and outlines future activities.

2. Background, context and state of the art

As the Spanish Data Protection Authority (AEPD) summarizes in its privacy by-design guide [2], the Privacy-by-Design concept [11] was developed in the 90s by Anne Cavoukian, Ontario's Data Protection Commissioner, and was presented and accepted internationally in the Conference of Data Protection and Privacy Commissioners of 2009 and 2010.[2] The concept of Privacy-by-Design involves seven different foundational principles regarding privacy and data protection: i) Proactive not Reactive; ii) Privacy as the Default Setting; iii) Privacy Embedded into Design; iv) Full Functionality - Positive-Sum, not Zero-Sum; v) End-to-End Security; vi) Visibility and Transparency; and vii) Respect for User Privacy.

The European Data Protection Supervisor defines Privacy by Default as *[...] the principle according to which an organization (the data controller) ensures that only data strictly necessary for each specific purpose of the processing are processed by default (without the intervention of the user)*. The concept also implies that no actions should be required of the user to maintain the highest levels of privacy since strict privacy-related settings are applied by default.

The Privacy-by-Default concept is closely aligned with the foundational principle #2 of the Privacy-by-Design approach, which also points out the importance of automatically protecting and guaranteeing privacy without user intervention.

Among the approaches for privacy, Cybersecurity for Privacy by Design, or C4P, is worth mentioning. C4P is a term used by the Secure Controls Framework Council to describe the concept of addressing the confidentiality and integrity of aspects of privacy, focusing on people, processes, and technology (PPT). As stated in the Secure Controls Framework, the goals of C4P are to: i) Enable privacy; ii) Pre-set security configuration settings so that they are secure by default; iii) "Bake in" security mechanisms, as compared to "bolt-on" protections as an afterthought; iv) Value keeping things simple to save resources and avoid negatively affecting users; v) Integrate throughout the lifecycle of projects/applications/systems; vi) Support a standard method to "trust but verify" projects/applications/systems; and vii) Set security up to be seen as an enabler through educating users, managing expectations, and supporting change.

As far as security is concerned, a security framework is a tool for achieving a common understanding among those involved in security-related processes. These frameworks typically contain contextual information as well as standards, guidelines, and recommendations for implementing security controls and analyzing and resolving security issues that may arise.

The selection of a security framework is not a technical task, but a decision that involves various stakeholders with different backgrounds (operational, technical, business-oriented, etc.), as it impacts the day-to-day activities of an organization or project.

Several different frameworks can be found in the literature, all of which share some similarities in terms of objectives and recommendations, but differ significantly in scope and coverage. In general, the objectives relate to understanding the current state and goals, performing the activities, continuously assessing the situation by understanding the risks and probabilities and developing a roadmap to reduce and mitigate the risks. This process is typically a cyclical one, with some steps overlapping and the boundaries between them blurring (Fig. 1)

[2] https://globalprivacyassembly.org/.

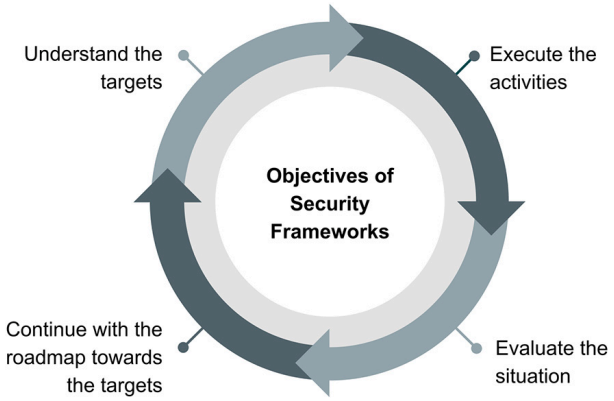


Fig. 1. Main Objectives of Security Frameworks.

In essence, the objectives are not far from the typical Plan-Do-Check-Act cycle used in continuous improvement, or even from the phases of ENISA's good practice guide on formulating security strategies [13].

Secure Control Framework (SCF)[3] is a free meta-framework that focuses on policies, procedures, and standards related to privacy and security. It is a meta-framework because it incorporates other cybersecurity frameworks (CSFs), including NIST CSF, ISO 27002, and several others, as well as a collection of laws, directives, and regulations. SCF takes a holistic approach to address strategic, operational, and tactical guidance, always with a strong focus on security and privacy. One of the key points of SCF is the mapping of all the controls to other frameworks, referencing the relevant section or principle, making SCF an interesting tool for assessing the status of the organization in relation to multiple frameworks.

Another notable framework is NIST CSF [7], a framework developed by the National Institute of Standards and Technology for understanding and managing cybersecurity risk. Although primarily focused on critical infrastructures, it also includes guidelines, standards, and best practices to address potential security threats and ensure that security incidents and risks do not impede organizational activities. The goal of the framework is to provide a common language and taxonomy for organizations to: i) describe their current cybersecurity situation; ii) describe their security goals; iii) identify and prioritize opportunities for improvement and provide a tool for that improvement to be continuous and repeatable; iv) assess progress toward goals; and v) communicate with stakeholders.

The FAIR framework (Factor Analysis of Information Risk) [16] is a cybersecurity and operational risk framework developed by the nonprofit FAIR Institute. The FAIR model assesses multiple factors associated with cybersecurity risks and enables organizations to analyze, measure, and monitor those risks. FAIR taxonomy shows that risk is divided into two concepts: Loss Event Frequency (LEF) and Loss Magnitude (LM). LEF refers to the frequency of occurrence, while LM refers to the impact and consequences. The LEF, and therefore the frequency, can be derived from the frequency of threat events (TEF) and vulnerability. The likely frequency with which threat agents may cause losses (e.g., the frequency of brute force login attempts can be considered a TEF, while if one attempt gains access to the system and compromises the data it would be an LEF). Losses are categorized as primary and secondary and are associated with a monetary cost. With this information, the organization can evaluate its assets, events, and threats, and has quantitative information about risks and the economic impact it faces so that the organization can make decisions based on this information.

Meanwhile, ISO /IEC 27002 [20] is an information security standard that is also used as a framework for good information security practices. ISO 27002 provides specific controls for implementing the information security management system defined by ISO /IEC 27001. ISO /IEC 27002 contains 14 domains, 35 objectives, and 114 controls.

In addition to those mentioned above, NIST Special Publication 800-53 [34] should be mentioned. NIST SP 800-53 contains a catalog of security and privacy controls for implementing secure information systems. This catalog is of particular importance to U.S. federal government agencies and contractors who must comply with these controls.

From a historical perspective, TARA [32] and OCTAVE [3] are also worth mentioning. TARA (Threat Agent Risk Assessment) is a methodology developed by Intel Information Technology in the early 2010s. It can be considered as a risk prediction methodology and allows, among other things, to detect the targets of threat agents, identify the possible attack methods, or determine the most important threats. OCTAVE (Operationally Critical Threat, Asset and Vulnerability Evaluation) is another information security risk management and assessment framework developed by Carnegie Mellon University and SEI (Software Engineering Institute). OCTAVE was extended to include OCTAVE Allegro [10]), a methodology published in 2007 to improve the security risk measurement process. In 2020, SEI published the OCTAVE FORTE model, which was aimed more at enterprises to help decision-makers understand and prioritize risks.

All of these tools and the above frameworks are the foundations of the approach for defining privacy and security for interoperable software components in manufacturing presented in this work. Many of the above controls apply only to organizations, requiring

[3] https://www.securecontrolsframework.com/.

adaptation to software environments and software platform infrastructures. The framework designed in this study also addresses this issue by providing adaptation to the specifics of platforms and middleware for the industry.

The impact of security on the digitization of manufacturing in particular, and industry in general, is an issue that is taking center stage as virtually all traditional manufacturing companies are evolving into nearly technology-driven enterprises [26]. Rao and Deebak [31] addressed some of the issues and challenges, detailing the need for clear requirements, authentication, and key agreement schemes and countermeasures, for IoT and CPS systems. In addition, Soori et al. [35] detailed the need for secure and privacy-protecting IoT architectures for smart factories to ensure the protection of sensitive data and intellectual property. In general, the literature shares concerns about the massive deployment of sensors and IoT devices. The communication of these devices and the subsequent storage and processing of data also require support for security. Enforcement of data access policies, data protection, secure service management, and risk management [6] are also examples of challenges in securing software components for use in manufacturing environments.

All of these security/privacy challenges lead to the need for platforms that support the development of innovative applications for the industry and make it easier for designers and developers to meet minimum security and privacy requirements. As far as these platforms for industry are concerned, there are various alternatives in the form of architectures, reference models, and middlewares for building digitally enabled industrial systems.

Reference Architecture Model Industrie (RAMI 4.0) [18] is one of these architectures. RAMI is a reference architecture that provides a common vocabulary and framework for Industry 4.0 systems. This model describes the main elements of an Industry 4.0 system in a three-dimensional coordinate system. It includes an axis called “Hierarchy Levels”, which includes levels that relate to international standards for enterprise IT and control systems. The second axis, labeled “Life Cycle & Value Stream”, represents the life cycle of an asset and its value creation process. Finally, an architectural layer is presented where different sub-layers detail the physical systems, layers of communication between components and integration, and layers related to the business and its objectives, among others.

The OpenFog Reference Architecture (OpenFog RA) [29] is a set of documents and a high-level framework developed by a consortium of high-tech companies and academic institutions to standardize and illustrate the network environment between the edge of a network and the cloud to facilitate complex systems and transactions. This architecture is based on eight pillars, including openness, security, and scalability, and aims to accelerate the adoption of Fog computing in the industry through a high-performance, cross-industry, and interoperable architecture.

The Industrial Internet Reference Architecture (IIRA) describes a common architecture for the development of IoT applications. It is defined as *a standards-based architectural template and methodology enabling industrial Internet of Things system architects to design their systems based on a common framework and concepts* [24] and focuses on, among other things, establishing high-level conceptual consensus and providing guidelines for Industrial IoT system functionality and features. IIRA also introduces the concept of trustworthiness, which includes resilience, reliability, safety, security, and privacy.

FIWARE, on the other hand, is not so much an architecture as a middleware, a set of tools, and a community that facilitates the development of IoT applications and digital twins in different sectors in the context of digital transformation. FIWARE provides a collection of freely usable API specifications and a context broker engine that enables exchange and discovery between components. All this together with a collection of services and enablers that enable the use of multiple third-party applications and devices on the FIWARE platform.

Some of these and other emerging architectures are conceptual or require software platforms to support them, like the one analyzed in this work. Others, like FIWARE, are an ecosystem that can work perfectly on a composable platform like KITT4SME, which is the main platform addressed in this article.

3. Meta-framework for security and privacy analysis

Among the different ways to perform risk analysis, this work focuses on two of them: risk-oriented and goal-oriented risk analysis. The former is the most well-known and commonly used approach, while the latter is common in cases where the analysis is performed during the requirements-gathering phase.

The risk-oriented method can be classified as a reactive approach, as it focuses on knowing the risks and reacting to them. This method is the best known and usually focuses on four phases:

1. Define the assets;
2. Define the threats;
3. Analyze the risks;
4. Define and implement the countermeasures;

The goal-oriented method is also relatively widespread and can be classified as preventive since the risks are already taken into account in the requirements phase. This method has three phases:

1. Define the functional requirements and potential requirements based on a regulatory framework or even security/privacy principles;
2. Study the security and privacy implications and extract new requirements if needed (e.g., non-functional requirements);
3. Implement the requirements (including the privacy/privacy requirements);

The herein proposed framework suggests a method that implements a hybrid approach, starting from both requirements and assets. On the one hand, it is key to analyze the security and privacy implications of the end-user requirements, as suggested by the goal-based methodology. Working with requirement templates and linking them to security and privacy implications provides relevant information in the early stages of infrastructure design.

On the other hand, traditional risk-based analysis provides relevant information about assets and threats. This information is especially important when designing a software platform that is intended to be generic and where it is impossible to cover all threats for all assets. For this reason, knowing the potential risks and performing risk analysis at an early stage can be of great benefit to companies and technology providers using the digital platform.

As mentioned earlier, the proposed approach draws on the elements of both methods to define a six-step framework:

1. Understand the assets;
2. Define the requirements and link them with security and privacy implications;
3. Explore and classify the risks and threats;
4. Perform a risk analysis;
5. Define countermeasures;
6. Implement countermeasures and security and privacy requirements;

The advantages of this approach against the conventional ones are, on the one hand, the opportunity to have a link between the requirements and security implications, which is not usually included in the risk-oriented methods. On the other hand, not focusing only on the requirements but having a view of the assets, is not a usual viewpoint in the goal-based approach, which is more focused on how risk can affect the achievement of goals. In general, the method provides a more holistic overview, linking software and hardware requirements with risks and goals, and thus not focusing only on assets or requirements, covering both cases, given that both are of utter importance. The approach also allows us to couple the more cybersecurity-oriented approaches (more usually related to risk-oriented) with those more related to business continuity (often related to goals). The most complex and important steps, of this hybrid framework, are explained below with examples and details.

3.1. Understanding and categorizing the assets

To understand assets, it is crucial to have a clear picture of the data handled and to know if these data are grouped in datasets. This enables to later categorize them together and even create a taxonomy of sensitive data.

Early discussions with end users and the development of a data management plan, a formal document that describes how data will be handled during the lifecycle of a project, can help define groups. Examples of data sets common to different types of industry projects are listed below.

- Skills and capacity building data: Data relating to the activities and tasks performed by the operator and his/her capabilities. Including potential growth paths and data related to their possible skill gap.
- Preferences and personal settings: User preferences and personal information. For example, application-specific or activity-specific settings for the user.
- Real-time activity execution data: Information obtained in real-time using sensors installed on the machines or in the environment, to monitor and evaluate the execution of activities.
- Control data: Data from quality control machines and AI models and algorithms related to quality control.
- Reconfiguration data: Settings of the machines used for quality control and modifications to these settings.
- KPIs: Key Performance Related Indicators used to monitor projects and activities.
- Data from 3rd party organizations: Data, tools, libraries coming external data sources.

These groups can be further refined and classified into four categories:

- Individual data: Operator profiles, that have to be shared with the platform to build customized service packages that include some human monitoring activity and to provide real-time support to enhance the human-machine interaction.
- Production system data: To offer real-time support for improved quality control, more agile reconfigurability, or a symbiotic interaction between operators and the production system, data from the production system have to be shared with the platform;
- Assessment data: The dashboard of KPIs collected from the activity and projects;
- Registry data: Data providing information about the SME and third parties, including third-party tools and components.

From these categories, it is possible to extract the initial sensitive data groups that will be analyzed in the further steps of the framework. It should be noted that these groups and sensitive data are quite high level and depend on the use cases and solutions that the platform will support.

Table 1
Template to attach preliminary security and privacy information to requirements.

#: **Title** (Referencing the requirement ID and title)			
Description	Brief description of the requirement, underlying the security and privacy needs		
Domain	Requirement domain(s) (e.g.: Data, Deployment...)	**Impact**	Preliminary impact levels (e.g., high, low)
Implications	Description of potential security and privacy implications		
Groups	Security and privacy group. e.g., Role management, storage, communication, etc.		

Table 2
Adapted taxonomy, based on Secure Controls Framework risks, and linked to NIST CSF functions.

Group	Risk	Proposed Adaptation	Function
Access Control	Inability to maintain individual accountability Improper assignment of privileged functions Privilege escalation Unauthorized access	Inability to maintain accountability Improper privilege management Improper privilege management Unauthorized access	Protect Protect Protect Protect
Asset Management	Loss of integrity through unauthorized changes	Loss of integrity	Protect
Business Continuity	Business interruption Data loss / corruption Reduction in productivity Information loss/corruption or system compromise due to technical attack	Functionality interruption Information loss/corruption or system compromise Reduction in performance Information loss/corruption or system compromise	Recover Recover Protect Protect
Exposure	Unmitigated vulnerabilities System compromise	Unmitigated vulnerabilities Information loss/corruption or system compromise	Protect Protect
Governance	Inadequate internal practices Inadequate third-party practices	Inadequate module integration practices Inadequate module integration practices	Protect Protect
Incident Response	Inability to investigate / prosecute incidents Improper response to incidents Ineffective remediation actions	Inability to detect, analyze and respond to incidents Inability to detect, analyze and respond to incidents Inability to detect, analyze and respond to incidents	Respond Respond Protect
Situational Awareness	Inability to maintain situational awareness	Inability to detect, analyze and respond to incidents	Detect

3.2. Defining the requirements and implications

There are different approaches for security and privacy requirements elicitation in digital production ecosystems. From a strict software engineering perspective, the security and privacy requirements can be considered non-functional requirements (NFR) and therefore analyzed using one of the well-known requirements elicitation methods.

Volere snow cards[4] is an example of a widely used template for requirements gathering and tracking. In this framework, the proposal is to use similar templates to capture security and privacy requirements and append the information to the main requirements. A suggested approach is to use templates inspired by the snow cards, in which the details of the usual requirement gathering can be extended with extra information to understand if any of the functional requirements may have non-functional implications related to security and privacy. Table 1 shows an example of the template used in this work to attach security and privacy information to the requirements.

3.3. Explore and classify the risks and threats

Risks of the Secure Controls Framework are considered relevant as a foundation for classifying the risks associated with the manufacturing platform ecosystem. This decision was made primarily because of the broad risk coverage of the framework and because it is well grouped and mapped to the functions of NIST CSF and therefore may be useful for SMEs looking to implement or deepen one of the frameworks.

In general, the security frameworks are closely related to organizational risks while the meta-framework we propose and this work focus on the issues and guidelines related to digital platforms and a specific collection of manufacturing use cases. Therefore, it is important to filter, by emphasizing the software rather than the organization, and regroup some of the risks, which although they apply to organizational risks do not apply to platforms and software. The applicable risks that align with the CSF and the proposed adaptation are shown in Table 2.

Table 2 shows the risks coming from the CSF, clustered by groups, as the CSF does. They are also grouped by function, as NIST CSF suggests. It is important to note that not all groups and risks in the previously described frameworks are represented, since some do not directly impact manufacturing platforms or because they are simply more focused on organizational security and business continuity.

[4] https://www.volere.org/.

Table 3
Mapping of three well-known definitions of privacy.

Proposed Adaptation	Privacy-by-Design	Privacy-by-Default	C4P
Privacy by default in the core	1) Proactive and preventive privacy 2) Privacy as default setting 3) Privacy Embedded into Design	a) only strictly necessary data for the specific purpose of the solution b) no actions needed to maintain privacy	1) Enable privacy 2) Pre-set security 3) Security and privacy mechanisms in the core
Simplicity and transparency	6) Visibility and Transparency 7) Respect for User Privacy	-	4) Keep things simple 6) Trust but verify 7) Educate users
Privacy throughout the lifecycle	4) Full Functionality 5) End-to-End Security	-	5) Integrate throughout the lifecycle

Table 4
Risk classification.

Likelihood	Severity				
	Insignificant	**Minor**	**Serious**	**Major**	**Catastrophic**
Rare	Low	Low	Low	Low	Medium
Unlikely	Low	Low	Low	Medium	Medium
Possible	Low	Low	Medium	Medium	High
Probable	Low	Medium	Medium	High	High
Certain	Medium	Medium	High	High	High

The table also includes the proposed adjustment and grouping of risks. As can be seen, the adjusted risks have been grouped into ten main categories. This adjustment was made by focusing on the risks related to the development and execution of the software rather than the continuity of the organization's business.

Concerning privacy and data protection, it is also considered the idea of relying on 3 well-known definitions of privacy. These are the definitions already detailed in Section 2: Privacy-by-Design, Privacy-by-Default, and C4P. This work proposes to group them into three meta-principles:

- Privacy and security by default in the core of the application and not as an "extra";
- Simplicity for the user and transparency;
- Privacy and security throughout the lifecycle of the solution;

Table 3 maps the general principles of this meta-framework with those of the different privacy terms above.

3.4. Perform a risk analysis

To carry out the risk analysis, it is suggested to use a scoring matrix based on likelihood and severity where the potential impact is indicated. This approach to risk analysis is easily understood by different stakeholders, is commonly used in different fields, and allows an individual view of each risk since the analysis is performed for each one of them.

The likelihood measures the probability that the problem will occur considering the conditions evaluated. It is measured in qualitative values ranging from a rare occurrence to a certain occurrence. Severity measures the damage the problem can cause. A qualitative scale ranging from insignificant to catastrophic damage is used. The intersection of the two values is the impact that the combination can have on operations. Impact is also measured qualitatively at three levels: low, medium, and high.

Table 4 shows this widely used risk classification matrix. The objective is to use the matrix to find, for each of the defined threats, the impact, based on the likelihood and severity values.

3.5. Define countermeasures

Once the requirements and risks are known, it is necessary to define the countermeasures and the software platform approach to privacy and security. There are multiple catalogs of countermeasures available in the literature. Notable among them are the aforementioned NIST CSF SP 800-53 *Security and Privacy Controls for Information Systems and Organizations* and the SCF controls. The first includes crosswalks with other NIST catalogs and with ISO 27001. On the other hand, the second is the result of a comprehensive and complex exercise that includes more than 1000 countermeasures along with crosswalks with an extensive number of catalogs and standards and includes a weight value for each control that can be used to prioritize and categorize them.

The selection of countermeasures may be based on these catalogs or previous experience and know-how. In the proposed meta-framework, as it is more oriented towards defining and adapting component-based platforms, the proposal is to group high-level countermeasures, which can later be used to deepen a group of specific countermeasures.

To make this definition of groups of countermeasures the focus has been put on the NIST functions and the groups defined in the taxonomy presented in our framework. Starting from these groups and functions, several of the most important countermeasures of SCF have been selected. These countermeasures are based on the requirements, the threats we define, and the risks we want

to mitigate. In this study, following the objectives of the meta-framework, the main focus is on the definition of platforms for the industry instead of focusing on the purely organizational ones. This enables to define reference controls such as the following:

- **Protection controls**: aimed at reducing the probability of occurrence of the problem. Examples of protection controls might be: i) Data encryption and cryptographic protection; ii) Identity management and access control; iii) Network segmentation; and iv) Input data validation and error handling
- **Detection controls**: targeted to recognizing the potential problem as early as possible. Examples of detection controls are: i) Tamper detection; ii) Continuous monitoring; and iii) Inbound and outbound communications
- **Identification controls**: focused on the identification and analysis of problems. Examples of identification controls can be: i) Traceability and accountability; ii) Asset governance and inventories; iii) Review of third-party components and services; and iv) Data Privacy
- **Recovery controls**: aimed at reducing damage or restoring systems. Examples of recovery controls are: i) Business continuity management; ii) Testing for reliability and integrity; and iii) Transaction recovery
- **Response controls**: oriented to the definition of actions to be taken following a security or privacy problem. Examples of response controls are: i) Incident handling; and ii) Incident response testing.

3.6. Implement countermeasures and security and privacy requirements

If in the previous step, countermeasures were defined based on the security and privacy risks and the impact associated with the functional and non-functional requirements, this last step is the practical implementation of the countermeasures. Based on countermeasures, several of the controls and solutions associated with critical risks can be selected to mitigate them. Since the objective is the deployment of a specific middleware or platform, a large part of these solutions will be components, software, configurations of the infrastructure, or the service mesh. Once the selection has been made, it can be re-evaluated to analyze how it would mitigate the security impact mentioned in the requirements.

This implementation step is the last step of the method for understanding risks, goals, and countermeasures, but as mentioned above the process is usually cyclical, and during the life of the platform and the solutions running on it, continuous monitoring, evaluation and refinement of the countermeasures applied will be necessary.

4. Case study: KITT4SME security and privacy analysis

As an evaluation method, the framework is applied to analyze the security and privacy of the KITT4SME platform. The information from the high-level description of the project pilots using the platform and the key asset information from a data management plan, together with this hybrid approach aided us in more reliably investigating the likelihood of incidents and their impact and defining the countermeasures to be taken.

It should be noted that the goal of the approach is not only to be useful for these use cases but to be as general as possible so that other SMEs can use the component-based platform to put together their AI systems for production environments.

4.1. Understand and categorize the assets

Based on the ideas in Section 3.1, the process started with analyzing the assets of the use cases. The original groups mentioned in the framework to create a taxonomy of sensitive data to be managed and monitored are then used. In this step the data groups are defined in more detail, examples of these data groups are provided, and it is analyzed how the data used and processed are categorized in the list of project assets.

As part of the project's data management plan, several assets were identified, most of which relate to the datasets originally identified in the project. In the following list, the datasets are shown, providing a description and contextual information, and indicating which group or groups in the taxonomy they can be placed in.

- Quality control and defect detection images
  - **Description:** A collection of images of defects in pieces to facilitate the AI training of the classification algorithms. The dataset includes a collection of images of defective pieces and Regions of Interest showing the defect.
  - **Type:** Images
  - **Relevant Groups:** Control data, Reconfiguration data, Real-time activity execution data

- Online survey data
  - **Description:** Dataset collected by partners through online surveys and online interviews, to make a qualitative and quantitative analysis of stakeholders' needs, requirements, and expectation, which has to be satisfied by the project ecosystem
  - **Type:** Tables
  - **Relevant Groups:** Skills, Preferences, Data from 3rd party organizations

- Competence and skill gap assessment data

  - **Description:** Data coming from the analysis of International Competences' database, such as ESCO (European Skills Competences qualifications and Occupations) and O*NET (Occupational Information Network), the analysis of technical databases (patents) and the processing of MOOCs (such as Coursera)
  - **Type:** Tables
  - **Relevant Groups:** Skills, Preferences, Data from 3rd party organizations

- Stakeholders information for dissemination and promotion
  - **Description:** Lists of possible stakeholders
  - **Type:** Tables and documents
  - **Relevant Groups:** Data from 3rd party organizations

- Performance Indicators
  - **Description:** KPIs from use cases. Data coming from use cases' machines and databases as well as devices used to monitor shopfloor parameters.
  - **Type:** Tables
  - **Relevant Groups:** KPIs (Project KPIs, Pilot KPIs)

- Shop Floor/machining data set
  - **Description:** Information to assess the accuracy/reliability/suitability of machine learning method to predict/estimate manufacturing process condition and quality.
  - **Type:** Tables
  - **Relevant Groups:** Skills, Preferences

- Registration data from the platform marketplace
  - **Description:** Registration data for companies that participate in the marketplace for robotic applications, are to be delivered together to the digital platform, including some personal data from representative personnel for verification purposes (contact points).
  - **Type:** Tables
  - **Relevant Groups:** Preferences (Marketplace registration information)

- Open call applicants
  - **Description:** The platform will be tested by a number of small manufacturing companies. AI technology providers are also invited to develop components that run on the platform. The management of this participation is done through open calls. This dataset includes proof that SMEs are legal entities and abide by the instructions of the open call, for example not declared bankruptcy. Description of the technical solutions, level of maturity, and potential for market use
  - **Type:** Tables, documents
  - **Relevant Groups:** Data from 3rd party organizations

- Time series data from welding machines
  - **Description:** Data extracted from machines for training the AI models
  - **Type:** Tables
  - **Relevant Groups:** Real-time activity execution data, Control data

4.2. Define privacy and security requirements and implications

From all the requirements defined in the project, those that could have security and privacy implications were extracted. For each of these requirements, the specific template defined in the method for linking security aspects and requirements was completed. As mentioned earlier, this template included a summary of the requirement description, the domain, the implications, and potential impact, as well as keywords to group them later, as described in Table 1. Two examples of such templates linking security implications to requirements are shown in Table 5 and Table 6.

4.3. Exploring risks and threats

As mentioned in [15], a threat model is an engineering technique used to identify the threats, attacks, vulnerabilities, and countermeasures affecting a system. In the section devoted to exploring and classifying the risks and threats, a security and privacy risk taxonomy is proposed. This taxonomy is based on CSF, NIST, and other frameworks' recommendations and guidelines for organizations. In this work, the focus was on digital ecosystems and platforms supporting modules and software components, therefore the adaptation of some of the risks and their application to this specific case was required. For the same reason, the number of risks was reduced because the CSF and other frameworks contained risks related to business continuity or governance that were not fully applicable to the digital platform middleware.

Table 5
Privacy and Security info for the requirement #F6.

#F6: Container Orchestration			
Description	The platform shall allow for deploying containerized solution architectures, which allow AI developers to easily integrate their solutions into the platform		
Domain	Deployment	**Impact**	Medium-low
Implications	From a security point of view, intercommunication between modules implies knowing the data exchange mechanisms and adapting a posture that allows encrypted communication between modules and managing their authentication. On the one hand, so that an unauthorized module cannot listen to messages between authorized modules. On the other hand, a module cannot impersonate another module to obtain some benefit		
Groups	Communication		

Table 6
Privacy and Security info for the requirement #NF9.

#NF9: GDPR compliance			
Description	Platform must provide an option for GDPR-compliant handling of personal data (e.g., permission opt-in)		
Domain	General Privacy and security	**Impact**	High
Implications	The modules and services that are developed on the platform must comply with all the requirements of the GDPR. At the organizational level using the KITT4SME ecosystem, the organizations should also study the implications of the GDPR and ensure that they comply with the principles of user information and the right to control data		
Groups	Compliance, Storage and persistence		

The following list shows the risks included in the proposed framework (as defined in Table 2) and briefly describes each of them in the context of the use case, providing examples of some associated threats.

- **Inability to maintain accountability:** Not having a record of actions, and even traceability of who, when, and how an action was performed, can make it difficult to manage non-repudiation and may further complicate the investigation and/or resolution of other security and privacy issues. CSF includes a similar risk more focused on individual accountability, which has been generalized so that it can be considered in different use cases, e.g., accountability in data access or even accountability in the modules through which certain information passes. Examples of threats are the following: auditing failures; loss of reliability and credibility; ineffective and slow responses to incidents; deficient decision support.
- **Improper privilege management:** Incorrect privilege management can lead to certain users being able to perform actions for which they are not authorized. It is also related to misconfiguration of privileges giving users more privileges than necessary to do the task they have to do. In this risk, two of the risks mentioned in the CSF have been grouped: Improper assignment of privileged functions and Privilege escalation. Incorrect privilege management is often related to the following risk: unauthorized access. Examples of threats are the following: privilege escalation; data breaches; access to sensitive data.
- **Unauthorized access:** Related to users or modules having access to data or actions for which they should not be authorized. Examples of threats are the following: access to unauthorized information; unauthorized changes; sniffing; eavesdropping.
- **Loss of integrity:** It is related to the fact that the data must be valid, accurate, and not modified by users or processes that do not have permission to do so. CSF includes integrity-related risks such as “Loss of integrity through unauthorized changes” which has been adapted as “Loss of integrity”. There are also several controls related to data corruption, but we have decided to separate them into two risks, focusing on deliberate user actions or processes to affect integrity. Examples of threats are the following: data corruption; lack of trustworthiness; difficulties recovering information;
- **Functionality interruption:** Involves the disruption of the processes or functionalities of the system. For example, attacks related to the degradation of the availability of certain modules, or even denial of service techniques can force an unplanned service stoppage.
- **Reduction in performance:** Related to the previous risk, this one focuses on the reduction of the functionality of the module or ecosystem element to perform the task for which it is programmed. This risk can be caused deliberately by an attacker or even unintentionally by the degradation of the network or certain components. Examples of threats are the following: downtime; delays; lack of availability.
- **Information loss/corruption or system compromise:** Includes risks related to the loss, leak, or unauthorized disclosure of information. As well as any violation of security and privacy policies and settings. Examples of threats are the following: Data inaccuracy; data leaks; permanent data access block.
- **Unmitigated vulnerabilities:** Is related to either lack of knowledge or neglect of system maintenance that may result in unmitigated vulnerabilities that can lead to security and privacy issues. Examples of threats are the following: System compromise; discovery and exploitation.
- **Inadequate module integration practices:** The original risk was related to inadequate internal practices and inadequate third-party practices. In this study, it has been decided to adapt it to the risks related to the integration of the modules, either between

Table 7
Quality control data - Risk classification.

A5: Quality control data			
Threats	**Likelihood**	**Severity**	**Risk**
Inability to maintain accountability	Possible	Insignificant	Low
Improper privilege management	Possible	Serious	Medium
Unauthorized access	Possible	Catastrophic	High
Loss of integrity	Possible	Minor	Low
Functionality interruption	Possible	Minor	Low
Reduction in performance	Probable	Insignificant	Low
Information loss/corruption or system compromise	Possible	Catastrophic	High
Unmitigated vulnerabilities	Possible	Serious	Medium
Inadequate module integration practices	Possible	Minor	Low
Inability to detect, analyze and respond to incidents	Possible	Minor	Low

themselves in the platform, or with the integration of the modules in the platform users' solutions. Examples of threats are the following: Data inconsistency; unreliable performance; system instability.

- **Inability to detect, analyze, and respond to incidents:** Related to the actions of detecting, studying, and solving incidents. This risk has also been adapted as the original risks separate detection, study, and solution. Examples of threats are the following: persistent vulnerabilities; non-reliable incident response.

4.4. Perform a risk analysis

As suggested in the framework, Table 4 was used to analyze the risks for each asset in relation to all the defined threats. The classification of risks is based on the likelihood and severity of each risk and indicates the potential impact: low, medium, and high. The analysis was made based on experience in other platforms and similar projects, together with a part of the team involved in the definition and concept of the KITT4SME idea. In Table 7 an example of the analysis for one of the assets of the project is provided. Specifically, the example shows the risk values related to each of the threats for the particular case of the quality control data.

4.5. Define countermeasures

After analyzing the tables for all assets (similar to the example in Table 7), the conclusion was that the most important security and privacy requirements and needs relate to i) unauthorized access and improper management of privileges; ii) information loss or system compromise; and iii) inability to detect and analyze incidents.

These key requirements can be categorized into three groups, depending on the countermeasures to be taken to address these requirements/needs:

- **Identity Management & Access Control**: Primarily, identity management, and a correct configuration of permissions, will allow to mitigate improper privilege management and reduce unauthorized access. On the other hand, access control and identity management mechanisms help us with accountability management, by identifying the users or processes performing the actions, or even mitigating the inability to detect, analyze, and respond to incidents, by studying who is accessing certain data or who is trying to access them without the necessary authorization levels.
- **Data encryption and secure communication**: Primarily, having encryption, signature, or hashing mechanisms in place allows for detecting loss of integrity. In turn, it also allows that if an information loss/corruption or system compromise occurs, the data that can be accessed has an extra level of protection because it is encrypted and cannot be read without effort. Secondarily, and in some ways, it also collaterally influences part of the unmitigated vulnerabilities. Encryption is not by any means a solution to the vulnerabilities but it can be an extra level of protection in case a vulnerability is exploited, for example, to obtain content from a database, or to address some of the privacy concerns and issues regarding the original industrial Big Data [36].
- **Traceability** of actions and data transactions: The objective of these controls is to be able to keep track of actions and to be able to determine which user or module was responsible for performing a given action, and at what point in time. Among the risk categories mentioned, traceability clearly covers the inability to maintain accountability, and in a certain sense, it also helps in the detection and analysis of incidents, so it can be useful to reduce the inability to detect, analyze, and respond to incidents. On a more secondary level, action logs can be used to detect unmitigated vulnerabilities, functionality interruption, or even reduction in performance, and take appropriate action.

4.6. Implement countermeasures and security and privacy requirements

The immediate result of applying the framework was that we not only gained an understanding of assets, risks, and countermeasures but also recognized the need to implement some concrete measures in the KITT4SME ecosystem. This led to the selection of several components and solutions that were integrated into the different layers of the digital platform.

In terms of privacy, during the application of the framework, we became aware of some concepts that should be taken into account in the implementation of the platform and that we had to communicate to the users of the platform. These concepts and recommendations were:

- Privacy and security by default in the core of the application and not as an “extra”;
- Simplicity for the user and transparency;
- Privacy throughout the lifecycle of the solution;

Regarding security, the results led to three main groups of controls that cover several of the risks of the project’s assets:

- Identity management & access control;
- Data encryption and secure communication;
- Traceability of actions and data transactions;

Concerning identity and resource access management, not only for controlling user access, but also to provide mechanisms for the different modules of the ecosystem to control which components can access certain data or resources, or even control access to the module.

Within the identity management solutions, and considering the KITT4SME ecosystem support for FIWARE middleware, the results led to the analysis and study of two alternatives for the OAuth2 framework and open standard protocol-based authentication:

- Keycloak is an open-source solution, compatible with FIWARE and an enabler of authentication in applications. Keycloak relies on standard protocols (e.g., OAuth2) and could be applied to users and modules to support single sign-on, social login, or identity federation, allowing centralized management of the identities.
- Keyrock as a FIWARE component for identity management enabling the support of OAuth2-based authentication and authorization for modules and applications. The first one was finally selected for implementation.

The detected need for data encryption and secure communication led us to deploy Istio[5] as a service mesh supporting the traffic encryption and enforcement of policies [30]. Istio is an extension or an abstraction layer for cluster management platforms (e.g., Kubernetes), an open-source solution that simplifies and harmonizes the integration of microservices and manages and standardizes communications and data flow between services. These two capabilities of Istio are key to providing an extra point of security to the project platform. The uniformity is obtained among other things thanks to the use of interception proxies. These sidecar proxies are a component that allows to integration of third-party services or services that were developed without security capabilities in mind. This is because the countermeasures defined above were managed in these sidecar proxies (e.g. two-way authentication between components).

The standardization of information flows is also a critical issue from a security point of view. As in the previous case, it enabled the integration of third-party services that were not developed following the KITT4SME data exchange rules, since proxies can be used for data conversion and formatting.

Istio can be also used for enforcing policies and telemetry data collection. The former with the use of an Open Policy Agent to define authorization rules. The latter is totally in line with the accountability and traceability objectives suggested by the framework. The capabilities of Istio proxies to obtain telemetry data, allowed us to keep track of actions performed in modules and of course the requests and exchanges of information between components. Fig. 2 depicts the console for the Istio service mesh (Kiali), showing in the upper part the workload detail with the request duration over a given period in the lower part of the heat maps. These heat maps are useful for monitoring request duration compared to other duration metrics over time.

Another mechanism provided by Istio is the access to proxy logs. These logs can be configured and can be used to obtain temporal information, protocols, durations, or response codes. This information is accessible from within the web but also through the configuration of sinks or destinations where this information can be stored and later monitored. Istio allows distributed tracing to centralize the logs of everything that happens in the mesh and provide a complete view of latencies or even the flow of interactions.

5. Case study evaluation

To evaluate the meta-framework and its application in the use case, this section first describes the results of applying the framework in practice. Then the challenges or limitations discovered in implementing the meta-framework are identified to provide specific

[5] https://istio.io/.

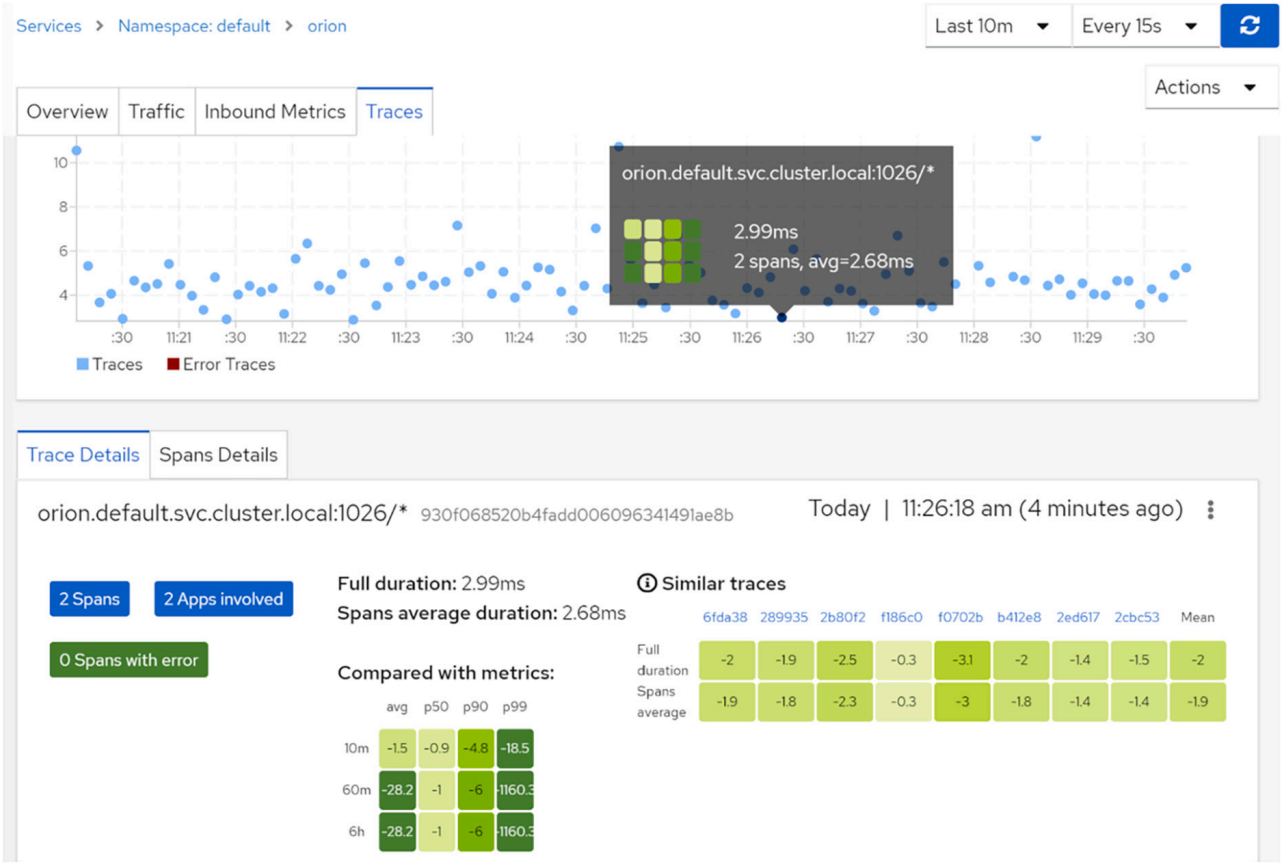


Fig. 2. Istio traceability interface.

examples of its effectiveness and lessons learnt. Finally, the potential of extending the proposed strategy to other architectures is analyzed.

Undoubtedly, the approach derived from applying the meta-framework has enabled the design and development of a platform that covers the main security and privacy needs detected in the asset understanding and requirements analysis phases. However, it has also allowed us to identify challenges and limitations. The first of these challenges is the implications derived from where the platform and components are deployed. From the beginning the idea of the platform and therefore the security analysis was more oriented to platforms, distributed or not, residing in the cloud. However, it is true that after analyzing the use cases, it is noticed the need for certain SMEs to run the platform on servers on their premises. This occurred in cases where the applications depended on large volumes of data stored on local disks and databases, and in cases where machines with sensors were connected directly to one of the local machines or were not prepared for connection to the cloud. This situation of dependencies and the need to adapt applications was also detailed by Abdollahzadegan et al. [1] in their study on the adoption of the cloud in SMEs.

In this case, the platform was designed with the idea that it would work in different cases and that it would be reasonably generic. This can also be another challenge, as the different layers added can increase complexity in a sector that usually suffers from a lack of resources and IT knowledge. For example, while authorization support provides ways to control traffic between components and restrict operations, they are rarely used by component developers and end-user SMEs. One of the main reasons for this was the lack of experience with declaring and using policies in the Rego language. This suggests that the implemented plug-and-play solutions were beneficial for security and privacy, but forcing SMEs to do additional work that is outside of their usual functions hurts the use of the platform.

In contrast, the platform's effectiveness was found to be due to the ease of integrating components and providing secure communication mechanisms between them by default. Mutual TLS (mTLS) enables such functionality and makes it possible for components that did not consider secure communication during their development to authenticate each other and establish a secure communication channel. Another observed benefit comes from the emphasis on observability. The traceability and observability information of the component network is effectively monitored on the Istio Service Mesh (Kiali) web console.

To conclude this use case evaluation, it is important to highlight some points of the meta-framework that can be applicable not only to the development of the KITT4SME software platform used as a reference but also to other architectures and platforms for use in industrial environments. Cases such as the inspection of wooden cabinets, inventory management, or waste management showed interest in using or adapting the results and fit with the approach of composability, security, and privacy adopted in the article.

The adapted taxonomy, based on the risks of the Secure Controls Framework and linked to the functions of NIST CSF, is useful for risk analysis and grouping. Especially since the customization focuses less on organizational risks and more on the software or

platform being developed. Similarly, the taxonomy of privacy principles is useful for developing new platforms that have three clear principles from design and by default.

In addition, the blended goal- and risk-oriented approach and the assessment of security and privacy requirements from two perspectives can be interesting for the development of component-based platforms covering organizational aspects, together with functional and non-functional requirements and asset and impact understanding. In this work, there were a limited number of specific applications to support, but the objective was to develop a generic platform that could serve as a reference. To this end, advanced templates for defining security requirements and implications are also important.

6. Discussion

In this section the objectives are revisited, followed by details of the highlights and major findings of the approach. It also aims to justify some of the decisions made and to detail the main limitations discovered. Finally, it is intended to analyze both the usefulness and limitations in a broader context, focusing on the applicability of the approach to other types of industrial cases.

As mentioned above, the main objective was to provide a method and a framework for manufacturing SMEs to customize or build interoperable component-based software platforms that take into account their minimum security and privacy needs. To this end, a journey via different frameworks and meta-frameworks has been carried out to define a fundamental meta-framework that allows us to identify and analyze critical security and privacy issues and to define and implement countermeasures.

This objective is coupled with other secondary goals, always aimed at the adoption of innovative technologies by industrial SMEs and always from the point of view of security and privacy. One of them is to reduce uncertainty when approaching the tasks of designing or creating a service deployment platform. It is not uncommon for the ICT ecosystem of industrial SMEs to integrate in-house developments, open-source solutions, third-party custom developments, or off-the-shelf components. Solutions such as Docker,[6] which offers support for containerized applications, have helped significantly at this point. Sometimes it is desired to go beyond component isolation towards automated deployments, scalability or observability, and security. In these cases, the design of a platform, for example, based on Kubernetes,[7] is undertaken. A meta-framework such as the one suggested adds the possibility of enabling privacy and security in the design and implementation of the platform.

Another secondary goal of our approach is to reduce the complexity that dealing with the current security frameworks and standards can pose to the technical staff of SMEs. The frameworks and standards mentioned in this paper are complex and extremely extensive. Most of them are designed for organization experts to address security and privacy needs and ensure business continuity after an incident. Some of these include, for example, more than a thousand countermeasures, or hundreds of potential risks. In the technical departments of SMEs, they often do not have this expert profile and focus more on application development or integration. Therefore, providing them with a practical and simplified framework that allows them to understand cybersecurity implications and to discover a basic set of countermeasures, based on objectives, requirements, and risks, was part of the interests of the article.

The most relevant contribution of the article to achieve these previous objectives can be summarized in the following three major highlights:

- **Meta-Framework for Security and Privacy**: The hybrid framework details six steps for identifying risks and defining countermeasures, and allows a better view of both, starting not only from the development objectives but also from the requirements. We believe the framework provides a not extremely strict but fundamental guide to facilitate the understanding of potential security issues to address when implementing platforms on which component-based solutions coexist.
- **Case study and demonstrative example**: Defining a framework is most meaningful when its use has been demonstrated in a case study and the examples provided can be used to illustrate how the approach can be replicated. In our case, the experimentation during the KITT4SME project has allowed us to learn the advantages and limitations of the approach, simplify complexities, and, above all, show particular aspects of how we approached each step and the technical countermeasures we arrived at. For us, the use within the KITT4SME project is of significant importance because it focuses on software platforms for small and medium-sized industries to adopt interoperable and containerized artificial intelligence solutions.
- **Adapted taxonomies:** As we have discussed, the analyzed frameworks are extraordinary but also complex and aimed at defining countermeasures at the organizational level. There are cases where a complete security and privacy analysis of the organization's entire business continuity is not desired, but the goal is simply an understanding of the fundamental security needs for the development or deployment of an application in an industrial environment. In these cases, the adapted taxonomies we introduce focus on specific aspects related to software development and allow to have a minimum common taxonomy to the most important frameworks. This also enables that if at some point more resources or time become available, further progress can be made towards the full implementation of a framework or even towards certification.

In a broader context linked to the applicability of the proposal, we consider that having started from the needs and requirements of four pilots from different industrial sectors, and the fact that it was applied to designing a platform adapted to the needs of those industrial pilots, confirms its suitability. This platform and its security features were also the basis for the open calls of cascade funding of the project in which parts of the platform were used by third parties. In general, the meta-framework has been designed

[6] https://www.docker.io.
[7] https://kubernetes.io/.

to meet any need to develop a platform and middleware. The hybrid approach, based on goals and risks, can be applied to various cases in which the developed software has to be integrated with other components. Also in a wider context, we believe that this approach can serve to have a fundamental understanding of security and privacy, without having to go into detail in more complex frameworks, but at the same time, it is not too restrictive to be extended in case the organization considers following a more complex methodology in the future.

Reflecting in a broader context, we have also identified some limitations and constraints that may require some additional considerations in the application of the framework to different case studies and industrial setups. For example, the framework focuses on the development of middleware or platforms where AI components can interoperate, but does not delve into the specific security of these components. In particular, the security of AI components and models is a hot topic that can impact platform security. Docker and Kubernetes-based platforms indeed offer component isolation, but to avoid the spread of potential issues, more control and specific analysis of potential risks and countermeasures for AI components may be necessary.

We believe that the risk adaptation suggested and the proposed framework can be helpful for industrial SMEs to understand and implement security controls and to facilitate replication in future developments, but another potential limitation indeed has to do with the impact of the approach on the technical staff of the SMEs. Moreover, our meta-framework does not require large departments or expert knowledge. Still, it does require technical staff with a minimum background because it defines a basis on which the technical staff has to make technical decisions, such as selecting which solution or approaches are going to be implemented for each countermeasure. This also has some relation with the limitation mentioned in the case study related to the overhead derived from adding to the software additional layers that may help in addressing some security issues but also add some complexity.

To conclude this discussion it is also fair to mention another limitation we discovered in the case study. The meta-framework does not go into detail about the deployment infrastructure, as each case is quite specific. But as in the case of the AI components, the deployment infrastructure may have an impact as more analysis may be needed to define extra countermeasures if the deployment is going to be in the cloud.

7. Conclusions

Manufacturing in particular and industry in general continue to evolve and are in the midst of a fourth revolution in which interoperability, data, and software platforms are essential elements. Add to this situation the pace and impact of artificial intelligence and the opportunities it can bring to industry, and the result is an industry that in many cases is having to deal address major changes without adequate resources and often without the necessary knowledge.

Software platforms based on component orchestration can be used to simplify interoperability requirements, facilitate integration, and provide the security and privacy mechanisms to support industrial applications.

This work is focused on these digital platforms for industry and how users and technology providers can assess their security and privacy needs to configure these systems accordingly while adhering to organizational security and privacy frameworks.

To this end, a novel meta-framework is presented. This meta-framework is based on security and privacy frameworks as well as well-known risk analysis methodologies, relying on and combining concepts from both: the goal-oriented and purely risk-based methodologies. To implement this meta-framework and associated methodology, some of these frameworks were adapted, as many of them focus on organizational aspects and not on software platforms or middleware.

In this case, this work is focused on a container orchestration platform that can help manufacturing companies develop innovative applications and even facilitate the integration of AI solutions.

A 6-step methodology was described in detail and a use case was presented and applied to evaluate the meta-framework. This use case study is based on how the KITT4SME project platform team assessed the security and privacy needs of four pilots and other future user cases to adapt the platform to their needs. The results of the application of the meta-framework allowed us to know the main security and privacy requirements and to select countermeasures that were integrated into the project platform. In the context of the described use case, the countermeasures were primarily related to i) identity management and access control, ii) secure data transmission and storage, iii) traceability and observability, iv) security and privacy, and v) security and privacy.

Security and privacy, along with interoperability, are some of the key elements of these platforms for Industry 4.0. It is also true that other non-functional requirements can make these platforms even more complete and make it even easier for SMEs in the manufacturing environment to integrate innovative AI solutions. Future work lies in these areas. Moreover, one of the requirements related to privacy, security, and verifiability, especially in the AI context, is explainability. A future direction will also focus on the impact that this explainability has on the design and development of software platforms

In general, Industry 4.0 offers myriad opportunities and AI offers new capabilities. It is therefore necessary to provide SMEs with tools and recommendations so that they can address them with confidence, knowing the challenges and opportunities they face. Methodologies and meta-frameworks, such as the one presented in this paper, along with the platforms on which the use case is implemented, can help facilitate the adoption of these new AI solutions in small and medium-sized manufacturing companies.

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CRediT authorship contribution statement

Rubén Alonso: Writing – original draft, Validation, Methodology, Investigation, Data curation. **Rodolfo E. Haber:** Writing – review & editing, Supervision, Project administration, Investigation. **Fernando Castaño:** Project administration, Investigation. **Diego Reforgiato Recupero:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no financial interest or personal relationship that could have influenced the work presented in this article.

Data availability

This work is based on defining, applying, and evaluating a framework for security and privacy analysis of software platforms for AI in the industry. New datasets were not generated. The used datasets were analyzed at a high level without the need to go deep into the details of the individual data. The specific content of these datasets is not necessary to replicate the content of the article.

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References

[1] A. Abdollahzadegan, A.R. Che Hussin, M. Moshfegh Gohary, M. Amini, The organizational critical success factors for adopting cloud computing in smes, J. Inf. Syst. Res. Innov. 4 (1) (2013) 67–74.
[2] AEPD, A Guide to Privacy by Design, Agencia Española de Protección de Datos, 2019.
[3] C.J. Alberts, S.G. Behrens, R.D. Pethia, W.R. Wilson, Operationally critical threat, asset, and vulnerability evaluation (octave) framework, version 1.0 (Tech. Rep.), Carnegie-Mellon University, Pittsburgh, Pa, 1999, Software Engineering Institute.
[4] K. Alexopoulos, N. Nikolakis, E. Xanthakis, Digital transformation of production planning and control in manufacturing smes-the mold shop case, Appl. Sci. 12 (21) (2022) 10788.
[5] V. Araujo, K. Mitra, S. Saguna, C. Åhlund, Performance evaluation of fiware: a cloud-based iot platform for smart cities, J. Parallel Distrib. Comput. 132 (2019) 250–261.
[6] D. Ayed, P.-A. Dragan, E. Félix, Z.A. Mann, E. Salant, R. Seidl, et al., Protecting sensitive data in the cloud-to-edge continuum: the fogprotect approach, in: 2022 22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid), 2022, pp. 279–288.
[7] M. Barrett, Framework for improving critical infrastructure cybersecurity version 1.1, in: NIST Cybersecurity Framework, 2018.
[8] S. Bhatt, T.K. Pham, M. Gupta, J. Benson, J. Park, R. Sandhu, Attribute-based access control for aws Internet of things and secure industries of the future, IEEE Access 9 (2021) 107200–107223.
[9] A. Bunte, F. Richter, R. Diovisalvi, Why it is hard to find ai in smes: a survey from the practice and how to promote it, Icaart (2) (2021) 614–620.
[10] R.A. Caralli, J.F. Stevens, L.R. Young, W.R. Wilson, Introducing octave allegro: Improving the information security risk assessment process, Tech. Rep., Carnegie-Mellon University, Pittsburgh, PA, 2007, Software Engineering Institute.
[11] A. Cavoukian, Privacy by design: the definitive workshop. A foreword by Ann Cavoukian, Ph.D, Ident. Inf. Soc. 3 (2) (2010) 247–251.
[12] F. Cirillo, G. Solmaz, E.L. Berz, M. Bauer, B. Cheng, E. Kovacs, A standard-based open source iot platform: fiware, IEEE Int. Things Mag. 2 (3) (2019) 12–18.
[13] N. ENISA, Good Practice Guide Designing and Implementing National Cyber Security Strategies, ENISA, Heraklion, Greece, 2016.
[14] B.R. Ferrer, W.M. Mohammed, J.L.M. Lastra, A. Villalonga, G. Beruvides, F. Castaño, R.E. Haber, Towards the adoption of cyber-physical systems of systems paradigm in smart manufacturing environments, in: 2018 IEEE 16th International Conference on Industrial Informatics (INDIN), 2018, pp. 792–799.
[15] F. Fraile, T. Tagawa, R. Poler, A. Ortiz, Trustworthy industrial iot gateways for interoperability platforms and ecosystems, IEEE Int. Things J. 5 (6) (2018) 4506–4514.
[16] J. Freund, J. Jones, Measuring and Managing Information Risk: a Fair Approach, Butterworth-Heinemann, 2014.
[17] B. Gladysz, D. Matteri, K. Ejsmont, D. Corti, A. Bettoni, R. Haber Guerra, Platform-based support for ai uptake by smes: guidelines to design service bundles, Cent. Eur. Manag. J. 31 (4) (2023) 463–478.
[18] M. Hankel, B. Rexroth, The reference architectural model industrie 4.0 (RAMI 4.0), ZVEI 2 (2) (2015) 4–9.
[19] E.B. Hansen, S. Bøgh, Artificial intelligence and Internet of things in small and medium-sized enterprises: a survey, J. Manuf. Syst. 58 (2021) 362–372.
[20] Information security, cybersecurity and privacy protection — information security controls (vol. 2022, Standard), International Organization for Standardization, Geneva, CH, 2022.
[21] A.A. Khan, A.A. Laghari, P. Li, M.A. Dootio, S. Karim, The collaborative role of blockchain, artificial intelligence, and industrial Internet of things in digitalization of small and medium-size enterprises, Sci. Rep. 13 (1) (2023) 1656.
[22] R. Kumar, R.K. Singh, Y.K. Dwivedi, Application of industry 4.0 technologies in smes for ethical and sustainable operations: analysis of challenges, J. Clean. Prod. 275 (2020) 124063.
[23] H. Lasi, P. Fettke, H.-G. Kemper, T. Feld, M. Hoffmann, Industry 4.0, Bus. Inf. Syst. Eng. 6 (2014) 239–242.
[24] S.-W. Lin, B. Miller, J. Durand, R. Joshi, P. Didier, A. Chigani, et al., Industrial Internet reference architecture, Tech. Rep., Industrial Internet Consortium (IIC), 2015.
[25] Z. Liu, P. Sampaio, G. Pishchulov, N. Mehandjiev, S. Cisneros-Cabrera, A. Schirrmann, et al., The architectural design and implementation of a digital platform for industry 4.0 sme collaboration, Comput. Ind. 138 (2022) 103623.
[26] C. Llopis-Albert, F. Rubio, F. Valero, Impact of digital transformation on the automotive industry, Technol. Forecast. Soc. Change 162 (2021) 120343.
[27] M. Mohamed, Challenges and benefits of industry 4.0: an overview, Int. J. Supply Oper. Manag. 5 (3) (2018) 256–265.
[28] W.M. Mohammed, R.E. Haber, J.L. Martinez Lastra, Ontology-driven guidelines for architecting digital twins in factory automation applications, Machines 10 (10) (2022) 861.
[29] A. OpenFog Consortium Architecture Working Group, et al., Openfog reference architecture for fog computing, in: OPFRA001, 2017, 20817.

[30] M. Pace, Zero Trust Networks with Istio (Unpublished Doctoral Dissertation), Politecnico di Torino, 2021.
[31] P.M. Rao, B. Deebak, Security and privacy issues in smart cities/industries: technologies, applications, and challenges, J. Ambient Intell. Humaniz. Comput. (2022) 1–37.
[32] M. Rosenquist, T. Casey, Prioritizing Information Security Risks with Threat Agent Risk Assessment (TARA), 2009.
[33] M. Sana, M.U. Farooq, S. Anwar, R. Haber, Predictive modelling framework on the basis of artificial neural network: a case of nano-powder mixed electric discharge machining, Heliyon 9 (12) (2023).
[34] Security and privacy controls for information systems and organizations (vol. 2020, Tech. Rep.), National Institute of Standards and Technology, 2020.
[35] M. Soori, B. Arezoo, R. Dastres, Internet of things for smart factories in industry 4.0, a review, in: Internet of Things and Cyber-Physical Systems, 2023.
[36] J. Wang, P. Zheng, Y. Lv, J. Bao, J. Zhang, Fog-ibdis: industrial big data integration and sharing with fog computing for manufacturing systems, Engineering 5 (4) (2019) 662–670.

3.3 A Novel Approach for Job Matching and Skill Recommendation Using Transformers and the O*NET Database

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- **Resumen:** Disponemos de toneladas de información publicada en la web sobre ofertas y demandas de trabajo. La tarea de evaluar los currículos de los candidatos y proporcionar recomendaciones automáticas sobre cuál se adapta mejor a un puesto en particular se ha vuelto esencial para acelerar el proceso de contratación y asignación a tareas específicas. Del mismo modo, es importante ayudar a los solicitantes a encontrar actividades adecuadas a sus habilidades y proporcionar recomendaciones sobre lo que los candidatos deben dominar para ser elegibles para ciertos puesto de trabajo. Este artículo propone un nuevo método para identificar las habilidades de los currículos de los candidatos y hacer coincidir estos con las descripciones de actividades y puestos de trabajo. Se emplea la base de datos O*NET de conceptos relacionados con habilidades y destrezas requeridas por las diferentes actividades laborales; además, se aprovechan las tecnologías de aprendizaje profundo para calcular la similitud semántica entre las entidades O*NET y parte del texto extraído de los currículums y de la interacción de los candidatos. El objetivo final es identificar el puesto de trabajo o actividad más adecuado de acuerdo con la información intercambiada. Se han definido dos escenarios: i) dado un currículum, identificar las principales ocupaciones de O*NET con la mayor coincidencia con el currículum, ii) dado un currículum de un candidato y un conjunto de descripciones de trabajo, identificar cuál de las ocupaciones es la más adecuada para el candidato. La evaluación que se ha llevado a cabo indica que el enfoque propuesto supera las líneas de base en los dos escenarios. Por último, se proporciona un caso de uso para los candidatos en el que es posible obtener recomendaciones de actividades formativas con el objetivo de cumplir con ciertas habilidades y cualificarlos para una determinada ocupación.

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A novel approach for job matching and skill recommendation using transformers and the O*NET database

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ABSTRACT

Today we have tons of information posted on the web every day regarding job supply and demand which has heavily affected the job market. The online enrolling process has thus become efficient for applicants as it allows them to present their resumes using the Internet and, as such, simultaneously to numerous organizations. Online systems such as Monster.com, OfferZen, and LinkedIn contain millions of job offers and resumes of potential candidates leaving to companies with the hard task to face an enormous amount of data to manage to select the most suitable applicant. The task of assessing the resumes of candidates and providing automatic recommendations on which one suits a particular position best has, therefore, become essential to speed up the hiring process. Similarly, it is important to help applicants to quickly find a job appropriate to their skills and provide recommendations about what they need to master to become eligible for certain jobs. Our approach lies in this context and proposes a new method to identify skills from candidates' resumes and match resumes with job descriptions. We employed the O*NET database entities related to different skills and abilities required by different jobs; moreover, we leveraged deep learning technologies to compute the semantic similarity between O*NET entities and part of text extracted from candidates' resumes. The ultimate goal is to identify the most suitable job for a certain resume according to the information there contained. We have defined two scenarios: i) given a resume, identify the top O*NET occupations with the highest match with the resume, ii) given a candidate's resume and a set of job descriptions, identify which one of the input jobs is the most suitable for the candidate. The evaluation that has been carried out indicates that the proposed approach outperforms the baselines in the two scenarios. Finally, we provide a use case for candidates where it is possible to recommend courses with the goal to fill certain skills and make them qualified for a certain job.

1. Introduction

The job market has been heavily influenced by the tons of information that are posted on the Web every day regarding job supply and demand. The online enrolling process has become efficient for applicants as it allows them to present their resumes using the Internet and, as such, simultaneously to numerous organizations (companies or research centers). One problem that arises from the application to multiple online systems is that there is no universal standard format to adopt when filling resume information (although systems such as Europass[1] or LinkedIn[2] have some methods to automatically generate structured profiles). It follows that resumes are all very different from each other in terms of structure, design, and format, hindering an efficient and fast analysis from people working within human resources divisions of

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the different organizations. Studies on job-resume matching have highlighted the challenges posed by the lack of standardization in resumes, which might lead to inconsistencies in candidate evaluation [1,2].

The simplicity and speed of sending online applications hide one more problem related to resume analysis. When an organization posts a certain job, there is a number of required job-specific and mandatory skills. However, several applicants who do not satisfy them keep sending their resume "just to give it a try". This happens because there are no risks in sending non-eligible applications (although, in certain situations, they might end up in some black list of low-quality applications[3]).

Online systems such as Monster.com[4], OfferZen[5] and LinkedIn contain millions of job offers and resumes of potential candidates leaving to companies the hard task to face enormous amounts of data with the goal of selecting the most suitable applicant for their needs[6]. Quickly assessing the resumes of candidates and providing automatic recommendations on which one suits best a particular position have, therefore, become essential to speed up the screening and hiring process for companies [3–5]. Similarly, it is important to help applicants to quickly find a job appropriate to their skills and provide recommendations about what they need to master to become eligible for their desired jobs [6].

The first problem to solve has been the text extraction from resumes represented by non-textual documents (e.g., images or PDF files). This problem has already been discussed and faced in other domains as well leading to the creation of systems with very accurate extraction of body text [7–9]. For instance, the study presented in [10] discusses methods for extracting information from documents like resumes, highlighting advancements in this area.

Once the body text has been extracted, the next challenge is to identify words or compound expressions that match the candidate's skills peculiar to the underlying job. One resource that has been created to support this problem is the O*NET database [11][7]. It contains a rich set of variables that describe work and worker characteristics, including skill requirements. It also contains hundreds of descriptors on almost a thousand occupations related to the United States. Thus, the O*NET database represents a valuable resource to identify skills in a candidate's resume and associate them with related jobs. Using O*NET, the challenge to find words indicating job skills is therefore narrowed to the identification of words or expressions within a certain resume which correspond to those described within the O*NET database. For example, a resume might contain the word *programming* which, in O*NET, corresponds to a skill for several jobs. Simple syntactic rules (i.e., string matching, approximate string matching using the Levenshtein distance, etc.) may not be enough; in fact, the presence of word synonyms and the different forms a sentence can be formulated hinder their use. As a further example, in O*NET the description associated with the technology skill *Google Drive*, which, in turn, is one of the skills required for the job *Marketing Managers*, is *Cloud-based data access and sharing software*. Hardly a resume will contain the same formulation; more likely it will include different expressions conveying the same meaning such as *use of cloud-based software*, or *use of sharing applications* to name a few.

With the advancements of Machine Learning technologies, we have today very powerful methods able to compute the semantic similarity of two pieces of text (documents, paragraphs, expressions). Furthermore, the evolution of Deep Learning transformers [12,13], enabled by the huge amount of publicly available annotated datasets and disposal of powerful graphical processing units (GPU) to run neural networks in parallel, have provided cutting-edge solutions for a great number of problems included those revolved around the Semantic domain [14]. Transformers are able to efficiently solve several tasks such as sequence classification, question answering, language modeling, text generation, named entity recognition, and most of all, semantic similarity between two texts. Therefore, thanks to the advancements in Natural Language Processing (NLP), Semantic Web, Artificial Intelligence, Deep Learning, and Information Extraction, we have a plethora of technologies (software and hardware) to efficiently extract and identify valuable and relevant information from candidates' resumes. For instance, recent work has proposed the use of Large Language Models (LLMs) models to produce personalized job descriptions given input CVs [15], leverages semantic-enhanced transformers to parse job descriptions and align them with user skills, improving the accuracy of recommendations [16], handling job recommendation as resource allocation tasks to study the fairness of machine learning-based models [17]. Despite these advancements, LLM- and deep learning-based recommenders face a significant challenge: they frequently introduce biases and fail to provide clear, interpretable justifications for why a particular CV is matched with a given job, raising concerns about fairness and accountability in the recommendation process.

Our paper lies in this context and proposes a new approach to identify skills from candidates' resumes and job postings and match the two of them. We consider the O*NET database information related to technology skills, skills, knowledge, abilities, work activities, task ratings, and tools used. Then, given the resume of a candidate, we run different NLP tools to extract information that is then matched against the O*NET entities and returns the most suitable job for the underlying resume. We consider two scenarios, one useful for the companies needed to screen several resumes and another one useful for candidates when looking for a job among several job postings:

- First scenario: a resume is provided as input. Then the goal is to identify the O*NET occupation with the highest match with the resume;
- Second scenario: a resume and a list of job postings are given in input. The goal is to find which of the input job postings matches the most candidate's skills present in the resume.

The contributions of our paper are therefore the following:

- we employ Deep Learning transformers to identify the pieces of text in a resume with high semantic similarity with the entries of corresponding O*NET entities;
- using ad-hoc metrics we identify the most related O*NET job to the resume of the underlying candidate;
- we provide two complementary scenarios where we applied our approach;
- for the first scenario, we tested our approach on a dataset of 105 resumes and outperformed two baseline methods we have defined; for such a purpose, three independent experts have annotated the dataset which we have publicly released;
- for the second scenario, we tested our approach on a dataset of 100 resumes where each of them was associated with 10 different job postings (the resume was qualified for only one of them); our approach outperformed two more baselines we have defined; again, for such a purpose, three independent experts have annotated the dataset which we have publicly released;
- we provide a use case where we apply our approach to recommend courses and lectures to candidates with the goal of acquiring certain skills and thus resulting qualified for certain jobs they had targeted;
- we release the source code of our approach for both the scenarios and the recommendation task in a public repository[8];

[3] https://www.monster.com/career-advice/article/things-that-will-blacklist-you-from-job.

[4] http://www.monster.com.

[5] https://www.offerzen.com/.

[6] https://economicgraph.linkedin.com/resources/linkedin-workforce-report-april-2022.

[7] https://www.onetcenter.org/database.html.

[8] Scenarios source code: https://gitlab.com/hri_lab1/using-transformers-and-o-net-to-match-jobs-to-applicants-resumes.

- we provide an interactive demo[9] and its source code[10] that implements our proposed approach to match applicants' resumes with the O*NET database jobs to measure the eligibility of a user for a given job.

The remainder of this paper is organized as follows. Section 2 discusses related works on resume extraction and job recommendation. Section 3 details the tools we have used in this paper as well as the O*NET database. The proposed approach is presented in Section 4 whereas Section 5 introduces the scenarios we have considered. Section 6 illustrates the evaluation we have carried out for the two scenarios. A use case we propose in the paper which can leverage the proposed approach to recommend skills to job seekers is shown in Section 7. Finally, Section 8 ends the paper with conclusions, limitations, and future works where we are headed.

2. Related work

The last two decades have seen the development of online recruiting platforms, a topic that has recently acquired increasing attention. Authors in [18,19] presented two surveys of existing recommendation approaches that have been proposed to create recommendation systems for job seekers and recruiters. On the one hand, extracting features from resumes and from job postings has always been challenging. In particular, user profiling deals with acquiring, extracting, and representing the features of users [20]. On the other hand, job profiling is a representation of job descriptions and their requirements. Usually, job descriptions come in unstructured text with no attribute names with well-defined values. It follows that the skill set for a particular job includes skills with a Boolean value: True if the skill is required for that job, False otherwise.

A kind of approach that achieved great success to recommend jobs is collaborative filtering. It is based on the assumption that if users A and B have similar behaviors they will rate other items similarly [21–23]. Authors in [23] developed a job recommendation system using a model-based collaborative algorithm with clustering algorithms. The Latent Semantic Analysis (LSA) and Singular Value Decomposition (SVD) have been adopted to create a lower-rank matrix with information about skills and positions. Moreover, the inverse cosine similarity was employed as distance to perform the agglomerative clustering to create clusters of positions. Authors in [22] created a system for job recommendation by making clusters of users that are based on skills extracted from different websites. The Euclidean distance has been leveraged as a measure of similarity between the skills of users. Then, identified skills with low occurrence were removed from the list and a classification using Naive Bayes was employed to rank the final set of recommendations for a user.

The Labor Market Explorer interactive dashboard for job seekers has been presented by authors in [24]. It was built with a careful user-centered design processing where both job seekers and job mediators were involved so that the matching process between jobs and job seekers could be optimized. The dashboard enables an exploration of the job market in a personalized way based on the skills and competencies of the applicants. Efforts related to career exploration and the detection of training needs have also been and are being carried out. For example, the STAR[11] project has the goal to design new technologies to enable the deployment of standard-based secure, safe, reliable, and trusted human-centric AI systems in manufacturing environment. There, the Workers' Training Platform is being developed. This platform allows workers to self-assess themselves and detect training needs related to skills or knowledge while offering them training recommendations for those skills. All in an anonymous way and based on public occupation databases such as O*NET or ESCO[12].

Other authors in [25] created a system to suggest jobs based on users' profiles. Users and jobs have been treated as text documents, and a model that incorporates job transitions trained on the career progressions of a set of users has been adopted. The authors also showed that combining career transitions with cosine similarity outperforms the system using just career transitions. The evaluation proving the statements above has been carried out on a dataset of 2,400 Linkedin users with the task of predicting users' current positions by looking at their profiles and their jobs history.

The works illustrated above do not leverage a widely recognized taxonomy of skills or other similar entities. Sometimes the set of the considered skills is too high and approaches to reduce the dimensional space need to be adopted with the drawback of losing precision. Differently from the previous approaches, we use the O*NET taxonomy by including not just skills but several other entities of a different kind (i.e., knowledge, abilities, technology skills, etc.). O*NET is one of the main occupational databases, and is almost a reference for occupation analysis and worker requirements. It is the primary source of occupational information in the United States and it includes regularly updated occupational characteristics based on questionnaires made to several hundred workers. In doing so, we inject into our approach cognitive, interpersonal, and physical skill knowledge representing worker and job requirements coming from a large number of workers and companies, thus making the person-job matching more robust. Finally, our work differs from previous studies [21–25] because it does not rely on previous applicants for a specific job position like in collaborative filtering approaches. Therefore, each resume is used as it is and is matched against jobs thus allowing us to avoid biases as well as the cold start problem [26]. Last but not least, to match each of these entities with applicants' resumes and job postings we make use of sentence transformers thus leveraging the semantics of each word, sentence, and context. This differs from the existing studies which often make use of only well-defined attributes mentioned in the applicants' resumes or job postings, thus making it possible for our approach to deal with the complexity of the natural language text contained in applicants' resumes.

3. Task definition and the used material

In this section, we provide a formal definition of the task addressed in this work, alongside details about the transformers model and the O*NET database leveraged in the proposed approach.

3.1. Task definition

The objective of our approach is to develop a system that automatically identifies and matches relevant skills and qualifications between candidates' resumes and job postings. Formally, given:

- **Resume Information (R):** A structured or semi-structured textual representation of a candidate's skills, experiences, education, and qualifications.
- **Job Information (J):** A set of job classes derived from O*NET, comprising multiple entities such as technology skills, general skills, knowledge, abilities, work activities, task ratings, and tools used.

The task is defined as a mapping function $f : R \rightarrow J$, where for each resume $r \in R$, the function f returns a ranked list of job classes $\{j_1, j_2, ..., j_n\} \subset J$, ordered by their relevance to the candidate's profile.

The relevance score is calculated by leveraging semantic similarity to align the extracted elements from the resume with the O*NET entities corresponding to each job. For each matched element, a job-specific

[9] Demo: http://192.167.149.11:8000.

[10] Demo source code: https://gitlab.com/hri_lab1/onet-db26-transformers-demo.

[11] https://www.star-ai.eu.

[12] https://ec.europa.eu/esco/.

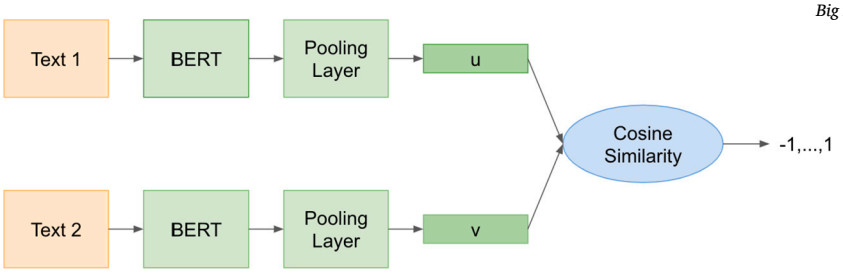


Fig. 1. The figure illustrates the structure of a Siamese network designed for text comparison. Each branch of the network consists of a text input processed through a BERT model that encodes the text into vectorial representations, and a pooling layer that aggregates the token-level features into a fixed-size vector representation capturing the most salient information. The resulting vectors from both branches u and v are then compared using cosine similarity to measure their similarity. .

score derived from O*NET is assigned across various categories, including "skills", "knowledge", "abilities", "work activities", and "tasks". Additionally, conventional scores are applied for "technology skills" and "tools used".

The approach operates in two specific scenarios:

1. **Resume-to-Job Matching (R2J):** Assisting companies in screening multiple resumes to identify the most suitable candidates for a specific job posting.
2. **Job-to-Resume Matching (J2R):** Helping candidates find the most relevant job postings based on their resumes.

By leveraging NLP advancements and semantic similarity computation, the proposed system aims to address challenges such as entity disambiguation, contextual understanding of skills and qualifications, and alignment between heterogeneous textual data from resumes and job descriptions.

3.2. Sentence transformers

SentenceTransformers[13] [27] is the state-of-the-art Python framework for text embeddings generation. The framework leverages models that are built on top of the original BERT model [28] or its further developments such as RoBERTa [29], MPNet [30], and ALBERT [31]. It provides transformer models that use siamese and triplet network structures to transform sentences into embeddings. Then, these can be compared by using metrics (e.g., the cosine similarity) to perform tasks such as information retrieval, clustering, and semantic search. The first models were trained on Natural Language Inference (NLI) datasets [32,33] and successfully became the state of the art for solving Semantic Textual Similarity (STS) tasks.

Today, *SentenceTransformers* pre-trained models are built as extensions of Huggingface Transformers models[14] by applying pooling layers within siamese structures. The reader can observe an example of a common siamese structure with two BERT models to compute the cosine similarity between two texts in Fig. 1. The pool of *SentenceTransformers* models can be found at https://huggingface.co/models.

Within the proposed system, we have chosen to embed the *all-mpnet-base-v2* model, which achieves a $\rho = 0.69$ on encoding sentences over 14 diverse tasks, and a $\rho = 0.57$ on 6 diverse tasks for semantic search such as encoding of queries and questions. According to the official Sentence Transformers documentation[15], this model is specifically noted for achieving higher performance compared to a range of other pre-trained models, including standard BERT-based and RoBERTa-based architectures. The documentation emphasizes that Sentence Transformers models are optimized for sentence-level tasks, such as semantic similarity, making them particularly well-suited for our application, which involves comparing resumes, job descriptions, and O*NET entities. In addition, our use of Sentence Transformers aligns with the nature of our task, where capturing nuanced semantic similarity between textual data is critical. These models are specifically designed to deliver superior results for tasks involving sentence embeddings and pairwise comparisons, which are central to our approach. The reader notices that ρ corresponds to the Spearman rank correlation coefficient [34] between the ranking of sentence pairs using the cosine similarity as score and the gold standard ranking for various semantic textual similarity tasks. The Spearman rank correlation coefficient gives a score in the continuous range $[-1, 1]$ where a value of 1 indicates a perfect correlation between the two rankings, a value of 0 indicates a weak correlation, and a value of -1 means a perfect negative correlation.

The *all-mpnet-base-v2* model has a size of 420 Mbytes, a Max Sequence Length of 384 tokens, and an embedding speed of $2,800$ sentences/sec on a V100 GPU. The generated embeddings are 768-dimensional vectors. The suitable score functions are the dot product, the cosine similarity, and the Euclidean distance.

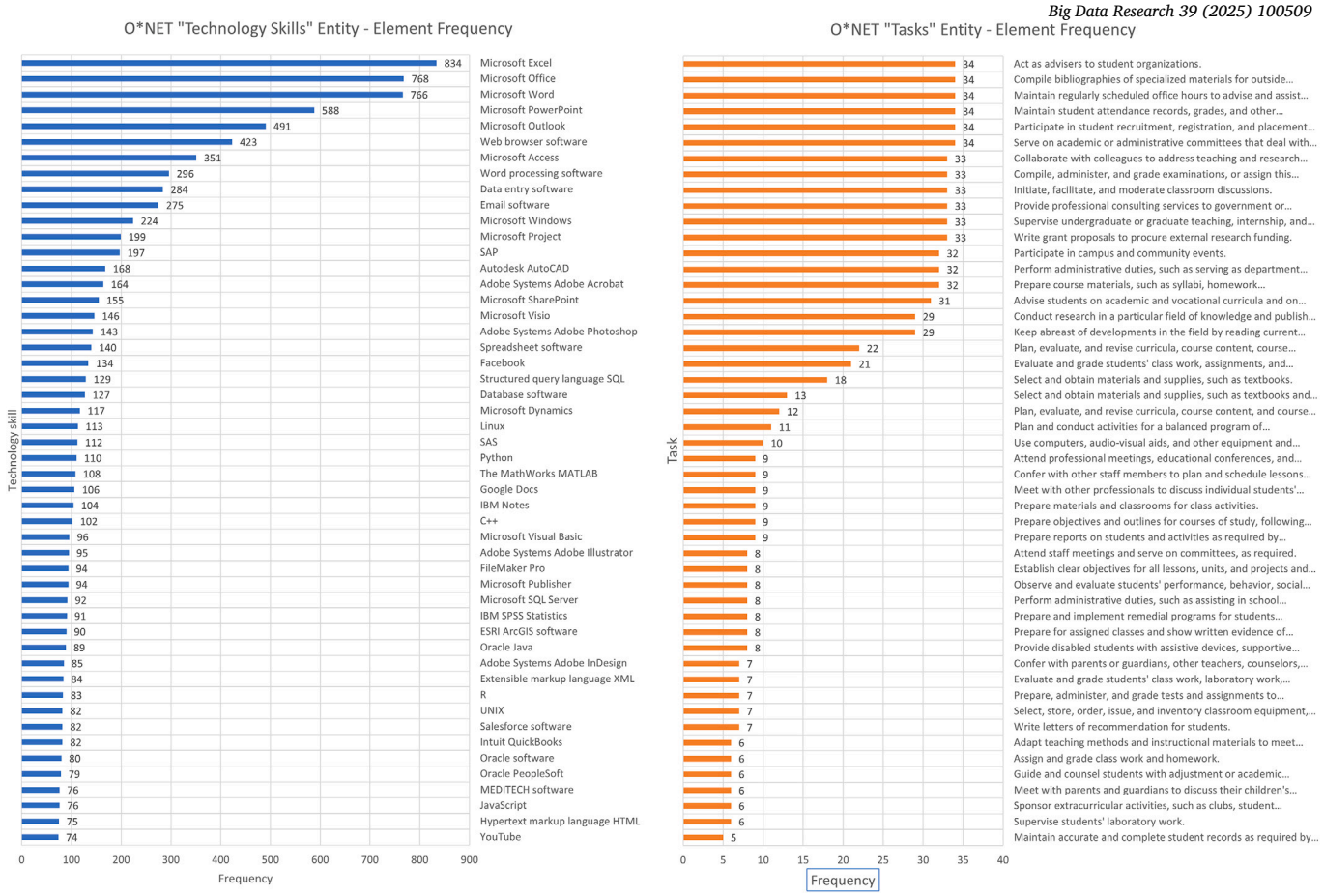
3.3. The O*NET database

The O*NET Program[16] is the primary source of occupational information for the United States. The goal of its creation was to understand the rapidly changing nature of work and how it impacts the workforce and the U.S. economy. One of the main outcomes of the program is the O*NET database, which includes hundreds of standardized and occupation-specific descriptors on almost a thousand occupations covering the entire U.S. economy. The database is freely available, under a Creative Commons license, and is continuously updated on a quarterly basis by different institutions. It has already been used by millions of persons for career exploration and to discover which training is necessary to be eligible for a position, and by employers to find skilled workers to better compete in the marketplace.

Each occupation included in the O*NET database requires a disparate mix of knowledge, skills, and abilities and is performed using a variety of activities and tasks. We have used the O*NET database version 26.2 which includes 1,016 occupations, 52 *Abilities*, 33 *Knowledge*, 35 *Skills*, 41 *Work Activities*, 4,127 *Tools Used*, 17,975 *Task Ratings*, and 8761 *Technology Skills* grouped into 135 categories. In the remainder of the paper, we will refer to *Abilities, Knowledge, Skills, Work Activities, Tools Used, Task Ratings* and *Technology Skills* as O*NET entities. Each occupation is associated with a unique identifier, a title, and a related description. For example, the occupation *Computer and Information Research Scientist* has description *Conduct research into fundamental computer and information science as theorists, designers, or inventors. Develop solutions to problems in the field of computer hardware and software*.

The *Abilities* consist of a set of capacities related to each occupation. One occupation may be associated with many abilities, each with a related score. A given ability (as for *Knowledge*, *Skills* and *Work Activities*) might be present in different occupations with different scores.

[13] https://www.sbert.net/.
[14] https://huggingface.co/docs/transformers/index.
[15] https://www.sbert.net/docs/pretrained_models.html.
[16] https://www.onetcenter.org/.



(a) O*NET "Technology Skills" Entity.

(b) O*NET "Tasks" Entity.

Fig. 2. O*NET "Technology Skills" and "Tasks" Entities - First 50 elements by frequency.

One example is *Oral Comprehension* which, among others, appears in the occupations *Chief Executives* and *Biostatisticians* with a score value of, respectively, 4.5 and 4. The score reflects the importance of that ability with respect to the associated occupation and has a value in the [1-6] continuous range. The higher the score, the more important that ability is for the job it refers to.

Knowledge represents the required area of expertise for the underlying job and has scores in the [1-7] continuous range. An example is provided by *Mathematics* which has a score of 6.83 for the occupation *Mathematicians*. Similarly to the abilities, also Knowledge has many-to-many relations with the occupations and each different pair (occupation, knowledge) has a different score value.

Skills are the competencies required for each occupation and have a score in the [1-6] continuous range. One example is *Programming*, associated with the job *Computer Systems Engineers/Architects* with a score value of 3.38.

For *Work Activities* (values range in the [1-7] continuous interval), an example is given by *Analyzing Data or Information*, with a score of 6.61 with respect to the occupation *Financial Quantitative Analysts*. Both Skills and Work Activities have many-to-many relations with different score values for each different job.

One example of *Task Ratings* is *Direct and coordinate activities involving sales of manufactured products, services, commodities, real estate or other subjects of sale*, related to the job *Sales Manager* with an importance score of 4.22 (scores of Task Ratings are in the interval [1-5]).

An example of *Tools Used* is *Personal Computer*, associated with several jobs (almost all of them).

Finally, an example of *Technology Skills* is *Atlassian JIRA*, related to the occupation *Administrative Services Managers*; Technology Skills have one more field, (*hot technology*), meaning whether the underlying technology skill is hot or not.

The reader notices that the last three described O*NET entities, Task Ratings, Tools Used, and Technology Skills, only occur when they are required for the underlying job. Moreover, Tools Used and Technology Skills do not have a score value associated with an occupation, Task Ratings does. Overall, as previously mentioned, there are 4,127 Tools Used, 17,975 Task Ratings, and 8761 distinct Technology Skills. A certain O*NET job will require a subset of each of the three entities. For example, the O*NET job *Chief Executives* requires 7 Tools Used, 31 Task Ratings, and 49 Technology Skills. Conversely, each O*NET occupation is always associated with a fixed number for each of the first four described entities. Hence, any O*NET occupation will be associated with a vector of 52 Abilities, a vector of 33 Knowledge, a vector of 35 Skills, and a vector of 41 Work Activities.

One more table worth to be mentioned is the content model reference which contains complete descriptions of all the elements included in the entities of the O*NET database previously described. For example, one of the Abilities' elements is *Cognitive Abilities* and its description is *Abilities that influence the acquisition and application of knowledge in problem solving*.

To provide the reader with some statistics, Figs. 2 and 3 show the first 50 elements, in decreasing order by the number of times occurring within the dataset's jobs, of the O*NET entities with a variable number of elements per job, that is "Technology skills", "Tasks", and "Tools Used". For example, *Microsoft Excel* is a technology skill required by 834 O*NET jobs whereas *Personal Computers* is a Tool required by 656 occupations. Similarly, Figs. 4 and 5 show the average values of the scores, calculated over all the O*NET jobs, of the elements of the O*NET entities

Table 1
Dimension of the vector space for the first four O*NET entities.

Name	Dimension	Example elements
Abilities	52	Oral Comprehension, Written Comprehension
Knowledge	33	Sales and Marketing, Computers and Electronics
Skills	35	Science, Active Learning, Service Orientation, Speaking
Work Activities	41	Analyzing Data or Information, Thinking Creatively

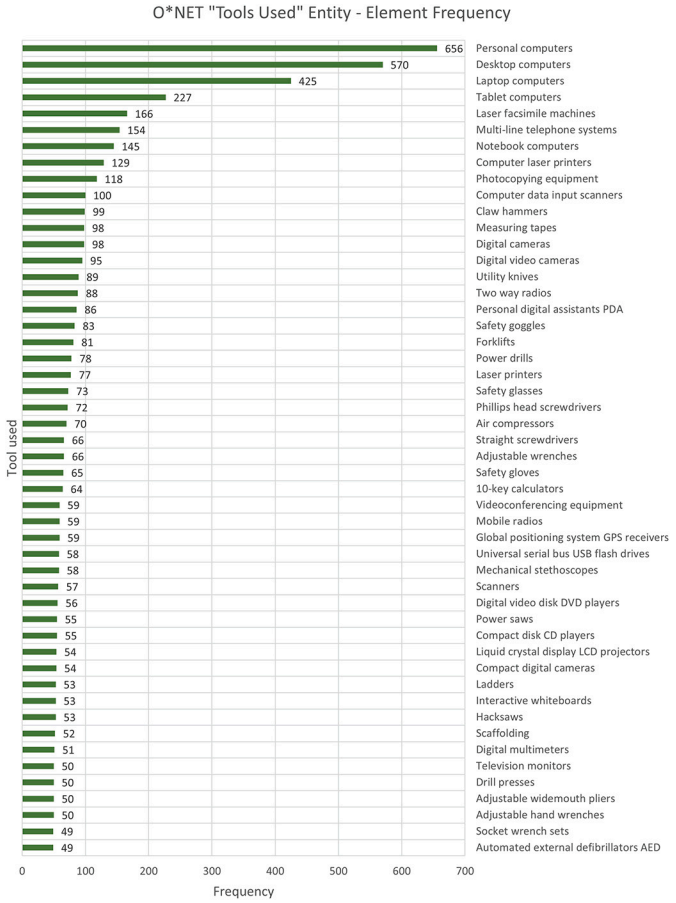


Fig. 3. O*NET "Tools Used" Entity - First 50 elements by frequency.

with a fixed number of elements per job. So, for example, *Active Listening* is the value of the entity Skills with the highest average score (3.60) whereas *English Language* is the value of the Knowledge entity with the highest average score (3.66). Similar considerations can be done for the values of the entities in Fig. 5.

4. The proposed approach

This section describes the proposed approach used to match resumes and jobs.

4.1. Adopted O*NET entities

We used the following O*NET entities:

- "Abilities",
- "Knowledge",
- "Skills",
- "Work Activities",
- "Task Ratings",
- "Technology Skills",
- "Tools Used".

We have mapped the first four entities to n-dimensional vectors in which the components are the scores (already mentioned in Section 3) attributed to the importance of the ith element of the underlying vector (whose size is reported in Table 1) of one of the entities with respect to the 1,016 different jobs present in the O*NET database. For sake of clarity, let us remark that, as discussed in Section 3, there is a difference among the entities. The first four entities have a fixed dimension for each job as indicated in Table 1. The other three entities have dimensions that vary depending on the job. For Task Ratings for example, the job *Sales Manager* includes 17 of them (each with a score or data value, as shown in Fig. 6) whereas the job *Wind Energy Engineers* includes 16 other different tasks.

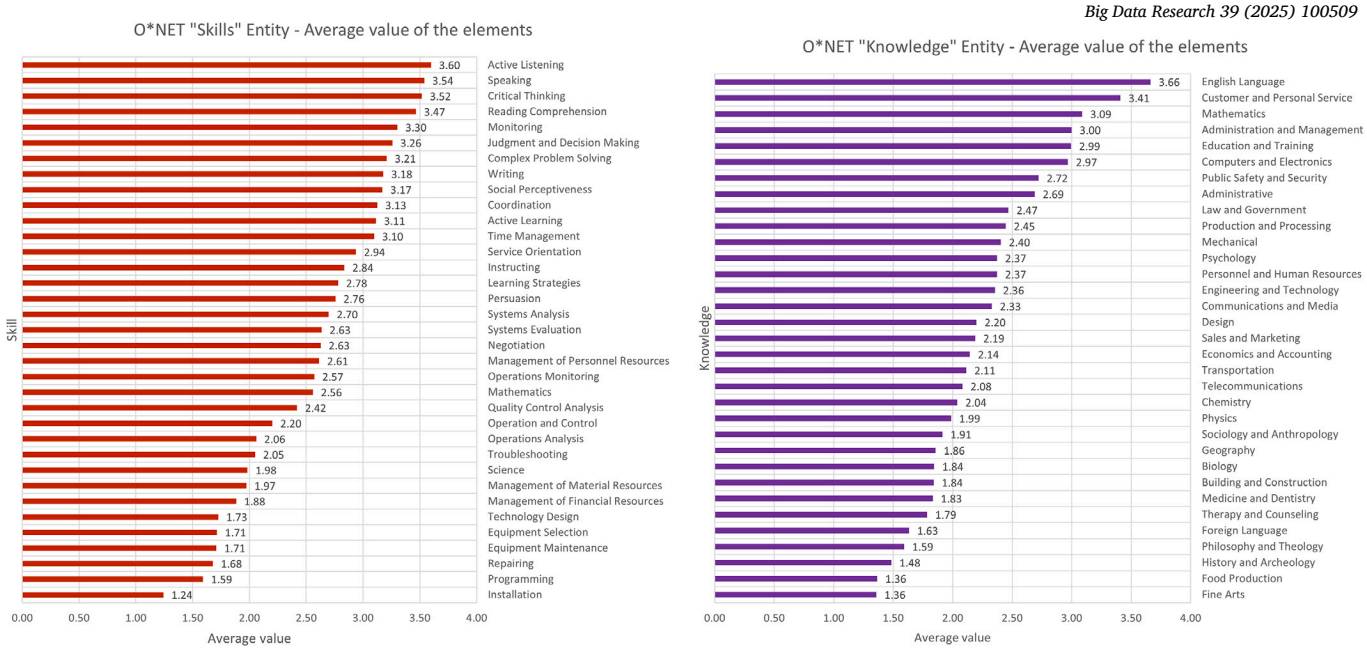
The reader notices that for tasks we have considered the scores associated with the scale importance which lies within the [1-5] interval. As mentioned in Section 3.3, in the O*NET database, the only elements that are not weighted differently based on their relative importance for job matching are technology skills and tools. Our system utilizes all the information provided by the O*NET database as it is, without altering or interpreting the design choices made by its developers, in order to preserve the consistency and structure of the database. Specifically, Technology Skills have no values relative to their importance but are classified by O*NET as "hot technology" or not. Hence, assuming we extract an element from the resume that is matched against a certain Technology Skill, we decided to assign a score of 1 if the recognized item is classified from O*NET as "hot" and 0.75 otherwise. Finally, for a certain *Tool Used* identified in a resume, we simply assign a score of 1.0 if it is identified and 0 if it has not been detected.

4.2. Transformer-based comparison

To collect information to compare against the O*NET entities, we extracted sentences, nouns (e.g., *network, mathematics, archaeology*), and noun phrases (e.g., *system administrator, server management, machine learning*) from the text of resumes or jobs descriptions using the *TextBlob* library[17]. It was selected for its simplicity, efficiency, and reliability in extracting nouns and noun phrases, which align well with the structured elements of O*NET entities. This library allowed us to effectively preprocess the textual data, ensuring compatibility with the downstream tasks of semantic similarity and job matching.

The nouns and noun phrases extracted are compared with the values of the elements of the O*NET entities, while the sentences are compared with the descriptions of these elements. The descriptions of elements have been introduced in Section 3 and exist for the entities Abilities, Knowledge, Skills, Work Activities, and Task Ratings. The reason why we chose nouns and noun phrases to be matched with the entities elements is the size of the latter (in general they consist of a few tokens). To not lose the information given by the descriptions of some of the entities we have considered, we have also compared the descriptions with the entire sentences extracted from the resumes or job descriptions. We used a pre-trained neural network model from Sentence Transformers *('all-mpnet-base-v2')* to get text embeddings and leverage the cosine similarity to generate, given a certain resume or job posting, one similarity matrix for each pair (job, entity). Given a similarity matrix for a pair (job, entity), the columns of the matrix are the information extracted from the

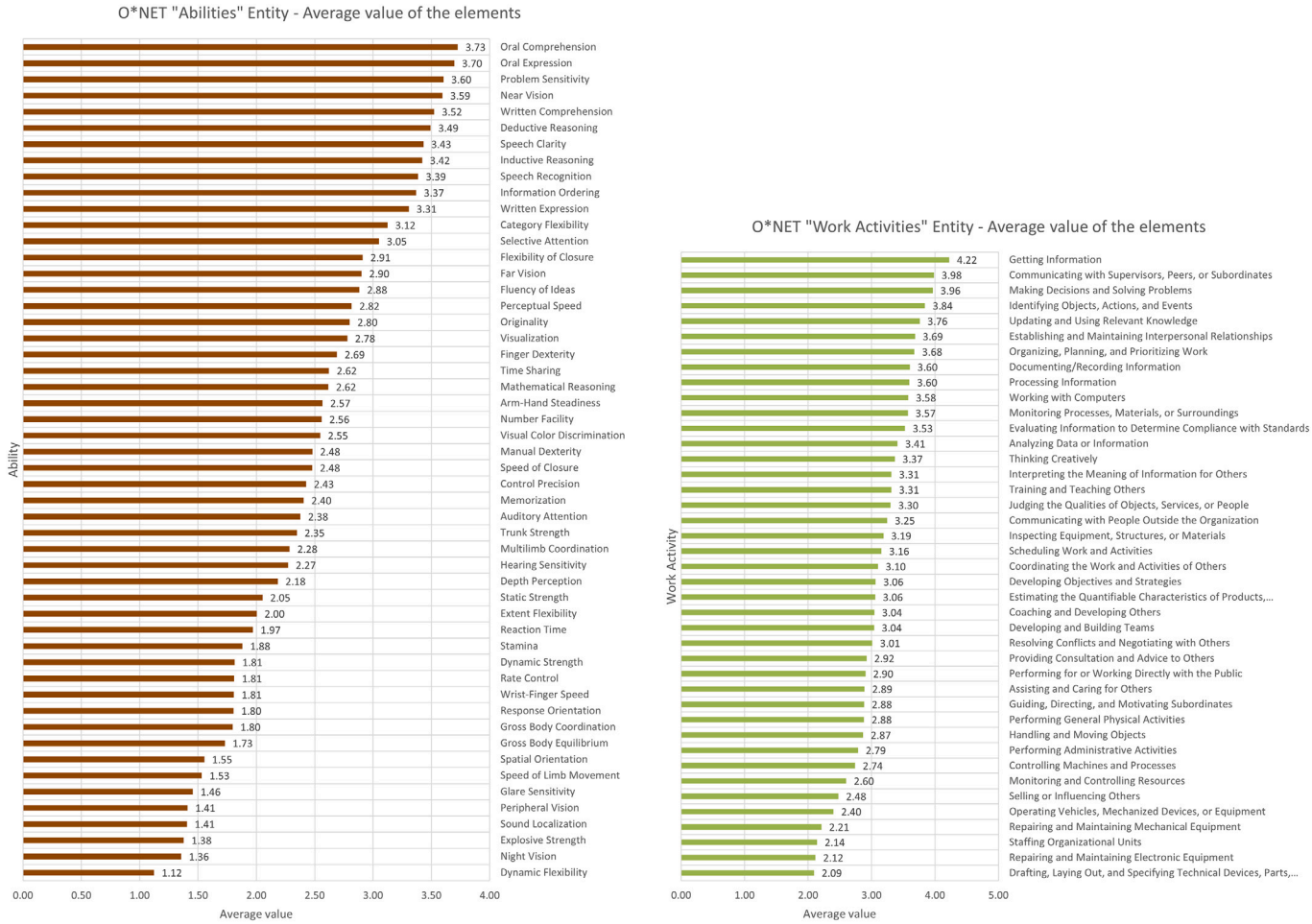
[17] https://textblob.readthedocs.io/en/dev/.



(a) O*NET "Skills" Entity.

(b) O*NET "Knowledge" Entity.

Fig. 4. O*NET “Skills” and “Knowledge” Entities - Average value of the elements.



(a) O*NET "Abilities" Entity.

(b) O*NET "Work Activities" Entity.

Fig. 5. O*NET “Abilities” and “Work Activities” Entities - Average value of the elements.

O*NET-SOC Code	Title	Task	Task	Scale ID	Data Va
11-2022.00	Sales Managers	1	Resolve customer complaints regarding sales and service.	IM	4,09
11-2022.00	Sales Managers	2	Monitor customer preferences to determine focus of sales efforts.	IM	3,57
11-2022.00	Sales Managers	3	Direct and coordinate activities involving sales of manufactured products, services, commodities, real estate or other subjects of sale.	IM	4,22
11-2022.00	Sales Managers	4	Determine price schedules and discount rates.	IM	3,61
11-2022.00	Sales Managers	5	Review operational records and reports to project sales and determine profitability.	IM	3,95
11-2022.00	Sales Managers	6	Direct, coordinate, and review activities in sales and service accounting and record-keeping, and in receiving and shipping operations.	IM	3,48
11-2022.00	Sales Managers	7	Confer or consult with department heads to plan advertising services and to secure information on equipment and customer specifications.	IM	3,25
11-2022.00	Sales Managers	8	Advise dealers and distributors on policies and operating procedures to ensure functional effectiveness of business.	IM	3,26
11-2022.00	Sales Managers	9	Prepare budgets and approve budget expenditures.	IM	3,59
11-2022.00	Sales Managers	10	Represent company at trade association meetings to promote products.	IM	3,14
11-2022.00	Sales Managers	11	Plan and direct staffing, training, and performance evaluations to develop and control sales and service programs.	IM	3,5
11-2022.00	Sales Managers	12	Visit franchised dealers to stimulate interest in establishment or expansion of leasing programs.	IM	3,44
11-2022.00	Sales Managers	13	Confer with potential customers regarding equipment needs and advise customers on types of equipment to purchase.	IM	4,07
11-2022.00	Sales Managers	14	Oversee regional and local sales managers and their staffs.	IM	3,7
11-2022.00	Sales Managers	15	Direct clerical staff to keep records of export correspondence, bid requests, and credit collections, and to maintain current information on tariffs, licenses, and restrictions.	IM	3,38
11-2022.00	Sales Managers	16	Direct foreign sales and service outlets of an organization.	IM	3,14
11-2022.00	Sales Managers	17	Assess marketing potential of new and existing store locations, considering statistics and expenditures.	IM	3,55

Fig. 6. Task Ratings for the job Sales Manager.

resume (or job posting) and the rows are the elements of the O*NET entity which might occur in the job being considered. For example, when analyzing the entity Abilities for a certain job, the associated similarity matrix will have 52 rows and a number of columns depending on the information (nouns, sentences, noun phrases) extracted from the resume or job posting. We fill each cell of the matrix with the similarity value between the extracted information from the resume/job posting (column c) and the elements of the O*NET entities (row r). From each column, the system extracts the element with the maximum value, which corresponds to the element of the O*NET entity with the highest similarity value with respect to the extracted element of the resume/job posting represented in the underlying column. If the similarity value is greater than an empirically found threshold, the score of the element of the O*NET entity is added to the resume/job posting score for the current job.

This threshold was determined after conducting numerous experiments, and it was adjusted to identify the value that maximized classification accuracy. We tested different values and found that 0.65 provided the best balance between correctly identifying relevant O*NET entities and minimizing misclassifications.

The system then calculates the maximum score obtainable for the current job j and normalizes the score obtained from the resume/job posting by applying the formula:

$$score(j) = \sum_{i=1}^{i=num_entities} \left(\frac{\sum_{k \in entity_i} score_k(j)}{score_max_i(j)} + 0.5 * \sum_{k \in entity_i} score_k(j) \right) \quad (1)$$

where $num_entities$ is equal to seven and corresponds to the entities that have been previously described, $score_k(j)$ is the sum of the scores of each element (identified in the resume/job posting) of the entity i of the job j that is being checked whereas $score_max_i(j)$ is the maximum score that the job j would obtain if all the related elements of the entity i were detected in a resume/job posting. The term $(0.5 * \sum_{k \in entity_i} score_k(j))$ acts as a corrective factor, empirically determined to prevent the misjudgment of job categories with a large number of entities associated in ONET. Without this correction, the formula would only normalize the scores by dividing by the maximum possible score, which could unfairly disadvantage jobs with many associated entities. For example, if a job category has 25 entities associated in O*NET and only 5 entities are matched in the resume, the score would be normalized by dividing by the sum of the 25 scores, yielding a lower score compared to a job with only 10 associated entities in O*NET. This imbalance could lead to the underrepresentation of jobs with a higher number of associated entities, such as IT-related jobs, which are linked to many tools and technical skills in O*NET, even if a significant portion of their relevant entities is detected in the resume. The corrective factor is necessary to address this issue, ensuring that jobs with a large number of associated entities do not receive disproportionately low scores due to the normalization process. We conducted several experiments adjusting this parameter (e.g., testing values like 0.4, 0.3, etc.), and found that 0.5 provided the best balance for correct classification across diverse job types. Changing this value would influence the classification results, as a smaller corrective factor could lead to the misclassification of jobs with many associated entities, while increasing the factor could cause underrepresentation of jobs with fewer associated entities. Therefore, the value of 0.5 was chosen based on its ability to yield the highest accuracy across various datasets. This

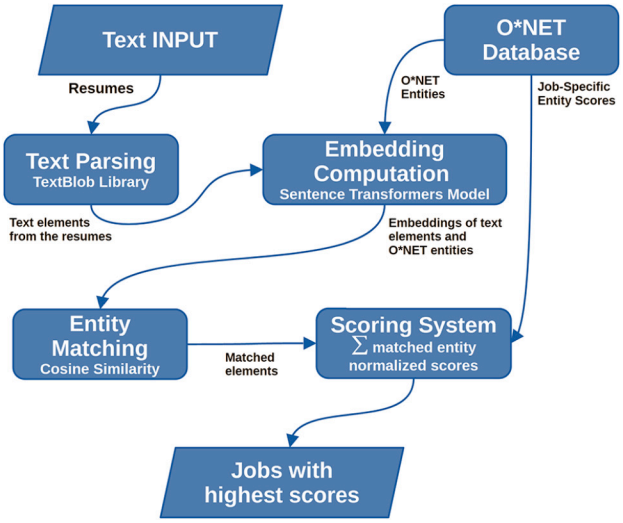


Fig. 7. System diagram.

SIMILARITY MATRIX

	Information from O-NET database job = ['15-1251.00', 'Java Programmer']		Information from the resume						
index	Knowledge element	value	computers	training	maths	english	sales	administrator	engineer
0	Administration and Management	3.26	0.3067	0.2420	0.3296	0.2536	0.4138	0.5764	0.3703
1	Administrative	2.74	0.3654	0.2423	0.2996	0.2819	0.3066	0.8404	0.4794
2	Economics and Accounting	2.47	0.2506	0.1430	0.4473	0.2936	0.4193	0.2083	0.3087
3	Sales and Marketing	2.07	0.3459	0.2934	0.3514	0.2702	0.7480	0.2688	0.3465
4	Customer and Personal Service	3.30	0.2763	0.2987	0.2691	0.1961	0.4606	0.2444	0.3122
5	Personnel and Human Resources	2.00	0.3245	0.3324	0.3236	0.2698	0.3637	0.3159	0.3729
6	Transportation	1.69	0.4229	0.3826	0.3341	0.3014	0.4288	0.1454	0.3436
7	Production and Processing	2.55	0.2568	0.2375	0.2337	0.2191	0.3406	0.1750	0.3123
8	Food Production	1.00	0.1712	0.2081	0.2334	0.1899	0.2773	0.1309	0.2384
9	Computers and Electronics	4.87	0.6389	0.2166	0.3923	0.2473	0.3481	0.2507	0.4653
10	Engineering and Technology	3.47	0.3844	0.2484	0.4444	0.3085	0.2537	0.2114	0.6642
11	Design	2.88	0.4339	0.2534	0.2906	0.2978	0.3643	0.2943	0.4662
12	Building and Construction	1.00	0.3259	0.3185	0.3882	0.2990	0.3212	0.2105	0.5303
13	Mechanical	1.53	0.4239	0.2479	0.3228	0.2206	0.3080	0.2707	0.4703
14	Mathematics	3.53	0.4126	0.2621	0.8136	0.3491	0.3429	0.1322	0.3572
15	Physics	2.11	0.3966	0.2368	0.5643	0.2787	0.2826	0.1021	0.3639
16	Chemistry	1.15	0.3293	0.2326	0.4350	0.3483	0.2403	0.1525	0.3985
17	Biology	1.00	0.2240	0.1572	0.3598	0.3042	0.1845	0.0838	0.2209
18	Psychology	2.04	0.3882	0.3322	0.4652	0.4113	0.3531	0.1604	0.3561
19	Sociology and Anthropology	2.00	0.1975	0.1069	0.3295	0.3473	0.2516	0.1557	0.2554
20	Geography	2.75	0.3247	0.2106	0.3521	0.4288	0.3259	0.1547	0.2702
21	Medicine and Dentistry	1.46	0.1917	0.1796	0.3602	0.2266	0.2339	0.1398	0.3669
22	Therapy and Counseling	1.09	0.1811	0.3672	0.2487	0.2593	0.2463	0.1096	0.1648
23	Education and Training	2.79	0.3007	0.6519	0.3434	0.3106	0.3646	0.2143	0.4727
24	English Language	3.41	0.3422	0.2747	0.4170	0.8104	0.2497	0.2903	0.3680
25	Foreign Language	1.64	0.1954	0.2331	0.2604	0.7266	0.1462	0.1997	0.2232
26	Fine Arts	1.55	0.3638	0.3355	0.4289	0.3827	0.3085	0.2595	0.4454
27	History and Archeology	1.06	0.2172	0.1555	0.3589	0.2983	0.1524	0.1375	0.3388
28	Philosophy and Theology	1.05	0.1351	0.1111	0.2554	0.2408	0.1781	0.0857	0.1532
29	Public Safety and Security	1.45	0.2866	0.2776	0.2625	0.1880	0.2914	0.1930	0.2989
30	Law and Government	1.39	0.2086	0.1231	0.3216	0.3171	0.2190	0.1833	0.2051
31	Telecommunications	2.40	0.5197	0.2846	0.3371	0.3025	0.3396	0.2226	0.3802
32	Communications and Media	2.43	0.4413	0.2283	0.3618	0.3693	0.3582	0.1617	0.2833

Max values		0.6389	0.6519	0.8136	0.8104	0.7480	0.8404	0.6642
Indices of max values		9	23	14	24	3	1	10
Theshold	**0.65**							
Values above threshold		0.0000	0.6519	0.8136	0.8104	0.7480	0.8404	0.6642
Job scores		0.00	2.79	3.53	3.41	2.07	2.74	3.47
Score	**18.01**							
Total for all knowledge elements	71.13							
Normalized and averaged score	**9.26**							

Fig. 8. Example of similarity-matrix for the Knowledge entity.

corrective factor can be considered a hyperparameter, and its optimal value was determined through extensive experimentation.

The described procedure is performed for each job j and returns the first five jobs that have obtained the highest scores. Fig. 7 shows the flow diagram of the algorithm just illustrated.

Fig. 8 shows an example of a similarity matrix for the O*NET entity Knowledge (with 33 rows) with *Java Programmer* as job and *Computers, training, maths, English, sales, administrator, engineer* as information extracted from an input resume. The reader should note that in the similarity matrix, for each piece of information extracted from the resume (column), we have a cell with the maximum value on the row corresponding to an element of the Knowledge entity (as previously mentioned, Knowledge has 33 rows). For example, for the information *Computers*, this maximum value corresponds to the Knowledge element *Computers and Electronics*, but the corresponding similarity value of 0.6389 is below the threshold (empirically fixed for all the entity values at 0.65). This represents a typical "failure case", where the match is incorrectly excluded despite being highly relevant. The O*NET score of 4.87 (associated with the value *Computers and Electronics* for the job *Java Programmer*) for the entity value *Computers and Electronics* will not, therefore, be added to the total score of the resume. The normalized and averaged score of 9.26 (at the bottom of the figure and corresponding to the computation of the member between the parenthesis of Equation (1) for the Knowledge entity) will be then summed with the other normalized and averaged scores of the other O*NET entities. Their similarity matrices used to calculate the other normalized and averaged scores, work in the same way. What changes is the number of rows and columns, according to the elements of the underlying O*NET entity and the information extracted from the underlying resume or job posting.

5. Scenarios

In this section, we will show the two scenarios that we have considered for job matching. They are related to the identification of the best match between a resume and a certain job description. They are also complementary, in the sense that the first one is company-oriented whereas the second one is candidate-oriented.

5.1. First scenario

The input of the first scenario is a resume. Here the task is to identify the O*NET occupations with the highest match with the given resume. This might be very useful for companies that need to quickly assess different resumes. With an automatic tool like this, companies would be able to categorize plenty of resumes with very high precision in a matter of seconds. In such a case, the application of our approach is straightforward. We would first compute the entities of the resume as described in Section 4. Next, we would compute the score of each job as indicated in Equation (1) and would consider the job whose score is the highest.

5.2. Second scenario

The input of the second scenario we have envisaged consists of a resume and a set of job descriptions. Here the goal is to return the job whose description matches the most with the candidate's skills present

Education Details
January 2016 B.Sc. Information Technology Mumbai, Maharashtra University of Mumbai
January 2012 HSC Allahabad, Uttar Pradesh Allahabad university
January 2010 SSC dot Net Allahabad, Uttar Pradesh Allahabad university
Web designer and Developer Trainer

Web designer and Developer
Skill Details
Web design- Exprience - 12 months
Php- Exprience - 12 monthsCompany Details
company - NetTech India
description - Working. (salary - 12k)
PERSONAL INTEREST

Listening to Music, Surfing net, Watching Movie, Playing Cricket.
company - EPI Center Academy
description - Working. (Salary Contract based)
company - Aptech Charni Road
description - Salary Contract based)

Fig. 9. A resume within the category Web Designing.

in his/her resume. If the first scenario was thought to help companies in the fast screening of resumes, the second scenario is more applicant-oriented and allows to quickly assess which jobs among a list of many are more suitable to the given resume. To adapt our method for this case we first apply our approach to each job description. For a given job description $jdesc$ present in the input set, after having collected its related entities vectors, it returns the O*NET occupation with the highest match with $jdesc$. Please note that at this stage the returned O*NET occupations OCC are less than or equal to the number of job descriptions present in our set (it might be less if multiple job descriptions are mapped to the same O*NET occupation). Finally, we apply once more our approach to the input resume with the difference that, instead of looking to the best possible match over all the O*NET occupations, we limit the search to the OCC set we identified in the step before. Therefore, one of the O*NET occupations present in OCC will be returned as the one that matches the most with the input resume.

6. Evaluation

In this section, we will illustrate the evaluation assessment we have performed for each scenario. The reader notices that the scoring mechanism that we have introduced for Scenario 1 has been used as well for Scenario 2.

6.1. Evaluating scenario 1

As far as the first scenario is concerned, to check the performances of our approach we used the Resume Dataset version 1[18]. The dataset consists of 963 resumes from 25 different categories. An example of a resume is shown in Fig. 9. Several resumes had duplicates and some categories contained only a few of them. To create the test dataset so that we could analyze the performances on its categories, we fixed as a constraint that at least five resumes should occur for each category. Only 21 categories satisfied such a constraint. Therefore, we selected a set of 105 resumes from 21 different categories (5 resumes per category) and run our method which, for each resume, returned the top five O*NET occupations with the highest match with the underlying resume. The 21 classes were “Data Science”, “Human Resources”, “Advocate”, “Mechanical Engineer”, “Sales”, “Health and fitness”, “Civil Engineer”, “Java Developer”, “Business Analyst”, “SAP Developer”, “Automation Testing”, “Electrical Engineering”, “Python Developer”, “DevOps Engineer”, “Network Security Engineer”, “Database”, “Hadoop”, “ETL Developer”, “DotNet Developer”, “Blockchain”, “Testing”.

To evaluate our method, we performed an extensive manual assessment. The relevance of each suggested job was determined by adopting a similar approach presented by [35]. We rely on a five-point relevance scale. As such, human assessors assign a relevance score to each suggested job. The score can be any of the following values:

- **non-relevant (score 1):** The predicted job is completely different from the original one.
- **ordinary (score 2):** The predicted job has an ambiguous or unfair match with the original one.
- **marginally relevant (score 3):** The predicted job is more generic or somewhat related with the original one or vice-versa.
- **relevant (score 4):** The predicted job fairly matches with the original one.
- **highly relevant (score 5):** The predicted job totally matches with the original one.

Table 2
Inter-agreement evaluation of the three experts on our approach and the two baselines.

	Annotation_1	Annotation_2
Our Approach	0.37	0.27
Baseline1	0.56	0.43
Baseline2	0.39	0.37

Although precision, recall, and F1-scores are widely used metrics for performance evaluation, we deliberately adopted the described methodology to ensure a more nuanced and context-aware assessment of job relevance. This approach allows human annotators to capture qualitative aspects of the predictions, which are not fully accounted for by traditional quantitative metrics.

We used the mentioned ratings for two different annotations ($annotation_1$ and $annotation_2$). $Annotation_1$ assesses the first predicted job. $Annotation_2$ assesses the five predicted jobs for a resume. More in detail, annotators were asked to give a score by picking the most matching job according to their expertise among the 5 proposed by our approach. Therefore, if the resume’s label was Data Science and the five returned O*NET occupations were, respectively, Models, Data Scientists, Logisticians, Computer Programmers, and Mechanical Engineers, one annotator should assign 1 to $annotation_1$ (because Models is not relevant to Data Science) and 5 to $annotation_2$ (because among the first five results, we have Data Scientists which completely matches the original Data Science label). It is straightforward to note that the following statement holds:

$$score(annotation_2) \geq score(annotation_1)$$

Three independent recruiters have annotated the two scores returned by our algorithm for the 105 resumes. To decide which annotation to pick, we applied the majority voting algorithm. If three annotations for a particular resume were all different from each other (so that the majority voting algorithm could not be applied) we computed the average and rounded it to the closest integer score (for example, if three scores were 1, 3, and 4 the final score would be $round(1+3+4)/3 = 3$). The inter-annotator agreement scores computed according to Fleiss’ kappa [36] were 0.37 and 0.27 for both $annotation_1$ and $annotation_2$ respectively as reported in Table 2, indicating a fair/moderate agreement among the annotators, despite the elevated number of classes to pick for each sample. Then, we applied the majority voting and scored the results of the proposed algorithm accordingly. In particular, we obtained an average score of 3.8 for $annotation_1$ and an average score of 4.2 for $annotation_2$, as indicated in Table 3. We compared these results against those obtained using two naive unsupervised approaches which would constitute our baselines.

The two baselines do not take into account any entity of the O*NET database and work as follows. First, each input resume is broken down in nouns, noun phrases, and sentences. For each element, we compute the cosine similarity against each of the 21 class names of jobs ($baseline1$) and against each of the 1,016 O*NET occupations names ($baseline2$).

[18] https://www.kaggle.com/datasets/gauravduttakiit/resume-dataset.

Table 3
Results of our approach and the two baselines.

	$Annotation_1$	$Annotation_2$
Our Approach	3.8	4.2
Baseline1	3.4	4.9
Baseline2	3.0	3.8

In both cases, we return the five job classes with the highest semantic similarity. Those classes have been annotated by the three experts with the same scoring approach mentioned above and a majority voting strategy has been applied. We computed the inter-annotator agreement scores [36] among the three annotators for the baselines as well. The inter-agreement values for the two baselines are reported in Table 2. The reader might notice a much higher inter-agreement value for the $baseline1$ than the others. The reason is that $baseline1$ returns elements from a set of 21 classes which are much easier to rate for the annotators with respect to 1,016 classes. Finally, as similarly performed for our approach, we consider the majority voting result that we report in Table 3. We can observe that for $baseline1$, for the annotation $annotation_2$, the obtained score is very high. This happens for the same reason as the high inter-annotator agreement value mentioned earlier. Because the set of considered classes for this baseline is 21, it is very likely that out of 21 candidate classes, one of the first five returned jobs is correct (equal to the resume label).

6.2. Evaluating scenario 2

For this second scenario, we employed a dataset of 10k job posts of the CareerBuilder job dataset from the UK in 2019[19]. A subset of 8948 of these 10k job posts are in English and belong to 299 distinct categories. Each category might contain one or more jobs. Each job belongs to one category only. To prepare the dataset to be used with our system, we selected the posts belonging to categories with cardinality greater than or equal to 5 (i.e., each job category contained at least five jobs). We then selected 100 resumes (5 for each category) from the dataset used in scenario 1, belonging to the following 20 categories: *Data Science, Human Resources, Advocate, Mechanical Engineer, Sales, Health and fitness, Pharmacists, Java Developer, Business Analyst, SAP Developer, Automation Testing, Electrical Engineering, Python Developer, DevOps Engineer, Network Security Engineer, Database, Hadoop, ETL Developer, DotNet Developer, Testing.* Then we selected the most semantically similar job post category for each of the 20 resume categories: *Computer and Information Research Scientists, Human Resources Specialists, Lawyers, Industrial Engineers, Sales Managers, Health Educators, Pharmacists, Computer Programmers, Management Analysts, Software Developers, Applications, Software Quality Assurance Engineers and Testers, Electrical Engineers, Computer Programmers, Computer Systems Engineers/Architects, Network and Computer Systems Administrators, Database Administrators, Computer Systems Analysts, Software Developers, Applications, Software Developers, Applications, Software Quality Assurance Engineers and Testers.*[20] Each of the 100 resumes has been associated with 10 different job posts, the last of which belongs to the corresponding resume category. The extraction of job posts was random and each extracted post could occur just once.

Therefore, each sample contained a resume, ten job postings and their related categories included in the CareerBuilder job dataset. For a given sample, the ten job postings included are always of different categories. The last job posting of each sample corresponds to the same category of the resume. The same job posting is never present more than once in the entire collection. Because the related categories of each job posting are not always matched with the occupations found within the O*NET database, we applied our method to each job posting in order to identify the O*NET occupation that matches the most the related job posting. Therefore, for each sample, we also added ten O*NET job categories. Each one is related to each job posting in the sample.

We defined two baselines. The first one works as follows. Given one sample, it starts by extracting noun phrases, nouns, and sentences from the resume. Then for each extracted element it computes the cosine similarity against the 20 job labels (the job posting categories of the CareerBuilder dataset and those from the O*NET database calculated by using our approach) and returns the five job classes whose semantic similarity with any of the extracted element is the highest.

The second baseline uses the same O*NET entities we have defined within our approach but without using the scores. From the resume, it identifies all the values for each O*NET entity and builds one binary vector for each entity. Thus, the entity *ent* will consist of a vector of 0s and 1s: 0 if the *i* feature of *ent* is not included in the resume, 1 otherwise. To note that we considered all the values of each entity so that vectors of a certain entity will always have the same size. We perform the same operation for the 20 jobs present in each sample (thus we obtain binary entity vectors for each job in the sample) and then compute the Euclidean distance between binary vectors. We take the 5 jobs with the smallest distance with respect to the resume.

Finally, in a given sample, our approach is applied to the resume by computing the entities using the method already illustrated in Section 4 and looking for the best O*NET job label among the twenty we have.

Three human annotators filled two annotations for the two baselines and our approach. The first annotation assesses the first predicted job post whereas the second annotation assesses the top-5 predicted job posts, similarly as defined within the first scenario. To note that a certain prediction is compared against the list of the 10 original job labels and against the list of the derived 10 O*NET job labels. So, for example, if a baseline contains “Architectural and Engineering Managers” as its first prediction, $annotation_1$ gets a 5 if that category is either the tenth among the original job categories or if that is the O*NET category corresponding to the tenth job description. We computed the final annotations by using a majority voting strategy on the three annotators’ scores similarly as performed within the first evaluation scenario. The inter-annotator agreement scores according to Fleiss’ kappa [36] among the three annotators are reported in Table 4 indicating a substantial agreement.

The reader notices that such values are much higher than those obtained within scenario 1. The reason is because, in this case, each method had to choose among 10 different jobs only. This low number and the fact that they are different from each other helped the role of the annotators that were in agreement more often. Table 5 shows the results we have obtained for the three methods where it

Table 4
Inter-agreement score of the three experts on our approach and the two baselines for the second scenario.

	$Annotation_1$	$Annotation_2$
Our Approach	0.69	0.66
Baseline1	0.81	0.63
Baseline2	0.66	0.65

Table 5
Results of our approach and the two baselines for the second scenario.

	$Annotation_1$	$Annotation_2$
Our Approach	4.17	4.95
Baseline1	3.8	4.87
Baseline2	2.03	3.75

[19] https://data.world/promptcloud/jobs-on-careerbuilder-uk.

[20] The reader notices that such a list is presented in order of similarity with the list of resume categories shown above. That is, the first element, *Computer and Information Research Scientists* is the most semantically similar job category to the resume category *Data Science*, etc.

is possible to note how ours outperforms the two baselines for both $annotation_1$ and $annotation_2$ showing again its efficiency. The data related to the evaluation that has been carried out for both scenarios is freely available at https://gitlab.com/hri_lab1/using-transformers-and-o-net-to-match-jobs-to-applicants-resumes.

7. Use case: recommending skills to job seekers

Our approach is well-suited for a recommendation task for job seekers. In this section, we will show an extension of our method in this direction. It works as follows. The user of the system, that is a person looking for a job, should just provide his/her resume and a list of job names he/she is interested in. First of all, the job names are matched against the O*NET occupations so that a map can be defined between input job names and O*NET occupations. We already performed a similar computation within the evaluation of the proposed second scenario. Then, for each identified O*NET job, the approach would compute a score of the resume against the input O*NET categories according to the Equation (1) introduced in Section 4.2.

7.1. Job seeking and skills recommendation

The reader can find an implementation of this use case at http://192.167.149.11:8000/ under the *first demo* link. Let us assume that the user gives as an input one job name only, and thus only one job from O*NET is selected. Then the user uploads his/her resume. The system elaborates on it. To assess whether the score obtained from the resume for the requested job makes the resume eligible, we compare it with an experimentally established threshold value of 34.62. The value of 34.62 for eligibility represents the average score that resumes from the correct job category obtained during the testing phase. It was chosen as a natural cutoff, distinguishing eligible candidates from ineligible ones based on the distribution of scores across different classes. This value reflects the typical score achieved by resumes correctly classified, ensuring that only resumes with a high likelihood of being relevant to the job are considered eligible. It is calculated by averaging the scores obtained from all elements of the Resume Dataset used in the previous scenarios, under the assumption (confirmed among the release notes of the dataset) that each resume in that dataset is eligible for the O*NET occupation that was selected.

When the user's resume is not eligible with the input job, the system returns the list of the O*NET entity elements not found within the input resume in decreasing order of importance. For the entities with scores (Abilities, Knowledge, Skills, Work Activities, Task Ratings), it means to return their elements in decreasing order of scores. For the other entities (Technical Skills and Tools Used), they are returned in alphabetical order (for the technological skills we first return the entries with hot technologies and then the others). For each element, the system indicates the O*NET score, the name of the element, a brief description (when available), and the value that would be added to the overall CV score if it was found in the resume (that is the value between the parenthesis of the Equation (1) for a fixed i, k and j). If the user notices that he/she has simply failed to include some of his/her skills or knowledge or some previous work activity, he/she can add it to the resume and resubmit it. Otherwise, he/she would need to get more competencies and, consequently, raise the score of his/her resume over the pre-established threshold. In addition to this, the demonstration also allows us to identify the skills, technical abilities, and other qualifications needed for a job and can recommend job opportunities that align with the user's characteristics (*second* and *third demo*).

7.2. Acquiring new skills

To acquire new skills, a system that embeds our approach can cover the different values present in the entities so that the candidate becomes eligible for that specific job.

There are several ways that can be employed for such a purpose. The first one is to leverage existing APIs of well-known massive open online course providers such as Udemy[21] or Coursera[22]. In such a case we would simply leverage the search engine of those providers by feeding them with the entities to be mastered by the applicant.

As an example, we submitted for the job *Mechanical Engineer* a resume from the Resume Dataset belonging to the category *Electrical Engineering*. Our system returns a score of 19.33, which is below the threshold (34.62) by 15.29 points. Hence, the system does not consider the resume eligible for the *Mechanical Engineer* job and lists the elements of the O*NET entities that have not been identified in the CV, including for example: *Reading comprehension*, *Active Listening*, *Mathematics* and *Critical Thinking*. For each item, up to 5 Udemy courses are listed, when available. In our example, the *Become Active Reader & Master Your Reading Comprehension Skills* course is suggested for the first element, *Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times* course for the second, *Math* and *Perfect your Mathematics skills* courses for the third, and *Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems* course for the fourth. Then the candidate should read or attend those lectures and improve his/her resume with the acquired competencies. Afterward, by adding the missing elements listed above to the resume and resubmitting the application to the system, the augmented resume obtains a score of 39.85 and passes the eligibility threshold.

Besides relying on online providers, we can also exploit dumps of resources with courses' information [37,38]. For example, one resource that can be leveraged is the COCO [38] dataset, a large collection of courses that includes lessons, teachers, instructors, and learners' ratings collected with the purpose of developing AI-based e-learning applications on top. For the courses, it provides metadata such as short and long descriptions, the necessary requirements, and the expected skills acquired by successfully attending them. These can be leveraged by our system to find out which courses would support an applicant to master new skills by matching the detected missing skills of a resume. Our system would suggest a list of courses from the COCO dataset similar to what is performed by e-learning providers' search engines. The advantage of using such a dataset lies in the opportunity to develop customized AI-based applications (for example leveraging transformers or other language models) giving freedom to the developers to build their own solutions to find the best matches with the missing skills.

Finally, with the aim to recommend lectures or articles, our approach can also be easily fed with modern state-of-the-art resources such as the "Academia/Industry DynAmic" (AIDA) [39] or "The Computer Science Knowledge Graph" (CS-KG) [40–43]. AIDA is a collection of metadata about $21M$ publications and $8M$ patents categorized within a taxonomy of computer science topics from the Computer Science Ontology (CSO) [44]. CS-KG is an automatically generated knowledge graph that describes the content of $6.7M$ scientific papers with $10M$ research entities and $41M$ relationships among them within the computer science domain. These resources can be exploited to recommend patents and research papers that might be of interest for the applicants to discover where and how state-of-the-art technologies are used to improve their skills and be ready for the job application. Considering the example above, if an applicant has to acquire the skill *Critical Thinking* for the *Mechanical Engineering* job, the system can be easily extended to look for research papers that study it within the CS-KG SparQL endpoint[23], suggesting to the applicant interesting reads to make stronger his/her application. For example, CS-KG suggests "Serious games on environ-

[21] https://www.udemy.com.
[22] https://www.coursera.org.
[23] https://scholkg.kmi.open.ac.uk/sparql/.

```
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX cskg: <http://scholkg.kmi.open.ac.uk/cskg/resource/>
PREFIX cskg-ont: <http://scholkg.kmi.open.ac.uk/cskg/ontology#>
PREFIX provo: <http://www.w3.org/ns/prov#>
PREFIX cso: <http://cso.kmi.open.ac.uk/schema/cso#>
PREFIX dcterm: <http://purl.org/dc/terms/>

SELECT ?sub ?pre ?obj?paperTitle ?doi  FROM
<http://scholkg.kmi.open.ac.uk/cskg>
WHERE { ?t rdf:subject cskg:critical_thinking .
        ?t rdf:subject ?sub .
        ?t rdf:predicate ?pre .
        ?t rdf:object ?obj .
        ?t provo:wasDerivedFrom ?paperID .
        ?paperID dcterm:title ?paperTitle ;
        cskg-ont:hasDOI ?doi .
        ?t cskg-ont:hasSupport ?sup}
ORDER BY desc (?sup)
```

Fig. 10. SparQL query to find papers related to the *Critical Thinking* skill.

mental management"[24] and "Teaching Flooding Attack to the SDN Data Plane with POGIL"[25] as relevant papers to master the *Critical Thinking* skill. Fig. 10 shows the SparQL query to be executed in the CS-KG SparQL endpoint to retrieve the papers related to the *Critical Thinking* skill.

8. Conclusions and future directions

In this paper, we have proposed an approach for job recommendation based on the information extracted from applicants' resumes and job postings. The information is then matched against different O*NET entities using semantic similarity of deep learning transformers in order to identify the most suitable O*NET job to the underlying resume or job posting. Two scenarios have been considered: one useful for companies with the goal of quickly screening several resumes and the other useful for applicants with the goal of quickly identifying the job posting more suitable to their skills.

For the first scenario, an extensive evaluation on the Resume Dataset version 1 has been carried out considering 105 resumes from 21 different categories. We have defined a scoring mechanism with five different values from 1 to 5. Moreover, we came up with two baselines (one returning the output over the 21 categories and the other returning the output over the 1,016 possible jobs) to compare our approach against and manually annotated the results of the three methods with the scoring mechanism we defined. The scoring of the three methods has been applied to two outputs (the job with the highest score and the first five jobs with the highest scores). Our approach outperformed the others in three cases out of four.

For the second scenario, we considered 1000 job postings from the CareerBuilder job dataset and 100 resumes from the Resume Dataset used within the first scenario. With similar scoring metrics as those used within the first scenario, we showed how our method obtained much better results than two more baselines that we defined.

Finally, as a use case of the proposed approach, we discussed a recommending task for job seekers where, given in input a resume and a list of occupations, the system returns, for each occupation, a list of courses that need to be attended to acquire the missing skills and become eligible for that specific job.

As future directions, our aim is to refine our prototype and create a real system that performs all the tasks we have illustrated in this work, including the use case discussed at the end of the manuscript. We will release the APIs so that everyone will be able to play with them and integrate them into existing systems. The API will also be employed to collect users' feedback and create a gold standard to evaluate the accuracy of the system at scale. We believe this system might be either used standalone or incorporated into well-known platforms for job seekers. One feature we would like to add is the possibility to include a generator of resumes starting from a list of skills and abilities: this feature would leverage new text generation models with the goal of preparing a complete resume in natural language. Furthermore, we would like to study the injection of O*NET database knowledge and transformer models into existing recommendation methodologies (e.g., collaborative filtering) with the goal of creating a full-fledged platform for job recommendation that takes benefits from domain knowledge using O*NET, transformers, and past candidate experiences.

One of the areas where the authors are continuing their research focuses on providing recommendations for courses and training materials. These recommendations aim to help job seekers acquire the skills or competencies they lack. Additionally, the focus will shift towards applying the lessons learned in the job-seeking domain to worker reskilling initiatives within organizations, especially under the context of Industry 5.0. In this context, LLMs and retrieval-augmented generation (RAG) systems will be explored for the development of study guides based on the organization's training documentation. Another case study will involve adapting the system for document classification. This adaptation will aim to identify the specific skills and abilities that a job seeker or student could improve through the study of the corresponding document. Finally, we plan to explore fine-tuning strategies and further improvements to the system's performance.

CRediT authorship contribution statement

Rubén Alonso: Conceptualization, Funding acquisition, Visualization, Writing – review & editing. **Danilo Dessí:** Formal analysis, Investigation, Validation, Writing – review & editing. **Antonello Meloni:** Data curation, Investigation, Software, Writing – original draft. **Diego Reforgiato Recupero:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

No conflict of interest exists.

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Data availability

Data will be made available on request.

References

[1] S.D. Risavy, C. Robie, P.A. Fisher, S. Rasheed, Resumes vs. application forms: why the stubborn reliance on resumes?, Frontiers in Psychology 13 (2022) 884205.
[2] S. Rojas-Galeano, J. Posada, E. Ordoñez, A bibliometric perspective on ai research for job-résumé matching, The Scientific World Journal 2022 (1) (2022) 8002363.
[3] U. Goyal, A. Negi, A. Adhikari, S.C. Gupta, T. Choudhury, Resume data extraction using nlp, in: J. Singh, S. Kumar, U. Choudhury (Eds.), Innovations in Cyber Physical Systems, Springer, Singapore, Singapore, 2021, pp. 465–474.
[4] T.M. Harsha, G.S. Moukthika, D.S. Sai, M.N.R. Pravallika, S. Anamalamudi, M. Enduri, Automated resume screener using natural language processing (nlp), in: 2022 6th International Conference on Trends in Electronics and Informatics (ICOEI), 2022, pp. 1772–1777.
[5] A. Deshmukh, A. Raut, Applying bert-based nlp for automated resume screening and candidate ranking, Annals of Data Science (2024) 1–13.
[6] S. Westman, J. Kauttonen, A. Klemetti, N. Korhonen, M. Manninen, A. Mononen, S. Niittymäki, H. Paananen, Artificial intelligence for career guidance–current requirements and prospects for the future, IAFOR Journal of Education 9 (4) (2021) 43–62.
[7] C. Yu, C. Zhang, J. Wang, Extracting body text from academic pdf documents for text mining, in: KDIR, 2020, pp. 229–236.
[8] C. Stahl, S. Young, D. Herrmannova, R. Patton, J. Wells, Deeppdf: a deep learning approach to extracting text from pdfs, in: Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018), European Language Resources Association (ELRA), Paris, France, 2018.
[9] J. Tiedemann, Improved text extraction from pdf documents for large-scale natural language processing, in: A. Gelbukh (Ed.), Computational Linguistics and Intelligent Text Processing, Springer Berlin Heidelberg, Berlin, Heidelberg, 2014, pp. 102–112.
[10] Z. Cheng, P. Zhang, C. Li, Q. Liang, Y. Xu, P. Li, S. Pu, Y. Niu, F. Wu, Trie++: towards end-to-end information extraction from visually rich documents, arXiv preprint, arXiv:2207.06744, 2022.
[11] M.J. Handel, The o*net content model: strengths and limitations, Journal for Labour Market Research 49 (2) (2016) 157–176, https://doi.org/10.1007/s12651-016-0199-8.
[12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17, Curran Associates Inc., Red Hook, NY, USA, 2017, pp. 6000–6010.
[13] C. Helwe, C. Clavel, F.M. Suchanek, Reasoning with transformer-based models: deep learning, but shallow reasoning, in: 3rd Conference on Automated Knowledge Base Construction, 2021, https://openreview.net/forum?id=Ozp1WrgtF5_.
[14] A. Meroño-Pe nuela, D. Spagnuelo, Can a transformer assist in scientific writing? Generating semantic web paper snippets with gpt-2, in: A. Harth, V. Presutti, R. Troncy, M. Acosta, A. Polleres, J.D. Fernández, J. Xavier Parreira, O. Hartig, K. Hose, M. Cochez (Eds.), The Semantic Web: ESWC 2020 Satellite Events, Springer International Publishing, Cham, 2020, pp. 158–163.
[15] Z. Zheng, Z. Qiu, X. Hu, L. Wu, H. Zhu, H. Xiong, Generative job recommendations with large language model, arXiv preprint, arXiv:2307.02157, 2023.
[16] Z. Guan, J.-Q. Yang, Y. Yang, H. Zhu, W. Li, H. Xiong, Jobformer: skill-aware job recommendation with semantic-enhanced transformer, ACM Transactions on Knowledge Discovery from Data (2024).
[17] Y. Li, M. Yamashita, H. Chen, D. Lee, Y. Zhang, Fairness in job recommendation under quantity constraints, in: AAAI-23 Workshop on AI for Web Advertising, 2023.
[18] J. Dhameliya, N. Desai, Job recommender systems: a survey, in: 2019 Innovations in Power and Advanced Computing Technologies (i-PACT), vol. 1, 2019, pp. 1–5.
[19] R. Mishra, S. Rathi, Efficient and scalable job recommender system using collaborative filtering, in: A. Kumar, M. Paprzycki, V.K. Gunjan (Eds.), ICDSMLA 2019, Springer, Singapore, Singapore, 2020, pp. 842–856.
[20] Y. Lu, S. El Helou, D. Gillet, A recommender system for job seeking and recruiting website, in: Proceedings of the 22nd International Conference on World Wide Web, WWW '13 Companion, Association for Computing Machinery, New York, NY, USA, 2013, pp. 963–966.
[21] W. Shalaby, B. AlAila, M. Korayem, L. Pournajaf, K. Aljadda, S. Quinn, W. Zadrozny, Help me find a job: a graph-based approach for job recommendation at scale, in: Proceedings of the International Conference on Big Data (BiG Data), 2017, pp. 1544–1553.
[22] S. Choudhary, S. Koul, S. Mishra, A. Thakur, R. Jain, Collaborative job prediction based on Naïve Bayes classifier using python platform, in: 2016 International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), 2016, pp. 302–306.
[23] G. Domeniconi, G. Moro, A. Pagliarani, K. Pasini, R. Pasolini, Job recommendation from semantic similarity of linkedln users' skills, in: Proceedings of the 5th International Conference on Pattern Recognition Applications and Methods, ICPRAM 2016, SCITEPRESS - Science and Technology Publications, Lda, Setubal, PRT, 2016, pp. 270–277.
[24] F. Gutiérrez, S. Charleer, R. De Croon, N.N. Htun, G. Goetschalckx, K. Verbert, Explaining and exploring job recommendations: a user-driven approach for interacting with knowledge-based job recommender systems, in: Proceedings of the 13th ACM Conference on Recommender Systems, RecSys '19, Association for Computing Machinery, New York, NY, USA, 2019, pp. 60–68.
[25] B. Heap, A. Krzywicki, W. Wobcke, M. Bain, P. Compton, Combining career progression and profile matching in a job recommender system, in: D.-N. Pham, S.-B. Park (Eds.), PRICAI 2014: Trends in Artificial Intelligence, Springer International Publishing, Cham, 2014, pp. 396–408.
[26] X. Zhao, Cold-start collaborative filtering, SIGIR Forum 50 (1) (2016) 99–100, https://doi.org/10.1145/2964797.2964819.
[27] N. Reimers, I. Gurevych, Sentence-bert: sentence embeddings using siamese bert-networks, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2019, https://arxiv.org/abs/1908.10084.
[28] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: pre-training of deep bidirectional transformers for language understanding, arXiv preprint, arXiv:1810.04805, 2018.
[29] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, Roberta: a robustly optimized bert pretraining approach, arXiv preprint, arXiv:1907.11692, 2019.
[30] K. Song, X. Tan, T. Qin, J. Lu, T.-Y. Liu, Mpnet: masked and permuted pre-training for language understanding, Advances in Neural Information Processing Systems 33 (2020) 16857–16867.
[31] Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma, R. Soricut, Albert: a lite bert for self-supervised learning of language representations, arXiv preprint, arXiv:1909.11942, 2019.
[32] S.R. Bowman, G. Angeli, C. Potts, C.D. Manning, A large annotated corpus for learning natural language inference, in: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Lisbon, Portugal, 2015, pp. 632–642, https://aclanthology.org/D15-1075.
[33] A. Williams, N. Nangia, S. Bowman, A broad-coverage challenge corpus for sentence understanding through inference, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, vol. 1 (Long Papers), Association for Computational Linguistics, New Orleans, Louisiana, 2018, pp. 1112–1122, https://aclanthology.org/N18-1101.
[34] C. Spearman, The proof and measurement of association between two things, The American Journal of Psychology 100 (3/4) (1987) 441–471, http://www.jstor.org/stable/1422689.
[35] A. Konjengbam, N. Kumar, M. Singh, Unsupervised tag recommendation for popular and cold products, Journal of Intelligent Information Systems 54 (2019) 545–566.
[36] J.L. Fleiss, J.C. Nee, J.R. Landis, Large sample variance of kappa in the case of different sets of raters, Psychological Bulletin 86 (5) (1979) 974.
[37] D. Dessì, G. Fenu, M. Marras, D. Reforgiato Recupero, Coco: semantic-enriched collection of online courses at scale with experimental use cases, in: Á. Rocha, H. Adeli, L.P. Reis, S. Costanzo (Eds.), Trends and Advances in Information Systems and Technologies, Springer International Publishing, Cham, 2018, pp. 1386–1396.
[38] D. Dessì, G. Fenu, M. Marras, D. Reforgiato Recupero, Bridging learning analytics and cognitive computing for big data classification in micro-learning video collections, Computers in Human Behavior 92 (2019) 468–477, https://doi.org/10.1016/j.chb.2018.03.004, https://www.sciencedirect.com/science/article/pii/S0747563218301092.
[39] S. Angioni, A.A. Salatino, F. Osborne, D.R. Recupero, E. Motta, AIDA: a knowledge graph about research dynamics in academia and industry, Quantitative Science Studies 2 (4) (2021) 1356–1398, https://doi.org/10.1162/qss_a_00162.
[40] D. Dessì, F. Osborne, D.R. Recupero, D. Buscaldi, E. Motta, H. Sack, AI-KG: an automatically generated knowledge graph of artificial intelligence, in: The Semantic Web - ISWC 2020 - 19th International Semantic Web Conference, in: Lecture Notes in Computer Science, vol. 12507, Springer, 2020, pp. 127–143.
[41] D. Dessì, F. Osborne, D.R. Recupero, D. Buscaldi, E. Motta, Generating knowledge graphs by employing natural language processing and machine learning techniques within the scholarly domain, CoRR, arXiv:2011.01103, 2020, https://arxiv.org/abs/2011.01103.
[42] D. Dessí, F. Osborne, D.R. Recupero, D. Buscaldi, E. Motta, Scicero: a deep learning and nlp approach for generating scientific knowledge graphs in the computer science domain, Knowledge-Based Systems 258 (2022) 109945.
[43] D. Dessí, F. Osborne, D. Reforgiato Recupero, D. Buscaldi, E. Motta, Cs-kg: a large-scale knowledge graph of research entities and claims in computer science, in: International Semantic Web Conference, Springer, 2022, pp. 678–696.
[44] A.A. Salatino, T. Thanapalasingam, A. Mannocci, F. Osborne, E. Motta, The computer science ontology: a large-scale taxonomy of research areas, in: International Semantic Web Conference, Springer, 2018, pp. 187–205.

3.4 Extending Asset Lifespan Through Data Augmentation-Assisted Quality Control

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- **Resumen:** El control de calidad es esencial para el ciclo de vida de los componentes y tiene un impacto en los costos y el tiempo de mantenimiento de toda la cadena de producción. En el control de calidad moderno, el poder de la inteligencia artificial podría desatarse si se dispusiera de suficiente información para entrenar modelos. En este trabajo se ha aplicado un enfoque basado en el aumento de datos al control de calidad de los tapones de los depósitos de aceite en automoción para mitigar la falta de imágenes de piezas defectuosas, lo que dificulta el entrenamiento de modelos de clasificación. El enfoque propuesto ha demostrado un impacto positivo en la inspección de calidad al mantener o reducir el porcentaje de errores de clasificación con respecto a una línea de base entrenada con imágenes de tapones de depósito de aceite reales.
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- **Relación con la tesis:** Resiliencia y mejora de calidad. Impacto de las piezas defectuosas en toda la cadena de producción. Reducción de residuos derivados de la generación de piezas defectuosas. Detección temprana de defectos. Caso demostrativo del uso de inteligencia artificial y generación de imágenes sintéticas en pequeñas y medianas empresas manufactureras.

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Extending Asset Lifespan Through Data Augmentation-Assisted Quality Control ⋆

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Abstract: Quality control is essential for the life cycle of components and has an impact on the costs and maintenance time of the entire production chain. In modern quality control, AI power could be unleashed if enough information for training AI models is always available. In this work, an approach based on data augmentation has been applied to oil plug quality control to mitigate the lack of images of defective parts, which hinders the training of classification models. The proposed approach has demonstrated a positive impact on quality inspection by maintaining or reducing the percentage of classification errors with respect to a baseline trained with images of real oil plugs.

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Keywords: Quality Assurance, Quality Maintenance, Industry 4.0, Machine Learning, Sorting Machines, Artificial Intelligence

1. INTRODUCTION

Although the ideas of Industry 5.0 are on the agendas of the automotive industry, e.g., sustainability and human-centricity, some key points of what came to be called Industry 4.0 are still being addressed, including the integration of the supply chain and interoperability (Frederico et al., 2023). Both approaches are not mutually exclusive; in fact, a solid foundation in cyber-physical systems facilitates the new paradigm of Industry 5.0, where more emphasis is placed on human-machine cooperation, customization, and long-term sustainability (Aheleroff et al., 2022).

Moreover, both paradigms emphasize the crucial integration of supply chains and the maintenance of high-quality standards throughout the production process. This integration underscores the significance of quality control across all components, materials, and parts involved in assembly and production. Ensuring quality throughout the integrated chain is paramount for optimizing the lifespan of each part or component, consequently impacting overall production costs and maintenance timelines. For instance, premature failure of any of these assets not only undermines reliability but also tarnishes the brand image indirectly.

In this paper, we approach asset reliability from the perspective of upstream quality control. We assume that thorough and accurate quality control of assets can impact the maintenance of machines in the production process and even improve the predictability of failures and downtime. In particular, starting from the assumption that the purpose of maintenance is *"to ensure equipment functions to its original optimal level and this can be accomplished by eliminating losses that affect the equipment"* (Mansor et al., 2012), our article focuses on the elimination of *breakdown losses* and *scrap and rework losses*.

In our investigation, we delve into the potential integration of machine learning techniques for data augmentation and classification within the realm of quality control for manufactured parts. Our primary objective is to ascertain the viability of these advanced methodologies in enhancing the overall quality and predictability across the entirety of the production chain. Furthermore, by leveraging the power of machine learning in conjunction with traditional quality control methodologies, we endeavor to pave the way for more agile, adaptive, and predictive manufacturing processes.

The implementation of the quality control plan is based on the customer's specifications. These specifications define the conformance characteristics that each part will have. These characteristics can be defined by variables, such as dimensions specified by a nominal value and a tolerance range. Additionally, there are features defined as discrete attributes, such as the presence of burrs or even color-

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based attributes. In terms of dimensions, these values and tolerances can be used to carry out efficient control. However, when the challenge is to detect other types of defects, such as the presence of stains or scratches, this detection becomes more challenging. It is in these cases that new approaches based on computer vision, machine learning, and artificial intelligence(AI) are proliferating (Fahle et al., 2020).

Aligned with these challenges and the idea that quality affects the entire process, the main objective of this paper is to present the approach and the results of improving quality control within an automotive assembly, focusing on oil plugs.

In our inquiry, we pose a fundamental question: in scenarios where images of defective parts are limited, can data augmentation serve as a viable strategy to enrich the dataset utilized for training oil plug quality inspection models? This question underscores a critical challenge faced by many in the realm of quality control: the scarcity of diverse and representative data. In such circumstances, traditional machine learning approaches may falter, hindered by the lack of sufficient training examples to effectively discern between acceptable and defective components. By leveraging data augmentation, a technique that addresses data scarcity by artificially expanding the dataset through various transformations such as rotation, scaling, and flipping, our goal is to introduce a broader spectrum of variations. This enables the model to generalize more effectively, accurately identifying defects even when authentic samples are limited.

To tackle one of the primary challenges in quality inspection—specifically, the scarcity of images depicting defective parts and the absence of references—we have developed an approach centered on synthesized image generation (data augmentation). We hypothesize that synthetic images could be used as a replacement or supplement to real images in environments where there are not enough defective parts to train an AI, and validated this idea in the specific case mentioned above by empirically evaluating the quality control results with a commercial-grade classification solution, demonstrating the effectiveness of the augmentation technique. We do not focus on the specific machine learning algorithm but rather on the analysis of the augmented dataset with synthetic images and how they affect the overall accuracy. Moreover, the methodology tailored for the intricacies of oil plug manufacturing presents a robust framework transferrable to various domains. Its adaptability extends beyond the confines of its original application, offering insights and strategies that can be effectively implemented across a spectrum of industries and scenarios. By embracing this methodology, organizations can unlock new avenues for problem-solving and efficiency improvements, fostering continuous advancement and competitive advantage in diverse fields.

These challenges and results are presented in the subsequent sections. Specifically, Section 2 summarises the state of the art. Section 3 presents the augmentation technique we have adopted in the proposed use case. Section 4 discusses the results. Finally, Section 5 presents conclusions and outlines future activities where we are headed.

2. STATE OF THE ART

Quality control and product life cycle management are very related to each other (Brettel et al., 2016). Specifically, the well-known Sand Cone Model (Ferdows and De Meyer, 1990) is already positioned as a reference element in the framework, as reduced rework and waste directly impact cost, efficiency, and dependability.

One of the benefits of an optimal quality control process is the reduction of waste generation, which in turn brings sustainability improvements and cost savings to industries. The future of asset management will be characterized, among other aspects, by extended asset lifespan, cost-effectiveness, and improved sustainable performance, adhering to the concepts of Quality 5.0 and Maintenance 5.0, which are closely related to Industry 5.0 (Frick, 2023). Asset life cycle management, besides being key to this Industry 5.0 approach, is fully in line with the ideas of sustainability and environmental improvement. A life cycle management approach focused on these aspects will result in a *"practical, and integrated, approach to minimize the environmental burdens associated with a product over its life cycle"* (Hunkeler et al., 2003).

Therefore, improvements in quality control can have a significant impact on the product life cycle and the entire production chain, and contribute to achieving the objectives of Industry 5.0. AI may represent a game changer for quality control, leading to outstanding improvements. While modern quality control approaches are embracing AI more and more, a huge uptake is hindered by the scarcity of a crucial piece of information needed to train powerful AI models, i.e., information about defective parts enabling balanced training for optimal results. This is especially true when the manufacturing process has not started yet, thus it is not possible to collect data about defective parts (Volkau et al., 2022).

When dealing with a data-lacking issue, data augmentation represents a powerful solution, as demonstrated by its application in different fields (Li et al., 2022; Mikołajczyk and Grochowski, 2018). Apart from expanding and balancing datasets, data augmentation capabilities provide the needed flexibility to adapt existing datasets to the requirements of new product families in initial phases or even to adapt to newly detected defects. This flexibility enables iterative improvement in the classification and quality control of the parts, showing appreciable performance boosts when compared with non-augmented datasets (Yang et al., 2022; Singh and Bruzzone, 2022). Similar augmentation techniques have been applied to the automotive industry, currently in full transition to Industry 5.0, where consistent improvements have been achieved on quality inspection of automobile parts (Lee et al., 2021) and structural adhesives for automotive part application (Peres et al., 2021).

In their comprehensive survey, Shorten and Khoshgoftaar (2019) offer a taxonomy that encapsulates these diverse strategies of data augmentation techniques, spanning from traditional image manipulation methods to more sophisticated deep learning approaches. Within the realm of image manipulation, techniques encompass a spectrum of transformations. Geometric alterations, such as rotation, translation, and scaling, alongside image blending

methods, afford simple yet effective means of augmenting datasets. Additionally, kernel filters for sharpening or blurring images provide further avenues for enhancing dataset variability and model robustness. On the other end of the spectrum lie techniques rooted in deep learning paradigms, which operate within the feature space rather than directly on the images themselves (Nanni et al., 2022). These methods leverage the intrinsic characteristics of the data to enact transformations that transcend mere pixel-level modifications, thereby enriching the dataset with more nuanced variations. Moreover, advancements in adversarial training and generative modeling have propelled the development of cutting-edge augmentation techniques, exemplified by the emergence of Generative Adversarial Networks (GANs) (Antoniou et al., 2018). These models employ a game-theoretic framework to generate synthetic data samples that closely mimic real-world instances, thereby expanding the dataset while preserving its underlying statistical properties.

3. DATA AUGMENTATION ASSISTED QUALITY CONTROL CASE

The sorting operation is the quality control step where defective parts are recognized and separated from non-defective parts. Nowadays, some sorting methods do exist and are also implemented in automatic sorting machines; however, these methods are traditionally limited and not extremely powerful, require a delicate fine-tuning of image analysis algorithms, and depend considerably on the stability/uniformity of lighting and stability of the position of the part under the lighting, the image quality and the variability of the reflective characteristics of the surfaces. To solve this issue, recent advancements in AI, especially in object detection and recognition, are increasingly employed for sorting operations. Unfortunately, and despite the sector's extensive knowledge about the possible defects that may occur, due to experiences with other similar parts, a critical requirement for the adoption of AI models for image recognition and detection of defections is the availability of large datasets of images for training models. In the context of quality control, in mechanical series production, this requirement could represent a big barrier, due to the shortage of images depicting representative defective parts. Indeed, defective parts may occur very rarely (defective parts are usually described using the unit "parts per million"), and randomly. Also, this challenge is compounded by the costs of effort, time, and waste required to generate such data.

The use case presented in this work fits into these circumstances. Specifically, our use case is focused on the sorting of oil plugs used in the automotive industry. These plugs constitute small elements of a specific component, yet they wield a significant influence on the overall quality of the final product. For this reason, having a quality process for these plugs is essential to correctly manage the life cycle of the final product. An oil plug is assembled with an O-ring to ensure sealing. However, the seal is only guaranteed if the outer metal surface of the plug as well as the O-ring grove are perfectly flat, to ensure the O-ring fits the plug perfectly and prevent O-ring damage. It follows that the quality of the outer metal surface and the O-ring groove are key decision points for sorting. Figure 1 exemplifies typical defects occurring on these two plug components. Figure 1 (left) depicts a defective O-ring groove, possibly due to the absence of surface coating or the formation of lumps due to excessive coating. Figure 1 (right) shows a plug with an irregular outer metal surface. Both defects, which the human eye can easily detect, are difficult to detect with conventional testing tools. In this scenario, AI image classification algorithms could be a viable solution if reference images of the defects were available for quality classification.

In this work, we propose an application of data augmentation techniques based on computer vision and image manipulation to generate artificial images of defective parts. Starting from images depicting non-defective plugs, the proposed solution artificially generates images of defective plugs, which can be in turn exploited for training classification algorithms. Expert knowledge is required however to identify which are the regions of interest where defects may occur (i.e., the outer metal surface and the O-ring groove), as well as how the defects may appear (in terms of shapes and sizes). This information is needed to prevent the solution from generating non-realistic defects.

Given an image of a non-defective plug, the region of interest can be described in two ways: i) through user interaction, where the user selects the area where the defect may appear; and ii) through the automatic generation of a mask that detects the surfaces of the reference part, for example by background removal. Once the region of interest is defined, we can employ one of two different techniques to introduce defects. The first one is based on image manipulation and blending (Zhang et al., 2020). This technique exploits different pixel combination functions to merge two distinct images into a new one. These operations enable the attainment of distinct effects, making them applicable to various use cases depending on different contexts. Effects may include integrating the defect region of interest into the original image while maintaining, darkening, or lightening the area. Another option is the subtraction of the effect representation from the reference image, for instance, to represent cracks or grooves. Figure 2 shows two examples of synthetic images of defective plugs generated by using the image manipulation and blending technique. In particular, the image on the left is generated with the dodge blending mode, whereas within the image on the right, the defect was blended by multiplying the defect data on the reference image. The second technique involves the automatic generation of spots or stains (Buslaev et al., 2020), allowing the creation of blemishes with varying numbers or sizes on the reference image.

Through the application of these two techniques, we managed to generate synthetic images depicting defective parts, starting from real images of similar parts containing correct parts and defects. The new synthetic images enable us to generate a new training dataset, with balanced samples of good and defective parts. The balanced dataset, in turn, allows us to train and validate different classification models, which can be exploited to automatize the sampling operation. Additionally, it facilitates the comparison of different off-the-shelf models, enabling the selection of the most suitable one for each family of parts. Finally, it streamlines the fine-tuning of these algorithms and models,

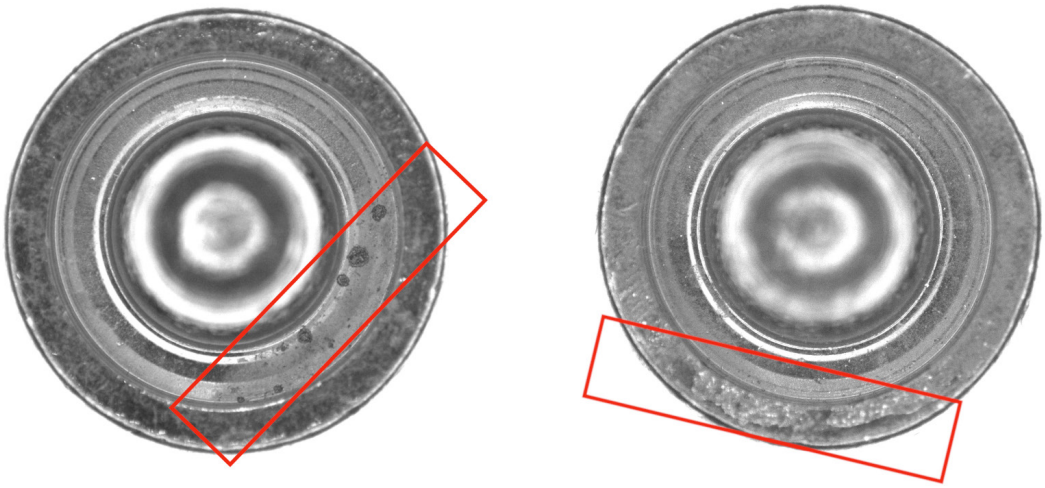


Fig. 1. Defects in the oil plug. On the left, a type of defect affecting the groove housing the O-ring. On the right, the defect is located on the outer metal surface, which is irregular.

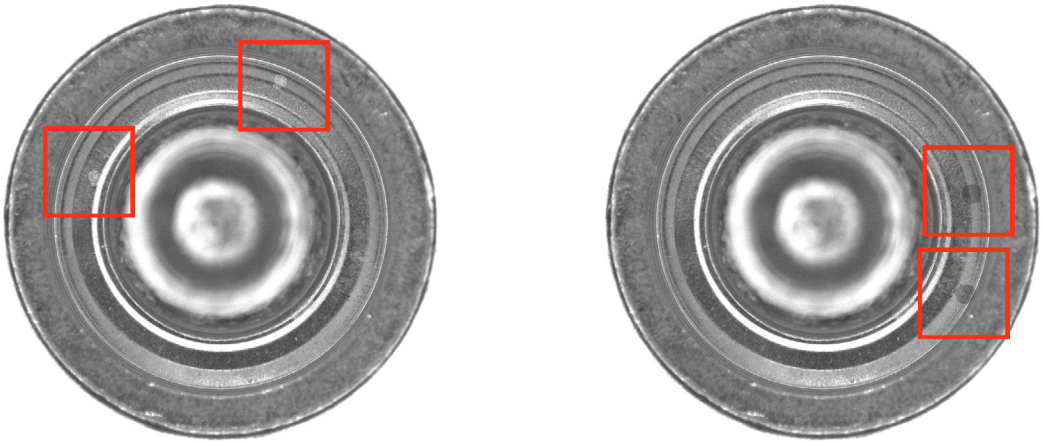


Fig. 2. Two images of defective plugs generated by applying the image manipulation and blending technique to an image of a good plug.

making them easily adaptable to newly identified defects within different product families.

4. VALIDATION AND RESULTS

To validate the quality of the generated images, as well as the performance of classification models for solving the sorting problem, we set up an experiment in the context of the KITT4SME project [1] . Within the experiment, DIMAC, a company of the KITT4SME consortium, adopted the solution by envisioning potential applications for its automatic sorting machines. An unbalanced dataset of real images (showing both non-defective and defective plugs) has been provided by DIMAC. This dataset has been augmented with new synthetic images of defective parts generated with the proposed solution. Then, an AI classification model was trained to solve a binary classification problem, i.e., to classify non-defective products (OK - positive class) and defective parts (NOK - negative class). For this experiment, we selected as a classification model a commercial product widely employed in inspection and quality control, i.e., Zebra Aurora Vision Deep Learning [2] . This solution is hardware-agnostic. It provides a dual Deep Learning and traditional machine vision system, thus addressing tasks from surface inspection to dimensional measurements or presence/absence checks. After some trials with different libraries that yielded comparable results, the final choice was mainly driven by the tool's versatility and ease of integration into DIMAC software.

To evaluate the utility of the augmented dataset, the proposed solution is compared against a baseline solution, where the same classification model is trained using the original dataset with real images only.

The proposed solution is evaluated concerning the following key performance indicators (KPI):

[1] https://kitt4sme.eu/

[2] https://www.zebra.com/gb/en/products/oem/software/aurora-deep-learning.html

- KPI #1 - False positives: the number of pieces classified as correct (OK) when they have a defect (NOK). This indicator is crucial as it has a direct impact on the life cycle of the assets that include the part and may lead to safety issues. The target value is set to not more than 100 parts per million (ppm) compared to the baseline[3], a threshold that is ten times the residual typical value for a mature commercial solution (ISO 16426). This is measured as a result of a comparison between classification algorithms trained with synthetically generated defective parts versus algorithms trained with only real images of defective parts.
- KPI #2 - False negatives: the percentage of pieces incorrectly classified as defective (NOK). This indicator is relevant from a business perspective (process inefficiencies and waste generation), even if it is not as critical as the previous one (related to safety). Based on DIMAC's experience in the field and the technological maturity level of the solution, the target value is set to a maximum increment of 3% when comparing the proposed solution against the baseline.

Table 1. Training and Test sets

Use	Set name	Number of images
Training	Real OK set	200
Training	Real NOK set	3
Training	Artificial NOK set	30
Test	Control Set, Real OK images	968
Test	Control Set, Real NOK images	920

To support our experiment, we created different image training and test sets, as reported in Table 1. We started with sets featuring images of defective (NOK) and non-defective (OK) oil plugs, both from real production processes or crafted deliberately. Then, due to the difficulty of obtaining samples of defective pieces, some defective parts have been artificially generated using image manipulation and blending techniques.

We proceeded to train three distinct models, each utilizing a training set comprised of both authentic and synthetic images, as illustrated in Table 2.

After training, each of the three models underwent testing using an identical set of 1888 control images, comprising 968 "OK" and 920 "NOK" images. The control set of NOK images has been generated from a small sample with about 40 real defective parts by running the 40 pieces through the sorting machine several times, changing the light and the angle slightly each time. The obtained results are detailed in Table 3.

The first model (baseline) has been trained on a dataset combining the OK set (200 real images of parts without defects) and the real NOK set (3 parts with defects). In this baseline scenario, no artificial defects are introduced. False-negative results were significant (10%) with one case of false positive. The second model (CF1) has been trained on a dataset combining the OK set (200 real OK images) and the artificial NOK set (30 artificially generated images of defective parts shown in the Synthetic NOK columnof Table 2). In this case, the images of real NOK images have been excluded. CF1 performance is better than the baseline mode, with a significant reduction in false negatives (4%), and no cases of false positives. Finally, a third model (CF2) has been trained on a dataset combining all the training sets (200 real OK images, 3 real NOK images, and 30 artificially generated NOK images). The performance of CF2 is identical to CF1, with 4% false negatives and no false positives.

[3] We used the ppm unit rather than a percentage because the number of defective parts among those classified as OK should be as low as possible (also for safety reasons).

Table 2. Model data composition

Method/Model	Real OK	Real NOK	Synthetic NOK
Baseline	200	3	0
CF1	200	0	30
CF2	200	3	30

Table 3. Results in terms of false negatives (FN) and false positives (FP).

Method	False Negatives	False Positives
Baseline	10%	1
CF1	4%	0
CF2	4%	0

Our results suggest that extending the training dataset with artificial images of defects leads to significant improvements (both with CF1 and CF2). More specifically, within the mentioned KPIs, the already minimal number of false positives is further decreased. Additionally, the percentage of false negatives not only remains below the 3% increase mentioned in KPI #2, but also decreases when considering images with artificial defects. Hence, training with a larger number of images containing artificial defects is preferable to training with a limited number of images depicting real defects.

5. CONCLUSION

This paper presents a case study on the quality control of automotive parts, focusing specifically on the quality of oil plugs. The scarcity of images depicting defective parts complicated the training of classification models. Consequently, data augmentation techniques were employed to generate synthetic images. Our approach has demonstrated a positive impact on quality inspection by either reducing or, at the very least, maintaining the number of classification errors, thus suggesting that artificially generated images with defects could replace or complement real images of defective pieces.

In our opinion, these results have a positive impact on maintenance engineering and lifecycle management of the assets where the evaluated parts are used. The potential reductions in costs and waste generation align the solution with the ideas of Industry 5.0. This enhanced control over part quality facilitates data-driven decision-making. It enables the selection of the most suitable quality inspection method and ensures the availability of quality data in subsequent stages of product assembly.On the other hand, it enables adaptation to changes. Upon detecting previously unidentified defects, these new findings can be utilized to synthetically generate images of parts from the same or other families, subsequently training the classification models accordingly. This ultimately affects the volume and requirement for spare parts, and, significantly, the consistency of these parts and, consequently, the overall consistency of production.

The results obtained and the advancements in the state of the art suggest continuing the research by applying other data augmentation techniques. Techniques based on Generative Adversarial Networks, Diffusion Models, and Variational Autoencoders are regarded as possible improvements over the more traditional techniques focused on image manipulation that we have applied. As classification algorithms, we aim to enhance efficiency by utilizing Large Language Models (LLMs) for classification once we successfully generate larger artificial datasets.

Furthermore, we are interested in applying the approach to other use cases within the automotive sector, such as different types of fasteners. We would also like to compare other KPIs, such as performance and the impact on sorting times. This data would give us a more comprehensive view and move toward the possible exploitation of a solution based on the presented approach.

Finally, we believe that further research into software platforms that can simplify the deployment of data documentation tools and their interoperability with classification models may be of interest. This will allow SMEs, in particular, to compose their solutions and adapt to the needs of quality control and asset management as they arise.

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REFERENCES

Aheleroff, S., Huang, H., Xu, X., and Zhong, R.Y. (2022). Toward sustainability and resilience with industry 4.0 and industry 5.0. *Frontiers in Manufacturing Technology*, 2, 951643.

Antoniou, A., Storkey, A., and Edwards, H. (2018). Augmenting image classifiers using data augmentation generative adversarial networks. In *Artificial Neural Networks and Machine Learning–ICANN 2018: 27th International Conference on Artificial Neural Networks, Rhodes, Greece, October 4-7, 2018, Proceedings, Part III 27*, 594–603. Springer.

Brettel, M., Klein, M., and Friederichsen, N. (2016). The relevance of manufacturing flexibility in the context of industrie 4.0. *Procedia Cirp*, 41, 105–110.

Buslaev, A., Iglovikov, V.I., Khvedchenya, E., Parinov, A., Druzhinin, M., and Kalinin, A.A. (2020). Albumentations: Fast and flexible image augmentations. *Information*, 11(2). doi:10.3390/info11020125. URL `https://www.mdpi.com/2078-2489/11/2/125`.

Fahle, S., Prinz, C., and Kuhlenkötter, B. (2020). Systematic review on machine learning (ml) methods for manufacturing processes–identifying artificial intelligence (ai) methods for field application. *Procedia CIRP*, 93, 413–418.

Ferdows, K. and De Meyer, A. (1990). Lasting improvements in manufacturing performance: in search of a new theory. *Journal of Operations management*, 9(2), 168–184.

Frederico, G.F., Kumar, V., Garza-Reyes, J.A., Kumar, A., and Agrawal, R. (2023). Impact of i4. 0 technologies and their interoperability on performance: future pathways for supply chain resilience post-covid-19. *The International Journal of Logistics Management*, 34(4), 1020–1049.

Frick, J. (2023). Future of industrial asset management: A synergy of digitalization, digital twins, maintenance 5.0/quality 5.0, industry 5.0 and iso55000. *International Journal of Business Marketing and Management (IJBMM)*, 8(4), 93–99.

Hunkeler, D., Saur, K., Stranddorf, H., Rebitzer, G., Schmidt, W., Jensen, A.A., and Christiansen, K. (2003). Life cycle management. *SETAC*.

ISO 16426 (2002). Fasteners — quality assurance system. Standard, International Organization for Standardization, Geneva, CH.

Lee, K.J., Kwon, J.W., Min, S., and Yoon, J. (2021). Deploying an artificial intelligence-based defect finder for manufacturing quality management. *AI Magazine*, 42(2), 5–18.

Li, W., Chen, J., Cao, J., Ma, C., Wang, J., Cui, X., and Chen, P. (2022). Eid-gan: Generative adversarial nets for extremely imbalanced data augmentation. *IEEE Transactions on Industrial Informatics*, 19(3).

Mansor, M.A., Ohsato, A., and Sulaiman, S. (2012). Knowledge management for maintenance activities in the manufacturing sector. *International Journal of Automotive and Mechanical Engineering*, 5, 612–621.

Mikołajczyk, A. and Grochowski, M. (2018). Data augmentation for improving deep learning in image classification problem. In *2018 international interdisciplinary PhD workshop (IIPhDW)*, 117–122. IEEE.

Nanni, L., Paci, M., Brahnam, S., and Lumini, A. (2022). Feature transforms for image data augmentation. *Neural Computing and Applications*, 34(24), 22345–22356.

Peres, R.S., Guedes, M., Miranda, F., and Barata, J. (2021). Simulation-based data augmentation for the quality inspection of structural adhesive with deep learning. *IEEE Access*, 9, 76532–76541.

Shorten, C. and Khoshgoftaar, T.M. (2019). A survey on image data augmentation for deep learning. *Journal of big data*, 6(1), 1–48.

Singh, A. and Bruzzone, L. (2022). Data augmentation through spectrally controlled adversarial networks for classification of multispectral remote sensing images. In *IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium*, 651–654. IEEE.

Volkau, I., Mujeeb, A., Dai, W., Erdt, M., and Sourin, A. (2022). The impact of a number of samples on unsupervised feature extraction, based on deep learning for detection defects in printed circuit boards. *Future Internet*, 14(1), 8.

Yang, S., Xiao, W., Zhang, M., Guo, S., Zhao, J., and Shen, F. (2022). Image data augmentation for deep learning: A survey. *arXiv preprint arXiv:2204.08610*.

Zhang, L., Wen, T., and Shi, J. (2020). Deep image blending. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 231–240.

3.5 Early Defect Detection in Thermo-Insulating Panels: A Case Study on Quality Inspection

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- **Resumen:** La Industria 4.0 se caracteriza por una integración creciente de tecnologías digitales y por una mayor expectativa de calidad, mientras que la Industria 5.0 introduce un nuevo paradigma centrado en la sostenibilidad y el factor humano. Tanto la calidad como la sostenibilidad pueden beneficiarse de modelos predictivos para la detección de defectos. En línea con estos factores de calidad y sostenibilidad, este trabajo tiene como objetivo demostrar cómo las plataformas de software interoperables, combinadas con técnicas de aumento de datos, segmentación y modelos de clasificación, mejoran el control de calidad y permiten la detección temprana de defectos. Se presenta un caso de estudio sobre la inspección de calidad de paneles termoaislantes, ampliamente utilizados en la construcción, el cual se valida en la línea piloto GAMHE 5.0, mostrando mejoras significativas tanto en tiempos de control como en la detección temprana de errores y en la reducción de piezas defectuosas al final de la línea de producción.
- **Estado:** Enviado a revista. En segunda fase de revisión.
- **Palabras clave:** Industria 5.0, Control de la calidad, Detección temprana de defectos, Fabricación sostenible, Líneas piloto
- **Relación con la tesis:** Mejora de calidad. Reducción de residuos derivados de la generación de piezas defectuosas. Detección temprana de defectos. Reducción de piezas defectuosas al final de la línea de producción. Caso demostrativo del uso de inteligencia artificial y generación de imágenes sintéticas en línea piloto.

Early Defect Detection in Thermo-Insulating Panels: A Case Study on Quality Inspection

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Abstract

Industry 4.0 is characterized by an increasing integration of digital technologies and higher quality expectations, whereas Industry 5.0 introduces a new paradigm focused on sustainability and human-centricity. Both, quality and sustainability, can benefit of predictive models for defect detection. Aligned with these quality and sustainability factors, this work aims at demonstrating how interoperable software platforms combined with data augmentation, segmentation and classification models improve quality control and enable early defect detection. A case study involving the quality inspection of thermo-insulating panels, widely used in construction, is presented and validated on the GAMHE 5.0 pilot line showing significant improvements both in control time and early error detection.

Keywords: industry 5.0, quality control, early defect detection, sustainable manufacturing, pilot lines

1 Introduction

Industry 4.0 has enabled shorter production times, better quality expectations and more integration more with suppliers and customers. The new Industry 5.0 intrinsically considers productivity and efficiency as main targets, placing the worker as a central element while adding a strong focus on sustainability [1].

Disruptive technologies, interconnectivity, robotics and human-machine interaction have evolved into human-machine cooperation, humans in the loop, and increased observability and explainability. This new revolution has come hand in hand with Artificial Intelligence (AI), which is contributing to various industry processes and operations, while at the same time highlighting a further need to adapt to new technological innovations. KPMG's study *Thriving in an AI World* [2], already reflected this trend in 2021 in different industries.

But not all of these innovations are having the expected impact on the industry. In many cases, it is being noticed that companies, especially small and medium-sized ones, cannot keep up with the pace of technological advances, and, above all, that these advances require knowledge that their

workforce do not necessarily have. The reasons are, among others, organizational and infrastructure obstacles and limitations, lack of workforce experience, problems with data collection and the complexity involved in implementing AI-based solutions [3].

Sustainability is not only focused on environmental protection issues, but also seeks to reduce waste, to promote BIO-based materials and to enable circular economy practices, which ultimately will lead to lower production costs. One way to achieve this is through digital solutions, improving production processes and optimizing production cycles.

In particular, quality control is one of the processes that can have a direct impact on sustainability. Not only at the process level, by making them faster and more efficient, but also at the results level, by avoiding waste generation and energy consumption in non-productive cycles [4]. To this end, approaches such as Zero Defects Manufacturing (ZDM) are key for achieving the highest quality and lowest waste, avoiding imperfections and failures [5]. In fact, ZDM-compatible platforms provide detection and repair features through preventive and predictive measures. The burst of AI is also collaborating in these zero-defect tasks, with the increasing availability of predictive or defect detection models [6].

Often, to implement defect detection, AI-based methods should be setup properly and, thus, it is necessary to perform multiple cycles of data generation, a time-consuming and costly task [7], that often generates waste. This is especially relevant in cases of computer vision based inspections, where training the models require having images of correct and faulty parts, and therefore, multiple classification cycles are required. This lack of data can be addressed through data augmentation and synthetic image generation techniques. Through various techniques, from image processing to the use of AI models, large volumes of data, comparable in quality to the original images, can be generated from a small set of source images. This reduces time and costs, and also facilitates other activities such as testing different models and comparing results.

All these elements and areas mentioned above, especially when they are combined, require evaluation in different laboratory environments, to be subsequently validated operationally. Industrial pilot lines serve as a effective tool for this purpose.

The research conducted in this article has been validated on the GAMHE 5.0 pilot line. GAMHE 5.0 is a pilot line of the "Industry and Society 5.0" initiative aiming at developing studies and solutions from social and economic up to high-level factory optimization.

This pilot line allowed the authors to evaluate a particular use case requiring a reliable classification of defective elements but lacking large data collections to train the models. This is the case of manufacturing of insulation panels for the construction sector.

In summary, advancing towards Industry 5.0 requires creating environments that enable manufacturing companies to deploy proven solutions. Additionally, there is a clear need to simplify the setup of quality control systems, and ensure they align with sustainability requirements. These gaps are especially significant in cases such as the quality inspection of thermo-insulating panels, with the goal of achieving zero defects in their production.

This paper presents a case study on the quality control of insulating materials, focusing on enhancing early defect detection, reducing waste from defective pieces, and minimizing rework time. The solution proposed is based on composable platforms and data augmentation, segmentation, and classification components. The study was validated using an Industry 5.0 pilot line. The results obtained, which show improvements in defect detection, make us be positive about the approach followed and open the possibility of further research in the different topics addressed in the article, from ZDM, to modern AI-based data generation techniques.

In this context, the objective of the article is to present an approach for thermo-insulating panel inspection, and to demonstrate and validate on a pilot line the operations and components that simplify quality control related tasks, as well as how these components can be used to improve defect detection. To achieve this objective, in this article these main points are addressed:

- The application of a real case study for inspecting thermo-insulating panels for modular façade cladding in buildings.

- Demonstration and validation of the case study on the GAMHE 5.0 pilot line.
- Conceptualization of the computational architecture and the software platform needed to support quality inspection components.
- Definition of a multi-technique approach to synthetic image generation and introduce a segmentation and classification approach for the quality inspection of thermo-insulating panels.
- Identification and discussion on main gaps and shortcomings according to the state of the art in the field of data augmentation, Zero-Defect Manufacturing and thermo-insulating panel inspection.

These points are treated in the subsequent sections. Specifically, the section 2 presents the state of the art in the multiple areas covered in this article. Section 3 details the architecture, the platform and the approaches followed for data augmentation and defect classification. Section 4 provides the information about the case study, while Section 5 summarizes the results and presents the evaluation. Finally, section 7 presents the conclusions and outlines future activities.

2 Related Work

The Zero Defects concept has been established for considerable time. This term was already mentioned in the 1960s in handbooks on quality and reliability [8]. Zero Defect Manufacturing is a disruptive concept linking production strategies and methods with innovative technologies. The main idea is based on the elimination of defects, errors, failures and even waste in manufacturing processes [5].

Psarommatis in [9] suggests an interesting methodology for implementing a ZDM-oriented strategy. Specifically, it is based on a Digital Twin that allows to configure different parameter-combination sets, so that the effect of the ZDM strategy can be monitored at different points of the process. Additionally, the Digital Twin generates a map that allows the selection of the most suitable strategy to be applied in each case.

According to [10], also as referenced by [11], ZDM can be implemented in two different ways. One, called Product-oriented ZDM, is focused on finding a solution to counteract defects in actual parts. The other one, Process-oriented ZDM, evaluates each product based on the presence of defects. Powell et al. in [12] add the People-oriented approach as a third dimension of Zero Defect Manufacturing, including continuous operator training as a method to improve the efficiency of production processes and to enhance the worker's ability to solve quality problems. This People-oriented approach is closely connected to the concept of human centricity, the focal point the Industry 5.0.

Human-centric Industry 5.0 focuses on more interconnected industries, with smart, resilient and sustainable factories, where the human operator is the reference. In particular, where the individual operator is empowered as a key element of production, and his or her well-being and individual skills and competences are prioritized when cooperating with innovative digital technologies [13]. This change of paradigm also implies a change in technologies, so the human operator is able to understand, trust and adapt to the new processes, especially in the rapidly changing environments associated with AI. This link between Human-Centricity and AI technologies is called Human-Centricity in AI (HCAI), and it is a key concept in AI governance [14], which also has an impact on the use of these technologies in industry.

This shift requires platforms that are reliable enough to meet resilience demands and flexible enough to support emerging technologies, particularly those related to artificial intelligence. In recent years, particularly since the introduction of containerization technologies, and especially with the widespread adoption of orchestration systems and deployment automation, these platforms have become increasingly relevant and essential.

An Ning, for example, presents in [15] a high-availability platform based on containerized applications, focusing on ease of application deployment and emphasizing the use of existing computational resources to make the system effective for small and medium-sized enterprises.

Another example is provided in [16] , where the authors introduce a platform for digital twins, aimed at facilitating the modeling, development, deployment and efficient operation of applications. Their work not only extends Ning's concept of effectiveness but also highlights cost reduction

compared to traditional overprovisioned scenarios. Similarly, [17] follow a comparable approach in their study of AIaaS functionalities and the required tech stack to support them, including the need of reliable software platforms that also avoid vendor lock-in.

Quality inspection is one of the most common activities in the manufacturing industry and, therefore, a good example of those that are recently being aligned with the idea of human centricity. Rožanec et al. proposed in [18] an architecture for human-centric AI and applied it to three use cases, namely demand forecasting, quality inspection, and intention recognition. Within the case of quality inspection, they achieved significant improvements in labelling accuracy and reduction of unidentified errors by more than 80%.

[19] analyze quality inspection from a human-centric and ZDM perspective, detecting that most systems tend to be machine-based, with in-line operators and mainly based on computer vision and image processing. Within ZDM, they also detail that detection and prevention are the main strategies used. Their research also echoes how the human component can increase efficiency in quality control, despite recent campaigns claiming the benefits of removing the human factor from production in order to reduce errors [20].

When it comes to quality control of thermo-insulating materials, dozens of technologies and methods are also found. From solutions based on neural networks [21], to image processing or mathematical calculations and data analysis based on thermal conductivity and diffusivity [22]. Focusing on those related to machine vision and image processing, there are several key subtasks that, in collaboration with the operator, can serve to improve quality control. These include segmentation and classification techniques that can be used to detect defects and data augmentation techniques to improve the training of AI systems by balancing or extending the input data.

Multiple approaches to automatic or semi-automatic defect detection can be found in the literature. In line with the ideas of ZDM. Defect detection and automatic error classification also allows the focus of the manufacturing industry to be shifted from reactive quality models to preventive models. As [23] mention, in the past, problems occurred to be corrected, and after a problem, machines would stop until the cause of the problem was known and solved. This approach caused issues and increased costs, not only in materials, but also in energy and personnel, and could impact the logistics chain and affect the image of the company. The most modern detection approaches benefit from a more prevention-oriented mindset, which allows prior knowledge, preparation for eventual production stoppages and a reduction of the costs mentioned above. The proposal presented in this work is focused on the early identification of problems and the detection of parameters that enables the configuration of the quality models. It also facilitates the monitoring and comparison of process quality at different moments of time.

Obtaining large amounts of data usually requires time-consuming and costly processes [7], especially in the manufacturing world. Therefore, it is often necessary to use techniques that generate new data and allow testing configurations or algorithms on production lines. Data augmentation is the concept that encompasses the generation of synthetic data, to artificially augment the data available to perform a specific task.

As far as images and pictures is concerned, several different data augmentation techniques can be found in the literature, from the most traditional ones based on basic image operations to machine learning approaches and the most innovative ones based on generative models. Including among others techniques to generate synthetic data from photorealistic renderings of CAD data [24] or even from 2D based engineering drawings of parts [25].

Overall, the use of data augmentation techniques reduces the effort and time required for the creation of model training datasets. Moreover, there are examples not only of results comparable to the real images, but even improvements in performance are achieved, for example, those of between 4% and 6% reported by [23]. Other examples are the 3%-3.5% improvements on image classification presented in [26], or the ones between 2% and 5% detailed in [27]. Also in [28] the authors present an improvement of 9.7% for segmentation and 7.1% for object detection, considering objects of small size.

In the context of segmentation and classification models, the literature is extensive and varies significantly depending on the specific use case and the required performance and accuracy. In the particular case of visual inspection, Konovalenko

et al. present a study focused on defect detection in metal structures [29] . In their work, they explore the use of U-Net neural networks for image segmentation of defective areas, analyzing different backbones and modifications (such as ResNet, SEResNet, DenseNet, InceptionV3, MobileNet, and EfficientNet, among others). ResNet152 produced the best results in their evaluation. As part of their study, they also highlight several use cases where U-Net-based segmentation systems have been successfully applied [30, 31].

Another relevant example is found in [32], which focuses on fabric defect detection using U-Net and related techniques. The authors report improvements in defect detection performance. In the construction domain, the study presented in [33] is also noteworthy. Although it focuses on window frame defect detection, it is representative of the construction sector. This work also employs U-Net, and data augmentation methods, achieving significant improvements, with F1 score gains of over 7%, even under real-world conditions.

All this state of the art converges in a specific use case, namely the reduction of waste and the assistance to the operator in the improvement of the quality control of thermal insulation panels, widely used in construction. Kalnæs and Jelle [34] have already stated that the quality assurance of insulation panels is a key factor to promote the use of insulation panels in the building sector. Is also of relevance the need of detecting when defects in panels are production defects, are due to ageing or due to other service failures [35].

2.1 State of the art: gaps and shortcomings

Our analysis of the state of the art has revealed some limitations in the reviewed articles and methods. For instance, many studies cite U-NET as the primary segmentation method. While common limitations of U-NET, such as high computational cost when processing large and complex images, and sensitivity to dataset quality and noise, are frequently noted, they did not significantly affect our specific case study, which relied on smaller not specially noise images. In our method, and beyond the use of U-NET, we introduced an additional step after segmentation. In this step, the segmented region was bounded by a convex hull, which was then simplified into a high-precision polygonal curve. If exactly two four-sided polygons were identified, the panel was classified as non-defective.

It is also important to note that, in the specific context of our study, another significant limitation lies in the lack of standardized datasets. Given the nature of the use case, there are no large, publicly available data collections. While this is generally a limitation, it also partially benefited our work by creating the need to develop data augmentation strategies and synthetic image generation methods, in order to ensure that the models were trained on balanced datasets.

In the state-of-the-art section, we have also discussed several platforms based on containerization and orchestration. A common limitation of these platforms is their complexity and the requirement for skilled personnel to manage and maintain them. The selected works focus on platforms either designed for SMEs or intended to be simple to implement. One of the most frequent gaps in the literature is the lack of a comprehensive view of the problem. It is common to find papers that focus on platforms, or on classification or segmentation, or on specific quality control use cases. However, it is much less common to find a complete end-to-end approach. In our article, we aimed to cover the entire pipeline, from the definition of a lightweight platform to the development of components that can run on the platform and interoperate to perform a specific quality control task.

Closely related to the previous point, validation in near-production environments, especially those where humans are involved in the cross-validation process, is not commonly addressed in the literature. For example, studies focusing on real-time inspection of parts in production settings, particularly on lines that integrate general purpose computing systems, robots and legacy systems, remain relatively scarce. The practical orientation of our work also aims to overcome this limitation by detailing the hardware-software system used and providing a concrete example of how to validate the solution in test-before-invest environments.

3 Methods

With the goal of evaluating and validating the approach presented in this article, an architecture

have been defined, including a core software platform and different components. This architecture is built on the lessons learned in the KITT4SME project when designing a minimal Kubernetes-based platform along with a modernized service networking layer, Istio. This platform is complemented with a deployment and management system for the control of the components, namely the data augmentation module, and the segmentation and classification subsystem. The platform is complemented with a provisioning system that allows to create, destroy, upgrade, and maintain production-grade solutions based on the recipes and the descriptors associated to the modules. The benefit of this approach is that it provides the fundamental elements so that both technology providers and manufacturing companies have a convenient and repeatable way for building their own AI platform in the cloud of their choice. In this manner, the organizations can use the components they need and rely on the underlying platform to manage part of the complexity related to security, scalability, or interoperability.

The KITT4SME reference platform is based on a 3-tier architecture on top of the hardware layer. The layer just above the hardware is the *"Mesh Infrastructure Layer"*. This layer, in addition to managing resources, is responsible for the communication and security. A layer called *"Platform Services Infrastructure Layer"* is placed on top of it. This is where a collection of services that support the user applications are located. These services include databases, connectors for IoT sensors or even support for other middleware used in the industry. Finally, as just mentioned, the top layer is the *"Application Layer"* where specific applications that cover particular use cases, such as deployed AI solutions, dashboards or user interfaces, are located.

The platform relies on industry standards for data transfer and open communication to facilitate service interoperability. For example, REST principles for microservices integration, and in cases necessary Next Generation Service Interfaces (NGSI) specifications for information modelling and data exchange.

In the particular implementation of this work, with the goal of making it lighter and reducing complexity, the defined platform includes only two tiers: infrastructure level and applications and services level. This approach is shown in the figure 1.

The *"Infrastructure Layer"* includes a container orchestration system that allows to manage deployments, scaling and load balancing if necessary, along with a continuous delivery tool and a service mesh to simplify the use of microservices and provide observability, traffic management and security capabilities. These three elements establish a flexible core without excessive overhead, especially with the use of lightweight Kubernetes distributions, and open up a wide range of possibilities for developing application modules and composing solutions such as the one mentioned later in the use case.

The *"Applications and Services Layer"* brings together third-party services and in-house applications in the same layer. Unlike other 3-tier platforms, the need to provide different levels of services or to define default services, have not been considered. This strategy, in which all services running on top of the infrastructure layer are considered optional, enables the definition of a lighter architecture and allows integrators to choose the best combination of microservices for their task.

Other architectures also include an extra layer dedicated to information presentation. This particular implementation have considered these presentation elements, such as user interfaces and dashboards, as applications that can be integrated into the applications and services layer. Ultimately, they will make use of data that are generated and components that are integrated in this same layer.

In general, and in relation to other similar architectures, this particular approach has been directed to the design and implementation of a light infrastructure and core, to cover the minimum needs of interoperability, security and scalability. Also observability, through monitoring and traceability tools, to facilitate ZDM detect and predict tasks. The lightweight optional services layer, without excessive default applications, enables extensibility by means of an application deployment system while maintaining the minimum complexity. The platform is based on industry standards for communication, and can optionally support FIWARE components for the interconnection with intelligent solutions of other

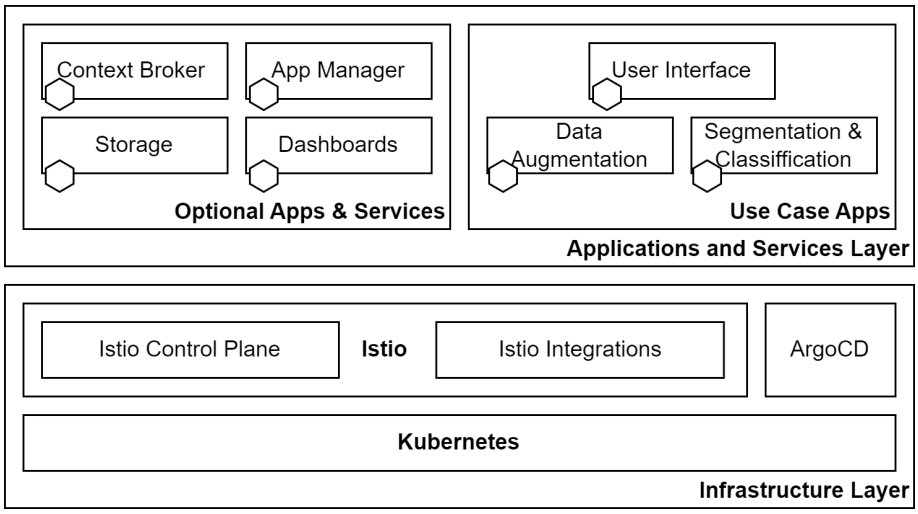


Fig. 1 Use Case Platform

industries. Finally, the platform perfectly facilitates the use case of synthetic data generation and image segmentation and classification, which allows the seamless integration of components and supports their reuse in different contexts.

The figure 2 shows the Business Process Model and Notation (BPMN) diagram of the use case where the architecture is instantiated. The process begins by gathering multiple reference images of correctly manufactured pieces, as they are the easiest to obtain. Simultaneously, images of defects that have appeared in similar parts are selected. Both collections of images are the source for the data augmentation module, which is detailed later. This module applies various synthetic data generation techniques to produce a large number of images. These images are usually reviewed, in case the operator wants to exclude some of them, and then are used for two tasks. One is to train image classification and quality control models. The other is to use a set of automatically generated images to evaluate proprietary or even third party models and select the most suitable one for a particular family of parts. Finally, the results are evaluated enabling a decision on whether to deploy the solution or to start over with other parameters.

The following subsections detail the approach followed to define the platform and to develop the data augmentation, segmentation and classification modules.

3.1 Software platform

Aligned with the aforementioned architecture, a platform that covers the infrastructure layer and adds proxies to the components of the applications and services layer have been implemented. Acting as as interceptors, these proxies manage communications, as well as support observability and load management. The platform is based on Kubernetes, as presented in the architecture. Lightweight versions have been used, specifically K3s and MicroK8s. Both are small, fast, production ready and compatible with the service mesh used in the platform. On top of this distributed containerized application management layer sits the Istio service mesh. With Istio the functionality of Kubernetes is extended to provide even more monitoring and traceability. The service mesh additionally enables the platform to support future applications with security or scalability requirements.

Istio's architecture is divided into a control plane and a data plane. The data plane is composed of a series of proxies that are attached to the services and applications. These proxies act as intermediaries in the communications, which in addition to managing these communications facilitate obtaining information and metrics. The proxies, which are represented in figure 1 by a hexagon, are Envoy lightweight proxies, and as mentioned above, they provide load balancing or health checking capabilities, among other features.

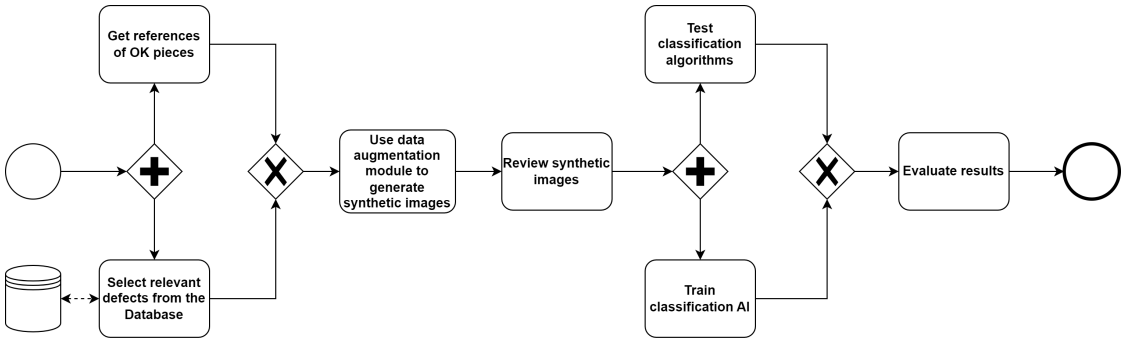


Fig. 2 Use Case - BPMN

In addition to these capabilities, Envoy proxies can manage inbound/outbound traffic to the components, and add security capabilities such as access control or policy enforcement.

The control layer is responsible for managing these proxies, configuring the mesh and managing communication between the workload instances within the mesh. It also manages certificate management and service discovery.

In turn, ArgoCD has been deployed on top of the container management layer. This component provides continuous delivery capabilities and behaves as a controller that allows continuous monitoring of the state of applications performing actions when one of these changes state or when there have been modifications in the source code. This allows us to easily roll out applications and components, while keeping them up to date.

In general, this lightweight platform set up has been helpful in facilitating the orchestration, the deployment and the integration of microservices, both in premises and the cloud. Applications and modules residing in top of this platform, such as synthetic data generation or image classification that are detailed next.

3.2 Data augmentation

Not having sufficient data to design and train models is a common problem in the manufacturing sector. The problem is even more serious in manufacturing SMEs, which often do not have the capacity, knowledge, or time to generate the data needed to train algorithms or to hand it over to technology providers to design AI solutions. That's where synthetic data generation comes into play. Specifically, there are three situations in which synthetic data generation can be beneficial.

The first one is when there is no balanced data for the training of models. For example, it is common in industrial classification systems to have, or to be able to generate, data for correct parts, but not so easy for defective parts. For the correct parts, references or specifications are usually available, for the faulty ones, in the best cases, some examples of similar parts and some knowledge of what has been happening in the past can be used. It is certainly true that not all classification algorithms require balanced data, and zero-shot or one-shot classification models are becoming increasingly common, but there are still multiple cases in which balanced data equals better overall results.

The second situation is related to the comparison of models, either proprietary or acquired from third parties. In this case having a large collection of data, in our case images, can allow the evaluation of the models and the selection of the most suitable for the final objective. Moreover, it can be useful, not only to compare different models, but also the same models with different parameters.

The last situation is related to the reconfiguration, fine tuning and adjustments of the quality control systems once new defects are detected. Over time new defects or irregularities will appear with which the system was not trained. It is likely that some of these errors will only occur in such rare cases that it is very difficult to have enough information to retrain or adapt the model. Here the generation of synthetic data is also beneficial. For example, in the particular case of this work, a defect that has originated in a similar part can serve to adjust the classification system of a new family of pieces.

In this case-study, the data gap is determined by the lack of relevant imagery to train the model. Therefore, the suggested data augmentation approach focuses on the generation of synthetic images, and specifically uses three fundamental techniques: i) image manipulation, ii) blending, and iii) pattern generation. Each of these techniques and our implementation is detailed in the use case section.

3.3 Segmentation and classification

In the case addressed in this work, the target was to detect manufacturing defects on the panels. Given the characteristics of the panel, the possible manufacturing defects can be theoretically detected with traditional image processing techniques by matching geometric shapes with the panel surfaces. However, in practice, due to the environmental conditions under which the inspection is carried out (i.e., lighting variability, frequent changes in the colors of panel materials, high background noise, etc.), directly applying the image processing techniques often result in erroneous outcomes. Consequently, the decision-making algorithm can select a wrong process workflow execution given that its premises were incorrect. For this reason, it was decided to add a segmentation stage at the beginning of the visual inspection task to separate the panel surface from the rest of the elements in the image, thus facilitating the execution of the defect detection algorithm.

The segmentation model developed for validating the ZDM concept in the GAMHE 5.0 manufacturing pilot line (Figure 3) attempts to separate the side surface of the panel from other elements within an image, allowing later decisions on the panel quality and modifications of the process workflow. This model is based on a U-net architecture, which consists of an encoder path that gradually downsamples the input image and a corresponding decoder path that upsamples the feature maps to produce a segmentation map of the same size as the input image. On the other side, the decoder path aims to reconstruct the input image's spatial dimensions while combining the contextual information captured by the encoder. For this purpose, a series of transposed convolution (up-sampling) and convolution layers are applied, resulting in a symmetric architecture [36]. This network also includes skip connections between the encoder and decoder paths that allow to retain and fuse both high-level and low-level features, facilitating accurate segmentation and object localization [37].

The U-net approach was selected since it was specifically designed for segmentation tasks. This type of network is very efficient in capturing local details, which is achieved via the skip connections. Initially, it was proposed for segmentation of elements in biomedical images; however, given its high performance in this task it was rapidly applied to a large number of different fields. In several studies, U-net has outperformed other segmentation neural networks such as SegNet or Fast Fully Convolutional Networks at a reasonable computational cost [38].

As previously described, the visual inspection task for detecting defects in the panels comprises two stages. After the image is segmented by the deep learning model, a second algorithm is used. Here, a convex hull is adjusted to each separate contour in the segmented image. Then, a polygonal curve is generated for each convex hull with a precision smaller than 1.5% of the perimeter of the segmented contour. Finally, if the polygonal curve has 4 sides, it is drawn over the original image. After this procedure, if two rectangles were drawn over the image it is assumed that the slotting was correct and the panel did not suffer any significant damage; thus, it can be sent to the next stage of the line. On the other hand, if only one rectangle was drawn, it is assumed that the slotting was not carried out or the panel was damaged during this process.

4 Case study

The pilot line chosen for the particular application case has two machine tools fully equipped with sensors, two manipulator robots, two conveyor belts, and an autonomous mobile robot as a pallet truck for re-manufactured parts. The machine tools are a Deckel Maho DMC 75V Linear high-speed machining center with a CNC Siemens 840D and an ultra-precision micromachining center equipped with a laser control Nano NT. In addition, two manipulator robots, Universal Robots UR5e and Stäubli RX90, are located next to the machines, for operational handling

of the work pieces. Finally, two conveyor belts are in charge of transporting the work pieces between machines that perform different cutting operations. Further down the two conveyor belts and the autonomous mobile robot ROSbot XL HUSARION with a LiDAR RPLIDAR A2 and a stereoscopic camera Intel RealSense D435 are in charge of the transfer between machines of manufactured parts.

Furthermore, it is important to highlight that the communication link, between the IoT nodes and the machine tools are implemented by two industrial standard protocols, i.e., OPC DA with the Deckel Maho, TCP-IP via Ethernet to Kern Evo, RX90 and UR5. Finally, the conveyor belts are directly connected to the DC motor controller DC-DC controller. The type of hardware running as IoT nodes are three raspberry pi 3B+ and two 4B+ models and two PC MSI core i3, with 4GB RAM. On the other hand, all these IoT nodes are interconnected with each other and with a higher layer through the MQTT protocol.

One key stage inside the pilot line is the quality control system that is composed of the artificial vision inspection camera Allied Vision Mako G-192 (Figure 4a), a GigE machine vision camera with an e2v CMOS color sensor and 1.9 Megapixels. This is located in front of the pick and place arm robot UR5. The robot picks up the panel from the conveyor belt and brings it in front of the camera for processing to inspect all faces of the panel.

The pilot line makes possible a wide variety of manufacturing operations of different parts. A real case of application is the manufacture of thermo-insulating panels for modular façade coatings in homes and buildings (Figure 4b). The manufacturing of panels/covers for facades and coating has a significant environmental and socioeconomic impact. Energy efficiency, fire spread and overheating are key issues and risks to the health and productivity of people and businesses. From a technical point of view, the main challenge is the productivity and quality of the covers manufactured following criteria such as dimensional accuracy and final finish. The manufacturing process of these covers based on metallic, ceramic, mosaic or natural stone cladding adhered to extruded polystyrene (XPS) panels consists of several phases: gluing, pressing and profiling.

Different types of defects can occur during the cutting of XPS panels in a machining center including irregular borders caused by degradation of the cutting tool or inadequate cutting parameters, material melting caused by excessive cutting speed or pressure, and cutting deviations caused by misalignment between the cutting tool and the work piece or excessive vibrations. In the GAMHE 5.0 pilot line the most frequent defects are irregular borders as result of processing batches of XPS panels with different mechanical properties. This difference between batches may cause that fixed cutting parameters are not adequate for all situations, resulting in non-compliant panels that should be discarded in a zero-defect manufacturing scenario.

The early detection of these defects in the production process allows to remove the panel from the line, saving time and energy, and ensuring that the products at the end of line do not have irregular borders. Additionally, when defective panels are identified, a timely decision can be made to modify the cutting parameters to avoid future defects, resulting in a reduction of waste.

4.1 Data augmentation

Image manipulation, in the most traditional sense, is the functionality of applying editing techniques to alter the original image. Specifically, image manipulation is referred as the collection of features that allows to obtain images derived from a reference image by applying functions that alter parameters of the original image. For example, the use of brightness and contrast to lighten or darken an image, hue or saturation to play with colors or simply rotate or mirror the image to generate a new one that is different but maintains the structure of the original image. In some cases, it is even interesting to fully alter the original image, for example, in cases where the classification model does not require structural context. For these cases, the particular approach of this work makes use of shear mapping to move points of the image to other positions, or even the displacement of the entire image or a scaled version of it. Based on these and other image modifications, an application percentage are used to each one, so that one or more operations are applied to each image. Each operation in turn is preset with a series of values so that a wide range of values is covered.



Fig. 3 GAMHE 5.0 pilot line for thermal insulation panels manufacturing

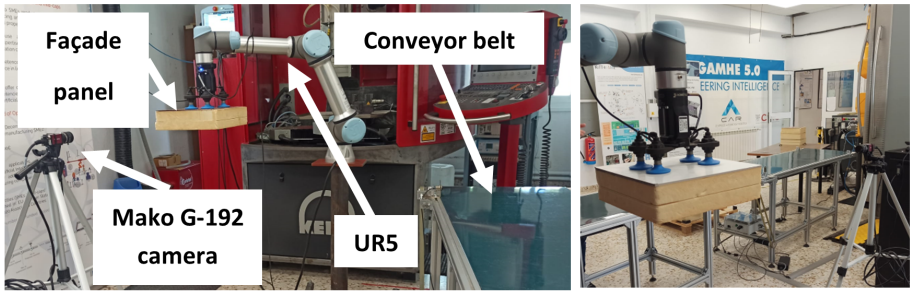


Fig. 4 (a) Quality control system and (b) inspection procedure of all faces of the panel.

For example, it may be interested in rotating the region of interest of the original image only by certain degrees. With this variety of parameters and functions is possible to generate as many different synthetic images as needed, starting from a few reference images.

Blending plays with the combination of pixels from two images to integrate the result. For this mix or blend of pixels, what is known as blending modes are used: a series of well-known operations that modify the way these images are joined. Examples of these blend modes are: darken (which compares the colors of both images and keeps the darker colors), subtract (which subtracts the colors of the image to be blended from the base image) or dodge (which lightens the base image colors and reduces the contrast). Figure 5 shows an example of the reference parts used. For this purpose, data augmentation by blending regions of interest with defects in random positions, sizes and shapes have been applied.

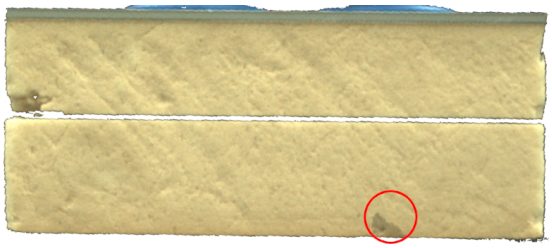


Fig. 5 Images generated by blending a defect in a reference image

Pattern generation is the technique used to add spots, drops, or stains to a reference image. This technique involves the automatic generation of these spots and stains randomly in a defined region, to which manipulation actions (e.g., scaling, rotation, shifts) are applied before proceeding with blending (e.g., lighting, subtracting) techniques. This process of combining automatically generated patterns with image manipulation and

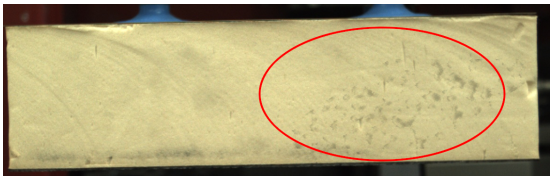


Fig. 6 Images generated by blending a pattern

subsequent blending allows the generation of large datasets. These images have a noise or blemish component, for example, such as those that occur in cases where the part to be evaluated is defective. Figure 6 includes an automatic generation of stains of different sizes, based on the pattern generation functionality just explained.

4.2 Segmentation and classification

A dataset containing images with their corresponding masks was prepared for training and evaluating the model. The image dataset was split into three subsets: training (70% of the data), validation (15% of the data), and testing (15% of the data). In this case, the validation subset serves the objective of facilitating early stopping during training. This means that if the model's performance evaluated on the validation subset fails to improve after a predetermined number of epochs, the training process is halted. By employing this technique, over-fitting can be effectively mitigated and the training time can be significantly reduced.

A second version of the dataset was prepared by applying data augmentation to the training set while keeping the validation and test sets unchanged. This helps increase the number of training examples, improving the model's prediction capability and making it more robust to noise. Using the two versions of the dataset, two models with the same architecture were trained. Figure 8 presents the output of the two models for two examples taken from the test set. As can be observed, the predictions obtained with the model trained on the augmented dataset are significantly better than those obtained with the model trained on the original dataset. This is also confirmed by the values obtained in several metrics, which are shown in Figure 7 and the Table 1.

Table 1 presents the results obtained using the different methods. The results are reported using well-known metrics: Accuracy, F1 Score, Jaccard

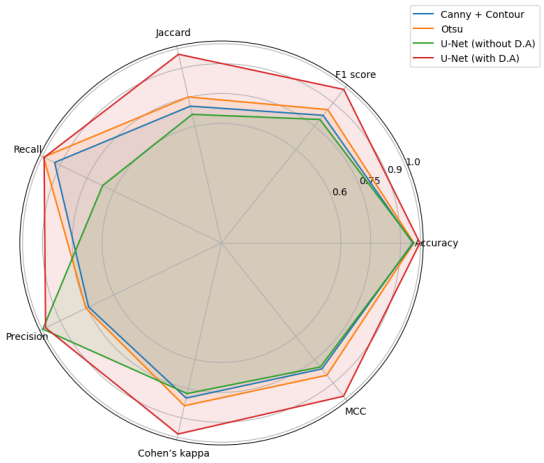


Fig. 7 Metrics - radar chart

Index [39], Recall, Precision, Cohen's Kappa [40], and MCC (Matthews Correlation Coefficient [41]). The methods evaluated include a technique based on Canny edge detection followed by contour filling, a thresholding method using Otsu's algorithm, and two deep learning approaches using U-Net, one without and one with data augmentation.

Canny edge detection and Otsu thresholding were selected due to their simplicity and computational efficiency, as they do not require large datasets or expensive training. Both methods deliver reasonably good results. For instance, Otsu's high Recall suggests it is capable of correctly identifying most regions of interest, although this comes at the expense of lower Precision.

In contrast, the U-Net based methods tend to achieve higher Precision, indicating that they classify the identified regions with greater confidence. Specially, the U-Net model trained with data augmentation outperforms all others across nearly all metrics, particularly in global metrics such as the F1 Score. This highlights the importance of training with synthetic data, which improves generalization and segmentation accuracy.

5 Case Study Evaluation

Figure 8 shows the results obtained for illustrative cases of a compliant panel and a damaged panel, respectively. If only one rectangle is drawn,

Table 1 Segmentation and classification metrics

Metric	Canny edge detection + contour filling	Otsu thresholding	U-Net trained without data augmentation	U-Net trained with data augmentation
Accuracy	0.96160	0.96691	0.96542	**0.99709**
F1 score	0.81950	0.85629	0.79309	**0.98552**
Jaccard	0.70443	0.75194	0.66211	**0.97151**
Recall	0.93118	**0.99305**	0.66355	0.98953
Precision	0.74134	0.75594	**0.99702**	0.98158
Cohen's kappa	0.79848	0.83816	0.77537	**0.98390**
MCC	0.80876	0.84936	0.79636	**0.98392**

depending on its size and location, the panel will be sent to the slotting stage again or removed from the line. This method was applied to the test set images and in all the cases the output produced matched the expected output.

In order to evaluate the benefits of the inspection system implemented in the GAMHE 5.0 laboratory, three batches of 1000 panels were processed. The first batch, used as baseline, was processed without using the inspection system, which means that defective panels where not removed from the line and the cutting parameters were not modified. On the other hand, the inspection system was used for the second and third batches. In these two cases, if a defect was detected, the panel was removed from the line and the operator was notified to modify the cutting parameters. For the second batch, the inspection system was based on the model trained on the original dataset, while for the third batch, it was based on the model trained on the augmented dataset.

It is important to compare the U-Net based segmentation method, with and without data augmentation (Figure 8), to the other models studied. Figure 9 presents the results of the Canny Edge detection and Otsu thresholding methods. Both of these techniques failed to accurately segment the images and were unable to remove background objects effectively. As a result, in the classification stage using the segmentation outputs from these methods, the system consistently misclassifies all parts as defective. Therefore, these approaches are not viable for practical application in our case study.

Table 2 summarizes the results obtained for the defects related to panels with irregular borders (Figure 10) which were the main defects considered in this study. The main Key Performance Indicators (KPIs) analyzed are as follows: i) Number of defective panels at the end of line. This indicator counts the number of panels that pass through the entire manufacturing process and end up with defects. It represents the effectiveness of the inspection system in reducing defects in the final product; ii) Number of defective panels removed after inspection. This KPI demonstrates the efficiency of identifying and removing defective parts before they reach the end of the line. Early removal of defective panels saves time, prevents waste, and ensures higher quality of the final product; iii) Time saved by removing defective panels: This metric evaluates the reduction in time spent processing defective panels, thereby improving overall process efficiency.

Table 2 Segmentation and classification results

	Batch 1	Batch 2	Batch 3
Number of defective panels at the end of line	26	7	3
Number of defective panels removed after inspection	-	12	13
Time saved by removing the defective panels	-	$\approx$ 1319 s	$\approx$ 1429 s

As it is shown in Table 2, the use of any of the two versions of the inspection system contribute to decrease the number of defective panels at the end of line when compared to the baseline. Moreover, by modifying the cutting parameters after detecting a defect, the total number of defective panels per batch was also reduced when the inspection

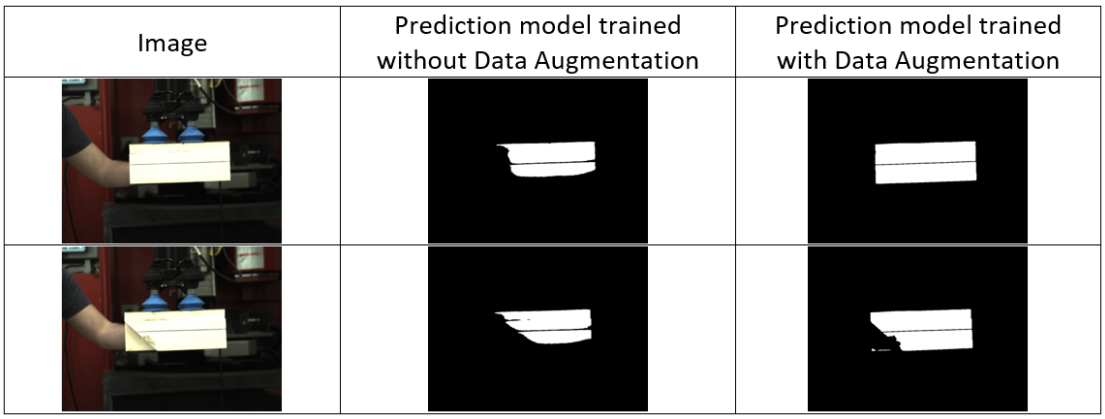


Fig. 8 Prediction model comparison

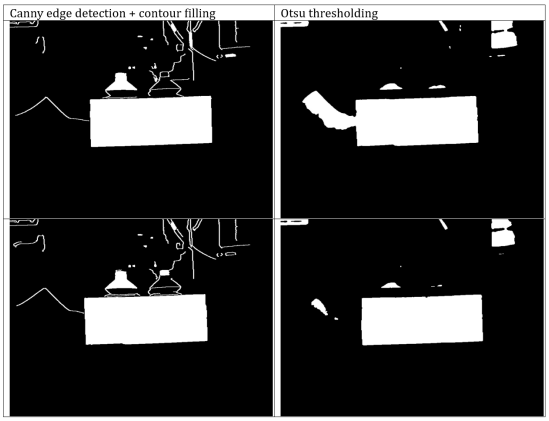


Fig. 9 Canny edge detection (left) and Otsu thresholding (right) based segmentation

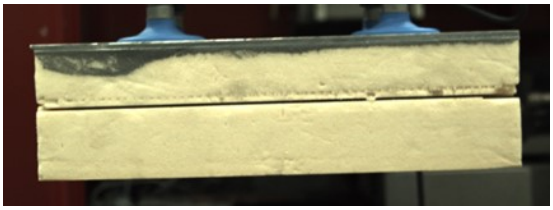


Fig. 10 Panel with irregular borders

system was used, with the best results obtained using the version of the system based on the model trained on the augmented data set. This result can also be observed comparatively in Figure 11, which shows the rate of defective panels at the end of the line.

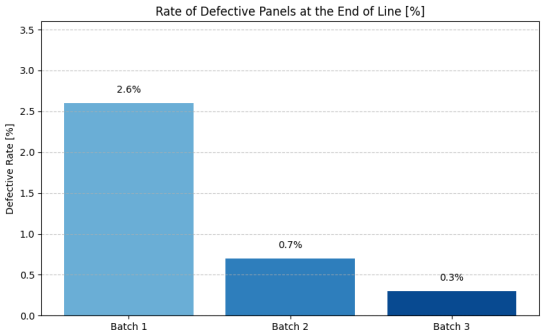


Fig. 11 Rate of defective panels at the end of line

Additionally, when the defective panels are early removed from the line thanks to the inspection system, the final machining operation is not applied to them, saving considerable time. The results also show that the classification model trained with synthetic data not only does not worsen the one trained with real data, but actually improves it in all three aspects evaluated. This is also evident in Figure 12, which illustrates the improvement in batch processing time for the models that use data augmentation, compared to the model without augmentation, and especially in comparison to the baseline.

6 Discussion

The work aimed to provide an overview of a complete case study on how to design a platform and the software components that support a quality control system. It also aimed to evaluate methods

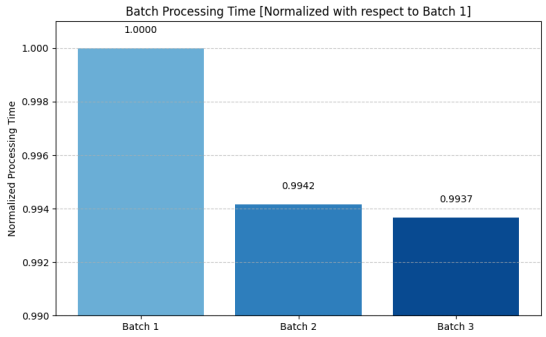


Fig. 12 Batch processing time [normalized with respect to Batch 1

used for data augmentation, segmentation, and classification. Additionally, it demonstrates how this system was practically implemented and validated in a pilot line. During this process, several important gaps and opportunities were identified in both the existing literature and current industrial practices regarding the use of data augmentation and synthetic data generation for quality inspection in the specific context of thermal insulation panel manufacturing.

One of the most notable gaps identified is the absence of relevant case studies applying data augmentation techniques to the specific use case addressed in this work. Although related efforts have been found in the domain of quality control for generic pre-manufactured building materials, these applications tend to be limited in scope. Specifically, [42] focuses on the detection of scratches on transparent glass surfaces, given that glass façade is one of the most relevant components in the building construction industry. To do this, they propose a pixel-level instance segmentation model based on Mask R-CNN [43], which allows them to generate bounding boxes and masks for each instance of the object in the analyzed image. All this is based on an dataset extended through the generation of synthetic images. Meanwhile, [44] base their method for the detection and classification of defects of the structure of facing bricks on a YOLOv5s convolutional neural network. Also assisted by a data augmentation algorithm, they report high accuracy in defect detection. Detecting and eliminating defects in early stages has the aforementioned benefits in costs and sustainability, and in turn allows to control and reduce the spread of defective materials to the rest of the production.

Another identified issue is the limited discussion on validation using real-world pilot lines. While many works evaluate model accuracy or compare augmentation strategies, few of them report on practical Key Performance Indicators in relevant or operational settings. In contrast, our approach places emphasis not just on the performance of the classification or segmentation models, but also on operational metrics such as piece rejection rates and time efficiency. The use of the GAMHE 5.0 pilot line has enabled a more holistic validation of our approach, providing valuable information regarding the practical implications of the system.

The complexity of these systems and their application in manufacturing environments, where companies still find it difficult to adapt to the new, more digitalized industry, has also been identified as a barrier. This is described in detail by Kumar et al., in their analysis of the fifteen main challenges for Industry 4.0 [45]. In their article, they refer, among others, to the lack of resources for research and development in Industry 4.0 technologies (Challenge 6), the lack of IT-based infrastructures, both software and hardware (Challenge 9), and the lack of knowledge and trained workforce (Challenge 10).

To address this barrier, reference architectures and platforms, such as RAMI 4.0 [46], OpenFog [47] and IIRA [48], are increasingly emerging. All of them aim to make the integration of systems much easier and the adoption of innovative technology based on artificial intelligence seamless. In particular, the implementation and deployment of these platforms and architectures can provide scalability, security and meet part of the non-functional requirements of industrial applications. Moreover, they can reduce the need for technical knowledge and even the initial costs. These component-based platforms are key to assist SMEs in adopting AI-based solutions by offering various added services, such as monitoring, traceability or by providing access to messaging protocols or context brokers that can manage context information, such as FIWARE's Orion Context Broker. All together, these platforms create an ecosystem on top of which data

augmentation functionalities can be easily integrated together with classification and segmentation in order to improve quality inspection solutions. Some of these architectures are either conceptual or server as reference architectures. The particular case of this work have been inspired by several of them to define a reduced complexity platform that provides the basic capabilities to integrate different components. At the same time, the platform provides observability and scalability to the use case, making its use in manufacturing SME environments more viable.

Regarding the methods, our work and analysis on data augmentation and segmentation and classification has provided some discoveries.

For example, with the advances in generative artificial intelligence models, data augmentation capabilities have also changed significantly. In this study, we focused on traditional and hybrid methods, although some basic tests using image manipulation techniques to generate defects were also conducted.

The main limitation of these approaches is the potential lack of diversity in the generated patterns, which can lead to overfitting in the segmentation or classification model. Another important limitation is the difficulty in generating rare or previously unseen defects. These challenges can be addressed by using models based on Generative Adversarial Networks, diffusion models or autoencoders with defect injection, as is in the roadmap of the authors.

On the other hand, the selected traditional methods offer advantages over deep learning-based approaches, mainly due to their simplicity, speed, and low computational requirements. This made the selected approaches suitable for developing lightweight data augmentation modules that can run on the previously described platform. A second advantage is the ease with which the generated images can be explained to a human operator, allowing for more straightforward validation of the synthetic data, and a alignment with the ideas of the human centricity of the Industry 5.0.

Regarding segmentation, one of the most interesting points in the literature is the suggestion to generalize the U-NET models combined with data augmentation, as proposed by [33]. Although the strategy and specific case differ, the goal of quality control and some of the underlying technologies are related. This connection allows us to further generalize the approach, confirming that it has also proven useful in our case.

In addition to addressing these gaps and supporting adaptable SMEs aligned with Industry 5.0 sustainability goals, the approach presented in this work contributes to the state of the art in several areas, including: i) containerized and modular software integration for scalable pilot lines; ii) application-specific early defect detection and quality control systems; and iii) enhanced flexibility in quality inspection through synthetic image generation techniques.

7 Conclusion

We are in the midst of what is being called the fifth industrial revolution. A revolution that inherits the needs of integration and reduction of time and costs, to which requirements of quality, sustainability and well-informed cooperation between technological solutions and human operators are added. A revolution that is attractive in terms of advantages, but which involves difficulties of adaptation, especially for small and medium-sized enterprises. These difficulties are not only related to budget and time constraints, but also to the complexity of the new technologies and above all of a workforce that has to adapt its know-how. One of the trends in this new evolution of the industry is Zero Defects Manufacturing. Processes such as quality control were already key elements, but now they have a sustainability derivative and the importance of getting things right the first time is even more stressed.

In line with these trends, this paper presents the study carried out to analyze the improvement of the quality control of thermo-insulating panels for modular façade cladding in buildings. The main objective of this work has been to demonstrate and validate on a pilot line the operations and components that simplify quality control tasks, while also showing how these components can facilitate defect detection and reduce waste at the end of the production line.

To achieve this, the state of the art has been investigated to design a lightweight platform that masks the complexities derived from interoperability, security and monitoring. On this platform, modules that allow the classification of the panels have been deployed. This classification is performed by means of computer vision and AI

techniques. A common problem in training these classification models is the lack of original data. In this case, the lack of images of damaged panels made model design and training more challenging. For this purpose, a data augmentation module has been designed to allow the generation of large volumes of synthetic training data.

This approach has been applied on a pilot line for quality control of thermal insulation panels. Some of these panels showed defects, being the most frequent defects the panels with irregular borders. Data augmentation techniques have allowed the development of models that have improved both detection time and accuracy of the results. Specifically, an improvement of about 8% in the time saved by removing the defective panels have been obtained, and in addition the detection of defective panels and therefore reduced the defective panels at the end of line have been increased. As mentioned above, the early detection of the defects in the production process allowed to remove the panel from the line, saving time and costs, and ensuring that the products at the end of line do not have irregular border, reducing potential waste.

These results have two main implications. The first is that composable platforms and the modular approach can simplify quality control, while also enabling flexibility when scaled up to production lines. The second is that the case study demonstrates that data augmentation techniques can help provide the necessary data to train models for accurate quality control. This helps to reduce training data generation efforts, which, in some cases, such as with thermo-insulating panels for buildings, can be complex, time-consuming and generate waste.

In addition, approaches like the one presented in this article enable SMEs to adopt new technologies and adapt to the requirements of Industry 5.0. For example, the modular design facilitates the compatibility with legacy systems and provides the flexibility to replace components as new solutions are developed or new requirements arise. This allows SMEs to adapt to future use cases. Additionally, the use of pilot lines enables the validation of solutions without the investment and disruptions that would come from deploying the solutions directly on the production line. Lastly, platforms like the one detailed mask the complexity and reduce the technical knowledge required to address functionalities such as security and interoperability.

The step towards validation in productive industrial environments is one of the future challenges derived from this article. Specially by applying these techniques to BIO-based insulating materials. New data augmentation and synthetic image generation techniques are also being studied, such as conditional generative adversarial networks (cGAN) or the increasingly common diffusion models. Finally, it is also in the interest of the authors to validate the approach with parts from other sectors, such as the automotive industry. This will allow to reuse the platform, components and models, and with an adaptation on the pilot line analyze the universality of the solution.

Declarations

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Conflict of interest

The authors declare no conflicts of interest.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Materials availability

Requests for further information and resources should be directed to and will be fulfilled by the lead contact.

Author contribution

Rubén Alonso: Conceptualization, Writing - Original Draft, Writing - Review & Editing, Software; Fernando Castaño: Resources, Supervision, Writing - Review & Editing; Yarens J. Cruz: Software, Validation; Alberto Villalonga: Software, Validation; Rodolfo E. Haber: Supervision, Writing - Review Editing.

References

[1] Xu, X., Lu, Y., Vogel-Heuser, B., Wang, L.: Industry 4.0 and industry 5.0—inception, conception and perception. Journal of Manufacturing Systems **61**, 530–535 (2021)

[2] Krishna, S., Campana, E., Chandrasekaran, S.: Thriving in an ai world. Technical report, KPMG (2021)

[3] Deloitte: Deloitte survey on ai adoption in manufacturing. Technical report, Deloitte (2020)

[4] Castaño, F., Haber, R.E., Mohammed, W.M., Nejman, M., Villalonga, A., Lastra, J.L.M.: Quality monitoring of complex manufacturing systems on the basis of model driven approach. Smart Structures and Systems (2020)

[5] Fragapane, G., Eleftheriadis, R., Powell, D., Antony, J.: A global survey on the current state of practice in zero defect manufacturing and its impact on production performance. Computers in Industry **148**, 103879 (2023)

[6] Cruz, Y.J., Rivas, M., Quiza, R., Haber, R.E., Castaño, F., Villalonga, A.: A two-step machine learning approach for dynamic model selection: A case study on a micro milling process. Computers in Industry **143**, 103764 (2022)

[7] Meister, S., Möller, N., Stüve, J., Groves, R.M.: Synthetic image data augmentation for fibre layup inspection processes: Techniques to enhance the data set. Journal of Intelligent Manufacturing **32**, 1767–1789 (2021)

[8] Foche, G.: A guide to zero defects, quality and reliability assurance handbook. Office of the assistant secretary of defence, US., Washington (1965)

[9] Psarommatis, F.: A generic methodology and a digital twin for zero defect manufacturing (zdm) performance mapping towards design for zdm. Journal of Manufacturing Systems **59**, 507–521 (2021)

[10] Psarommatis, F., May, G., Dreyfus, P.-A., Kiritsis, D.: Zero defect manufacturing: state-of-the-art review, shortcomings and future directions in research. International journal of production research **58**(1), 1–17 (2020)

[11] Caiazzo, B., Di Nardo, M., Murino, T., Petrillo, A., Piccirillo, G., Santini, S.: Towards zero defect manufacturing paradigm: A review of the state-of-the-art methods and open challenges. Computers in Industry **134**, 103548 (2022)

[12] Powell, D., Eleftheriadis, R., Myklebust, O.: Digitally enhanced quality management for zero defect manufacturing. Procedia CIRP **104**, 1351–1354 (2021)

[13] Alves, J., Lima, T.M., Gaspar, P.D.: Is industry 5.0 a human-centred approach? a systematic review. Processes **11**(1), 193 (2023)

[14] Sigfrids, A., Leikas, J., Salo-Pöntinen, H., Koskimies, E.: Human-centricity in ai governance: A systemic approach. Frontiers in artificial intelligence **6**, 976887 (2023)

[15] Ning, A.: A platform of containers based on high availability kubernetes cluster. In: 2022 International Conference on Applied Physics and Computing (ICAPC), pp. 278–281 (2022). IEEE

[16] Wermann, A.G., Wickboldt, J.A.: Ktwin: A serverless kubernetes-based digital twin platform. Computer Networks, 111095 (2025)

[17] Fortuna, C., Mušić, D., Cerar, G., Čampa, A., Kapsalis, P., Mohorčič, M.: On-premise artificial intelligence as a service for small and medium size setups. In: Advances in Engineering and Information Science Toward Smart City and Beyond, pp. 53–73. Springer, ??? (2023)

[18] Rožanec, J.M., Novalija, I., Zajec, P., Kenda, K., Tavakoli Ghinani, H., Suh, S., Veliou, E., Papamartzivanos, D., Giannetsos, T., Menesidou, S.A., et al.: Human-centric artificial intelligence architecture for industry 5.0 applications. International Journal of Production Research, 1–26 (2022)

[19] Azamfirei, V., Psarommatis, F., Lagrosen, Y.: Application of automation for in-line quality inspection, a zero-defect manufacturing approach. Journal of Manufacturing Systems **67**, 1–22 (2023)

[20] Pacaux-Lemoine, M.-P., Trentesaux, D., Rey, G.Z., Millot, P.: Designing intelligent manufacturing systems through human-machine cooperation principles: A human-centered approach. Computers & Industrial Engineering **111**, 581–595 (2017)

[21] Sun, S.-q.: Quality detection method of phase change energy storage and thermal insulation building materials based on neural network. In: Multimedia Technology and Enhanced Learning: Second EAI International Conference, ICMTEL 2020, Leicester, UK, April 10-11, 2020, Proceedings, Part I 2, pp. 41–53 (2020). Springer

[22] Hammerschmidt, U., Hameury, J., Strnad, R., Turzó-Andras, E., Wu, J.: Critical review of industrial techniques for thermal-conductivity measurements of thermal insulation materials. International journal of thermophysics **36**, 1530–1544 (2015)

[23] Silva, B., Marques, R., Faustino, D., Ilheu, P., Santos, T., Sousa, J., Rocha, A.D.: Enhance the injection molding quality prediction with artificial intelligence to reach zero-defect manufacturing. Processes **11**(1) (2023) https://doi.org/10.3390/pr11010062

[24] Ruediger-Flore, P., Glatt, M., Hussong, M., Aurich, J.C.: Cad-based data augmentation and transfer learning empowers part classification in manufacturing. The International Journal of Advanced Manufacturing Technology **125**(11-12), 5605–5618 (2023)

[25] Villena Toro, J., Wiberg, A., Tarkian, M.: Optical character recognition on engineering drawings to achieve automation in production quality control. Frontiers in Manufacturing Technology **3**, 1154132 (2023)

[26] Shorten, C., Khoshgoftaar, T.M.: A survey on image data augmentation for deep learning. Journal of big data **6**(1), 1–48 (2019)

[27] Mumuni, A., Mumuni, F.: Data augmentation: A comprehensive survey of modern approaches. Array, 100258 (2022)

[28] Kisantal, M., Wojna, Z., Murawski, J., Naruniec, J., Cho, K.: Augmentation for small object detection (2019)

[29] Konovalenko, I., Maruschak, P., Brezinová, J., Prentkovskis, O., Brezina, J.: Research of u-net-based cnn architectures for metal surface defect detection. Machines **10**(5), 327 (2022)

[30] Guan, S., Khan, A.A., Sikdar, S., Chitnis, P.V.: Fully dense unet for 2-d sparse photoacoustic tomography artifact removal. IEEE journal of biomedical and health informatics **24**(2), 568–576 (2019)

[31] Enshaei, N., Ahmad, S., Naderkhani, F.: Automated detection of textured-surface defects using unet-based semantic segmentation network. In: 2020 Ieee International Conference on Prognostics and Health Management (icphm), pp. 1–5 (2020). IEEE

[32] Rong-qiang, L., Ming-hui, L., Jia-chen, S., Yi-bin, L.: Fabric defect detection method based on improved u-net. In: Journal of Physics: Conference Series, vol. 1948, p. 012160 (2021). IOP Publishing

[33] Vasquez, J., Furuhata, T., Shimada, K.: Image-enhanced u-net: optimizing defect

detection in window frames for construction quality inspection. Buildings **14**(1), 3 (2023)

[34] Kalnæs, S.E., Jelle, B.P.: Vacuum insulation panel products: A state-of-the-art review and future research pathways. Applied Energy **116**, 355–375 (2014)

[35] Brunner, S., Stahl, T., Wakili, K.G.: An example of deteriorated vacuum insulation panels in a building façade. Energy and Buildings **54**, 278–282 (2012)

[36] Chaudhary, V., Buttar, P.K., Sachan, M.K.: Satellite imagery analysis for road segmentation using u-net architecture. The Journal of Supercomputing **78**(10), 12710–12725 (2022)

[37] Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18, pp. 234–241 (2015). Springer

[38] Wang, H., Miao, F.: Building extraction from remote sensing images using deep residual u-net. European Journal of Remote Sensing **55**(1), 71–85 (2022)

[39] Fletcher, S., Islam, M.Z., et al.: Comparing sets of patterns with the jaccard index. Australasian Journal of Information Systems **22** (2018)

[40] Vieira, S.M., Kaymak, U., Sousa, J.M.: Cohen's kappa coefficient as a performance measure for feature selection. In: International Conference on Fuzzy Systems, pp. 1–8 (2010). IEEE

[41] Chicco, D., Jurman, G.: The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation. BMC genomics **21**, 1–13 (2020)

[42] Pan, Z., Yang, J., Wang, X.-e., Wang, F., Azim, I., Wang, C.: Image-based surface scratch detection on architectural glass panels using deep learning approach. Construction and Building Materials **282**, 122717 (2021) https://doi.org/10.1016/j.conbuildmat.2021.122717

[43] He, K., Gkioxari, G., Dollár, P., Girshick, R.B.: Mask R-CNN. CoRR **abs/1703.06870** (2017) 1703.06870

[44] Beskopylny, A.N., Shcherban', E.M., Stel'makh, S.A., Mailyan, L.R., Meskhi, B., Razveeva, I., Kozhakin, A., El'shaeva, D., Beskopylny, N., Onore, G.: Discovery and classification of defects on facing brick specimens using a convolutional neural network. Applied Sciences **13**(9) (2023) https://doi.org/10.3390/app13095413

[45] Kumar, R., Singh, R.K., Dwivedi, Y.K.: Application of industry 4.0 technologies in smes for ethical and sustainable operations: Analysis of challenges. Journal of cleaner production **275**, 124063 (2020)

[46] Hankel, M., Rexroth, B.: The reference architectural model industrie 4.0 (rami 4.0). Zvei **2**(2), 4–9 (2015)

[47] OpenFog Consortium Architecture Working Group, A., *et al.*: Openfog reference architecture for fog computing. OPFRA001 **20817**, 162 (2017)

[48] Lin, S.-W., Miller, B., Durand, J., Joshi, R., Didier, P., Chigani, A., Torenbeek, R., Duggal, D., Martin, R., Bleakley, G., et al.: Industrial internet reference architecture. Industrial Internet Consortium (IIC), Tech. Rep (2015)

Capítulo 4

Discusión

4.1 Los 3 pilares de la Industria 5.0: Tecnología, humano y sostenibilidad

Como se ha comentado anteriormente, la Industria 5.0 se define como un nuevo paradigma o modelo de industria con el foco en el humano y en la reducción de residuos. En algunos casos, se introduce como evolución de la Industria 4.0, sobre la cual se prioriza el humano y la sostenibilidad. En cualquiera de ambos casos, es cierto que mantiene conceptos de la Industria 4.0 como la calidad, los tiempos cortos de I+D, la automatización o la digitalización, los cuales se complementan con nuevas tecnologías, una centricidad humana e ideas relacionadas con la sostenibilidad.

Se puede matizar esta idea de sostenibilidad. En principio, el concepto de Industria 5.0 estaba más relacionado con la reducción de residuos, pero esto ha derivado en la reducción de consumo y la eficiencia energética, la productividad e incluso la trazabilidad de materiales para su reciclado, reúso u otras actividades relacionadas con la economía circular.

También es matizable la idea de tecnología, puesto que la digitalización y la colaboración humano máquina ya se consideraba en modelos de industria anteriores. En este, con el foco en la resiliencia y el humano, y la evolución rápida de la inteligencia artificial, la idea es que las tecnologías den un soporte mayor al trabajador y éste esté continuamente informado, pueda colaborar con la tecnología y entender sus acciones, e incluso pueda tomar el control cuando sea necesario.

Alves et al., 2023 presentan una imagen del concepto de Industria 5.0 a modo de representación esquemática de una factoría inteligente en la era de la Industria 5.0. La imagen muestra la mano de un operador 5.0, que representa la centricidad humana y la resiliencia, y sujeta una fábrica inteligente rodeada por un arco que representa la sostenibilidad. Esta imagen es ilustrativa del concepto y plasma la idea que detalla esta tesis: la idea de una industria inteligente, basada en tecnologías y máquinas innovadoras y colaborativas, soportada por trabajadores capacitados y resilientes, apoyada por objetivos de competitividad, sostenibilidad, escalabilidad y calidad.

4.2 Síntesis e interpretación de los resultados

Esta sección trata de resumir los principales resultados obtenidos en los artículos incluidos en el compendio. A su vez, enlaza estos resultados con los métodos aplicados, detallados en el capítulo 2.

4.2.1 Resultados relacionados con las líneas piloto

El artículo relacionado con las líneas piloto para la Industria 5.0 presenta un extenso análisis del estado del arte de las líneas para validación y formación. El resumen y análisis del estado del arte se ha realizado siguiendo el método presentado en 2.2. Este análisis es de por sí un interesante resultado, pues permite a la comunidad investigadora disponer de un punto de partida a la hora de iniciar investigaciones relacionadas con las líneas de automática y robótica.

Además de eso, quizás el resultado principal es el mapeo entre las varias docenas de artículos y actividades relevantes y los conceptos relacionados con la Industria 5.0. Así se han descubierto líneas piloto totalmente enfocadas a la formación y capacitación del humano, y otras mucho más centradas en la tecnología. También ciertos casos en los cuales las líneas piloto se han utilizado para tratar de inferir potenciales impactos ambientales, consumos energéticos y características de circularidad de sistemas de producción. En este mapeo también se ha analizado el enlace con tecnologías relacionadas con la inteligencia artificial y con la automática y robótica.

Más que un resultado, una contribución a valorar, es la descripción de la línea piloto GAMHE, presentando sus componentes y como interoperan entre ellos. También los ejemplos de artículos y actividades que se han realizado en la línea piloto y como estos están relacionados con las ideas de Industria 5.0. De hecho, el apartado de ejemplos tuvo que ser extendido a petición de los revisores del artículo, que solicitaron más casos y actividades realizados en la línea piloto y cómo estás estaban relacionadas con la tecnología, la sostenibilidad y especialmente con la centricidad humana.

También por una petición del equipo de revisión se incluyó información sobre sistemas heredados y su integración en entornos de robótica colaborativa. Una de las características de la línea piloto presentada en el artículo es que cuenta con máquinas de control numérico de más de una veintena de años, que no estaban pensadas para interoperar con tecnologías modernas. El enfoque, detallado en el artículo, de integración de sistemas heredados mediante puentes MQTT puede servir de inspiración para la elaboración de otras líneas piloto similares.

La parte más crítica, en lo que al artículo sobre líneas piloto se refiere, está relacionada con dos aspectos. El primero es el solapamiento del concepto línea piloto con otros conceptos similares. Es cierto que, si bien, teóricamente hay cierta distinción, en la práctica se diluyen las diferencias entre un laboratorio, un centro de competencia o incluso con algunos ejemplos de salas de exposición. La segunda parte crítica es la escalabilidad de los resultados, sobre todo en los puntos relacionados con la sostenibilidad. El artículo trata de detallar este punto en la sección de discusión, dado que es un tema recurrente en la literatura.

4.2.2 Resultados relacionados con el metamarco de seguridad

En lo relativo al metamarco de seguridad y privacidad, como se ha mencionado anteriormente, el objetivo principal estaba alineado con la idea de que las PYMEs manufactureras pudieran personalizar o construir plataformas de software interoperables basadas en componentes, y que estas tuvieran un soporte de seguridad y privacidad. Para ello, se ha llevado a cabo un recorrido por diferentes marcos para definir un metamarco fundamental que permita identificar y analizar cuestiones críticas de resiliencia en cuanto a seguridad y privacidad y definir e implementar contramedidas.

Este objetivo se combina con otros secundarios, siempre encaminados a la adopción de tecnologías innovadoras por parte de las PYMEs industriales, y siempre desde el punto de vista de la seguridad y la privacidad. Uno de ellos es reducir la incertidumbre a la hora de abordar las tareas de diseño o creación de una plataforma de despliegue de servicios. No es raro que el ecosistema TIC de las PYMEs industriales integre desarrollos internos, soluciones de código abierto, desarrollos personalizados de terceros y/o componentes externos. Las soluciones basadas en contenedores ayudan significativamente en este punto. En ocasiones se desea ir más allá del aislamiento de componentes hacia implementaciones automatizadas, escalabilidad u observabilidad y seguridad. En estos casos, se lleva a cabo el diseño de una plataforma, por ejemplo, basada en Kubernetes. Un metamarco como el sugerido añade la posibilidad de habilitar la privacidad y la seguridad en el diseño e implementación de la plataforma desde fases iniciales.

Otro objetivo secundario del enfoque presentado es reducir la complejidad que los marcos y estándares de seguridad actuales pueden suponer para el personal técnico de las PYMEs. Los marcos y normas mencionados en este documento son complejos y extremadamente extensos. La mayoría de ellos están diseñados para que los expertos de la organización aborden las necesidades de seguridad y privacidad y garanticen la continuidad del negocio después de un incidente. Algunos de estos incluyen, por ejemplo, más de mil contramedidas, o cientos de riesgos potenciales. En los departamentos técnicos de las PYMEs, a menudo no tienen este perfil de experto y se centran más la integración y uso de aplicaciones. Por lo tanto, proporcionarles un marco práctico y simplificado que les permita comprender las implicaciones de la ciberseguridad y descubrir un conjunto básico de contramedidas, basadas en objetivos, requisitos y riesgos, era parte de los intereses del artículo relacionado con el metamarco de seguridad.

Se considera que al haber partido de las necesidades y requerimientos de cuatro pilotos de diferentes sectores industriales, y el hecho de que se haya aplicado al diseño de una plataforma adaptada a las necesidades de dichos pilotos industriales, confirma su aplicabilidad. La plataforma original del proyecto KITT4SME y sus características de seguridad también fueron la base para las convocatorias abiertas de financiación en cascada en las que varios elementos de la plataforma fueron utilizados por terceros. En general, el marco ha sido diseñado para satisfacer cualquier necesidad de desarrollar una plataforma y middleware. El enfoque híbrido, basado en objetivos y riesgos, se puede aplicar a diversos casos en los que el software desarrollado tiene que integrarse con otros componentes.

También en un contexto más amplio, este enfoque puede servir para tener una comprensión

fundamental de la seguridad y la privacidad, sin tener que entrar en detalles en marcos más complejos. Al mismo tiempo, el marco no es demasiado restrictivo y puede ser extendido en caso de que la organización considere seguir una metodología más compleja en el futuro.

Entrando en detalles, los resultados más relevantes se pueden resumir en los siguientes tres puntos:

- **Metamarco para la Seguridad y la Privacidad**: El marco híbrido detalla seis pasos para identificar los riesgos y definir las contramedidas, y permite una mejor visión de ambos, empezando no solo por los objetivos de desarrollo sino también por los requisitos. El marco ha proporcionado una guía no extremadamente estricta, pero fundamental, para facilitar la comprensión de los posibles problemas de seguridad que se deben abordar al implementar plataformas en las que coexisten soluciones basadas en componentes.
- **Caso de estudio y ejemplo demostrativo en la plataforma KITT4SME**: La definición de un marco de referencia es más significativa cuando su uso se ha demostrado en un caso de estudio. También cuando los ejemplos proporcionados se pueden utilizar como base para ilustrar la replicación del enfoque. La experimentación durante el proyecto KITT4SME ha permitido conocer las ventajas y limitaciones del enfoque, simplificar complejidades y, sobre todo, mostrar aspectos particulares de cómo se puede abordar cada paso. Además del análisis, es interesante observar las decisiones finales y las contramedidas técnicas a las que se ha llegado. El uso dentro del proyecto KITT4SME ha sido de gran importancia porque se centra en plataformas de software para que las pequeñas y medianas industrias adopten soluciones de inteligencia artificial.
- **Taxonomías adaptadas:** Como se ha comentado, los marcos analizados son complejos y orientados a definir contramedidas a nivel organizacional. Hay casos en los que no se desea un análisis completo de seguridad y privacidad a nivel de continuidad de negocio de la organización, sino que el objetivo es simplemente conocer las necesidades de seguridad fundamentales para la implementación de una aplicación en un entorno industrial. La taxonomía adaptada que se ha presentado en la sección 2.2.4 se centra en aspectos específicos relacionados con el desarrollo de plataformas para la industria y permite tener una taxonomía mínima común a los marcos más importantes. Esto también permite que, si en algún momento se dispone de más recursos o tiempo, se pueda avanzar más hacia la plena aplicación de un marco o incluso hacia la certificación.

La aplicación del metamarco en la plataforma del proyecto KITT4SME incluyó un análisis de riesgos para cada grupo de datos críticos. Este análisis de riesgos estaba alineado con la taxonomía adaptada presentada en la sección 2.2.4. Un ejemplo de este análisis para uno de los grupos de datos se presenta en la tabla 4.1, en ella se pueden observa como la columna "Amenazas" está alineada con la taxonomía de riesgos definida en 2.2.4. El análisis de riesgos de todos y cada uno de los grupos de datos del proyecto derivó en las siguientes recomendaciones de privacidad:

- Privacidad y seguridad por defecto en el núcleo de la aplicación y no como un "extra";
- Simplicidad para el usuario y transparencia;

Tabla 4.1: Análisis de riesgos de los datos de control de calidad

Amenazas	**Probabilidad**	**Gravedad**	**Riesgo**
Incapacidad para mantener la responsabilidad	Posible	Insignificante	Bajo
Gestión inadecuada de privilegios	Posible	Serio	Medio
Acceso no autorizado	Posible	Catastrófica	Alto
Pérdida de integridad	Posible	Menor	Bajo
Interrupción de la funcionalidad	Posible	Menor	Bajo
Reducción del rendimiento	Probable	Insignificante	Bajo
Pérdida de información / corrupción o compromiso del sistema	Posible	Catastrófica	Alto
Vulnerabilidades no mitigadas	Posible	Serio	Medio
Prácticas inadecuadas de integración de módulos	Posible	Menor	Bajo
Incapacidad para detectar, analizar y responder a incidentes	Posible	Menor	Bajo

- Privacidad a lo largo del ciclo de vida de la solución;

En cuanto a las recomendaciones de seguridad, los resultados incluyeron contramedidas en tres grupos principales de controles. Estos grupos abarcan los riesgos fundamentales de los activos de los casos de uso del proyecto:

- **Gestión de identidades y control de acceso**: Tanto la gestión de identidades como el control de accesos tienen por objetivo garantizar que los usuarios, herramientas o procesos que acceden a datos, aplicaciones o servicios cuenten con los niveles de autorización necesarios. Estos controles y contramedidas abarcan el espectro de la autenticación, autorización e identificación de usuarios y procesos.

  Respecto a los riesgos detectados en el proyecto, se consideró que implementar mecanismos de gestión de identidad y control de acceso permitía mitigar varios riesgos de forma principal, e incluso otros de manera secundaria.

  Por otro lado, los mecanismos de control de accesos y gestión de identidades ayudan en la trazabilidad, identificando a los usuarios o procesos que realizan las acciones. También pueden servir para mitigar la incapacidad de detectar, analizar y responder a incidencias, por ejemplo, estudiando quién accede a determinados datos o quién intenta hacerlo sin los niveles de autorización necesarios.

- **Cifrado de datos y comunicación segura**: Los objetivos de los controles relacionados con el cifrado de datos son que los datos, ya sea en reposo, en transición o en uso, solo puedan ser legibles por aquellas entidades que realmente tienen acceso a ellos. Por lo tanto, por defecto, tanto los datos como la comunicación entre módulos deben estar cifrados, evitando que los módulos intermedios, que intervengan en la comunicación, visualicen o incluso alteren el contenido o la forma de los datos.

En cuanto a las contramedidas, contar con mecanismos de cifrado o firma permite detectar la pérdida de integridad. A su vez, también garantiza que si se produce una pérdida/corrupción de información o se compromete el sistema, los datos a los que se puede acceder tengan un nivel extra de protección. De alguna manera, también influye en parte de las vulnerabilidades no mitigadas. El cifrado no es ni mucho menos una solución a las vulnerabilidades, pero como se acaba de mencionar, puede ser un nivel extra de protección en caso de que se explote una vulnerabilidad.

- **Trazabilidad de acciones y transacciones de datos**: El objetivo de estos controles es poder realizar un seguimiento de las acciones y determinar qué usuario o módulo fue responsable de una acción específica y cuándo se realizó.

  Este grupo de contramedidas abarca claramente los riesgos relacionados con la incapacidad de mantener la responsabilidad y, en cierto sentido, también ayudan en la detección y análisis de incidentes, lo que puede ser útil para reducir la incapacidad de detectar, analizar y responder a estos incidentes. En un nivel secundario, los registros de acciones se pueden utilizar para detectar vulnerabilidades no mitigadas, interrupciones de la funcionalidad o incluso disminuciones en el rendimiento, y tomar las medidas adecuadas.

4.2.3 Resultados relacionados con la detección de necesidades formativas

La investigación relacionada con el ser humano y las necesidades formativas muestran unos resultados más cuantitativos. En concreto, que el método (Sección 2.2.2) utilizado para intentar mapear la información obtenida del usuario y las bases de datos ocupacionales demostró mejoras significativas en los dos escenarios evaluados.

Al realizar la validación se partía de dos premisas. La primera basada en el cálculo de la similitud semántica entre los currículos y un set de datos de categorías laborales de CareerBuilder. La segunda basada en entidades O*NET pero sin tener en cuenta ningún tipo de ponderación de cada habilidad en el puesto de trabajo y sin utilizar transformadores para evaluar la similitud semántica.

El primero de los escenarios evaluados trataba de analizar los datos de un currículo, o de una interacción entre el agente conversacional y el trabajador, con una ocupación laboral según las características de la base de datos O*NET. Este escenario está diseñado para que empresas puedan clasificar automáticamente currículums y determinar qué tipo de trabajos son más adecuados para cada candidato o trabajador. En este escenario se obtuvieron mejoras de un $+11.76\,\%$ con respecto a la referencia uno y un $+26.67\,\%$ para la detección de las coincidencias entre la ocupación laboral y la información proporcionada por el trabajador en la referencia dos.

El segundo escenario tenía como objetivo identificar que puestos de trabajo eran la mejor coincidencia para el candidato, partiendo de un currículum y una lista de descripciones de trabajo. Con ello se pretendía ayudar a solicitantes de empleo o trabajadores con interés en capacitarse a encontrar el trabajo más adecuado en una lista de ofertas o descripciones de puestos de trabajo disponibles. En este caso se obtuvieron mejoras que van desde el $+9.74\,\%$

en la referencia uno, hasta casos en los que se duplica el desempeño de la referencia dos en las predicciones.

Es cierto que estos casos dependían de una evaluación manual y que el número de evaluadores era limitado. También que los sets de datos y las referencias eran muy orientados a actividades relacionadas con las tecnologías software y no había excesivos ejemplos de puestos en planta de producción. Pero los resultados obtenidos demostraron la utilidad del enfoque y fueron importantes en investigaciones subsecuentes. Por ejemplo, facilitaron los desarrollos en los proyectos STAR y DENIM y se publicaron sendos artículos (Alonso, Dessí, Meloni, Murgia et al., 2024; Alonso, Dessí, Meloni y Recupero, 2024; Alonso et al., 2023; Fareri et al., 2023) que partían de ideas incluidas es este caso de estudio.

La principal contribución del enfoque era que, en un momento en el que la inteligencia artificial tendía a las alucinaciones y resultados poco deterministas, los modelos de lenguaje podían complementarse con información valida y globalmente aceptada como son las bases de datos ocupacionales.

El caso de estudio presentado se orienta principalmente al mapeo de perfiles y puestos de trabajo. Si bien en el artículo la búsqueda de empleo y la búsqueda de actividades asociadas a un perfil tienen más peso, el concepto se diseñó con una aplicación directa a entornos y tecnologías cambiantes como es el caso de la Industria 5.0. En concreto, gran parte de la utilidad está relacionada con la recapacitación de trabajadores. Estableciendo de esta forma las bases que permitan detectar las necesidades formativas para adecuarse a nuevos retos en la industria, derivados principalmente de la digitalización o de la irrupción de nuevas tecnologías.

4.2.4 Resultados relacionados con el control de calidad basado en aumento de datos

Otros de los casos de estudio que muestran resultados cuantitativos interesantes son los dos relacionados con el uso de técnicas de aumento de datos para el control de calidad. No tener suficientes datos para diseñar y entrenar un algoritmo de inteligencia artificial es un problema común en el sector manufacturero. El problema es aún más grave en las PYMEs, que a menudo no tienen la capacidad, el conocimiento o el tiempo para generar los datos necesarios para entrenar algoritmos o para entregarlos a los proveedores de tecnología para diseñar soluciones de inteligencia artificial. Es precisamente ahí es donde entra en juego la generación de datos sintéticos. En concreto, se identificaron en reuniones con empresas involucradas en el proyecto KITT4SME tres casos en las que la generación de datos sintéticos puede ser beneficiosa.

El primer caso se da cuando no hay datos equilibrados para entrenar los algoritmos de inteligencia artificial. Por ejemplo, es común en los sistemas de clasificación industrial tener, o poder generar, datos para piezas correctas. Sin embargo, esto no es tan fácil para piezas defectuosas. Para las piezas correctas, generalmente se dispone de referencias o especificaciones. Para las defectuosas, en el mejor de los casos, se pueden utilizar algunos ejemplos de piezas similares, o es necesario hacer trabajo manual para fabricar la pieza defectuosa. Es cierto que no todos los algoritmos de clasificación requieren datos equilibrados, y los modelos de clasificación *one-shot* o *zero-shot* son cada vez más comunes, pero todavía hay múltiples casos en los que los datos equilibrados equivalen a mejores resultados.

El segundo caso está relacionado con la comparación de modelos, ya sean propios o adquiridos a terceros. En este caso disponer de una gran colección de datos, imágenes en el caso de estudio presentado, puede permitir la evaluación de los modelos y la selección de los más adecuados para el objetivo final. Además, puede ser útil, no solo para comparar diferentes modelos, sino también los mismos modelos con diferentes parámetros.

El último caso está relacionado con la reconfiguración, puesta a punto y ajustes de los sistemas de control de calidad una vez que se detectan nuevos defectos. Con el tiempo irán apareciendo nuevos defectos o irregularidades con los que el sistema de inteligencia artificial no fue entrenado. Es probable que algunos de estos errores ocurran solo en casos tan extremadamente raros que sea muy difícil tener suficiente información para volver a entrenar o adaptar el modelo. Aquí la generación de datos sintéticos también es beneficiosa. Por ejemplo, en el caso particular de este trabajo, un defecto que se haya originado en una pieza similar puede servir para ajustar el sistema de clasificación de la pieza que se está evaluando.

Estos tres casos no solo tienen la derivada de la calidad, también tienen implicaciones significativas en temas de generación de residuos y sostenibilidad. Todas las piezas defectuosas generadas a propósito para entrenar los modelos son básicamente residuos, no tienen valor productivo y requieren de unos esfuerzos y coste en personal, maquinaria y energía. Por tanto, si unimos la detección temprana de piezas defectuosas antes de llegar al final de la línea de producción, la posibilidad de reprocesado de algunas de las piezas y que se pueden entrenar los modelos y obtener referencias sin necesidad de fabricar las piezas, logramos un avance también en reducción de residuos.

Uno de los artículos incluidos en esta tesis muestra uno de los tres casos de estudio que se realizaron en colaboración con una empresa de fabricación de maquinaria de control de calidad. Es el caso más relevante y complejo, debido a las características únicas de la pieza: un tapón de depósito de aceite para automoción. El objetivo era probar, en colaboración con el fabricante de máquinas clasificadoras, si las técnicas de aumento de datos y generación de imágenes sintéticas podrían ser una alternativa sostenible y económica a la fabricación manual de referencias defectuosas.

En este caso, el aumento de datos se basa en los métodos de mezcla de imágenes detallados en la sección 2.2.3. La combinación de píxeles para fusionar dos imágenes distintas en una nueva permite la consecución de diversos efectos, lo que las hace aplicables a diferentes casos de uso. Los efectos incluyen la integración de la región defectuosa de interés en la imagen original mientras se mantiene, oscurece o aclara el área. Por ejemplo, en este contexto se utilizó la sustracción de la representación del defecto de la imagen de referencia para representar grietas o ranuras. Como segundo método, se utilizó la generación automática de manchas, permitiendo la creación de imperfecciones en diferentes cantidades y tamaños en la imagen de referencia.

El gráfico 4.1 muestra los resultados de las pruebas con los tapones del depósito de aceite. En concreto, se muestran los falsos negativos, que si bien no son tan críticos en cuanto a seguridad, sí tienen un impacto en costes y generación de residuos. Se comparan el modelo de referencia entrenado solo con imágenes reales, el modelo CF1 entrenado con imágenes reales de piezas sin defectos y con imágenes sintéticas de piezas defectuosas, y el modelo CF2, en las

que las imágenes defectuosas eran una mezcla de piezas reales y piezas defectuosas sintéticas. Como se puede observar, se redujo el número de falsos positivos al entrenar el modelo con imágenes sintéticas.

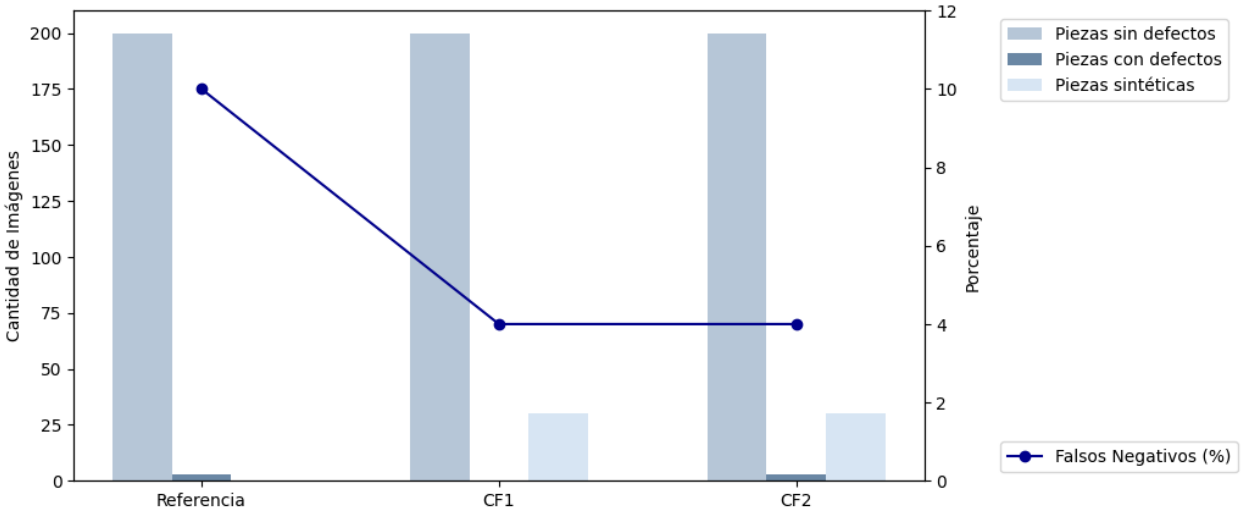


Figura 4.1: Comparación de Modelos: Composición de Datos y Resultados

Es importante indicar que el volumen de estas imágenes está alineado con los requisitos de la solución que la empresa donde se realizó el piloto utiliza para probar los modelos. También es importante mencionar que el conjunto de validación constaba de 1.888 imágenes, 968 de las cuales eran piezas correctas y 920 defectuosas.

En general, los resultados demostraron mejoras del 60 % cuando el sistema de clasificación comercial que utiliza la empresa era entrenado con imágenes más allá de las piezas defectuosas disponibles. La conclusión fue que era preferible un entrenamiento que incrementara el volumen de datos con piezas sintéticas al uso exclusivo de piezas defectuosas fabricadas a mano. El caso de estudio presentado para tapones de depósito de aceite fue replicado con otras piezas, principalmente arandelas, obteniendo resultados similares.

En un contexto ligeramente diferente, y que se detalla en los experimentos incluidos en el artículo 3.5, se realizaron pruebas de generación de datos sintéticos para tratar de mejorar el entrenamiento de modelos de clasificación de paneles termoaislantes para el sector de la construcción. Este casos de estudió se valido en el entorno de robótica colaborativa de la línea piloto GAMHE 5.0. Los resultados fueron similares a lo comentado en los párrafos anteriores, consiguiendo una detección temprana de paneles defectuosos y una reducción del número de paneles con errores al final de la línea de producción.

En el caso de estudio de los paneles termoaislantes se procesaron 1.000 paneles en una línea piloto que incluía brazos robóticos y un sistema de inspección. Este sistema de inspección había sido entrenado con imágenes de piezas reales (Modelo B1), mientras que en el otro caso se utilizó un grupo de datos que empleaban las técnicas de aumento de datos detalladas en 2.2.3 (Modelo B2).

Como se puede observar en el gráfico 4.2, los resultados del modelo entrenado con datos

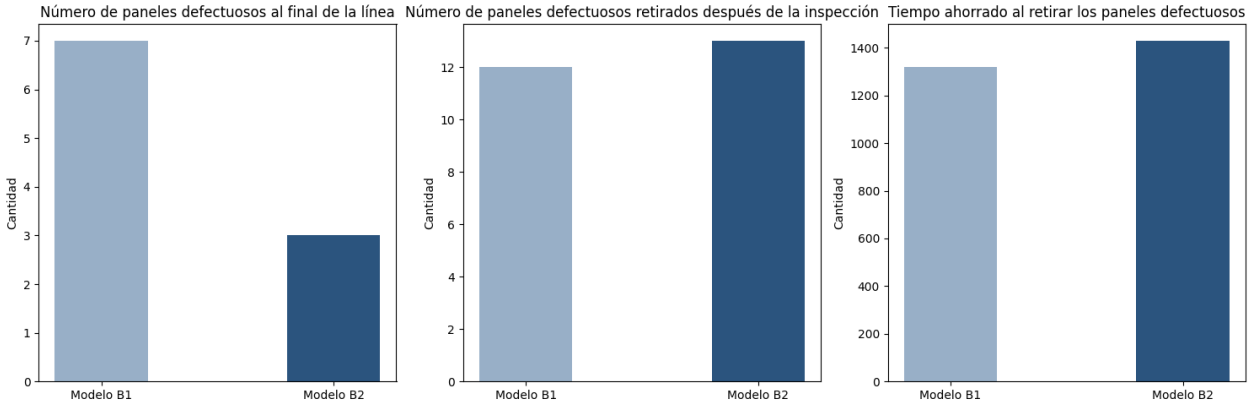


Figura 4.2: Comparación de Resultados: Paneles Termoaislantes

sintéticos fueron mejores en los tres indicadores que se evaluaron. En concreto, el número de paneles defectuosos al final de la línea se redujo de 7 a 3. Por su parte, el número de paneles defectuosos detectados y retirados después de la inspección pasó de 12 en el modelo entrenado con el grupo de datos original a 13 en el modelo con aumento de datos. La capacidad temprana de detectar y retirar paneles permite ahorrar costes de operación y procesamiento, siendo por lo tanto este valor positivo. Por último, el tiempo ahorrado al retirar los paneles defectuosos también se incrementó con el modelo entrenado con datos sintéticos, pasando de aproximadamente 1319 segundos a 1429 segundos. Este aumento en el ahorro del tiempo de procesado y del uso de las máquinas impacta también en la eficiencia del proceso.

Al igual que en el caso del control de calidad de piezas para el sector automoción, los resultados sugieren que el aumento de datos no solo no empeora la clasificación, sino que puede ser un complemento interesante en entornos donde es costoso disponer de datos más balanceados.

4.2.5 Limitaciones

Una de las principales limitaciones detectadas en el artículo que analiza las líneas piloto está relacionada con la falta de información cuantitativa para entender si es posible escalar los indicadores de sostenibilidad y ciclo de vida desde líneas piloto a entornos productivos. La información encontrada al respecto era específica de casos de uso concretos en sectores particulares y no era generalizable.

Además, varios de los artículos que se analizaron, sobre cómo los análisis de ciclo de vida se escalan desde proyectos piloto a producción completa, mencionaban que su validez era limitada "hasta cierto punto" debido al escalado no lineal de algunos factores (von Drachenfels et al., 2021), o simplemente concluían que "no existe un enfoque estándar para el escalado de análisis del ciclo de vida ex ante" (Tsoy et al., 2020). En algunos casos, incluso sugerían directamente que "las escalas de investigación, como las de laboratorio y piloto, no son adecuadas para evaluaciones significativas del impacto ambiental" (Muñoz et al., 2019). Por esta razón, se decidió citar a von Drachenfels et al., quienes señalan esta dificultad en la escalabilidad sin proporcionar información cuantitativa.

En lo relativo a las tecnologías y las plataformas interoperables, también se han identificado

limitaciones y restricciones que pueden requerir algunas consideraciones adicionales en la aplicación del marco a diferentes casos y configuraciones industriales. Por ejemplo, el marco se centra en el desarrollo de middleware o plataformas en las que los componentes de la inteligencia artificial puedan interoperar, pero no profundiza en la seguridad específica de estos componentes. En particular, la seguridad de los componentes y modelos de inteligencia artificial es un tema candente que puede afectar a la seguridad de la plataforma. De hecho, las plataformas basadas en Docker y Kubernetes ofrecen aislamiento de componentes, pero para evitar la propagación de posibles problemas, puede ser necesario un mayor control y un análisis específico de los riesgos potenciales y las contramedidas para los componentes de inteligencia artificial.

La adaptación al riesgo sugerida y el marco propuesto pueden ser útiles para que las PYMEs industriales comprendan e implementen controles de seguridad y faciliten la replicación en desarrollos futuros. Sin embargo, otra limitación potencial se relaciona con el impacto del enfoque en el personal técnico de las PYMEs. Se ha mencionado que el metamarco no requiere grandes departamentos ni conocimientos especializados, pero aun así, exige personal técnico con una formación mínima, ya que define una base sobre la cual el personal debe tomar decisiones técnicas, como la selección de soluciones o enfoques a implementar para cada contramedida. Esto también tiene cierta relación con la limitación mencionada en el caso de estudio relacionada con la sobrecarga derivada de agregar al software capas adicionales que pueden ayudar a abordar algunos problemas de seguridad, pero también añaden cierta complejidad.

Para concluir esta discusión relacionada con el tema de seguridad, también es justo mencionar otra limitación descubierta en los casos de estudio. El metamarco no entra en detalles sobre la infraestructura de implementación, ya que cada caso es bastante específico. Sin embargo, al igual que en el caso de los componentes de inteligencia artificial, la infraestructura de implementación puede tener un impacto, ya que se pueden necesitar más análisis para definir contramedidas adicionales si la implementación va a ser en la nube o en un entorno de riesgo.

En el caso de estudio relativo al humano, se ha comentado con antelación que gran parte de la información disponible, especialmente en cuanto a currículos y puestos de trabajo, está centrada en las nuevas tecnologías. Esto no implica que el enfoque no funcione en otros sectores, pero sí que, dado el volumen de datos, es más sencillo realizar mapeos adecuados cuando se dispone de más información.

También es interesante comentar que, debido al detalle de la información disponible en O*NET, se ha utilizado principalmente esta base de datos para realizar el análisis. O*NET es una base de datos orientada al empleo en Estados Unidos y puede haber diferencias significativas en cuanto a características de algunas ocupaciones con respecto a otras regiones y países. Esto puede generar un sesgo en ciertas ocupaciones, habilidades o tareas.

Otra limitación es que el sistema prioriza el mapeo semántico y no tiene en cuenta otros aspectos que pueden ser clave, como la experiencia laboral y otra información contextual. Añadir esta información podría enriquecer el sistema en futuras investigaciones.

En términos más concretos, es interesante mencionar que algunos de los umbrales utilizados para la similitud semántica y la elegibilidad fueron fijados empíricamente. Esto implica la

necesidad de ajustarlos para casos en sectores específicos, por lo que se podría considerar el estudio de valores dinámicos o incluso técnicas de aprendizaje automático para establecer estos valores.

Por último, en el caso del control de calidad de los tapones del depósito de aceite, quizás la mayor limitación está relacionada con una validación muy centrada en los modelos comerciales de control de calidad que se utilizan en la empresa de fabricación de maquinaria donde se realizó el experimento. Es cierto que en el caso de estudio de los paneles termo-aislantes, el algoritmo ad hoc de segmentación, basado en U-net, junto con el de clasificación, basado en YOLOv5s, dieron resultados similares, pero el avance en inteligencia artificial hace pensar que técnicas *one-shot* o *zero-shot* podrían ser la solución en algunos casos para la clasificación sin entrenamiento con piezas defectuosas.

También en lo que respecta al sistema de aumento de datos, la mayoría de las operaciones de generación de datos sintéticos han estado relacionadas con la manipulación de imágenes. El avance en Redes Generativas Adversarias y Modelos de Difusión puede ser una fuente de piezas defectuosas aún más realistas.

4.3 Evaluación de las hipótesis

Se parte de cuatro hipótesis alineadas con los artículos presentados en esta tesis. A continuación, se evaluarán una por una.

1. **Las líneas piloto mejoran significativamente la transición de la investigación a la producción a escala industrial y facilitan una industria más adaptable, centrada en el ser humano y sostenible.**

   Para evaluar esta hipótesis se diseñó un estudio que analiza el estado actual de las líneas piloto y su relación con las industrias 4.0 y 5.0. Este estudio parte de la falta de una definición clara y de la ausencia de un catálogo a nivel europeo de actividades de las líneas piloto. También tiene su origen en la promoción y puesta en valor de las líneas piloto para la industria llevada a cabo por la Comisión Europea dentro de sus programas Horizon 2020 y Horizon Europe. Este estudio, de revisión y en cierto modo comparativo, analizó mediante una metodología de análisis del estado del arte, más de una veintena de artículos relevantes, cubriendo diversos sectores y enfoques.

   Los resultados más interesantes demuestran cierto solapamiento entre conceptos, especialmente entre líneas piloto, laboratorios y centros de competencia. También un foco significativo dentro del área de los semiconductores, y un cada vez mayor énfasis en la robótica colaborativa, la inteligencia artificial, la automatización y la interoperabilidad entre componentes. Los aspectos humanos y de sostenibilidad son más habituales en los artículos de los últimos años, mostrando por tanto la alineación de las líneas piloto con las ideas de Industria 5.0. Por último, el estudio detecta ciertos problemas en el escalado de métricas e indicadores de sostenibilidad entre los estudios a escala de piloto y a nivel de línea de producción.

   En consecuencia, se considera que los datos respaldan la hipótesis. En concreto, no solo

se muestra el interés constante en los últimos años en el concepto de líneas piloto, sino que se constata su especial valor como herramienta de prueba y validación antes de invertir. Esto último se confirma mediante ejemplos que van desde la producción de lotes pequeños de productos muy específicos que no justifican la necesidad de una línea completa, hasta ejemplos de líneas para la formación sobre seguridad laboral. También se observa cómo estas líneas se van adaptando a los cambios de paradigma, teniendo en cuenta conceptos de interoperabilidad, robótica colaborativa, centricidad humana o sostenibilidad.

2. **Las pequeñas y medianas empresas manufactureras pueden acelerar su transformación digital adoptando arquitecturas de software basadas en componentes y entendiendo de manera temprana los posibles riesgos de seguridad y privacidad.**

   Para la evaluación de esta segunda hipótesis se realizó un estudio dentro del proyecto KITT4SME. Este proyecto, muy centrado en la transformación digital y la adopción de la inteligencia artificial en las pequeñas y medianas empresas manufactureras, presentó cuatro diferentes pilotos y una colección de más de una decena de activos. Como parte de este proyecto, se pretendía definir una arquitectura y una plataforma que soportaran la interoperabilidad, el despliegue continuo, la recuperación de fallos, la seguridad y la privacidad, para que las empresas industriales interesadas no tuvieran que centrarse en esos aspectos y pudieran concentrarse solo en el despliegue de las soluciones de aprendizaje automático útiles para su producción.

   Los resultados demuestran la necesidad de adaptar una taxonomía de riesgos, tradicionalmente centrada en la continuidad del negocio, hacia un enfoque centrado en plataformas y componentes de software. También evidencian que es posible mantener un enfoque híbrido que tenga en cuenta los riesgos y contramedidas desde una visión basada en objetivos y a su vez basada en riesgos. La aplicación al caso de estudio de la plataforma industrial KITT4SME, además de ser una referencia de la metodología aplicada, detectó la necesidad de implementar una gestión de identidades y acceso, sistemas de trazabilidad y herramientas de cifrado de datos y comunicación segura. Por último, en lo que respecta a la parte más práctica, el resultado fue una versión de la plataforma KITT4SME con soporte integrado para elementos de privacidad y seguridad.

   El hecho de haber aplicado y validado la metodología, utilizando como referencia los activos del proyecto, y definir dentro de la plataforma del proyecto las herramientas básicas de seguridad y privacidad, confirma la hipótesis. Más aún cuando la plataforma está diseñada como una solución flexible para diferentes sectores y casos de uso. La plataforma y las recomendaciones de seguridad fueron también ofrecidas a más de una decena de proyectos que validaron las funcionalidades del proyecto.

3. **Integrar datos estructurados de ocupaciones laborales con modelos basados en aprendizaje profundo para extraer habilidades semánticamente relevantes de textos no estructurados permite el descubrimiento de necesidades formativas y la alineación de las capacidades de los trabajadores con las actividades laborales.**

Para evaluar la tercera hipótesis, se diseñó un experimento que enlazaba modelos de aprendizaje profundo y bases de datos ocupacionales. El experimento es relevante en un momento histórico en el que los modelos de lenguaje todavía tienden a la alucinación; y cuando, disponer de una base de conocimiento considerada evaluada y fiable, permite dotar de cierto determinismo en los resultados. Este experimento, eminentemente práctico, parte de la necesidad de facilitar a los trabajadores o futuros trabajadores información sobre cómo pueden formarse o adaptar su carrera. Además, en un momento temporal de cambios tecnológicos, se pretendía experimentar cómo las empresas podrían descubrir qué perfiles son los más adecuados para una determinada tarea.

La validación de la hipótesis se benefició de pruebas en dos diferentes escenarios. En el primer escenario se proporciona un currículum vitae como entrada. El objetivo es identificar la ocupación O*NET con la mayor alineación con el currículo. En el segundo escenario se parte de un currículum vitae y una lista de ofertas de empleo o actividades. El objetivo es encontrar cuál de las actividades dadas coincide más con las habilidades del candidato presentes en el currículo.

Los resultados demuestran que la unión de las bases de datos ocupacionales y los modelos de aprendizaje profundo basados en transformadores mejoran el mapeo entre los datos no estructurados de los currículos y las descripciones de actividad, permitiendo así una mayor determinismo de los resultados. El resultado fue validado como parte de las herramientas de asistencia al trabajador en el proyecto STAR. Estas herramientas permiten detectar necesidades formativas tanto en habilidades técnicas como interpersonales. También demuestran que, una vez obtenidos los datos de la distancia entre ocupaciones y currículos, es posible diseñar sistemas de recomendación que ofrezcan planes formativos.

4. **Las imágenes sintéticas podrían utilizarse como reemplazo o complemento en entornos donde no hay suficientes piezas defectuosas para entrenar una inteligencia artificial, reduciendo, a su vez, tiempos, materiales y residuos derivados de fabricar las piezas físicas defectuosas.**

   Para validar la última hipótesis se realizaron dos experimentos de generación de imágenes sintéticas: uno en una línea piloto y otro en una empresa dedicada a la fabricación de máquinas para el control de calidad. Ambos estudios parten de la idea de que, para un control de calidad basado en modelos de inteligencia artificial, en ocasiones es necesario entrenar los modelos con datos balanceados de piezas correctas y defectuosas. Para ello se utilizaron varias técnicas de generación sintética de imágenes con el objetivo de generar datos de piezas defectuosas, que por esfuerzo y coste son las más complejas de generar a mano. Estos conjuntos de datos se utilizaron para entrenar modelos de clasificación, que a su vez se emplearon para validar la calidad de las piezas.

   Los resultados demostraron que los modelos entrenados con piezas generadas sintéticamente no solo no se comportaban peor que los modelos con piezas reales, sino que, a falta de ejemplos de piezas defectuosas, entrenar los modelos con piezas sintéticas era preferible a hacerlo con datos reducidos. Estos resultados se basaron tanto en algoritmos de clasificación desarrollados para validar el caso de estudio como en modelos de clasificación comerciales utilizados en las máquinas de la empresa que facilitó los datos

y las pruebas.

En consecuencia, los experimentos y los resultados respaldan la hipótesis, al menos en los casos en los que los datos equilibrados son una necesidad. Estos resultados permiten enfocar el caso de estudio desde un punto de vista no solo técnico, sino también en lo relativo a eficiencia y sostenibilidad. Esto se demuestra en los experimentos en la reducción de tiempos y materiales para generar piezas reales para el entrenamiento de los modelos y también en la detección temprana de piezas defectuosas.

Junto con la hipótesis también se presentaba una idea general que se resumía en: "El avance hacia una industria 5.0 depende significativamente de la comprensión del contexto de colaboración entre tecnología y humano, y del análisis del impacto de esta cooperación en la sostenibilidad ambiental, económica y el bienestar laboral".

En esta tesis se ha tratado de entender este contexto de una forma práctica, entendiendo los diferentes elementos que componen ésta Industria 5.0 y cómo impactan en las pequeñas y medianas empresas. Para ello se ha presentado la metodología y los materiales, se han presentado sendos artículos que cubren los tres principales aspectos de este nuevo paradigma industrial y se han analizado sus fortalezas y limitaciones.

Las líneas piloto han demostrado ser un elemento clave también para la nueva industria, permitiendo no sólo la validación de tecnologías, sino también la formación y capacitación de los trabajadores. Asimismo, en cierto modo, la evaluación de actividades e indicadores de sostenibilidad, si bien, la literatura coincide en que el escalado de estos sigue siendo un punto que requiere más investigación.

Las plataformas interoperables favorecen el despliegue y uso de nuevas tecnologías en entornos industriales, facilitando a los equipos técnicos la composición de soluciones alineadas con sus necesidades. A su vez, contribuyen a que los diseñadores de estas plataformas tengan en cuenta características de seguridad y privacidad desde el momento de su definición, lo que facilita la resiliencia tecnológica y operacional.

El ámbito humano ha sido tratado desde el punto de vista de la búsqueda de relaciones entre las capacidades de los trabajadores y las actividades laborales. Esta investigación tiene importancia en una industria del futuro que parece ser rápidamente cambiante y que deberá adaptarse a una falta de fuerza laboral con conocimientos específicos sobre estas nuevas tecnologías.

Por último, los casos de estudio sobre control de calidad demuestran cómo los tres conceptos (tecnología-humano-sostenibilidad) convergen en un caso de estudio particular de resiliencia operativa en el que, gracias a la aplicación de técnicas de aumento de datos y clasificación basada en inteligencia artificial, se puede llegar a una reducción del esfuerzo del trabajador, un mayor control sobre el sistema de clasificación mediante una detección temprana de piezas defectuosas y una reducción en costes y generación de residuos.

Los artículos presentados en esta tesis confirman que el contexto de la Industria 5.0 es amplio y que intervienen múltiples factores, y que además se sitúa en un momento social y tecnológico cambiante. También confirman que si bien se sigue hablando de Industria 4.0, la idea de complementarla con conceptos de centricidad humana y sostenibilidad está cada vez más

cerca de ser una necesidad.

4.3.1 Aportación al campo

La primera aportación al campo tiene que ver con lo que se acaba de mencionar en la sección anterior. La tesis proporciona una visión amplia de la Industria 5.0, cubriendo sus principales aspectos desde una perspectiva práctica. Las investigaciones específicas sobre una tecnología o un concepto concreto son más habituales en la literatura, por lo que esta visión que abarca los tres aspectos principales de la industria 5.0 y los interrelaciona representa una perspectiva distinta.

El análisis de las líneas piloto es exhaustivo y tiene en cuenta la automática y robótica, la inteligencia artificial y los conceptos de las nuevas industrias. Por tanto, proporciona a la comunidad científica una referencia sobre el estado de las líneas para validación y formación, sus posibles beneficios y desafíos, y sus aplicaciones actuales.

La metodología híbrida de análisis de seguridad y privacidad no solo conceptualiza los pasos necesarios, sino que proporciona un ejemplo de cómo se llevó a cabo el análisis para una plataforma utilizada en un proyecto de investigación orientado a las empresas manufactureras. A su vez, proporciona un marco para que equipos técnicos estén alineados sobre las capacidades de las arquitecturas software y hardware, y podría ser aplicado al diseño de líneas piloto.

Por su parte, los casos de estudio más centrados en el humano, la formación, el desarrollo del talento y la capacitación de los trabajadores han sido un punto de partida para diferentes estudios, tal y como se ha comentado anteriormente, y han contribuido siendo la base sobre la que se han sustentado las aplicaciones de entrevistas virtuales, el análisis de capacidades de los trabajadores y el sistema de recomendación formativa desarrollados en los proyectos STAR y DENIM.

Por último, los casos de estudio sobre aumento de datos han confirmado que el uso de imágenes sintéticas es equivalente o complementario al uso de imágenes reales, y lo ha situado en un contexto de ingeniería de mantenimiento donde la calidad de las piezas tiene un impacto en el producto final y donde la detección temprana reduce costos y residuos.

Capítulo 5

Conclusión

La industria manufacturera, y de forma más amplia, la industria en general, siguen evolucionando y se encuentran en un momento en el que convergen una cuarta revolución en la que la interoperabilidad, los datos y las plataformas de software son elementos esenciales, junto con una quinta revolución basada en la resiliencia, la centricidad humana y la sostenibilidad.

Si a esta situación le añadimos el ritmo y el impacto de la inteligencia artificial y las oportunidades que puede aportar a la industria, el resultado es una industria que, en muchos casos, está teniendo que hacer frente a grandes cambios sin los recursos adecuados y, a menudo, sin los conocimientos necesarios.

Las líneas piloto son un concepto que ha acompañado a la industria desde que se comenzaron a considerar aspectos de eficiencia y calidad. Estas líneas han servido como elemento para la investigación y desarrollo, como facilitador de la validación y como herramienta prueba antes de llevar a cabo una inversión. Además son de utilidad como entorno productivo, cuando no es económicamente rentable dedicar una línea de producción de mayor escala. Este es el caso, por ejemplo, de la fabricación de lotes pequeños de algunos productos. Estas líneas tienen también su sitio en el nuevo paradigma industrial, más si cabe cuando la automática y robótica y las tecnologías de aprendizaje automático se unen con la formación y capacitación de los trabajadores, los estudios de análisis ergonómico y las evaluaciones de carga de trabajo y fatiga, que también se pueden investigar y probar en estas líneas.

Esta tesis incluye elementos que ponen en valor las líneas piloto y las analiza desde los enfoques de la nueva industria, dotando de material de referencia para entender y diseñar líneas piloto alineadas con las necesidades futuras asociadas a los cambios de paradigma industrial.

Las plataformas de software basadas en la orquestación de componentes se pueden utilizar para simplificar los requisitos de interoperabilidad, facilitar la integración y proporcionar mecanismos de resiliencia operativa para respaldar las aplicaciones industriales. Este trabajo también se ha centrado en cómo los usuarios de estas plataformas digitales para la industria e incluso los proveedores de tecnología pueden evaluar sus necesidades de seguridad y privacidad para configurar estos sistemas en consecuencia, al tiempo que se adhieren a los marcos de análisis de riesgos y objetivos de la organización.

La seguridad y la privacidad, junto con la interoperabilidad, son algunos de los elementos clave en el software de la Industria 4.0. En la industria 5.0, con el uso masivo de modelos de inteligencia artificial y las ideas de centricidad humana, cobran aún mayor relevancia.

Por este motivo, se ha presentado un nuevo metamarco. Este metamarco se basa en marcos de seguridad y privacidad, así como en metodologías de análisis de riesgos bien conocidas, combinando conceptos pertenecientes a ambos: las orientadas a objetivos y las puramente basadas en riesgos. Para implementar este metamarco y la metodología asociada, se adaptaron algunos de estos marcos y metodologías, ya que muchos de ellos se centran en aspectos organizativos y no en plataformas de software o middleware.

Se describió en detalle una metodología de seis pasos y se presentó y aplicó un caso de uso para evaluar el metamarco. Este caso de estudio se basa en cómo el equipo de la plataforma del proyecto KITT4SME evaluó las necesidades de seguridad y privacidad de cuatro pilotos para adaptar la plataforma a estas necesidades. Los resultados de la aplicación del metamarco permitieron conocer los principales requisitos de seguridad y privacidad y seleccionar contramedidas que se integraron en la plataforma del proyecto. En el contexto del caso de uso descrito, las contramedidas estaban relacionadas principalmente con i) la gestión de identidad y el control de acceso, ii) la transmisión y el almacenamiento seguro de datos, y iii) la trazabilidad y la observabilidad.

En lo relativo al humano y cómo este se sitúa en la nueva industria, el punto de vista ha estado focalizado en el concepto de que los entornos cambiantes y las nuevas tecnologías van a requerir, por un lado, capacitación y, por otro, comprensión sobre qué capacidades son necesarias para abordar nuevas tareas. Esto se ha abordado no solo desde un punto de vista relacionado con las habilidades técnicas, sino también desde la consideración de elementos como las competencias interpersonales, emocionales y de comunicación (englobadas dentro del término *soft skills*). Para ello se han utilizado técnicas de aprendizaje automático y bases de datos ocupacionales que incluyen información sobre ocupaciones, tareas, tecnologías, actividades y *soft skills* para poder encontrar un mapeo entre descripciones de actividades y capacidades de los trabajadores. Esto se ha complementado con sistemas de recomendación de cursos y herramientas de aprendizaje que puedan capacitar al trabajador y dotarle de la formación necesaria.

En los casos de estudio y los artículos relacionados con la calidad en la producción de piezas convergen de alguna manera los elementos de la Industria 5.0 mencionados en esta tesis. Por un lado, la plataforma presentada anteriormente ha estado presente soportando la ejecución de algunos de los componentes de aumento de datos o visualización de resultados. El humano es responsable de la generación de piezas y está involucrado en el proceso de generación de imágenes, en la colaboración con las máquinas y robots, y en el ajuste de modelos de inteligencia artificial, lo que implica una necesidad de capacitación para la cual las líneas piloto pueden ser útiles. Por último, uno de los objetivos trata de reducir tiempos, esfuerzos, costes y piezas defectuosas fabricadas para el entrenamiento y al final del control de calidad. Teniendo esto un cierto impacto en la generación de residuos y uso energético.

En general, la Industria 5.0 ofrece innumerables oportunidades y las nuevas tecnologías aportan capacidades adicionales. Por lo tanto, es necesario brindar a las PYMEs herramientas

y recomendaciones para que puedan abordar los retos con confianza, conociendo los desafíos y oportunidades a los que se enfrentan. Las líneas piloto, las metodologías y marcos teóricos, como el presentado en esta tesis, junto con las plataformas y soluciones implementadas en los casos de estudio, pueden ayudar en la adopción de estas nuevas tecnologías en pequeñas y medianas empresas manufactureras. Si a esto se añade un enfoque que permita al factor humano tener más capacidades, facilitando su cooperación con la tecnología, proporcionándole explicabilidad e información suficiente para la toma de decisiones, y lo extendemos con un enfoque centrado en la calidad de procesos y productos, ideas de cero defectos y reducción de residuos, eficiencia energética y compromiso medioambiental, el resultado estará alineado con la idea de industria resiliente del futuro que está promoviendo la Unión Europea.

Esta tesis ha partido del objetivo de explorar los elementos clave del paradigma Industria 5.0, mediante el estudio de casos centrados en el factor humano, la tecnología y la sostenibilidad. Para ello ha realizado un recorrido teórico-práctico sobre los elementos fundamentales de la Industria 5.0. Tras una introducción al campo y un detalle de los materiales y métodos, se ha presentado un compendio de artículos que abordan diferentes aspectos de este nuevo modelo industrial. Posteriormente se han discutido los resultados, mencionando las contribuciones pero también las limitaciones.

De estas limitaciones nacen algunos de los pasos siguientes y potenciales ideas de investigación futura. El énfasis en la centricidad humana, las estrategias del humano en el bucle y gemelos digitales con información del estado del operador son algunas de las posibilidades más claras de investigación. Los avances en inteligencia artificial permiten abordar de formas diversas el control de calidad. Nuevas técnicas de aumento de datos, modelos de clasificación que no requieren entrenamiento específico y técnicas de reconstrucción de imágenes son claramente temas que pueden mejorar los procesos y controles de calidad en las empresas industriales.

Relacionada con los procesos y productos, en 2023, la Unión Europea llegó a un acuerdo para una regulación denominada “Ecodesign for Sustainable Products Regulation”, con la idea de que los productos comercializados sean más sostenibles, fomentar el reciclado y reducir el impacto climático. Esta regulación exige que gran parte de los productos incluyan información detallada sobre ellos y sus procesos de fabricación. Esta información estará almacenada en lo que se conoce como un pasaporte digital y será accesible electrónicamente. La relación de estos pasaportes con los procesos industriales y los productos fabricados, así como cómo se integran los datos de los pasaportes digitales en los ecosistemas de automática y robótica son también temas a abordar en un futuro.

Es cierto que, sin haber dejado de hablar de la Industria 4.0 y 5.0, se empezará a hablar de una sexta, séptima y octava revolución donde la automática y robótica, de la mano de la inteligencia artificial, tendrán un papel predominante y en la que aún está por determinar cuáles elementos se priorizan, en qué momento y si esto es dependiente de la industria o de la región donde se lleven a cabo. Por el momento, el paradigma de la Industria 5.0 está siendo promovido por diferentes instituciones y tiene implicaciones profundas, sobre todo para pequeñas y medianas empresas manufactureras.

En la sección de discusión se ha presentado el concepto que resume la visión de esta tesis sobre la Industria 5.0: la idea de una industria inteligente, basada en tecnologías innovadoras

y máquinas colaborativas, respaldada por trabajadores capacitados y resilientes, apoyada por objetivos de competitividad, sostenibilidad, escalabilidad y calidad. Este trabajo ha tratado de abordar esta industria inteligente, a través de casos de estudio tanto teóricos como prácticos sobre esas tecnologías, máquinas, trabajadores, escalabilidad, sostenibilidad y calidad que se mencionan en esa idea conceptual de las industrias del futuro.

Referencias

Alexopoulos, K., Nikolakis, N., & Xanthakis, E. (2022). Digital Transformation of Production Planning and Control in Manufacturing SMEs-The Mold Shop Case. *Applied Sciences*, *12*(21), 10788.

Alonso, R., Dessí, D., Meloni, A., Murgia, M., & Diego, R. R. (2023). Integrating Knowledge into Conversational Agents for Worker Upskilling. En *Artificial Intelligence in Manufacturing: Enabling Intelligent, Flexible and Cost-Effective Production Through AI* (pp. 265-280). Springer Nature Switzerland Cham.

Alonso, R., Dessí, D., Meloni, A., Murgia, M., Recupero, D. R., & Scarpi, G. (2024). A Seamless ChatGPT Knowledge Plug-in for the labour market. *IEEE Access*.

Alonso, R., Dessí, D., Meloni, A., & Recupero, D. R. (2024). A Novel Knowledge Plug-in for Incorporating Information about Employability from the O* NET Database into Conversational Agents. *IEEE Access*.

Alves, J., Lima, T. M., & Gaspar, P. D. (2023). Is Industry 5.0 a Human-Centred Approach? A Systematic Review. *Processes*, *11*(1), 193.

Araujo, V., Mitra, K., Saguna, S., & Åhlund, C. (2019). Performance evaluation of FIWARE: A cloud-based IoT platform for smart cities. *Journal of Parallel and Distributed Computing*, *132*, 250-261.

Barrett, M. (2018). Framework for Improving Critical Infrastructure Cybersecurity Version 1.1. https://doi.org/https://doi.org/10.6028/NIST.CSWP.04162018

Breque, M., De Nul, L., Petridis, A., et al. (2021). Industry 5.0: towards a sustainable, human-centric and resilient European industry. *Luxembourg, LU: European Commission, Directorate-General for Research and Innovation*, *46*.

Bunte, A., Richter, F., & Diovisalvi, R. (2021). Why It is Hard to Find AI in SMEs: A Survey from the Practice and How to Promote It. *ICAART (2)*, 614-620.

Castaño, F., Haber, R. E., Mohammed, W. M., Nejman, M., Villalonga, A., & Lastra, J. L. M. (2020). Quality monitoring of complex manufacturing systems on the basis of model driven approach.

Cimino, A., Elbasheer, M., Longo, F., Nicoletti, L., & Padovano, A. (2023). Empowering field operators in manufacturing: a prospective towards industry 5.0. *Procedia Computer Science*, *217*, 1948-1953.

Cirillo, F., Solmaz, G., Berz, E. L., Bauer, M., Cheng, B., & Kovacs, E. (2019). A standard-based open source IoT platform: FIWARE. *IEEE Internet of Things Magazine*, *2*(3), 12-18.

Deloitte. (2020). *Deloitte Survey on AI Adoption in Manufacturing* (inf. téc.). Deloitte.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

ESCO. (2022). The crosswalk between ESCO and O*NET (Technical Report). *ESCO Publications, European Commission*.

Eurostat. (2022). *Use of artificial intelligence in enterprises*.

Fareri, S., Apreda, R., Mulas, V., & Alonso, R. (2023). The worker profiler: Assessing the digital skill gaps for enhancing energy efficiency in manufacturing. *Technological Forecasting and Social Change*, *196*, 122844.

Ferrer, B. R., Mohammed, W. M., Lastra, J. L. M., Villalonga, A., Beruvides, G., Castaño, F., & Haber, R. E. (2018). Towards the adoption of cyber-physical systems of systems paradigm in smart manufacturing environments. *2018 IEEE 16th International Conference on Industrial Informatics (INDIN)*, 792-799.

Gladysz, B., Matteri, D., Ejsmont, K., Corti, D., Bettoni, A., & Haber Guerra, R. (2023). Platform-based support for AI uptake by SMEs: guidelines to design service bundles. *Central European Management Journal*, *31*(4), 463-478.

Hansen, E. B., & Bøgh, S. (2021). Artificial intelligence and internet of things in small and medium-sized enterprises: A survey. *Journal of Manufacturing Systems*, *58*, 362-372.

IBM. (2022). IBM Global AI Adoption Index.

Khan, A. A., Laghari, A. A., Li, P., Dootio, M. A., & Karim, S. (2023). The collaborative role of blockchain, artificial intelligence, and industrial internet of things in digitalization of small and medium-size enterprises. *Scientific Reports*, *13*(1), 1656.

Krishna, S., Campana, E., & Chandrasekaran, S. (2021). *Thriving in an AI World* (inf. téc.). KPMG.

Kumar, R., Singh, R. K., & Dwivedi, Y. K. (2020). Application of industry 4.0 technologies in SMEs for ethical and sustainable operations: Analysis of challenges. *Journal of cleaner production*, *275*, 124063.

Kung, H. (1987). Useful things to know about Ph. D. thesis research. *Computer Science Department, Carnegie Mellon University, Tech. Rep.*

Lasi, H., Fettke, P., Kemper, H.-G., Feld, T., & Hoffmann, M. (2014). Industry 4.0. *Business & information systems engineering*, *6*, 239-242.

Liu, Z., Sampaio, P., Pishchulov, G., Mehandjiev, N., Cisneros-Cabrera, S., Schirrmann, A., Jiru, F., & Bnouhanna, N. (2022). The architectural design and implementation of a digital platform for Industry 4.0 SME collaboration. *Computers in Industry*, *138*, 103623.

Mohamed, M. (2018). Challenges and benefits of industry 4.0: An overview. *International Journal of Supply and Operations Management*, *5*(3), 256-265.

Mohammed, W. M., Haber, R. E., & Martinez Lastra, J. L. (2022). Ontology-driven guidelines for architecting digital twins in factory automation applications. *Machines*, *10*(10), 861.

Muñoz, I., Portillo, F., Rosiek, S., Batlles, F. J., Martínez-Del-Río, J., Acasuso, I., Piergrossi, V., De Sanctis, M., Chimienti, S., & Di Iaconi, C. (2019). Prospective environmental and economic assessment of solar-assisted thermal energy recovery from wastewater through a sequencing batch biofilter granular reactor. *Journal of Cleaner Production*, *212*, 1300-1309.

Rada, M. (2017). Industry 5.0 Definition.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*.

Sana, M., Farooq, M. U., Anwar, S., & Haber, R. (2023). Predictive modelling framework on the basis of artificial neural network: A case of nano-powder mixed electric discharge machining. *Heliyon*, *9*(12).

Tsoy, N., Steubing, B., van der Giesen, C., & Guinée, J. (2020). Upscaling methods used in ex ante life cycle assessment of emerging technologies: a review. *The International Journal of Life Cycle Assessment*, *25*, 1680-1692.

von Drachenfels, N., Engels, P., Husmann, J., Cerdas, F., & Herrmann, C. (2021). Scale-up of pilot line battery cell manufacturing life cycle inventory models for life cycle assessment. *Procedia CIRP*, *98*, 13-18.

Xu, X., Lu, Y., Vogel-Heuser, B., & Wang, L. (2021). Industry 4.0 and Industry 5.0—Inception, conception and perception. *Journal of Manufacturing Systems*, *61*, 530-535.